

U. S. Department of Justice

Federal Bureau of Investigation

Criminal Justice Information Services Division



Criminal Justice Information Services (CJIS) Security Policy

Version 6.1

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Prepared by:
FBI CJIS Information Security Officer

Approved by:
CJIS Advisory Policy Board

EXECUTIVE SUMMARY

Criminal and noncriminal justice agencies need timely and secure access to services that provide data wherever and whenever for stopping and reducing crime. In response to these needs, the Advisory Policy Board (APB) recommended to the Federal Bureau of Investigation (FBI) that the Criminal Justice Information Services (CJIS) Division authorize the expansion of the existing security management structure in 2019. Administered through a shared management philosophy, the CJIS Security Policy (CJISSECPOL) contains information security requirements, guidelines, and agreements reflecting the will of law enforcement and criminal justice agencies for protecting the systems that process, store, and transmit Criminal Justice Information (CJI). The Federal Information Security Management Act of 2014 provides further legal basis for the APB approved management, operational, and technical security requirements mandated to protect CJI and by extension the hardware, software and infrastructure required to enable the services provided to the criminal and noncriminal justice communities.

The essential premise of the CJISSECPOL is to provide appropriate controls to protect the full lifecycle of CJI, whether at rest or in transit. The CJISSECPOL provides guidance for the creation, viewing, modification, transmission, dissemination, storage, or destruction of CJI. This Policy applies to every individual—contractor, private entity, noncriminal justice agency representative, or member of a criminal justice entity—with access to, or who operate in support of, criminal and noncriminal justice services and information.

The CJISSECPOL integrates presidential directives, federal laws, FBI directives and the criminal and noncriminal justice community decisions along with guidance from the National Crime Prevention and Privacy Compact Council (Compact Council). The Policy is presented at both strategic and tactical levels and is periodically updated to reflect the security requirements of evolving technology and business models. The Policy features modular sections enabling more frequent updates to address emerging threats and new security measures. The provided security criteria assists agencies with designing and implementing systems to meet a minimum level of risk and security protection while enabling agencies the latitude to institute more stringent security requirements and controls based on their business model and local needs.

The CJISSECPOL strengthens the partnership between the FBI and CJIS Systems Agencies (CSA), including, in those states with separate authorities, the State Identification Bureaus (SIB), and Interface Agencies (IA). Further, as use of criminal history record information (CHRI) for noncriminal justice purposes continues to expand, the CJISSECPOL becomes increasingly important in guiding the National Crime Prevention and Privacy Compact Council and State Compact Officers in the secure exchange of CHRI.

The Policy describes the vision and captures the security concepts that set the policies, protections, roles, and responsibilities with minimal impact from changes in technology. The Policy empowers CSAs, SIBs, and IAs with the insight and ability to tune their security programs according to their risks, needs, budgets, and resource constraints while remaining compliant with the baseline level of security set forth in this Policy. The CJISSECPOL provides a security framework based on laws, standards, and elements of published and vetted policies for accomplishing the mission across the broad spectrum of the criminal justice and noncriminal justice communities.

CHANGE MANAGEMENT

Revision	Change Description	Created/Changed by	Date	Approved By
6.0	Policy Modernization Completion	Security Policy Modernization Task Force	12/27/2024	APB & Compact Council
6.1	Incorporate Calendar Year 2025 Changes	CJIS ISO Program Office	06/25/2026	APB & Compact Council

SUMMARY OF CHANGES

Version 6.1

APB Approved Changes

1. **Addressing Omissions, Corrections, and Additions to the Modernized *CJIS Security Policy (CJISSECPOL)* Part 1, Spring 2025, APB#16, SA#1:** update sections with approved changes.
2. **Addressing Omissions, Corrections, and Additions to the Modernized *CJIS Security Policy (CJISSECPOL)* Part 2, Spring 2025, APB#16, SA#2:** update sections with approved changes.
3. **Addressing Omissions, Corrections, and Additions to the Modernized *CJIS Security Policy (CJISSECPOL)* Part 3, Spring 2025, APB#16, SA#3:** update sections with approved changes.

Administrative Changes

1. **Administrative Changes approved by the Security and Access Subcommittee on November 14, 2025.**

Note: Administrative changes are vetted through the Security and Access Subcommittee and not the entire APB process.

KEY TO APB APPROVED CHANGES (e.g., “**Modernizing Personnel Security (PS) in the *CJISSECPOL*, Spring 2024, APB#11, SA#7:** modernize the CJIS Security Policy requirements for:”)

Topic Paper Title

Spring 2024 – Advisory Policy Board cycle and year

APB# – Advisory Policy Board Topic number

SA# – Security and Access Subcommittee Topic number

Summary of change

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LIST OF PRIORITIES

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No.	Control Name	Enhancements	Priority
AC-1	POLICY AND PROCEDURES	AC-1	P2
AC-2	ACCOUNT MANAGEMENT	AC-2 (1) (2) (3) (4) (5) (13)	P1
AC-3	ACCESS ENFORCEMENT	AC-3 (14)	P1*
AC-4	INFORMATION FLOW ENFORCEMENT	AC-4	P1*
AC-5	SEPARATION OF DUTIES	AC-5	P1*
AC-6	LEAST PRIVILEGE	AC-6 (1) (2) (5) (7) (9) (10)	P1*
AC-7	UNSUCCESSFUL LOGON ATTEMPTS	AC-7	P3*
AC-8	SYSTEM USE NOTIFICATION	AC-8	P2*
AC-11	DEVICE LOCK	AC-11 (1)	P4*
AC-12	SESSION TERMINATION	AC-12	P3
AC-14	PERMITTED ACTIONS WITHOUT IDENTIFICATION OR AUTHENTICATION	AC-14	P4
AC-17	REMOTE ACCESS	AC-17 (1) (2) (3) (4)	P1
AC-18	WIRELESS ACCESS	AC-18 (1) (3)	P2*
AC-19	ACCESS CONTROL FOR MOBILE DEVICES	AC-19 (5)	P2*
AC-20	USE OF EXTERNAL SYSTEMS	AC-20 (1) (2)	P1
AC-21	INFORMATION SHARING	AC-21	P3*
AC-22	PUBLICLY ACCESSIBLE CONTENT	AC-22	P4
AT-1	POLICY AND PROCEDURES	AT-1	P2
AT-2	LITERACY TRAINING AND AWARENESS	AT-2 (2) (3)	P2*
AT-3	ROLE-BASED TRAINING	AT-3	P2*
AT-3	ROLE-BASED TRAINING PROCESSING PERSONALLY IDENTIFIABLE INFORMATION	AT-3 (5)	P2
AT-4	TRAINING RECORDS	AT-4	P4*
AU-1	POLICY AND PROCEDURES	AU-1	P2
AU-2	EVENT LOGGING	AU-2	P2*
AU-3	CONTENT OF AUDIT RECORDS	AU-3 (1)	P2*
AU-3	CONTENT OF AUDIT RECORDS LIMIT PERSONALLY IDENTIFIABLE INFORMATION ELEMENTS	AU-3 (3)	P2
AU-4	AUDIT LOG STORAGE CAPACITY	AU-4	P2
AU-5	RESPONSE TO AUDIT LOGGING PROCESS FAILURES	AU-5	P2*
AU-6	AUDIT RECORD REVIEW, ANALYSIS, AND REPORTING	AU-6 (1) (3)	P2*
AU-7	AUDIT RECORD REDUCTION AND REPORT GENERATION	AU-7 (1)	P3
AU-8	TIME STAMPS	AU-8	P2*
AU-9	PROTECTION OF AUDIT INFORMATION	AU-9 (4)	P2*
AU-11	AUDIT RECORD RETENTION	AU-11	P4*
AU-12	AUDIT RECORD GENERATION	AU-12	P2
CA-1	POLICY AND PROCEDURES	CA-1	P2
CA-2	CONTROL ASSESSMENTS	CA-2 (1)	P3
CA-3	INFORMATION EXCHANGE	CA-3	P2*
CA-5	PLAN OF ACTION AND MILESTONES	CA-5	P4

No.	Control Name	Enhancements	Priority
CA-6	AUTHORIZATION	CA-6	P3
CA-7	CONTINUOUS MONITORING	CA-7 (1) (4)	P1
CA-9	INTERNAL SYSTEM CONNECTIONS	CA-9	P3
CM-1	POLICY AND PROCEDURES	CM-1	P2
CM-2	BASELINE CONFIGURATION	CM-2 (2) (3) (7)	P1
CM-3	CONFIGURATION CHANGE CONTROL	CM-3 (2) (4)	P2
CM-4	IMPACT ANALYSES	CM-4 (2)	P3
CM-5	ACCESS RESTRICTIONS FOR CHANGE	CM-5	P1
CM-6	CONFIGURATION SETTINGS	CM-6	P1
CM-7	LEAST FUNCTIONALITY	CM-7 (1) (2) (5)	P1
CM-8	SYSTEM COMPONENT INVENTORY	CM-8 (1) (3)	P1
CM-9	CONFIGURATION MANAGEMENT PLAN	CM-9	P2
CM-10	SOFTWARE USAGE RESTRICTIONS	CM-10	P3
CM-11	USER-INSTALLED SOFTWARE	CM-11	P2
CM-12	INFORMATION LOCATION	CM-12 (1)	P2
CP-1	POLICY AND PROCEDURES	CP-1	P2
CP-2	CONTINGENCY PLAN	CP-2 (1) (3) (8)	P2
CP-3	CONTINGENCY TRAINING	CP-3	P3
CP-4	CONTINGENCY PLAN TESTING	CP-4 (1)	P3
CP-6	ALTERNATE STORAGE SITE	CP-6 (1) (3)	P2
CP-7	ALTERNATE PROCESSING SITE	CP-7 (1) (2) (3)	P2
CP-8	TELECOMMUNICATIONS SERVICES	CP-8 (1) (2)	P2
CP-9	SYSTEM BACKUP	CP-9 (1) (8)	P2
CP-10	SYSTEM RECOVERY AND RECONSTITUTION	CP-10 (2)	P2
IA-1	POLICY AND PROCEDURES	IA-1	P2
IA-2	IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS)	IA-2 (1) (2) (8) (12)	P1
IA-3	DEVICE IDENTIFICATION AND AUTHENTICATION	IA-3	P2
IA-4	IDENTIFIER MANAGEMENT	IA-4 (4)	P2*
IA-5	AUTHENTICATOR MANAGEMENT	IA-5 (1) (2) (6)	P1
IA-6	AUTHENTICATION FEEDBACK	IA-6	P3*
IA-7	CRYPTOGRAPHIC MODULE AUTHENTICATION	IA-7	P2
IA-8	IDENTIFICATION AND AUTHENTICATION (NON-ORGANIZATIONAL USERS)	IA-8 (1) (2) (4)	P2
IA-11	RE-AUTHENTICATION	IA-11	P2
IA-12	IDENTITY PROOFING	IA-12 (2) (3) (5)	P2
IR-1	POLICY AND PROCEDURES	IR-1	P2*
IR-2	INCIDENT RESPONSE TRAINING	IR-2	P3*
IR-2	INCIDENT RESPONSE TRAINING BREACH	IR-2 (3)	P3
IR-3	INCIDENT RESPONSE TESTING	IR-3 (2)	P3
IR-4	INCIDENT HANDLING	IR-4 (1)	P2*
IR-5	INCIDENT MONITORING	IR-5	P2*
IR-6	INCIDENT REPORTING	IR-6 (1) (3)	P2*
IR-7	INCIDENT RESPONSE ASSISTANCE	IR-7 (1)	P3*

No.	Control Name	Enhancements	Priority
IR-8	INCIDENT RESPONSE PLAN	IR-8	P2*
IR-8	INCIDENT RESPONSE PLAN BREACHES	IR-8 (1)	P2
MA-1	POLICY AND PROCEDURES	MA-1	P2
MA-2	CONTROLLED MAINTENANCE	MA-2	P3
MA-3	MAINTENANCE TOOLS	MA-3 (1) (2) (3)	P4
MA-4	NONLOCAL MAINTENANCE	MA-4	P3
MA-5	MAINTENANCE PERSONNEL	MA-5	P3
MA-6	TIMELY MAINTENANCE	MA-6	P3
MP-1	POLICY AND PROCEDURES	MP-1	P2*
MP-2	MEDIA ACCESS	MP-2	P2*
MP-3	MEDIA MARKING	MP-3	P3*
MP-4	MEDIA STORAGE	MP-4	P2*
MP-5	MEDIA TRANSPORT	MP-5	P2*
MP-6	MEDIA SANITIZATION	MP-6	P2*
MP-7	MEDIA USE	MP-7	P2*
PE-1	POLICY AND PROCEDURES	PE-1	P2
PE-2	PHYSICAL ACCESS AUTHORIZATIONS	PE-2	P2*
PE-3	PHYSICAL ACCESS CONTROL	PE-3	P2*
PE-4	ACCESS CONTROL FOR TRANSMISSION	PE-4	P2*
PE-5	ACCESS CONTROL FOR OUTPUT DEVICES	PE-5	P3*
PE-6	MONITORING PHYSICAL ACCESS	PE-6 (1)	P2*
PE-8	VISITOR ACCESS RECORDS	PE-8 (3)	P4
PE-9	POWER EQUIPMENT AND CABLING	PE-9	P2
PE-10	EMERGENCY SHUTOFF	PE-10	P2
PE-11	EMERGENCY POWER	PE-11	P2
PE-12	EMERGENCY LIGHTING	PE-12	P2
PE-13	FIRE PROTECTION	PE-13 (1)	P2
PE-14	ENVIRONMENTAL CONTROLS	PE-14	P2
PE-15	WATER DAMAGE PROTECTION	PE-15	P2
PE-16	DELIVERY AND REMOVAL	PE-16	P3
PE-17	ALTERNATE WORK SITE	PE-17	P3*
PL-1	POLICY AND PROCEDURES	PL-1	P2
PL-2	SYSTEM SECURITY AND PRIVACY PLANS	PL-2	P2
PL-4	RULES OF BEHAVIOR	PL-4 (1)	P3
PL-8	SECURITY AND PRIVACY ARCHITECTURES	PL-8	P2
PL-9	CENTRAL MANAGEMENT	PL-9	P4
PL-10	BASELINE SELECTION	PL-10	P3
PL-11	BASELINE TAILORING	PL-11	P3
PS-1	POLICY AND PROCEDURES	PS-1	P2
PS-2	POSITION RISK DESIGNATION	PS-2	P2
PS-3	PERSONNEL SCREENING	PS-3	P2*
PS-4	PERSONNEL TERMINATION	PS-4	P2

No.	Control Name	Enhancements	Priority
PS-5	PERSONNEL TRANSFER	PS-5	P3*
PS-6	ACCESS AGREEMENTS	PS-6	P4
PS-7	EXTERNAL PERSONNEL SECURITY	PS-7	P2*
PS-8	PERSONNEL SANCTIONS	PS-8	P4
PS-9	POSITION DESCRIPTIONS	PS-9	P4
RA-1	POLICY AND PROCEDURES	RA-1	P2
RA-2	SECURITY CATEGORIZATION	RA-2	P2
RA-3	RISK ASSESSMENT	RA-3 (1)	P2
RA-5	VULNERABILITY MONITORING AND SCANNING	RA-5 (2) (5) (11)	P1
RA-7	RISK RESPONSE	RA-7	P2
RA-9	CRITICALITY ANALYSIS	RA-9	P2
SA-1	POLICY AND PROCEDURES	SA-1	P2
SA-2	ALLOCATION OF RESOURCES	SA-2	P2
SA-3	SYSTEM DEVELOPMENT LIFE CYCLE	SA-3	P2
SA-4	ACQUISITION PROCESS	SA-4 (1) (2) (9) (10)	P2
SA-5	SYSTEM DOCUMENTATION	SA-5	P3
SA-8	SECURITY AND PRIVACY ENGINEERING PRINCIPLES	SA-8 (33)	P2
SA-9	EXTERNAL SYSTEM SERVICES	SA-9 (2)	P2*
SA-10	DEVELOPER CONFIGURATION MANAGEMENT	SA-10	P2
SA-11	DEVELOPER TESTING AND EVALUATION	SA-11	P2
SA-15	DEVELOPMENT PROCESS, STANDARDS, AND TOOLS	SA-15 (3)	P3
SA-22	UNSUPPORTED SYSTEM COMPONENTS	SA-22	P2
SC-1	POLICY AND PROCEDURES	SC-1	P2
SC-2	SEPARATION OF SYSTEM AND USER FUNCTIONALITY	SC-2	P2*
SC-4	INFORMATION IN SHARED SYSTEM RESOURCES	SC-4	P2*
SC-5	DENIAL-OF-SERVICE PROTECTION	SC-5	P2
SC-7	BOUNDARY PROTECTION	SC-7 (3) (4) (5) (7) (8) (24)	P1
SC-8	TRANSMISSION CONFIDENTIALITY AND INTEGRITY	SC-8 (1)	P2*
SC-10	NETWORK DISCONNECT	SC-10	P3
SC-12	CRYPTOGRAPHIC KEY ESTABLISHMENT AND MANAGEMENT	SC-12	P2*
SC-13	CRYPTOGRAPHIC PROTECTION	SC-13	P2*
SC-15	COLLABORATIVE COMPUTING DEVICES AND APPLICATIONS	SC-15	P2
SC-17	PUBLIC KEY INFRASTRUCTURE CERTIFICATES	SC-17	P2*
SC-18	MOBILE CODE	SC-18	P3
SC-20	SECURE NAME/ADDRESS RESOLUTION SERVICE (AUTHORITATIVE SOURCE)	SC-20	P2
SC-21	SECURE NAME/ADDRESS RESOLUTION SERVICE (RECURSIVE OR CACHING RESOLVER)	SC-21	P2
SC-22	ARCHITECTURE AND PROVISIONING FOR NAME/ADDRESS RESOLUTION SERVICE	SC-22	P2
SC-23	SESSION AUTHENTICITY	SC-23	P2
SC-28	PROTECTION OF INFORMATION AT REST	SC-28 (1)	P2*
SC-39	PROCESS ISOLATION	SC-39	P2*
SI-1	POLICY AND PROCEDURES	SI-1	P2
SI-2	FLAW REMEDIATION	SI-2 (2)	P1

No.	Control Name	Enhancements	Priority
SI-3	MALICIOUS CODE PROTECTION	SI-3	P1
SI-4	SYSTEM MONITORING	SI-4 (2) (4) (5)	P1
SI-5	SECURITY ALERTS, ADVISORIES, AND DIRECTIVES	SI-5	P2*
SI-7	SOFTWARE, FIRMWARE, AND INFORMATION INTEGRITY	SI-7 (1) (7)	P1
SI-8	SPAM PROTECTION	SI-8 (2)	P3*
SI-10	INFORMATION INPUT VALIDATION	SI-10	P1
SI-11	ERROR HANDLING	SI-11	P3
SI-12	INFORMATION MANAGEMENT AND RETENTION	SI-12 (1) (2) (3)	P3
SI-16	MEMORY PROTECTION	SI-16	P2
SR-1	POLICY AND PROCEDURES	SR-1	P2
SR-2	SUPPLY CHAIN RISK MANAGEMENT PLAN	SR-2 (1)	P3
SR-3	SUPPLY CHAIN CONTROLS AND PROCESSES	SR-3	P3
SR-5	ACQUISITION STRATEGIES, TOOLS, AND METHODS	SR-5	P2
SR-6	SUPPLIER ASSESSMENTS AND REVIEWS	SR-6	P2
SR-8	NOTIFICATION AGREEMENTS	SR-8	P3
SR-10	INSPECTION OF SYSTEMS OR COMPONENTS	SR-10	P3
SR-11	COMPONENT AUTHENTICITY	SR-11 (1) (2)	P2
SR-12	COMPONENT DISPOSAL	SR-12	P3

1 INTRODUCTION

This section details the purpose of this document, its scope, relationship to other information security policies, and its distribution constraints.

1.1 Purpose

The CJISSECPOL provides Criminal Justice Agencies (CJA), Noncriminal Justice Agencies (NCJA), and Interface Agencies (IA) with a minimum set of security requirements for access to Federal Bureau of Investigation (FBI) Criminal Justice Information Services (CJIS) Division systems and information and to protect and safeguard Criminal Justice Information (CJI). The essential premise of the CJISSECPOL is to provide the appropriate controls to ensure the continuity of the protection of CJI, from creation through dissemination or destruction, whether at rest or in transit.

1.2 Scope

At the consent of the advisory process, and taking into consideration federal law and state statutes, the CJISSECPOL applies to all entities with access to, or that operate systems which are used to process, store, or transmit CJI. The CJISSECPOL provides minimum security requirements associated with the creation, viewing, modification, transmission, dissemination, storage, or destruction of CJI.

Entities engaged in the interstate exchange of CHRI for noncriminal justice purposes are also governed by the standards and rules promulgated by the Compact Council.

1.3 Relationship to Local Security Policy and Other Policies

The CJISSECPOL may be used as the sole security policy for the agency. The local agency may complement the CJISSECPOL with a local policy, or the agency may develop their own stand-alone security policy; however, the CJISSECPOL shall always be the minimum standard. State, local, Tribal, territorial (SLTT), and Federal agencies may augment or implement more stringent policies or requirements.

The agency shall develop, disseminate, and maintain formal, documented policy and procedures to facilitate the implementation of the CJISSECPOL and, where applicable, the local security policy. The policies and procedures shall be consistent with applicable laws, executive orders, directives, policies, regulations, standards, and guidance. Procedures developed for CJISSECPOL areas can be developed for the security program in general, and for a particular information system, when required.

1.4 Terminology Used in This Document

Effective in version 5.9.5, priority and implementation markings have been added to the modernized controls. Based on the FBI Director approved APB recommendation, beginning October 1, 2024, requirements existing prior to the CJISSECPOL modernization (i.e., version 5.9) and those identified as Priority 1 ([Priority 1]) will be the set of sanctionable requirements.

- Non-modernized sections do not have markings but are considered “existing” requirements and continue to be auditable and sanctionable.

- “Existing” modernized requirements and indicated by the [Existing] marking.
- Priority 1 modernized requirements are indicated by the [Priority 1] marking.
- All [Priority 2], [Priority 3], and [Priority 4] modernized requirements fall into a zero-cycle status. The zero-cycle begins October 1, 2024 and ends September 30, 2027.

The following terms are used interchangeably throughout this document:

- **Agency and Organization:** The two terms in this document refer to any entity that submits or receives information, by any means, to/from FBI CJIS systems or services.
- **Organizational Personnel:** Individuals belonging to an agency or organization (see above) or contracted to an agency or organization to provide services under contract.
- **Information and Data:** Both terms refer to CJI.
- **System, Information System, Service, or named applications like NCIC:** all refer to connections to the FBI’s criminal justice information repositories and the equipment used to establish said connections.
- **Requirement and Control:** These terms refer to a condition or capability that must be met or possessed by a system or system element to satisfy an agreement, standard, specification, or other formally imposed document.

Appendix A and B provide extensive lists of the terms and acronyms.

Appendix I contains all of the references used in this Policy and may contain additional sources that could apply to any section.

1.5 Distribution of the CJIS Security Policy

The CJISSECPOL, version 6.0 and later, is a publicly available document and may be posted and shared without restrictions.

2 CJIS SECURITY POLICY APPROACH

The CJISSECPOL represents the shared responsibility among FBI CJIS, CJIS Systems Agency (CSA), State Identification Bureaus (SIB), and Interface Agency (IA) for the appropriate protection of CJI.

2.1 CJIS Security Policy Vision Statement

The executive summary of this document describes the vision in terms of business needs for confidentiality, integrity, and availability of information. The APB collaborates with the FBI CJIS Division to ensure that the Policy remains updated to meet evolving business, technology and security needs.

2.2 Architecture Independent

The CJISSECPOL looks at the data and protection controls that apply regardless of the implementation architecture. Architectural independence is not intended to lessen the importance of systems, but provide for the replacement of one technology with another while ensuring the controls required to protect the information remain constant. This objective and conceptual focus on security policy areas provide the guidance and standards while avoiding the impact of the constantly changing landscape of technical innovations. The architectural independence of the Policy provides agencies with the flexibility for tuning their information security infrastructure and policies to reflect their own environments.

2.3 Risk Versus Realism

Each agency faces risk unique to that agency. It is quite possible that several agencies could encounter the same type of risk however depending on resources would mitigate that risk differently. In that light, a risk-based approach can be used when implementing requirements.

3 ROLES AND RESPONSIBILITIES

3.1 Shared Management Philosophy

In the scope of information security, the FBI CJIS Division employs a shared management philosophy with SLTT and Federal criminal justice agencies. Although an advisory policy board for NCIC has existed since 1969, the Director of the FBI established the CJIS APB in March 1994 to enable appropriate input and recommend policy with respect to CJIS services. Through the APB and its Subcommittees and Working Groups, consideration is given to the needs of the criminal justice and law enforcement community regarding public policy, statutory and privacy aspects, as well as national security relative to CJIS systems and information. The APB, including a representative of the Compact Council, represents criminal and noncriminal justice agencies throughout the United States, its territories, and Canada.

The FBI has a similar relationship with the Compact Council, which governs the interstate exchange of criminal history records for noncriminal justice purposes. The Compact Council is mandated by federal law to promulgate rules and procedures for the use of the Interstate Identification Index (III) for noncriminal justice purposes. To meet that responsibility, the Compact Council depends on the CJISSECPOL as the definitive source for standards defining the security and privacy of records exchanged with noncriminal justice practitioners.

3.2 Roles and Responsibilities for Agencies and Parties

It is the responsibility of all agencies covered under this Policy to ensure the protection of CJI between the FBI CJIS Division and its user community. The following figure provides an abstract representation of the strategic functions and roles such as governance and operations.

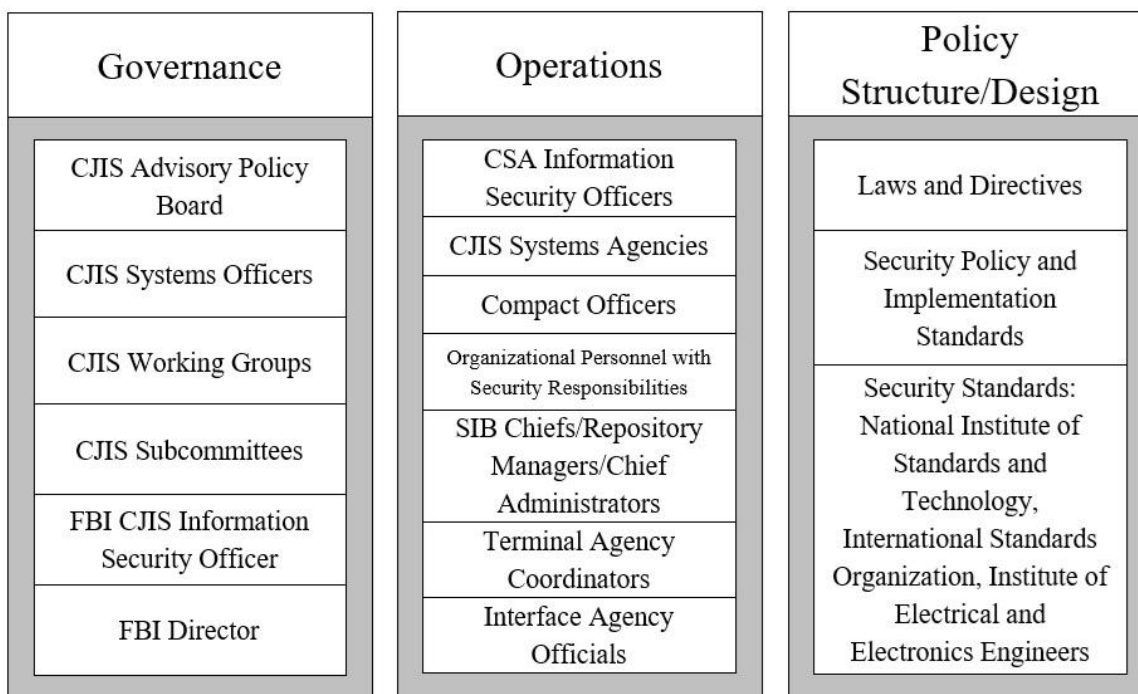


Figure 1 – Overview Diagram of Strategic Functions and Policy Components

This section provides a description of the following entities and roles:

1. CJIS Systems Agency (CSA).
2. CJIS Systems Officer (CSO).
3. Terminal Agency Coordinator (TAC).
4. Criminal Justice Agency (CJA).
5. Noncriminal Justice Agency (NCJA).
6. Contracting Agency (CA).
7. Agency Coordinator (AC).
8. CJIS Systems Agency (CSA) Information Security Officer (ISO).
9. Organizational Personnel with Security Responsibilities.
10. FBI CJIS Division Information Security Officer (ISO).
11. SIB Chief/Repository Manager/Chief Administrator.
12. Interface Agency (IA) Official
13. State Compact Officer (SCO).

3.2.1 CJIS Systems Agencies (CSA)

The CSA is responsible for establishing and administering an information technology security program throughout the CSA's user community, to include the local levels. The head of each CSA shall appoint a CJIS Systems Officer (CSO). The CSA may impose more stringent or additional protection measures than outlined in this document. Such decisions shall be documented and kept current.

3.2.2 CJIS Systems Officer (CSO)

The CSO is an individual located within the CSA responsible for the administration of the CJIS network for the CSA. Pursuant to the Bylaws for the CJIS Advisory Policy Board and Working Groups, the role of CSO shall not be outsourced. The CSO may delegate responsibilities to subordinate agencies. The CSO shall set, maintain, and enforce the following:

1. Standards for the selection, supervision, and separation of personnel who have access to CJI.
2. Policy governing the operation of computers, access devices, circuits, hubs, routers, firewalls, and other components that comprise and support a telecommunications network and related CJIS systems used to process, store, or transmit CJI, guaranteeing the priority of service required by the law enforcement community, confidentiality, integrity, and availability of CJI.
 - a. Ensure appropriate use, enforce system discipline, and ensure CJIS Division operating procedures are followed by all users of the respective services and information.
 - b. Ensure state/federal agency compliance with policies approved by the APB and adopted by the FBI.

- c. Ensure the appointment of the CSA ISO and determine the extent of authority to the CSA ISO.
- d. Ensure the designation of a Terminal Agency Coordinator (TAC) within each agency with devices accessing CJIS systems.
- e. Ensure each agency having access to CJI has someone designated as the Organizational Personnel with Security Responsibilities.
- f. Ensure each Organizational Personnel with Security Responsibilities receives enhanced security awareness training (ref. Awareness and Training (AT)).
- g. Approve access to FBI CJIS systems.
- h. Assume ultimate responsibility for managing the security of CJIS systems within their state and/or agency.
- i. Perform other related duties outlined by the user agreements with the FBI CJIS Division.

3. External System Services of Criminal Justice Functions

Responsibility for the management and control of security requirements for information systems which process, store, or transmit CJI shall remain with the CJA. Management and control includes the authority to:

- a. Set and enforce standards for the selection, supervision, and termination of access to CJI;
- b. Set and enforce policy governing the operation of computers, circuits, and telecommunications terminals,
- c. Guarantee the priority service as determined by the CSA.

3.2.3 Terminal Agency Coordinator (TAC)

The TAC serves as the point-of-contact at the local agency for matters relating to CJIS information access. The TAC administers CJIS systems programs within the local agency and oversees the agency's compliance with CJIS systems policies.

3.2.4 Criminal Justice Agency (CJA)

A CJA is defined as a court, a governmental agency, or any subunit of a governmental agency which performs the administration of criminal justice pursuant to a statute or executive order and which allocates a substantial part of its annual budget to the administration of criminal justice. State and federal Inspectors General Offices are included.

3.2.5 Noncriminal Justice Agency (NCJA)

A NCJA is defined (for the purposes of access to CJI) as an entity or any subunit thereof that provides services primarily for purposes other than the administration of criminal justice.

3.2.6 Contracting Agency (CA)

A CA is an agency, whether a CJA, NCJA (public), or NCJA (private), that enters into an agreement with a private contractor subject to the CJIS Security Addendum, the Security and Management Control Outsourcing Standard for Non-Channeling, or the Security and Management Control Outsourcing Standard for Channeling. The CA entering into an agreement with a contractor shall appoint an Agency Coordinator.

3.2.7 Agency Coordinator (AC)

An AC is a staff member of the CA who manages the agreement between the Contractor and agency. The AC shall be responsible for the supervision and integrity of the system, training and continuing education of employees and operators, scheduling of initial training and testing, and certification testing and all required reports by NCIC. The AC shall:

1. Understand the communications, records capabilities, and needs of the Contractor which is accessing federal and state records through or because of its relationship with the CA.
2. Participate in related meetings and provide input and comments for system improvement.
3. Receive information from the CA (e.g., system updates) and disseminate it to appropriate Contractor employees.
4. Maintain and update manuals applicable to the effectuation of the agreement, and provide them to the Contractor.
5. Maintain up-to-date records of Contractor's employees who access the system, including name, date of birth, social security number, date fingerprint card(s) submitted, date security clearance issued, and date initially trained, tested, certified or recertified (if applicable).
6. Train or ensure the training of Contractor personnel. If Contractor personnel access NCIC, schedule the operators for testing or a certification exam with the CSA staff, or AC staff with permission from the CSA staff as consistent with the NCIC policy.
7. The AC will not permit an untrained/untested or non-certified Contractor employee to access CJI or systems supporting CJI where access to CJI can be gained.
8. Where appropriate, ensure compliance by the Contractor with NCIC validation requirements.
9. Provide completed applicant fingerprint cards on each Contractor employee who accesses the system to the CA (or, where appropriate, CSA) for criminal background investigation prior to such employee accessing the system.
10. Any other responsibility for the AC promulgated by the CA.

3.2.8 CJIS Systems Agency Information Security Officer (CSA ISO)

The CSA ISO shall:

1. Serve as the security point of contact (POC) to the FBI CJIS Division ISO.
2. Document technical compliance with the CJIS Security Policy with the goal to assure the confidentiality, integrity, and availability of criminal justice information to the user community throughout the CSA's user community, to include the local level.

3. Document and provide assistance for implementing the security-related controls for the Interface Agency and its users.
4. Establish a security incident response and reporting procedure to discover, investigate, document, and report to the CSA, the affected criminal justice agency, and the FBI CJIS Division ISO major incidents that significantly endanger the security or integrity of CJI.

3.2.9 Organizational Personnel with Security Responsibilities

Each Organizational Personnel with Security Responsibilities (i.e., LASO, ARSO, Security Point of Contact, etc.) shall:

1. Identify who is using the CSA approved hardware, software, and firmware and ensure no unauthorized individuals or processes have access to the same.
2. Identify and document how the equipment is connected to the state system.
3. Ensure that personnel security screening procedures are being followed as stated in this Policy.
4. Ensure the approved and appropriate security measures are in place and working as expected.
5. Support policy compliance and ensure the CSA ISO is promptly informed of security incidents.

3.2.10 FBI CJIS Division Information Security Officer (FBI CJIS ISO)

The FBI CJIS ISO shall:

1. Maintain the CJIS Security Policy.
2. Disseminate the FBI Director approved CJIS Security Policy.
3. Serve as a liaison with the CSA's ISO and with other personnel across the CJIS community and in this regard provide technical guidance as to the intent and implementation of operational and technical policy issues.
4. Serve as a point-of-contact (POC) for computer incident notification and distribution of security alerts to the CSOs and ISOs.
5. Assist with developing audit compliance guidelines as well as identifying and reconciling security-related issues.
6. Develop and participate in information security training programs for the CSOs and ISOs, and provide a means by which to acquire feedback to measure the effectiveness and success of such training.
7. Maintain a security policy resource center (SPRC) on FBI.gov and keep the CSOs and ISOs updated on pertinent information.

3.2.11 Repository Manager

The State Identification Bureau (SIB) Chief, i.e., Repository Manager or Chief Administrator, is the designated manager of the agency having oversight responsibility for a state's fingerprint

identification services. If both state fingerprint identification services and CJIS systems control are managed within the same state agency, the SIB Chief and CSO may be the same person.

3.2.12 Interface Agency (IA) Official

The IA Official is an employee of the IA responsible for planning necessary hardware, software, funding, and training for the IA's authorized access to the CJIS Division's systems. The IA Official shall not be a contract employee.

3.2.13 State Compact Officer (SCO)

Pursuant to the National Crime Prevention and Privacy Compact, each party state shall appoint a Compact Officer who shall ensure that Compact provisions and rules, procedures, and standards established by the Compact Council are complied with in their respective state.

4 CRIMINAL JUSTICE INFORMATION AND PERSONALLY IDENTIFIABLE INFORMATION

4.1 Criminal Justice Information (CJI)

Criminal Justice Information is the term used to refer to all of the FBI CJIS provided data necessary for law enforcement and civil agencies to perform their missions including, but not limited to biometric, identity history, biographic, property, and case/incident history data. The following categories of CJI describe the various data sets housed by the FBI CJIS architecture:

1. **Biometric Data**—data derived from one or more intrinsic physical or behavioral traits of humans typically for the purpose of uniquely identifying individuals from within a population. Used to identify individuals, to include: fingerprints, palm prints, iris scans, and facial recognition data.
2. **Identity History Data**—textual data that corresponds with an individual’s biometric data, providing a history of criminal and/or civil events for the identified individual.
3. **Biographic Data**—information about individuals associated with a unique case, and not necessarily connected to identity data. Biographic data does not provide a history of an individual, only information related to a unique case.
4. **Property Data**—information about vehicles and property associated with crime when accompanied by any personally identifiable information (PII).
5. **Case/Incident History**—information about the history of criminal incidents.

The following type of data are exempt from the protection levels required for CJI: transaction control type numbers (e.g., ORI, NIC, UCN, etc.) when not accompanied by information that reveals CJI or PII.

The intent of the CJIS Security Policy is to ensure the protection of the aforementioned CJI until the information is: released to the public via authorized dissemination (e.g., within a court system; presented in crime reports data; released in the interest of public safety); purged or destroyed in accordance with applicable record retention rules. CJI introduced into the court system pursuant to a judicial proceeding that can be released to the public via a public records request is not subject to the CJIS Security Policy.

4.1.1 Criminal History Record Information (CHRI)

Criminal History Record Information (CHRI), sometimes informally referred to as “restricted data”, is a subset of CJI. Due to its comparatively sensitive nature, additional controls are required for the access, use and dissemination of CHRI. In addition to the dissemination restrictions outlined below, Title 28, Part 20, Code of Federal Regulations (CFR), defines CHRI and provides the regulatory guidance for dissemination of CHRI. While the CJIS Security Policy attempts to be architecturally independent, the III and the NCIC are specifically identified in Title 28, Part 20, CFR, and the NCIC Operating Manual, as associated with CHRI.

4.2 Access, Use and Dissemination of Criminal History Record Information (CHRI), NCIC Restricted Files Information, and NCIC Non-Restricted Files Information

This section describes the requirements for the access, use and dissemination of CHRI, NCIC restricted files information, and NCIC non-restricted files information.

4.2.1 Proper Access, Use, and Dissemination of CHRI

Information obtained from the III is considered CHRI. Rules governing the access, use, and dissemination of CHRI are found in Title 28, Part 20, CFR. The III shall be accessed only for an authorized purpose. Further, CHRI shall only be used for an authorized purpose consistent with the purpose for which III was accessed. Dissemination to another agency is authorized if (a) the other agency is an Authorized Recipient of such information and is being serviced by the accessing agency, or (b) the other agency is performing personnel and appointment functions for criminal justice employment applicants.

4.2.2 Proper Access, Use, and Dissemination of NCIC Restricted Files Information

The NCIC hosts restricted files and non-restricted files. NCIC restricted files are distinguished from NCIC non-restricted files by the policies governing their access and use. Proper access to, use, and dissemination of data from restricted files shall be consistent with the access, use, and dissemination policies concerning the III described in Title 28, Part 20, CFR, and the NCIC Operating Manual. The restricted files, which shall be protected as CHRI, are as follows:

1. Gang Files
2. Threat Screening Center Files
3. Supervised Release Files
4. National Sex Offender Registry Files
5. Historical Protection Order Files of the NCIC
6. Identity Theft Files
7. Protective Interest Files
8. Person With Information (PWI) data in the Missing Person Files
9. Violent Person File
10. NICS Denied Transactions File

The remaining NCIC files are considered non-restricted files.

4.2.3 Proper Access, Use, and Dissemination of NCIC Non-Restricted Files Information

4.2.3.1 For Official Purposes

NCIC non-restricted files are those not listed as restricted files in Section 4.2.2. NCIC non-restricted files information may be accessed and used for any authorized purpose consistent with

the inquiring agency's responsibility. Information obtained may be disseminated to (a) other government agencies or (b) private entities authorized by law to receive such information for any purpose consistent with their responsibilities.

4.2.3.2 For Other Authorized Purposes

NCIC non-restricted files may be accessed for other purposes consistent with the resources of the inquiring agency; however, requests for bulk data are discouraged. Information derived from NCIC non-restricted files for other than law enforcement purposes can be used by authorized criminal justice personnel only to confirm the status of a person or property (i.e., wanted or stolen). An inquiring agency is authorized to charge a nominal administrative fee for such service. Non-restricted files information shall not be disseminated commercially.

A response to a NCIC person inquiry may include NCIC restricted files information as well as NCIC non-restricted files information. Agencies shall not disseminate restricted files information for purposes other than law enforcement.

4.2.3.3 CSO Authority in Other Circumstances

If no federal, state or local law or policy prohibition exists, the CSO may exercise discretion to approve or deny dissemination of NCIC non-restricted file information.

4.2.4 Storage

When CHRI is stored, agencies shall establish appropriate administrative, technical and physical safeguards to ensure the security and confidentiality of the information. These records shall be stored for extended periods only when they are key elements for the integrity and/or utility of case files and/or criminal record files. See Physical and Environmental Protection (PE) for physical security controls.

4.2.5 Justification and Penalties

4.2.5.1 Justification

In addition to the use of purpose codes and logging information, all users shall provide a reason for all III inquiries whenever requested by NCIC System Managers, CSAs, local agency administrators, or their representatives.

4.2.5.2 Penalties

Improper access, use or dissemination of CHRI and NCIC Non-Restricted Files information is serious and may result in administrative sanctions including, but not limited to, termination of services and state and federal criminal penalties.

4.3 Personally Identifiable Information (PII)

For the purposes of this document, PII is information which can be used to distinguish or trace an individual's identity, such as name, social security number, or biometric records, alone or when combined with other personal or identifying information which is linked or linkable to a specific individual, such as date and place of birth, or mother's maiden name. Any FBI CJIS provided data maintained by an agency, including but not limited to, education, financial transactions, medical

history, and criminal or employment history may include PII. A criminal history record for example inherently contains PII as would a Law Enforcement National Data Exchange (N-DEx) case file.

PII shall be extracted from CJI for the purpose of official business only. Agencies shall develop policies, based on state and local privacy rules, to ensure appropriate controls are applied when handling PII extracted from CJI. Due to the expansive nature of PII, this Policy does not specify auditing, logging, or personnel security requirements associated with the life cycle of PII.

Figure 2 – Dissemination of restricted and non-restricted NCIC data

A citizen of Springfield went to the Springfield Police Department to request whether his new neighbor, who had been acting suspiciously, had an outstanding warrant. The Springfield Police Department ran an NCIC persons inquiry, which produced a response that included a Wanted Person File (non-restricted file) record and a Threat Screening Center File (restricted file) record. The Springfield Police Department advised the citizen of the outstanding warrant but did not disclose any information concerning the subject being on the Threat Screening Center File.

5 POLICY AND IMPLEMENTATION

The policy areas focus upon the data and services that the FBI CJIS Division exchanges and provides to the criminal justice community and its partners. Each policy area provides both strategic reasoning and tactical implementation requirements and standards.

While the major theme of the policy areas is concerned with electronic exchange directly with the FBI, it is understood that further dissemination of CJI to Authorized Recipients by various means (hard copy, e-mail, web posting, etc.) constitutes a significant portion of CJI exchanges. Regardless of its form, use, or method of dissemination, CJI requires protection throughout its life.

Not every consumer of FBI CJIS services will encounter all of the policy areas therefore the circumstances of applicability are based on individual agency/entity configurations and usage. Use cases within each of the policy areas will help users relate the Policy to their own agency circumstances. The policy areas are:

- Information Exchange Agreements
- Access Control (AC)
- Awareness and Training (AT)
- Auditing and Accountability (AU)
- Assessment, Authorization, and Monitoring (CA)
- Configuration Management (CM)
- Contingency Planning (CP)
- Identification and Authentication (IA)
- Incident Response (IR)
- Maintenance (MA)
- Media Protection (MP)
- Physical and Environmental Protection (PE)
- Planning (PL)
- Personnel Security (PS)
- Risk Assessment (RA)
- System and Services Acquisition (SA)
- System and Communications Protection (SC)
- System and Information Integrity (SI)
- Supply Chain Risk Management (SR)
- Mobile Devices

5.1 Policy Area 1: Information Exchange Agreements

The information shared through communication mediums shall be protected with appropriate security safeguards. The agreements established by entities sharing information across systems and communications mediums are vital to ensuring all parties fully understand and agree to a set of security standards.

5.1.1 Information Exchange

Before exchanging CJI, agencies shall put formal agreements in place that specify security controls. The exchange of information may take several forms including electronic mail, instant messages, web services, facsimile, hard copy, and information systems sending, receiving and storing CJI.

Information exchange agreements outline the roles, responsibilities, and data ownership between agencies and any external parties. Information exchange agreements for agencies sharing CJI data that is sent to and/or received from the FBI CJIS shall specify the security controls and conditions described in this document.

Information exchange agreements shall be supported by documentation committing both parties to the terms of information exchange. As described in subsequent sections, different agreements and policies apply, depending on whether the parties involved are CJAs or NCJAs. See Appendix D for examples of Information Exchange Agreements.

5.1.2 Monitoring, Review, and Delivery of Services

As specified in the interagency agreements, MCAs, and contractual agreements with private contractors, the services, reports and records provided by the service provider shall be regularly monitored and reviewed. The CJA, authorized agency, or FBI shall maintain sufficient overall control and visibility into all security aspects to include, but not limited to, identification of vulnerabilities and information security incident reporting/response. The incident reporting/response process used by the service provider shall conform to the incident reporting/response specifications provided in this Policy.

5.1.2.1 Managing Changes to Service Providers

Any changes to services provided by a service provider shall be managed by the CJA, authorized agency, or FBI. This includes provision of services, changes to existing services, and new services. Evaluation of the risks to the agency shall be undertaken based on the criticality of the data, system, and the impact of the change.

Figure 3 – Information Exchange Agreements Implemented by a Local Police Department

A local police department executed a Memorandum of Understanding (MOU) for the interface with their state CSA. The local police department also executed an MOU (which included an MCA) with the county information technology (IT) department for the day-to-day operations of their criminal-justice infrastructure. The county IT department, in turn, outsourced operations to a local vendor who signed the CJIS Security Addendum.

ACCESS CONTROL (AC)

Access control provides the planning and implementation of mechanisms to restrict reading, writing, processing, and transmission of CJI and the modification of information systems, applications, services and communication configurations allowing access to CJI.

Refer to Section 5.20.6 for additional access control requirements related to mobile devices used to access CJI.

AC-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to: organizational personnel with access control responsibilities
 1. Agency-level access control policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the access control policy and the associated access controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the access control policy and procedures; and
- c. Review and update the current access control:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Access control policy and procedures address the controls in the AC family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of access control policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies reflecting the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to access control policy and procedures

include assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: IA-1, PS-8, SI-12.

AC-2 ACCOUNT MANAGEMENT

[Existing] [Priority 1]

Control:

- a. Define and document the types of accounts allowed and specifically prohibited for use within the system;
- b. Assign account managers;
- c. Require conditions for group and role membership;
- d. Specify:
 1. Authorized users of the system;
 2. Group and role membership; and
 3. Access authorizations (i.e., privileges) and attributes listed for each account;

Attribute Name

Email Address Text

Employer Name

Federation Id

Given Name

Identity Provider Id

Sur Name

Telephone Number

Unique Subject Id

Counter Terrorism Data Self Search Home Privilege Indicator

Criminal History Data Self Search Home Privilege Indicator

Criminal Intelligence Data Self Search Home Privilege Indicator

Criminal Investigative Data Self Search Home Privilege Indicator

Display Name

Government Data Self Search Home Privilege Indicator

Local Id

NCIC Certification Indicator

N-DEx Privilege Indicator

PCII Certification Indicator

28 CFR Certification Indicator

Employer ORI

Employer Organization General Category Code

Employer State Code

Public Safety Officer Indicator

Sworn Law Enforcement Officer Indicator

Authenticator Assurance Level

Federation Assurance Level

Identity Assurance Level

Intelligence Analyst Indicator

- e. Require approvals by organizational personnel with account management responsibilities for requests to create accounts;
- f. Create, enable, modify, disable, and remove accounts in accordance with agency policy;
- g. Monitor the use of accounts;
- h. Notify account managers and system/network administrators within:
 - 1. One day when accounts are no longer required;
 - 2. One day when users are terminated or transferred; and
 - 3. One day when system usage or need-to-know changes for an individual;
- i. Authorize access to the system based on:
 - 1. A valid access authorization;
 - 2. Intended system usage; and
 - 3. Attributes as listed in AC-2(d)(3);

- j. Review accounts for compliance with account management requirements at least annually;
- k. Establish and implement a process for changing shared or group account authenticators (if deployed) when individuals are removed from the group; and
- l. Align account management processes with personnel termination and transfer processes.

Discussion: Examples of system account types include individual, shared, group, system, guest, anonymous, emergency, developer, temporary, and service. Identification of authorized system users and the specification of access privileges reflect the requirements in other controls in the security plan. Users requiring administrative privileges on system accounts receive additional scrutiny by organizational personnel responsible for approving such accounts and privileged access, including system owner, mission or business owner, senior agency information security officer, or senior agency official for privacy. Types of accounts that organizations may wish to prohibit due to increased risk include shared, group, emergency, anonymous, temporary, and guest accounts. Where access involves personally identifiable information, security programs collaborate with the senior agency official for privacy to establish the specific conditions for group and role membership; specify authorized users, group and role membership, and access authorizations for each account; and create, adjust, or remove system accounts in accordance with organizational policies. Policies can include such information as account expiration dates or other factors that trigger the disabling of accounts. Organizations may choose to define access privileges or other attributes by account, type of account, or a combination of the two. Examples of other attributes required for authorizing access include restrictions on time of day, day of week, and point of origin. In defining other system account attributes, organizations consider system-related requirements and mission/business requirements. Failure to consider these factors could affect system availability.

Temporary and emergency accounts are intended for short-term use. Organizations establish temporary accounts as part of normal account activation procedures when there is a need for short-term accounts without the demand for immediacy in account activation. Organizations establish emergency accounts in response to crisis situations and with the need for rapid account activation. Therefore, emergency account activation may bypass normal account authorization processes. Emergency and temporary accounts are not to be confused with infrequently used accounts, including local logon accounts used for special tasks or when network resources are unavailable (may also be known as accounts of last resort). Such accounts remain available and are not subject to automatic disabling or removal dates. Conditions for disabling or deactivating accounts include when shared/group, emergency, or temporary accounts are no longer required and when individuals are transferred or terminated. Changing shared/group authenticators when members leave the group is intended to ensure that former group members do not retain access to the shared or group account. Some types of system accounts may require specialized training.

Related Controls: AC-3, AC-5, AC-6, AC-17, AC-18, AC-20, AU-2, AU-12, CM-5, IA-2, IA-4, IA-5, IA-8, MA-3, MA-5, PE-2, PL-4, PS-2, PS-4, PS-5, SC-7, SC-12, SC-13.

Control Enhancements:

(1) ACCOUNT MANAGEMENT | AUTOMATED SYSTEM ACCOUNT MANAGEMENT

[Priority 1]

Control:

Support the management of system accounts using automated mechanisms including email, phone, and text notifications.

Discussion: Automated system account management includes using automated mechanisms to create, enable, modify, disable, and remove accounts; notify account managers when an account is created, enabled, modified, disabled, or removed, or when users are terminated or transferred; monitor system account usage; and report atypical system account usage.

Automated mechanisms can include internal system functions and email, telephonic, and text messaging notifications.

Related Controls: None.

(2) ACCOUNT MANAGEMENT | AUTOMATED TEMPORARY AND EMERGENCY ACCOUNT MANAGEMENT

[Priority 1]

Control:

Automatically remove temporary and emergency accounts within 72 hours.

Discussion: Management of temporary and emergency accounts includes the removal or disabling of such accounts automatically after a predefined time period rather than at the convenience of the system administrator. Automatic removal or disabling of accounts provides a more consistent implementation.

Related Controls: None.

(3) ACCOUNT MANAGEMENT | DISABLE ACCOUNTS

[Priority 1]

Control:

Disable accounts within one (1) week when the accounts:

- a. Have expired;
- b. Are no longer associated with a user or individual;
- c. Are in violation of organizational policy; or
- d. Have been inactive for 90 calendar days.

Discussion: Disabling expired, inactive, or otherwise anomalous accounts supports the concepts of least privilege and least functionality which reduce the attack surface of the system.

Related Controls: None.

(4) ACCOUNT MANAGEMENT | AUTOMATED AUDIT ACTIONS

[Existing] [Priority 1]

Control:

Automatically audit account creation, modification, enabling, disabling, and removal actions.

Discussion: Account management audit records are defined in accordance with AU-2 and reviewed, analyzed, and reported in accordance with AU-6.

Related Controls: AU-2, AU-6.

(5) ACCOUNT MANAGEMENT | INACTIVITY LOGOUT

[Priority 1]

Control:

Require that users log out when a work period has been completed.

Discussion: Inactivity logout is behavior- or policy-based and requires users to take physical action to log out when they are expecting inactivity longer than the defined period. Automatic enforcement of inactivity logout is addressed by AC-11.

Related Controls: AC-11.

(13) ACCOUNT MANAGEMENT | DISABLE ACCOUNTS FOR HIGH-RISK INDIVIDUALS

[Priority 1]

Control:

Disable accounts of individuals within 30 minutes of discovery of direct threats to the confidentiality, integrity, or availability of CJI.

Discussion: Users who pose a significant security and/or privacy risk include individuals for whom reliable evidence indicates either the intention to use authorized access to systems to cause harm or through whom adversaries will cause harm. Such harm includes adverse impacts to organizational operations, organizational assets, individuals, other organizations, or the Nation. Close coordination among system administrators, legal staff, human resource managers, and authorizing officials is essential when disabling system accounts for high-risk individuals.

Related Controls: AU-6, SI-4.

AC-3 ACCESS ENFORCEMENT

[Existing] [Priority 1]

Control:

Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.

Discussion: Access control policies control access between active entities or subjects (i.e., users or processes acting on behalf of users) and passive entities or objects (i.e., devices, files, records, domains) in organizational systems. In addition to enforcing authorized access at the system level and recognizing that systems can host many applications and services in support of mission and business functions, access enforcement mechanisms can also be employed at the application and service level to provide increased information security and privacy. In contrast to logical access controls that are implemented within the system, physical access controls are addressed by the controls in the Physical and Environmental Protection (PE) family.

Related Controls: AC-2, AC-4, AC-5, AC-6, AC-17, AC-18, AC-19, AC-20, AC-21, AC-22, AT-2, AT-3, AU-9, CA-9, CM-5, CM-11, IA-2, IA-5, IA-6, IA-7, IA-11, MA-3, MA-4, MA-5, MP-4, PS-3, SC-2, SC-4, SC-12, SC-13, SC-28, SI-4, SI-8.

Control Enhancements:

(14) ACCESS ENFORCEMENT | INDIVIDUAL ACCESS

[Existing] [Priority 1]

Control:

Provide automated or manual processes to enable individuals to have access to elements of their personally identifiable information.

Discussion: Individual access affords individuals the ability to review personally identifiable information about them held within organizational records, regardless of format. Access helps individuals to develop an understanding about how their personally identifiable information is being processed. It can also help individuals ensure that their data is accurate. Access mechanisms can include request forms and application interfaces. For federal agencies, [PRIVACT] processes can be located in systems of record notices and on agency websites. Access to certain types of records may not be appropriate (e.g., for federal agencies, law enforcement records within a system of records may be exempt from disclosure under the [PRIVACT]) or may require certain levels of authentication assurance.

Organizational personnel consult with the senior agency official for privacy and legal counsel to determine appropriate mechanisms and access rights or limitations.

Related Controls: IA-8.

AC-4 INFORMATION FLOW ENFORCEMENT

[Existing] [Priority 1]

Control:

Enforce approved authorizations for controlling the flow of information within the system and between connected systems by preventing CJI from being transmitted unencrypted across the public network, blocking outside traffic that claims to be from within the agency, and not passing any web requests to the public network that are not from the agency-controlled or internal boundary protection devices (e.g., proxies, gateways, firewalls, or routers).

Discussion: Information flow control regulates where information can travel within a system and between systems (in contrast to who is allowed to access the information) and without regard to subsequent accesses to that information. Flow control restrictions include blocking external traffic that claims to be from within the organization, keeping export-controlled information from being transmitted in the clear to the Internet, restricting web requests that are not from the internal web proxy server, and limiting information transfers between organizations based on data structures and content. Transferring information between organizations may require an agreement specifying how the information flow is enforced (see CA-3). Transferring information between systems in different security or privacy domains with different security or privacy policies introduces the risk that such transfers violate one or more domain security or privacy policies. In such situations, information owners/stewards provide guidance at designated policy enforcement points between connected systems. Organizations consider mandating specific architectural solutions to enforce

specific security and privacy policies. Enforcement includes prohibiting information transfers between connected systems (i.e., allowing access only), verifying write permissions before accepting information from another security or privacy domain or connected system, employing hardware mechanisms to enforce one-way information flows, and implementing trustworthy regrading mechanisms to reassign security or privacy attributes and labels.

Organizations commonly employ information flow control policies and enforcement mechanisms to control the flow of information between designated sources and destinations within systems and between connected systems. Flow control is based on the characteristics of the information and/or the information path. Enforcement occurs, for example, in boundary protection devices that employ rule sets or establish configuration settings that restrict system services, provide a packet-filtering capability based on header information, or provide a message-filtering capability based on message content. Organizations also consider the trustworthiness of filtering and/or inspection mechanisms (i.e., hardware, firmware, and software components) that are critical to information flow enforcement. Control enhancements 3 through 32 primarily address cross-domain solution needs that focus on more advanced filtering techniques, in-depth analysis, and stronger flow enforcement mechanisms implemented in cross-domain products, such as high-assurance guards. Such capabilities are generally not available in commercial off-the-shelf products. Information flow enforcement also applies to control plane traffic (e.g., routing and DNS).

Related Controls: AC-3, AC-6, AC-17, AC-19, AC-21, CA-3, CA-9, CM-7, SC-4, SC-7.

AC-5 SEPARATION OF DUTIES

[Existing] [Priority 1]

Control:

- a. Identify and document separation of duties based on specific duties, operations, or information systems, as necessary, to mitigate risk to CJI; and
- b. Define system access authorizations to support separation of duties.

Discussion: Separation of duties addresses the potential for abuse of authorized privileges and helps to reduce the risk of malevolent activity without collusion. Separation of duties includes dividing mission or business functions and support functions among different individuals or roles, conducting system support functions with different individuals, and ensuring that security personnel who administer access control functions do not also administer audit functions.

Because separation of duty violations can span systems and application domains, organizations consider the entirety of systems and system components when developing policy on separation of duties. Separation of duties is enforced through the account management activities in AC-2, access control mechanisms in AC-3, and identity management activities in IA-2, IA-4, and IA-12.

Related Controls: AC-2, AC-3, AC-6, AU-9, CM-5, CM-11, CP-9, IA-2, IA-4, IA-5, IA-12, MA-3, MA-5, PS-2, SA-8.

AC-6 LEAST PRIVILEGE

[Existing] [Priority 1]

Control:

Employ the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) that are necessary to accomplish assigned organizational tasks.

Discussion: Organizations employ least privilege for specific duties and systems. The principle of least privilege is also applied to system processes, ensuring that the processes have access to systems and operate at privilege levels no higher than necessary to accomplish organizational missions or business functions. Organizations consider the creation of additional processes, roles, and accounts as necessary to achieve least privilege. Organizations apply least privilege to the development, implementation, and operation of organizational systems.

Related Controls: AC-2, AC-3, AC-5, CM-5, CM-11, PL-2, SA-8, SA-15.

Control Enhancements:

(1) LEAST PRIVILEGE | AUTHORIZE ACCESS TO SECURITY FUNCTIONS

[Existing] [Priority 1]

Control:

Authorize access for personnel including security administrators, system and network administrators, and other privileged users with access to system control, monitoring, or administration functions (e.g., system administrators, information security personnel, maintainers, system programmers, etc.) to:

- a. Established system accounts, configured access authorizations (i.e., permissions, privileges), set events to be audited, set intrusion detection parameters, and other security functions; and
- b. Security-relevant information in hardware, software, and firmware.

Discussion: Security functions include establishing system accounts, configuring access authorizations (i.e., permissions, privileges), configuring settings for events to be audited, and establishing intrusion detection parameters. Security-relevant information includes filtering rules for routers or firewalls, configuration parameters for security services, cryptographic key management information, and access control lists. Authorized personnel include security administrators, system administrators, system security officers, system programmers, and other privileged users.

Related Controls: AC-17, AC-18, AC-19, AU-9, PE-2.

(2) LEAST PRIVILEGE | NON-PRIVILEGED ACCESS FOR NONSECURITY FUNCTIONS

[Existing] [Priority 1]

Control:

Require that users of system accounts (or roles) with access to privileged security functions or security-relevant information (e.g., audit logs), use non-privileged accounts or roles, when accessing non-security functions.

Discussion: Requiring the use of non-privileged accounts when accessing non-security functions limits exposure when operating from within privileged accounts or roles. The inclusion of roles addresses situations where organizations implement access control policies, such as role-based

access control, and where a change of role provides the same degree of assurance in the change of access authorizations for the user and the processes acting on behalf of the user as would be provided by a change between a privileged and non- privileged account.

Related Controls: AC-17, AC-18, AC-19, PL-4.

(5) LEAST PRIVILEGE | PRIVILEGED ACCOUNTS

[Existing] [Priority 1]

Control:

Restrict privileged accounts on the system to privileged users.

Discussion: Privileged accounts, including super user accounts, are typically described as system administrator for various types of commercial off-the-shelf operating systems. Restricting privileged accounts to specific personnel or roles prevents day-to-day users from accessing privileged information or privileged functions. Organizations may differentiate in the application of restricting privileged accounts between allowed privileges for local accounts and for domain accounts provided that they retain the ability to control system configurations for key parameters and as otherwise necessary to sufficiently mitigate risk.

Related Controls: IA-2, MA-3, MA-4.

(7) LEAST PRIVILEGE | REVIEW OF USER PRIVILEGES

[Existing] [Priority 1]

Control:

- a. Review annually the privileges assigned to non-privileged and privileged users to validate the need for such privileges; and
- b. Reassign or remove privileges, if necessary, to correctly reflect organizational mission and business needs.

Discussion: The need for certain assigned user privileges may change over time to reflect changes in organizational mission and business functions, environments of operation, technologies, or threats. A periodic review of assigned user privileges is necessary to determine if the rationale for assigning such privileges remains valid. If the need cannot be revalidated, organizations take appropriate corrective actions.

Related Controls: CA-7.

(9) LEAST PRIVILEGE | LOG USE OF PRIVILEGED FUNCTIONS

[Existing] [Priority 1]

Control:

Log the execution of privileged functions.

Discussion: The misuse of privileged functions, either intentionally or unintentionally by authorized users or by unauthorized external entities that have compromised system accounts, is a serious and ongoing concern and can have significant adverse impacts on organizations. Logging and analyzing the use of privileged functions is one way to detect such misuse and, in doing so, help mitigate the risk from insider threats and the advanced persistent threat.

Related Controls: AU-2, AU-3, AU-12.

(10) LEAST PRIVILEGE | PROHIBIT NON-PRIVILEGED USERS FROM EXECUTING PRIVILEGED FUNCTIONS

[Existing] [Priority 1]

Control:

Prevent non-privileged users from executing privileged functions.

Discussion: Privileged functions include disabling, circumventing, or altering implemented security or privacy controls, establishing system accounts, performing system integrity checks, and administering cryptographic key management activities. Non-privileged users are individuals who do not possess appropriate authorizations. Privileged functions that require protection from non-privileged users include circumventing intrusion detection and prevention mechanisms or malicious code protection mechanisms. Preventing non-privileged users from executing privileged functions is enforced by AC-3.

Related Controls: None.

AC-7 UNSUCCESSFUL LOGON ATTEMPTS

[Existing] [Priority 3]

Control:

- a. Enforce a limit of five (5) consecutive invalid logon attempts by a user during a 15-minute time period; and
- b. Automatically lock the account or node until released by an administrator when the maximum number of unsuccessful attempts is exceeded.

Discussion: The need to limit unsuccessful logon attempts and take subsequent action when the maximum number of attempts is exceeded applies regardless of whether the logon occurs via a local or network connection. Due to the potential for denial of service, automatic lockouts initiated by systems are usually temporary and automatically release after a predetermined, organization-defined time period. If a delay algorithm is selected, organizations may employ different algorithms for different components of the system based on the capabilities of those components. Responses to unsuccessful logon attempts may be implemented at the operating system and the application levels. Organization-defined actions that may be taken when the number of allowed consecutive invalid logon attempts is exceeded include prompting the user to answer a secret question in addition to the username and password, invoking a lockdown mode with limited user capabilities (instead of full lockout), allowing users to only logon from specified Internet Protocol (IP) addresses, requiring a CAPTCHA to prevent automated attacks, or applying user profiles such as location, time of day, IP address, device, or Media Access Control (MAC) address. If automatic system lockout or execution of a delay algorithm is not implemented in support of the availability objective, organizations consider a combination of other actions to help prevent brute force attacks. In addition to the above, organizations can prompt users to respond to a secret question before the number of allowed unsuccessful logon attempts is exceeded. Automatically unlocking an account after a specified period of time is generally not permitted. However, exceptions may be required based on operational mission or need.

Related Controls: AC-2, AU-2, AU-6, IA-5.

AC-8 SYSTEM USE NOTIFICATION

[Existing] [Priority 2]

Control:

- a. Display a system use notification message to users before granting access to the system that provides privacy and security notices consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines and state that:
 1. Users are accessing a restricted information system;
 2. System usage may be monitored, recorded, and subject to audit;
 3. Unauthorized use of the system is prohibited and subject to criminal and civil penalties; and
 4. Use of the system indicates consent to monitoring and recording.
- b. Retain the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access the system; and
- c. For publicly accessible systems:
 1. Display system use information consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines, before granting further access to the publicly accessible system;
 2. Display references, if any, to monitoring, recording, or auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities; and
 3. Include a description of the authorized uses of the system.

Discussion: System use notifications can be implemented using messages or warning banners displayed before individuals log in to systems. System use notifications are used only for access via logon interfaces with human users. Notifications are not required when human interfaces do not exist. Based on an assessment of risk, organizations consider whether or not a secondary system use notification is needed to access applications or other system resources after the initial network logon. Organizations consider system use notification messages or banners displayed in multiple languages based on organizational needs and the demographics of system users. Organizations consult with the privacy office for input regarding privacy messaging and the Office of the General Counsel or organizational equivalent for legal review and approval of warning banner content.

Related Controls: AC-14, PL-4, SI-4.

AC-11 DEVICE LOCK

[Existing] [Priority 4]

Control:

- a. Prevent further access to the system by initiating a device lock after a maximum of 30 minutes of inactivity and requiring the user to initiate a device lock before leaving the system unattended.

NOTE: In the interest of safety, devices that are: (1) part of a criminal justice conveyance; or (2) used to perform dispatch functions and located within a physically secure location; or (3) terminals designated solely for the purpose of receiving alert notifications (i.e., receive only terminals or ROT) used within physically secure location facilities that remain staffed when in operation, are exempt from this requirement.

- b. Retain the device lock until the user reestablishes access using established identification and authentication procedures.

Discussion: Device locks are temporary actions taken to prevent logical access to organizational systems when users stop work and move away from the immediate vicinity of those systems but do not want to log out because of the temporary nature of their absences. Device locks can be implemented at the operating system level or at the application level. A proximity lock may be used to initiate the device lock (e.g., via a Bluetooth-enabled device or dongle). User-initiated device locking is behavior or policy-based and, as such, requires users to take physical action to initiate the device lock. Device locks are not an acceptable substitute for logging out of systems, such as when organizations require users to log out at the end of workdays.

Related Controls: AC-2, AC-7, IA-11, PL-4.

Control Enhancements:

(1) DEVICE LOCK | PATTERN-HIDING DISPLAYS

[Existing] [Priority 4]

Control:

Conceal, via the device lock, information previously visible on the display with a publicly viewable image.

Discussion: The pattern-hiding display can include static or dynamic images, such as patterns used with screen savers, photographic images, solid colors, clock, battery life indicator, or a blank screen with the caveat that controlled unclassified information is not displayed.

Related Controls: None.

AC-12 SESSION TERMINATION

[Priority 3]

Control:

Automatically terminate a user session after a user has been logged out.

Discussion: Session termination addresses the termination of user-initiated logical sessions (in contrast to SC-10, which addresses the termination of network connections associated with communications sessions [i.e., network disconnect]). A logical session (for local, network, and

remote access) is initiated whenever a user (or process acting on behalf of a user) accesses an organizational system. Such user sessions can be terminated without terminating network sessions. Session termination ends all processes associated with a user's logical session except for those processes that are specifically created by the user (i.e., session owner) to continue after the session is terminated. Conditions or trigger events that require automatic termination of the session include organization-defined periods of user inactivity, targeted responses to certain types of incidents, or time-of-day restrictions on system use.

Related Controls: MA-4, SC-10, SC-23.

AC-14 PERMITTED ACTIONS WITHOUT IDENTIFICATION OR AUTHENTICATION

[Priority 4]

Control:

- a. Identify any specific user actions that can be performed on the system without identification or authentication consistent with organizational mission and business functions; and
- b. Document and provide supporting rationale in the security plan for the system, user actions not requiring identification or authentication.

Discussion: Specific user actions may be permitted without identification or authentication if organizations determine that identification and authentication are not required for the specified user actions. Organizations may allow a limited number of user actions without identification or authentication, including when individuals access public websites or other publicly accessible federal systems, when individuals use mobile phones to receive calls, or when facsimiles are received. Organizations identify actions that normally require identification or authentication but may, under certain circumstances, allow identification or authentication mechanisms to be bypassed. Such bypasses may occur, for example, via a software-readable physical switch that commands bypass of the logon functionality and is protected from accidental or unmonitored use. Permitting actions without identification or authentication does not apply to situations where identification and authentication have already occurred and are not repeated but rather to situations where identification and authentication have not yet occurred. Organizations may decide that there are no user actions that can be performed on organizational systems without identification and authentication, and therefore, the value for the assignment operation can be "none."

Related Controls: AC-8, IA-2, PL-2.

AC-17 REMOTE ACCESS

[Existing] [Priority 1]

Control:

- a. Establish and document usage restrictions, configuration/connection requirements, and implementation guidance for each type of remote access allowed; and
- b. Authorize each type of remote access to the system prior to allowing such connections.

Discussion: Remote access is access to organizational systems (or processes acting on behalf of users) that communicate through external networks such as the Internet. Types of remote access include dial-up, broadband, and wireless. Organizations use encrypted virtual private networks (VPNs) to enhance confidentiality and integrity for remote connections. The use of encrypted VPNs provides sufficient assurance to the organization that it can effectively treat such connections as internal networks if the cryptographic mechanisms used are implemented in accordance with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Still, VPN connections traverse external networks, and the encrypted VPN does not enhance the availability of remote connections. VPNs with encrypted tunnels can also affect the ability to adequately monitor network communications traffic for malicious code. Remote access controls apply to systems other than public web servers or systems designed for public access. Authorization of each remote access type addresses authorization prior to allowing remote access without specifying the specific formats for such authorization. While organizations may use information exchange and system connection security agreements to manage remote access connections to other systems, such agreements are addressed as part of CA-3. Enforcing access restrictions for remote access is addressed via AC-3.

Related Controls: AC-2, AC-3, AC-4, AC-18, AC-19, AC-20, CA-3, CM-10, IA-2, IA-3, IA-8, MA-4, PE- 17, PL-2, PL-4, SC-10, SC-12, SC-13, SI-4.

Control Enhancements:

(1) REMOTE ACCESS | MONITORING AND CONTROL

[Existing] [Priority 1]

Control:

Employ automated mechanisms to monitor and control remote access methods.

Discussion: Monitoring and control of remote access methods allows organizations to detect attacks and help ensure compliance with remote access policies by auditing the connection activities of remote users on a variety of system components, including servers, notebook computers, workstations, smart phones, and tablets. Audit logging for remote access is enforced by AU-2. Audit events are defined in AU-2a.

Related Controls: AU-2, AU-6, AU-12.

(2) REMOTE ACCESS | PROTECTION OF CONFIDENTIALITY AND INTEGRITY USING ENCRYPTION

[Existing] [Priority 1]

Control:

Implement cryptographic mechanisms to protect the confidentiality and integrity of remote access sessions.

Discussion: Virtual private networks can be used to protect the confidentiality and integrity of remote access sessions. Transport Layer Security (TLS) is an example of a cryptographic protocol that provides end-to-end communications security over networks and is used for Internet communications and online transactions.

Related Controls: SC-8, SC-12, SC-13.

(3) REMOTE ACCESS | MANAGED ACCESS CONTROL POINTS

[Existing] [Priority 1]

Control:

Route remote accesses through authorized and managed network access control points.

Discussion: Organizations consider the Trusted Internet Connections (TIC) initiative requirements for external network connections since limiting the number of access control points for remote access reduces attack surfaces.

Related Controls: SC-7.

(4) REMOTE ACCESS | PRIVILEGED COMMANDS AND ACCESS

[Existing] [Priority 1]

Control:

- a. Authorize the execution of privileged commands and access to security-relevant information via remote access only in a format that provides assessable evidence and for the following needs: compelling operational needs; and
- b. Document the rationale for remote access in the security plan for the system.

Discussion: Remote access to systems represents a significant potential vulnerability that can be exploited by adversaries. As such, restricting the execution of privileged commands and access to security-relevant information via remote access reduces the exposure of the organization and the susceptibility to threats by adversaries to the remote access capability.

Related Controls: AC-6, SC-12, SC-13.

AC-18 WIRELESS ACCESS

[Existing] [Priority 2]

Control:

- a. Establish configuration requirements, connection requirements, and implementation guidance for each type of wireless access; and
- b. Authorize each type of wireless access to the system prior to allowing such connections.

Discussion: Wireless technologies include microwave, packet radio (ultra-high frequency or very high frequency), 802.11x, and Bluetooth. Wireless networks use authentication protocols that provide authenticator protection and mutual authentication.

Related Controls: AC-2, AC-3, AC-17, AC-19, CA-9, CM-7, IA-2, IA-3, IA-8, PL-4, SI-4.

Control Enhancements:

(1) WIRELESS ACCESS | AUTHENTICATION AND ENCRYPTION

[Existing] [Priority 2]

Control:

Protect wireless access to the system using authentication of authorized users and agency-controlled devices, and encryption.

Discussion: Wireless networking capabilities represent a significant potential vulnerability that can be exploited by adversaries. To protect systems with wireless access points, strong authentication of users and devices along with strong encryption can reduce susceptibility to threats by adversaries involving wireless technologies.

Related Controls: SC-8, SC-12, SC-13.

(3) WIRELESS ACCESS | DISABLE WIRELESS NETWORKING

[Existing] [Priority 2]

Control:

Disable, when not intended for use, wireless networking capabilities embedded within system components prior to issuance and deployment.

Discussion: Wireless networking capabilities that are embedded within system components represent a significant potential vulnerability that can be exploited by adversaries. Disabling wireless capabilities when not needed for essential organizational missions or functions can reduce susceptibility to threats by adversaries involving wireless technologies.

Related Controls: None.

AC-19 ACCESS CONTROL FOR MOBILE DEVICES

[Existing] [Priority 2]

Control:

- a. Establish configuration requirements, connection requirements, and implementation guidance for organization-controlled mobile devices, to include when such devices are outside of controlled areas; and
- b. Authorize the connection of mobile devices to organizational systems.

Discussion: A mobile device is a computing device that has a small form factor such that it can easily be carried by a single individual; is designed to operate without a physical connection; possesses local, non-removable or removable data storage; and includes a self-contained power source. Mobile device functionality may also include voice communication capabilities, on-board sensors that allow the device to capture information, and/or built-in features for synchronizing local data with remote locations. Examples include smart phones and tablets. Mobile devices are typically associated with a single individual. The processing, storage, and transmission capability of the mobile device may be comparable to or merely a subset of notebook/desktop systems, depending on the nature and intended purpose of the device. Protection and control of mobile devices is behavior or policy-based and requires users to take physical action to protect and control such devices when outside of controlled areas. Controlled areas are spaces for which organizations provide physical or procedural controls to meet the requirements established for protecting information and systems. Due to the large variety of mobile devices with different characteristics

and capabilities, organizational restrictions may vary for the different classes or types of such devices. Usage restrictions and specific implementation guidance for mobile devices include configuration management, device identification and authentication, implementation of mandatory protective software, scanning devices for malicious code, updating virus protection software, scanning for critical software updates and patches, conducting primary operating system (and possibly other resident software) integrity checks, and disabling unnecessary hardware.

Usage restrictions and authorization to connect may vary among organizational systems. For example, the organization may authorize the connection of mobile devices to its network and impose a set of usage restrictions, while a system owner may withhold authorization for mobile device connection to specific applications or impose additional usage restrictions before allowing mobile device connections to a system. Adequate security for mobile devices goes beyond the requirements specified in AC-19. Many safeguards for mobile devices are reflected in other controls. AC-20 addresses mobile devices that are not organization-controlled.

Related Controls: AC-3, AC-4, AC-7, AC-11, AC-17, AC-18, AC-20, CA-9, CM-2, CM-6, IA-2, IA-3, MP-2, MP-4, MP-5, MP-7, PL-4, SC-7, SI-3, SI-4.

Control Enhancements:

(5) ACCESS CONTROL FOR MOBILE DEVICES | FULL DEVICE OR CONTAINER-BASED ENCRYPTION

[Existing] [Priority 2]

Control:

Employ full-device encryption to protect the confidentiality and integrity of information on full- and limited-feature operating system mobile devices authorized to process, store, or transmit CJI.

Discussion: Container-based encryption provides a more fine-grained approach to data and information encryption on mobile devices, including encrypting selected data structures such as files, records, or fields.

Related Controls: SC-12, SC-13, SC-28.

AC-20 USE OF EXTERNAL SYSTEMS

[Existing] [Priority 1]

Control:

- a. Establish agency-level policies governing the use of external systems consistent with the trust relationships established with other organizations owning, operating, and/or maintaining external systems, allowing authorized individuals to:
 1. Access the system from external systems; and
 2. Process, store, or transmit organization-controlled information using external systems; or
- b. Prohibit the use of personally-owned information systems including mobile devices (i.e., bring your own device [BYOD]) and publicly accessible systems for accessing, processing, storing, or transmitting CJI.

Discussion: External systems are systems that are used by but not part of organizational systems, and for which the organization has no direct control over the implementation of required controls or the assessment of control effectiveness. External systems include personally-owned systems, components, or devices; privately owned computing and communications devices in commercial or public facilities; systems owned or controlled by third-party organizations; systems managed by contractors; and information systems that are not owned by, operated by, or under the direct supervision or authority of the organization. External systems also include systems owned or operated by other components within the same organization and systems within the organization with different authorization boundaries. Organizations have the option to prohibit the use of any type of external system or prohibit the use of specified types of external systems, (e.g., prohibit the use of any external system that is not organizationally owned or prohibit the use of personally-owned systems).

For some external systems (i.e., systems operated by other organizations), the trust relationships that have been established between those organizations and the originating organization may be such that no explicit terms and conditions are required. Systems within these organizations may not be considered external. These situations occur when, for example, there are pre-existing information exchange agreements (either implicit or explicit) established between organizations or components or when such agreements are specified by applicable laws, executive orders, directives, regulations, policies, or standards. Authorized individuals include organizational personnel, contractors, or other individuals with authorized access to organizational systems and over which organizations have the authority to impose specific rules of behavior regarding system access. Restrictions that organizations impose on authorized individuals need not be uniform, as the restrictions may vary depending on trust relationships between organizations.

Therefore, organizations may choose to impose different security restrictions on contractors than on state, local, or tribal governments.

External systems used to access public interfaces to organizational systems are outside the scope of AC-20. Organizations establish specific terms and conditions for the use of external systems in accordance with organizational security policies and procedures. At a minimum, terms and conditions address the specific types of applications that can be accessed on organizational systems from external systems and the highest security category of information that can be processed, stored, or transmitted on external systems. If the terms and conditions with the owners of the external systems cannot be established, organizations may impose restrictions on organizational personnel using those external systems.

Related Controls: AC-2, AC-3, AC-17, AC-19, CA-3, PL-2, PL-4, SA-9, SC-7.

Control Enhancements:

(1) USE OF EXTERNAL SYSTEMS | LIMITS ON AUTHORIZED USE

[Priority 1]

Control:

Permit authorized individuals to use an external system to access the system or to process, store, or transmit organization-controlled information only after:

- a. Verification of the implementation of controls on the external system as specified in the organization's security and privacy policies and security and privacy plans; or
- b. Retention of approved system connection or processing agreements with the organizational entity hosting the external system.

Discussion: Limiting authorized use recognizes circumstances where individuals using external systems may need to access organizational systems. Organizations need assurance that the external systems contain the necessary controls so as not to compromise, damage, or otherwise harm organizational systems. Verification that the required controls have been implemented can be achieved by external, independent assessments, attestations, or other means, depending on the confidence level required by organizations.

Related Controls: CA-2.

(2) USE OF EXTERNAL SYSTEMS | PORTABLE STORAGE DEVICES — RESTRICTED USE

[Priority 1]

Control:

Restrict the use of organization-controlled portable storage devices by authorized individuals on external systems.

Discussion: Limits on the use of organization-controlled portable storage devices in external systems include restrictions on how the devices may be used and under what conditions the devices may be used.

Related Controls: MP-7.

AC-21 INFORMATION SHARING

[Existing] [Priority 3]

Control:

- a. Enable authorized users to determine whether access authorizations assigned to a sharing partner match the information's access and use restrictions as defined in an executed information exchange agreement; and
- b. Employ attribute-based access control (see AC-2(d)(3)) or manual processes as defined in information exchange agreements to assist users in making information sharing and collaboration decisions.

Discussion: Information sharing applies to information that may be restricted in some manner based on some formal or administrative determination. Examples of such information include, contract-sensitive information, classified information related to special access programs or compartments, privileged information, proprietary information, and personally identifiable information. Security and privacy risk assessments as well as applicable laws, regulations, and policies can provide useful inputs to these determinations. Depending on the circumstances, sharing partners may be defined at the individual, group, or organizational level. Information may be defined by content, type, security category, or special access program or compartment. Access restrictions may include non-disclosure agreements (NDA). Information flow techniques and

security attributes may be used to provide automated assistance to users making sharing and collaboration decisions.

Related Controls: AC-3, AC-4, RA-3, SC-15.

AC-22 PUBLICLY ACCESSIBLE CONTENT

[Priority 4]

Control:

- a. Designate individuals authorized to make information publicly accessible;
- b. Train authorized individuals to ensure that publicly accessible information does not contain nonpublic information;
- c. Review the proposed content of information prior to posting onto the publicly accessible system to ensure that nonpublic information is not included; and
- d. Review the content on the publicly accessible system for nonpublic information quarterly and remove such information, if discovered.

Discussion: In accordance with applicable laws, executive orders, directives, policies, regulations, standards, and guidelines, the public is not authorized to have access to nonpublic information, including information protected under the [PRIVACT] and proprietary information. Publicly accessible content addresses systems that are controlled by the organization and accessible to the public, typically without identification or authentication. Posting information on non-organizational systems (e.g., non-organizational public websites, forums, and social media) is covered by organizational policy. While organizations may have individuals who are responsible for developing and implementing policies about the information that can be made publicly accessible, publicly accessible content addresses the management of the individuals who make such information publicly accessible.

Related Controls: AC-3, AT-2, AT3.

AWARENESS AND TRAINING (AT)

Security training is key to the human element of information security. All users with authorized access to CJI should be made aware of their individual responsibilities and expected behavior when accessing CJI and the systems which process CJI. LASOs require enhanced training on the specific duties and responsibilities of those positions and the impact those positions have on the overall security of information systems.

AT-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to all personnel when their unescorted logical or physical access to any information system results in the ability, right, or privilege to view, modify, or make use of unencrypted CJI:
 1. Organization-level awareness and training policy that:
 - a. Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - b. Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the awareness and training policy and the associated awareness and training controls;
- b. Designate organizational personnel with information security awareness and training responsibilities to manage the development, documentation, and dissemination of the awareness and training policy and procedures; and
- c. Review and update the current awareness and training:
 1. Policy annually and following changes in the information system operating environment, when security incidents occur, or when changes to the CJIS Security Policy are made; and
 2. Procedures annually and following changes in the information system operating environment, when security incidents occur, or when changes to the CJIS Security Policy are made.

Discussion: Awareness and training policy and procedures address the controls in the AT family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of awareness and training policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or

role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to awareness and training policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

AT-2 LITERACY TRAINING AND AWARENESS

[Existing] [Priority 2]

Control:

- a. Provide security and privacy literacy training to system users (including managers, senior executives, and contractors):
 - 1. As part of initial training for new users prior to accessing CJI and annually thereafter; and
 - 2. When required by system changes or within 30 days of any security incident for individuals involved in the incident;
- b. Employ one or more of the following techniques to increase the security and privacy awareness of system users:
 - 1. Displaying posters
 - 2. Offering supplies inscribed with security and privacy reminders
 - 3. Displaying logon screen messages
 - 4. Generating email advisories or notices from organizational officials
 - 5. Conducting awareness events
- c. Update literacy training and awareness content annually and following changes in the information system operating environment, when security incidents occur, or when changes are made in the CJIS Security Policy; and
- d. Incorporate lessons learned from internal or external security incidents or breaches into literacy training and awareness techniques.

Discussion: Organizations provide basic and advanced levels of literacy training to system users, including measures to test the knowledge level of users. Organizations determine the content of literacy training and awareness based on specific organizational requirements, the systems to which personnel have authorized access, and work environments (e.g., telework). The content includes an understanding of the need for security and privacy as well as actions by users to maintain security and personal privacy and to respond to suspected incidents. The content addresses the need for operations security and the handling of personally identifiable information.

Awareness techniques include displaying posters, offering supplies inscribed with security and privacy reminders, displaying logon screen messages, generating email advisories or notices

from organizational officials, and conducting awareness events. Literacy training after the initial training described in AT-2a.1 is conducted at a minimum frequency consistent with applicable laws, directives, regulations, and policies. Subsequent literacy training may be satisfied by one or more short ad hoc sessions and include topical information on recent attack schemes, changes to organizational security and privacy policies, revised security and privacy expectations, or a subset of topics from the initial training. Updating literacy training and awareness content on a regular basis helps to ensure that the content remains relevant. Events that may precipitate an update to literacy training and awareness content include, but are not limited to, assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Related Controls: AC-3, AC-17, AC-22, AT-3, AT-4, CP-3, IA-4, IR-2, IR-7, PL-4, PS-7, SA-8.

Control Enhancements:

(2) LITERACY TRAINING AND AWARENESS | INSIDER THREAT

[Existing] [Priority 2]

Control:

Provide literacy training on recognizing and reporting potential indicators of insider threat.

Discussion: Potential indicators and possible precursors of insider threat can include behaviors such as inordinate, long-term job dissatisfaction; attempts to gain access to information not required for job performance; unexplained access to financial resources; bullying or harassment of fellow employees; workplace violence; and other serious violations of policies, procedures, directives, regulations, rules, or practices. Literacy training includes how to communicate the concerns of employees and management regarding potential indicators of insider threat through channels established by the organization and in accordance with established policies and procedures. Organizations may consider tailoring insider threat awareness topics to the role. For example, training for managers may be focused on changes in the behavior of team members, while training for employees may be focused on more general observations.

(3) LITERACY TRAINING AND AWARENESS | SOCIAL ENGINEERING AND MINING

[Existing] [Priority 2]

Control:

Provide literacy training on recognizing and reporting potential and actual instances of social engineering and social mining.

Discussion: Social engineering is an attempt to trick an individual into revealing information or taking an action that can be used to breach, compromise, or otherwise adversely impact a system. Social engineering includes phishing, pretexting, impersonation, baiting, quid pro quo, thread-jacking, social media exploitation, and tailgating. Social mining is an attempt to gather information about the organization that may be used to support future attacks.

Literacy training includes information on how to communicate the concerns of employees and management regarding potential and actual instances of social engineering and data mining through organizational channels based on established policies and procedures.

AT-3 ROLE-BASED TRAINING

[Existing] [Priority 2]

Control:

- a. Provide role-based security and privacy training to personnel with the following roles and responsibilities:
 - All individuals with unescorted access to a physically secure location;
 - General User: A user, but not a process, who is authorized to use an information system;
 - Privileged User: A user that is authorized (and, therefore, trusted) to perform security-relevant functions that general users are not authorized to perform;
 - Organizational Personnel with Security Responsibilities: Personnel with the responsibility to ensure the confidentiality, integrity, and availability of CJI and the implementation of technology in a manner compliant with the CJISSECPOL.
1. Before authorizing access to the system, information, or performing assigned duties, and annually thereafter; and
2. When required by system changes.

NOTE: The CSO/SIB Chief may accept the documentation of the completion of role-based security and privacy training from another agency. Accepting such documentation from another agency means that the accepting agency assumes the risk that the training may not meet a particular requirement or process required by federal, state, or local laws.

- b. Update role-based training content annually and following audits of the CSA and local agencies; changes in the information system operating environment; security incidents; or when changes are made to the CJIS Security Policy;
- c. Incorporate lessons learned from internal or external security incidents or breaches into role-based training;
- d. Incorporate the minimum following topics into the appropriate role-based training content:
 1. All individuals with unescorted access to a physically secure location
 - a. *Section 4.2 Access, Use and Dissemination of Criminal History Record Information (CHRI), NCIC Restricted Files Information, and NCIC Non-Restricted Files Information Penalties*
 - b. *IR-2 Incident Response Training*
 - c. *IR-6 Incident Reporting*
 - d. *PE-2 Physical Access Authorizations*
 - e. *PE-3 Physical Access Control*
 - f. *PE-6 Monitoring Physical Access*

- g. *PE-8 Visitor Access Records*
 - h. *PS-8 Personnel Sanctions*
2. General User: A user, but not a process, who is authorized to use an information system. In addition to AT-3 (d) (1) above, include the following topics:
- a. *Section 4.1 Criminal Justice Information*
 - b. *Section 4.2.3 Proper Access, Use, and Dissemination of NCIC Non-Restricted Files Information*
 - c. *Section 4.3 Personally Identifiable Information*
 - d. *AC-3 Access Enforcement*
 - e. *AC-6 Least Privilege*
 - f. *AC-8 System Use Notification*
 - g. *AC-11 Device Lock*
 - h. *AC-18 Wireless Access*
 - i. *AC-20 Use of External Systems*
 - j. *AT-2(3) Literacy Training and Awareness/Social Engineering and Mining*
 - k. *AU-6 Audit Record Review, Analysis and Reporting*
 - l. *CA-3 Information Exchange*
 - m. *IA-2 Identification and Authentication (Organizational Users)*
 - n. *IA-5(1)(a) Memorized Secret Authenticators and Verifiers*
 - o. *MP Media Protection*
 - p. *MP-2 Media Access*
 - q. *MP-4 Media Storage*
 - r. *PE-5 Access Control for Output Devices*
 - s. *SC-13 Cryptographic Protection*
 - t. *SC-28 Protection of Information at Rest*
 - u. *SI-3, 5.20.4.2 Malicious Code Protection*
 - v. *SI-8 Spam Protection*
 - w. *Section 5.20.1.2 Cellular Devices*
 - x. *Section 5.20.2 Mobile Device Management*
3. Privileged User: A user that is authorized (and, therefore, trusted) to perform security-relevant functions that general users are not authorized to perform. In addition to AT-3 (d) (1) and (2) above, include the following topics:
- a. *AC Access Control*
 - b. *CP-6 Alternate Storage Site*
 - c. *CP-9 System Backup*
 - d. *SC System and Communications Protection*
 - e. *SI-2 Flaw Remediation*
 - f. *Most recent changes to the CJIS Security Policy*
4. Organizational Personnel with Security Responsibilities: Personnel with the responsibility to ensure the confidentiality, integrity, and availability of CJI and the implementation of technology in a manner compliant with the CJISSECPOL. In addition to AT-3 (d) (1), (2), and (3) above, include the following topics:
- a. *Section 3.2.9 Organizational Personnel with Information Security Responsibilities*
 - b. *Authorized Recipient Security Officer Role*
 - c. *Additional state/local/tribal/territorial or federal agency roles and responsibilities*

- d. Summary of audit findings from previous state audits of local agencies
- e. Findings from the last FBI CJIS Division audit

Discussion: Organizations determine the content of training based on the assigned roles and responsibilities of individuals as well as the security and privacy requirements of organizations and the systems to which personnel have authorized access, including technical training specifically tailored for assigned duties. Roles that may require role-based training include senior leaders or management officials (e.g., head of agency/chief executive officer, chief information officer, senior accountable official for risk management, senior agency information security officer, senior agency official for privacy), system owners; authorizing officials; system security officers; privacy officers; acquisition and procurement officials; enterprise architects; systems engineers; software developers; systems security engineers; privacy engineers; system, network, and database administrators; auditors; personnel conducting configuration management activities; personnel performing verification and validation activities; personnel with access to system-level software; control assessors; personnel with contingency planning and incident response duties; personnel with privacy management responsibilities; and personnel with access to personally identifiable information.

Comprehensive role-based training addresses management, operational, and technical roles and responsibilities covering physical, personnel, and technical controls. Role-based training also includes policies, procedures, tools, methods, and artifacts for the security and privacy roles defined. Organizations provide the training necessary for individuals to fulfill their responsibilities related to operations and supply chain risk management within the context of organizational security and privacy programs. Role-based training also applies to contractors who provide services to federal agencies. Types of training include web-based and computer-based training, classroom-style training, and hands-on training (including micro-training). Updating role-based training on a regular basis helps to ensure that the content remains relevant and effective. Events that may precipitate an update to role-based training content include, but are not limited to, assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Related Controls: AC-3, AC-17, AC-22, AT-2, AT-4, CP-3, IR-2, IR-4, IR-7, PL-4, PS-7, PS-9, SA-3, SA-8, SA-11, SR-5, SR-6, SR-11.

Control Enhancements:

(5) ROLE-BASED TRAINING | PROCESSING PERSONALLY IDENTIFIABLE INFORMATION

[Priority 2]

Control:

Provide all personnel when their unescorted logical or physical access to any information system results in the ability, right, or privilege to view, modify, or make use of unencrypted CJI with initial and annual training in the employment and operation of personally identifiable information processing and transparency controls.

Discussion: Personally identifiable information processing and transparency controls include the organization's authority to process personally identifiable information and personally identifiable information processing purposes. Role-based training for federal agencies addresses the types of

information that may constitute personally identifiable information and the risks, considerations, and obligations associated with its processing. Such training also considers the authority to process personally identifiable information documented in privacy policies and notices, system of records notices, computer matching agreements and notices, privacy impact assessments, [PRIVACT] statements, contracts, information sharing agreements, memoranda of understanding, and/or other documentation.

AT-4 TRAINING RECORDS

[Existing] [Priority 4]

Control:

- a. Document and monitor information security and privacy training activities, including security and privacy awareness training and specific role-based security and privacy training; and
- b. Retain individual training records for a minimum of three years.

Discussion: Documentation for specialized training may be maintained by individual supervisors at the discretion of the organization. The National Archives and Records Administration provides guidance on records retention for federal agencies. Retention of records for three (3) years accounts for a triennial audit cycle.

Related Controls: AT-2, AT-3, CP-3, IR-2, SI-12.

Figure 4 – Security Awareness Training Use Cases

Use Case 1 - Awareness and Training Program Implementation by a Local Police Department

A local police department with a staff of 20 sworn criminal justice professionals and 15 support personnel worked with a vendor to develop role-specific -awareness training, and required all staff to complete this training upon assignment and every year thereafter. The vendor maintained the training records for the police department's entire staff, and provided reporting to the department to help it ensure compliance with the CJIS Security Policy.

Use Case 2 – All individuals with unescorted access to a physically secure location

A local police department hires custodial staff that will have physical access throughout the PD (a physically secure location) after normal business hours to clean the facility. These personnel have unescorted access to a physically secure location and therefore must be given the awareness training on all the topics identified in CJISSECPOL AT-3 d 1.

Use Case 3 – General User Awareness and Training

A school district maintains a locked file cabinet with hard copies of background check results of all teachers and employees which may include CJI (CHRI). Only authorized personnel who have the ability to open the cabinet are required to be given the baseline security awareness training on all the topics identified in CJISSECPOL AT-3 d 1 and 2.

Use Case 4 – General User Awareness and Training

A County Sheriff's Office has employed a number of dispatchers. Part of the function of these dispatchers is to run CJI queries at the request of the Sheriff and deputies. As part of their daily duties, the dispatchers have access to CJI both logically (running queries) and physically (printed copies of reports containing CJI). These dispatchers are entrusted with direct access to CJI and are therefore required to be given the awareness training on all the topics identified in CJISSECPOL AT-3 d 1 and 2.

Use Case 5 – Privileged User Awareness and Training

The State Police has hired a number of system and network administrator personnel to help bolster security of the state network. Part of their daily duties may include creating accounts for new personnel, implementing security patches for existing systems, creating backups of existing systems, and implementing access controls throughout the network. These administrators have privileged access to CJI and CJI-processing systems and are therefore required to be given the awareness training on all the topics identified in CJISSECPOL AT-3 d 1, 2, and 3.

AUDIT AND ACCOUNTABILITY (AU)

AU-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with audit and accountability responsibilities:
 1. Agency and system-level audit and accountability policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the audit and accountability policy and the associated audit and accountability controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the audit and accountability policy and procedures; and
- c. Review and update the current audit and accountability:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Audit and accountability policy and procedures address the controls in the AU family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of audit and accountability policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to audit and accountability policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

AU-2 EVENT LOGGING

[Priority 2]

Control:

- a. Identify the types of events that the system is capable of logging in support of the audit function: authentication, file use, user/group management, events sufficient to establish what occurred, the sources of events, outcomes of events, and operational transactions (e.g., NCIC, III);
- b. Coordinate the event logging function with other organizational entities requiring audit-related information to guide and inform the selection criteria for events to be logged;
- c. Specify the following event types for logging within the system:

All successful and unsuccessful:

1. System log-on attempts
2. Attempts to use:
 - a. Access permission on a user account, file, directory, or other system resource;
 - b. Create permission on a user account, file, directory, or other system resource;
 - c. Write permission on a user account, file, directory, or other system resource;
 - d. Delete permission on a user account, file, directory, or other system resource;
 - e. Change permission on a user account, file, directory, or other system resource.
3. Attempts to change account passwords
4. Actions by privileged accounts (i.e., root, Oracle, DBA, admin, etc.)
5. Attempts for users to:
 - a. Access the audit log file;
 - b. Modify the audit log file;
 - c. Destroy the audit log file.
- d. Provide a rationale for why the event types selected for logging are deemed to be adequate to support after-the-fact investigations of incidents; and
- e. Review and update the event types selected for logging annually.

Discussion: An event is an observable occurrence in a system. The types of events that require logging are those events that are significant and relevant to the security of systems and the privacy of individuals. Event logging also supports specific monitoring and auditing needs. Event types include password changes, failed logons or failed accesses related to systems, security or privacy attribute changes, administrative privilege usage, PIV credential usage, data action changes, query parameters, or external credential usage. In determining the set of event types that require logging, organizations consider the monitoring and auditing appropriate for

each of the controls to be implemented. For completeness, event logging includes all protocols that are operational and supported by the system.

To balance monitoring and auditing requirements with other system needs, event logging requires identifying the subset of event types that are logged at a given point in time. For example, organizations may determine that systems need the capability to log every file access successful and unsuccessful, but not activate that capability except for specific circumstances due to the potential burden on system performance. The types of events that organizations desire to be logged may change. Reviewing and updating the set of logged events is necessary to help ensure that the events remain relevant and continue to support the needs of the organization.

Organizations consider how the types of logging events can reveal information about individuals that may give rise to privacy risk and how best to mitigate such risks. For example, there is the potential to reveal personally identifiable information in the audit trail, especially if the logging event is based on patterns or time of usage.

Event logging requirements, including the need to log specific event types, may be referenced in other controls and control enhancements. These include AC-2(4), AC-3(10), AC-6(9), AC-17(1), CM-3f, CM-5(1), IA-3(3.b), MA-4(1), MP-4(2), PE-3, PM-21, PT-7, RA-8, SC-7(9), SC-7(15), SI-3(8), SI-4(22), SI-7(8), and SI-10(1). Organizations include event types that are required by applicable laws, executive orders, directives, policies, regulations, standards, and guidelines. Audit records can be generated at various levels, including at the packet level as information traverses the network. Selecting the appropriate level of event logging is an important part of a monitoring and auditing capability and can identify the root causes of problems. When defining event types, organizations consider the logging necessary to cover related event types, such as the steps in distributed, transaction-based processes and the actions that occur in service-oriented architectures.

Related Controls: AC-2, AC-3, AC-6, AC-7, AC-8, AC-17, AU-3, AU-4, AU-5, AU-6, AU-7, AU-11, AU-12, CM-3, CM-5, CM-6, IA-3, MA-4, MP-4, PE-3, SA-8, SC-7, SC-18, SI-3, SI-4, SI-7, SI-10, SI-11.

AU-3 CONTENT OF AUDIT RECORDS

[Existing] [Priority 2]

Control:

Ensure that audit records contain information that establishes the following:

- a. What type of event occurred;
- b. When the event occurred;
- c. Where the event occurred;
- d. Source of the event;
- e. Outcome of the event; and
- f. Identity of any individuals, subjects, or objects/entities associated with the event.

Discussion: Audit record content that may be necessary to support the auditing function includes event descriptions (item a), time stamps (item b), source and destination addresses (item c), user or process identifiers (items d and f), success or fail indications (item e), and filenames involved (items a, c, e, and f). Event outcomes include indicators of event success or failure and event-specific results, such as the system security and privacy posture after the event occurred.

Organizations consider how audit records can reveal information about individuals that may give rise to privacy risks and how best to mitigate such risks. For example, there is the potential to reveal personally identifiable information in the audit trail, especially if the trail records inputs or is based on patterns or time of usage.

Related Controls: AU-2, AU-8, AU-12, MA-4, SA-8, SI-7, SI-11.

Control Enhancements:

(1) CONTENT OF AUDIT RECORDS | ADDITIONAL AUDIT INFORMATION

[Existing] [Priority 2]

Control:

Generate audit records containing the following additional information:

- a. Session, connection, transaction, and activity duration;
- b. Source and destination addresses;
- c. Object or filename involved; and
- d. Number of bytes received and bytes sent (for client-server transactions) in the audit records for audit events identified by type, location, or subject.
- e. The III portion of the log shall clearly identify:
 1. The operator
 2. The authorized receiving agency
 3. The requestor
 4. The secondary recipient

Discussion: The ability to add information generated in audit records is dependent on system functionality to configure the audit record content. Organizations may consider additional information in audit records including, but not limited to, access control or flow control rules invoked and individual identities of group account users. Organizations may also consider limiting additional audit record information to only information that is explicitly needed for audit requirements. This facilitates the use of audit trails and audit logs by not including information in audit records that could potentially be misleading, make it more difficult to locate information of interest, or increase the risk to individuals' privacy.

Related Controls: None.

(3) CONTENT OF AUDIT RECORDS | LIMIT PERSONALLY IDENTIFIABLE INFORMATION ELEMENTS

[Priority 2]

Control:

Limit personally identifiable information contained in audit records to the following elements identified in the privacy risk assessment: minimum PII necessary to achieve the purpose for which it is collected (see Section 4.3).

Discussion: Limiting personally identifiable information in audit records when such information is not needed for operational purposes helps reduce the level of privacy risk created by a system.

Related Controls: RA-3.

AU-4 AUDIT LOG STORAGE CAPACITY

[Priority 2]

Control:

Allocate audit log storage capacity to accommodate the collection of audit logs to meet retention requirements (AU-11).

Discussion: Organizations consider the types of audit logging to be performed and the audit log processing requirements when allocating audit log storage capacity. Allocating sufficient audit log storage capacity reduces the likelihood of such capacity being exceeded and resulting in the potential loss or reduction of audit logging capability.

Related Controls: AU-2, AU-5, AU-6, AU-7, AU-9, AU-11, AU-12, SI-4.

AU-5 RESPONSE TO AUDIT LOGGING PROCESS FAILURES

[Existing] [Priority 2]

Control:

- a. Alert organizational personnel with audit and accountability responsibilities and system/network administrators within one (1) hour in the event of an audit logging process failure; and
- b. Take the following additional actions: restart all audit logging processes and verify system(s) are logging properly.

Discussion: Audit logging process failures include software and hardware errors, failures in audit log capturing mechanisms, and reaching or exceeding audit log storage capacity. Organization- defined actions include overwriting oldest audit records, shutting down the system, and stopping the generation of audit records. Organizations may choose to define additional actions for audit logging process failures based on the type of failure, the location of the failure, the severity of the failure, or a combination of such factors. When the audit logging process failure is related to storage, the response is carried out for the audit log storage repository (i.e., the distinct system component where the audit logs are stored), the system on which the

audit logs reside, the total audit log storage capacity of the organization (i.e., all audit log storage repositories combined), or all three.

Related Controls: AU-2, AU-4, AU-7, AU-9, AU-11, AU-12, SI-4, SI-12.

AU-6 AUDIT RECORD REVIEW, ANALYSIS, AND REPORTING

[Existing] [Priority 2]

Control:

- a. Review and analyze system audit records weekly for indications of inappropriate or unusual activity and the potential impact of the inappropriate or unusual activity;
- b. Report findings to organizational personnel with audit review, analysis, and reporting responsibilities and organizational personnel with information security and privacy responsibilities; and
- c. Adjust the level of audit record review, analysis, and reporting within the system when there is a change in risk based on law enforcement information, intelligence information, or other credible sources of information.

Discussion: Audit record review, analysis, and reporting covers information security- and privacy- related logging performed by organizations, including logging that results from the monitoring of account usage, remote access, wireless connectivity, mobile device connection, configuration settings, system component inventory, use of maintenance tools and non-local maintenance, physical access, temperature and humidity, equipment delivery and removal, communications at system interfaces, and use of mobile code or Voice over Internet Protocol (VoIP). Findings can be reported to organizational entities that include the incident response team, help desk, and security or privacy offices. If organizations are prohibited from reviewing and analyzing audit records or unable to conduct such activities, the review or analysis may be carried out by other organizations granted such authority. The frequency, scope, and/or depth of the audit record review, analysis, and reporting may be adjusted to meet organizational needs based on new information received.

Related Controls: AC-2, AC-3, AC-5, AC-6, AC-7, AC-17, AU-7, CA-2, CA-7, CM-2, CM-5, CM-6, CM-10, CM-11, IA-2, IA-3, IA-5, IA-8, IR-5, MA-4, MP-4, PE-3, PE-6, RA-5, SA-8, SC-7, SI-3, SI-4, SI-7.

Control Enhancements:

(1) AUDIT RECORD REVIEW, ANALYSIS, AND REPORTING | AUTOMATED PROCESS INTEGRATION

[Priority 2]

Control:

Integrate audit record review, analysis, and reporting processes using automated mechanisms.

Discussion: Organizational processes that benefit from integrated audit record review, analysis, and reporting include incident response, continuous monitoring, contingency planning, investigation and response to suspicious activities, and Inspector General audits.

(3) AUDIT RECORD REVIEW, ANALYSIS, AND REPORTING | CORRELATE AUDIT RECORD REPOSITORIES

[Priority 2]

Control:

Analyze and correlate audit records across different repositories to gain organization-wide situational awareness.

Discussion: Organization-wide situational awareness includes awareness across all three levels of risk management (i.e., organizational level, mission/business process level, and information system level) and supports cross-organization awareness.

Related Controls: AU-12, IR-4.

AU-7 AUDIT RECORD REDUCTION AND REPORT GENERATION

[Priority 3]

Control:

Provide and implement an audit record reduction and report generation capability that:

- a. Supports on-demand audit record review, analysis, and reporting requirements and after-the-fact investigations of incidents; and
- b. Does not alter the original content or time ordering of audit records.

Discussion: Audit record reduction is a process that manipulates collected audit log information and organizes it into a summary format that is more meaningful to analysts. Audit record reduction and report generation capabilities do not always emanate from the same system or from the same organizational entities that conduct audit logging activities. The audit record reduction capability includes modern data mining techniques with advanced data filters to identify anomalous behavior in audit records. The report generation capability provided by the system can generate customizable reports. Time ordering of audit records can be an issue if the granularity of the timestamp in the record is insufficient.

Related Controls: AC-2, AU-2, AU-3, AU-4, AU-5, AU-6, AU-12, CM-5, IA-5, IR-4, SI-4.

Control Enhancements:

(1) AUDIT RECORD REDUCTION AND REPORT GENERATION | AUTOMATIC PROCESSING

[Priority 3]

Control:

Provide and implement the capability to process, sort, and search audit records for events of interest based on the following content: information included in AU-3.

Discussion: Events of interest can be identified by the content of audit records, including system resources involved, information objects accessed, identities of individuals, event types, event locations, event dates and times, Internet Protocol addresses involved, or event success or failure. Organizations may define event criteria to any degree of granularity required, such as locations selectable by a general networking location or by specific system component.

AU-8 TIME STAMPS

[Existing] [Priority 2]

Control:

- a. Use internal system clocks to generate time stamps for audit records;
- b. Record time stamps for audit records that meet hundredths of a second (i.e., hh:mm:ss:00) interval and that use Coordinated Universal Time, have a fixed local time offset from Coordinated Universal Time, or that include the local time offset as part of the time stamp.

Discussion: Time stamps generated by the system include date and time. Time is commonly expressed in Coordinated Universal Time (UTC), a modern continuation of Greenwich Mean Time (GMT), or local time with an offset from UTC. Granularity of time measurements refers to the degree of synchronization between system clocks and reference clocks (e.g., clocks synchronizing within hundreds of milliseconds or tens of milliseconds). Organizations may define different time granularities for different system components. Time service can be critical to other security capabilities such as access control and identification and authentication, depending on the nature of the mechanisms used to support those capabilities.

Related Controls: AU-3, AU-12.

AU-9 PROTECTION OF AUDIT INFORMATION

[Existing] [Priority 2]

Control:

- a. Protect audit information and audit logging tools from unauthorized access, modification, and deletion; and
- b. Alert organizational personnel with audit and accountability responsibilities, organizational personnel with information security and privacy responsibilities, and system/network administrators upon detection of unauthorized access, modification, or deletion of audit information.

Discussion: Audit information includes all information needed to successfully audit system activity, such as audit records, audit log settings, audit reports, and personally identifiable information. Audit logging tools are those programs and devices used to conduct system audit and logging activities. Protection of audit information focuses on technical protection and limits

the ability to access and execute audit logging tools to authorized individuals. Physical protection of audit information is addressed by both media protection controls and physical and environmental protection controls.

Related Controls: AC-3, AC-6, AU-6, AU-11, MP-2, MP-4, PE-2, PE-3, PE-6, SA-8, SC-8, SI-4.

Control Enhancements:

(4) PROTECTION OF AUDIT INFORMATION | ACCESS BY SUBSET OF PRIVILEGED USERS

[Priority 2]

Control:

Authorize access to management of audit logging functionality to only organizational personnel with audit and accountability responsibilities, organizational personnel with information security and privacy responsibilities, and system/network administrators.

Discussion: Individuals or roles with privileged access to a system and who are also the subject of an audit by that system may affect the reliability of the audit information by inhibiting audit activities or modifying audit records. Requiring privileged access to be further defined between audit-related privileges and other privileges limits the number of users or roles with audit-related privileges.

Related Controls: AC-5.

AU-11 AUDIT RECORD RETENTION

[Existing] [Priority 4]

Control:

Retain audit records for a minimum of one (1) year or until it is determined they are no longer needed for administrative, legal, audit, or other operational purposes to provide support for after-the-fact investigations of incidents and to meet regulatory and organizational information retention requirements.

Discussion: Organizations retain audit records until it is determined that the records are no longer needed for administrative, legal, audit, or other operational purposes. This includes the retention and availability of audit records relative to Freedom of Information Act (FOIA) requests, subpoenas, and law enforcement actions. Organizations develop standard categories of audit records relative to such types of actions and standard response processes for each type of action.

Related Controls: AU-2, AU-4, AU-5, AU-6, AU-9, MP-6, RA-5, SI-12.

AU-12 AUDIT RECORD GENERATION

[Priority 2]

Control:

- a. Provide audit record generation capability for the event types the system is capable of auditing as defined in AU-2a on all systems generating required audit logs;
- b. Allow organizational personnel with audit record generation responsibilities, organizational personnel with information security and privacy responsibilities, and system/network administrators to select the event types that are to be logged by specific components of the system; and
- c. Generate audit records for the event types defined in AU-2c that include the audit record content defined in AU-3.

Discussion: Audit records can be generated from many different system components. The event types specified in AU-2c are the event types for which audit logs are to be generated and are a subset of all event types for which the system can generate audit records.

Related Controls: AC-6, AC-17, AU-2, AU-3, AU-4, AU-5, AU-6, AU-7, CM-5, MA-4, MP-4, SA-8, SC-18, SI-3, SI-4, SI-7, SI-10.

ASSESSMENT, AUTHORIZATION, AND MONITORING (CA)

CA-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with assessment, authorization, and monitoring policy responsibilities:
 1. An assessment, authorization, and monitoring policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the assessment, authorization, and monitoring policy and the associated assessment, authorization, and monitoring controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the assessment, authorization, and monitoring policy and procedures; and
- c. Review and update the current assessment, authorization, and monitoring:
 1. Policy annually and following changes to the assessment criteria and
 2. Procedures annually and following changes to the assessment criteria.

Discussion: Assessment, authorization, and monitoring policy and procedures address the controls in the CA family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of assessment, authorization, and monitoring policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to assessment, authorization, and monitoring policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

CA-2 CONTROL ASSESSMENTS

[Priority 3]

Control:

- a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;
- b. Develop a control assessment plan that describes the scope of the assessment including:
 1. Controls and control enhancements under assessment;
 2. Assessment procedures to be used to determine control effectiveness; and
 3. Assessment environment, assessment team, and assessment roles and responsibilities;
- c. Ensure the control assessment plan is reviewed and approved by the authorizing official or designated representative prior to conducting the assessment;
- d. Assess the controls in the system and its environment of operation and any controls that have been impacted by evolving threats at least once every three years to determine the extent to which the controls are implemented correctly, operating as intended, and producing the desired outcome with respect to meeting established security and privacy requirements;
- e. Produce a control assessment report that documents the results of the assessment; and
- f. Provide the results of the control assessment report to the individual who executed the CJIS User Agreement or is in contract with the FBI.

Discussion: Organizations ensure that control assessors possess the required skills and technical expertise to develop effective assessment plans and to conduct assessments of system-specific, hybrid, common, and program management controls, as appropriate. The required skills include general knowledge of risk management concepts and approaches as well as comprehensive knowledge of and experience with the hardware, software, and firmware system components implemented.

Organizations assess controls in systems and the environments in which those systems operate as part of initial and ongoing authorizations, continuous monitoring, system design and development, systems security engineering, privacy engineering, and the system development life cycle. Assessments help to ensure that organizations meet information security and privacy requirements, identify weaknesses and deficiencies in the system design and development process, provide essential information needed to make risk-based decisions as part of authorization processes, and comply with vulnerability mitigation procedures. Organizations conduct assessments on the implemented controls as documented in security and privacy plans. Assessments can also be conducted throughout the system development life cycle as part of systems engineering and systems security engineering processes. The design for controls can be assessed as RFPs are developed, responses assessed, and design reviews conducted. If a design to implement controls and subsequent implementation in accordance with the design are assessed during development, the final control testing can be a simple confirmation utilizing previously completed control assessment and aggregating the outcomes.

Organizations may develop a single, consolidated security and privacy assessment plan for the system or maintain separate plans. A consolidated assessment plan clearly delineates the roles

and responsibilities for control assessment. If multiple organizations participate in assessing a system, a coordinated approach can reduce redundancies and associated costs.

Organizations can use other types of assessment activities, such as vulnerability scanning and system monitoring, to maintain the security and privacy posture of systems during the system life cycle. Assessment reports document assessment results in sufficient detail, as deemed necessary by organizations, to determine the accuracy and completeness of the reports and whether the controls are implemented correctly, operating as intended, and producing the desired outcome with respect to meeting requirements. Assessment results are provided to the individuals or roles appropriate for the types of assessments being conducted. For example, assessments conducted in support of authorization decisions are provided to authorizing officials, senior agency officials for privacy, senior agency information security officers, and authorizing official designated representatives (i.e., CJIS Systems Officer [CSO], Interface Agency [IA] Official, or FBI Compact Officer [CO]).

To satisfy assessment requirements, organizations can use assessment results from the following sources: initial or ongoing system authorizations, continuous monitoring, systems engineering processes, or system development life cycle activities. Organizations ensure that assessment results are current, relevant to the determination of control effectiveness, and obtained with the appropriate level of assessor independence. Existing control assessment results can be reused to the extent that the results are still valid and can also be supplemented with additional assessments as needed. After the initial authorizations, organizations assess controls during continuous monitoring. Organizations also establish the frequency for ongoing assessments in accordance with organizational continuous monitoring strategies. External audits, including audits by external entities such as regulatory agencies, are outside of the scope of CA-2.

Related Controls: AC-20, CA-5, CA-6, CA-7, RA-5, SA-11, SI-3, SI-12, SR-2, SR-3.

Control Enhancements:

(1) CONTROL ASSESSMENTS | INDEPENDENT ASSESSORS

[Priority 3]

Control:

Employ independent assessors or assessment teams to conduct control assessments.

Discussion: Independent assessors or assessment teams are individuals or groups who conduct impartial assessments of systems. Impartiality means that assessors are free from any perceived or actual conflicts of interest regarding the development, operation, sustainment, or management of the systems under assessment or the determination of control effectiveness. To achieve impartiality, assessors do not create a mutual or conflicting interest with the organizations where the assessments are being conducted, assess their own work, act as management or employees of the organizations they are serving, or place themselves in positions of advocacy for the organizations acquiring their services.

Independent assessments can be obtained from elements within organizations or be contracted to public or private sector entities outside of organizations. Authorizing officials determine the required level of independence based on the security categories of systems and/or the risk to organizational operations, organizational assets, or individuals. Authorizing officials also determine if the level of assessor independence provides sufficient assurance that the results are

sound and can be used to make credible, risk-based decisions. Assessor independence determination includes whether contracted assessment services have sufficient independence, such as when system owners are not directly involved in contracting processes or cannot influence the impartiality of the assessors conducting the assessments. During the system design and development phase, having independent assessors is analogous to having independent SMEs involved in design reviews.

When organizations that own the systems are small or the structures of the organizations require that assessments be conducted by individuals that are in the developmental, operational, or management chain of the system owners, independence in assessment processes can be achieved by ensuring that assessment results are carefully reviewed and analyzed by independent teams of experts to validate the completeness, accuracy, integrity, and reliability of the results.

Assessments performed for purposes other than to support authorization decisions are more likely to be useable for such decisions when performed by assessors with sufficient independence, thereby reducing the need to repeat assessments.

Related Controls: None.

CA-3 INFORMATION EXCHANGE

[Existing] [Priority 2]

Control:

- a. Approve and manage the exchange of information between the agency system and external systems using the following agreements when applicable;
 1. Executed CJIS User Agreements
 - a. Each CSA, SIB, or IA shall execute a signed written agreement (see Appendix D.1) with the FBI CJIS Division stating their willingness to demonstrate conformity with the CJISSECPOL before accessing and consuming CJIS systems and services as set forth in the agreement.
 - b. The agreement shall include the standards, audit, and sanctions governing utilization of CJIS systems and services.
 - c. The FBI CJIS Division is authorized to periodically test the ability to penetrate the FBI's network through the external connection or system upon proper notification of all signatories in the user agreement.
 2. Criminal Justice Agency User Agreements
 - a. Any CJA receiving access to CJI shall enter into a signed written agreement with the appropriate signatory authority of the CSA providing the access.
 - b. The written agreement shall specify the FBI CJIS systems and services to which the agency will have access, and the FBI CJIS Division policies to which the agency must adhere. These agreements shall include:
 - i. Audit
 - ii. Dissemination
 - iii. Hit confirmation
 - iv. Logging

- v. Quality Assurance (QA)
- vi. Screening (Criminal Justice Employment)
- vii. Security
- viii. Timeliness
- ix. Training
- x. Use of the system
- xi. Validation

3. Agreements for Noncriminal Justice Use of CHRI

A CJA, NCJA (public), or NCJA (private) designated to request civil fingerprint-based background checks, with full consent of the individual to whom the background check is taking place, for noncriminal justice functions, shall be eligible for access to CHRI. Access shall be permitted when such designation is authorized pursuant to federal law or state statute approved by the U.S. Attorney General. The CJA, NCJA (public), or NCJA (private) receiving access to CHRI shall enter into signed written agreements with the appropriate signatory authority of the CSA, SIB, or authorized agency providing the access. The written agreement shall specify the policies to which the agency must adhere, which includes all pertinent areas of the CJISSECPOL. Each NCJA that directly access FBI systems shall allow the FBI to periodically test the ability to penetrate the FBI's network through the external connection or system. A CJA, NCJA (public), or NCJA (private) authorized to access CHRI for noncriminal justice functions pursuant to federal law or state statute approved by the U.S. Attorney General (defined by the Compact Council as an Authorized Recipient), cannot make CHRI available to another governmental agency, nongovernmental agency, or private contractor to perform noncriminal justice administrative functions without implementation of one of the following:

- a. Security and Management Control Outsourcing Standard for Non-Channeling. Implementation is applicable to noncriminal justice administrative functions that do not require a direct connection to the FBI for submission of fingerprints and receipt of CHRI. Examples include making fitness determinations, processing, storing, or destroying documents, and maintaining IT platforms that do not connect to CJIS systems. Prior to implementation, Authorized Recipients must request and receive written permission from the State Compact Officer, Chief Administrator of the state's criminal history record repository, or the FBI Compact Officer, as applicable.
- b. Security and Management Control Outsourcing Standard for Channeling. Implementation is applicable only to Channeling functions performed by an FBI-approved Channeler that require a direct connection to the FBI for submission of fingerprints and receipt of CHRI. Prior to implementation, Authorized Recipients must request and receive written permission from the State Compact Officer, Chief Administrator of the state's criminal history record repository, or the FBI Compact Officer, as applicable.
- c. Management Control Agreement or Security Addendum (see Appendix H) pursuant to Title 28, C.F.R., Section 20.33 (a) (6) or (7). Although by regulation

implementation of a Management Control Agreement or the Security Addendum is applicable to the administration of criminal justice pursuant to that agreement performed on behalf of CJAs, under very limited circumstances, implementation may also be applicable to CJAs that obtain and use CHRI for noncriminal justice purposes. Implementation for noncriminal justice purposes is only applicable when another governmental agency or private contractor performs both criminal justice and noncriminal justice administrative functions involving access to CHRI on behalf of the CJA. It is important to note that if the servicing governmental agency or private contractor solely performs noncriminal justice administrative functions, then the CJA would be required to implement the Security and Management Control Outsourcing Standard for Non-Channeling.

- b. Document, as part of each exchange agreement, the interface characteristics, security and privacy requirements, controls, and responsibilities for each system, and the impact level of the information communicated; and
- c. Review and update the agreements at least triennially or when responsibilities or signatories change.
- d. Secondary Dissemination
 - 1. Log the dissemination of CHRI when released to another authorized agency, and that agency was not part of the releasing agency's primary information exchange agreement(s). If CJI does not contain CHRI and is not part of an information exchange agreement, then it does not need to be logged.
 - 2. Validate the requestor of CJI in conformance with the local policy as an employee and/or contractor of a law enforcement agency or civil agency requiring the CJI to perform their mission; or a member of the public receiving CJI via authorized dissemination.

Discussion: System information exchange requirements apply to information exchanges between two or more systems. System information exchanges include connections via leased lines or virtual private networks, connections to internet service providers, database sharing or exchanges of database transaction information, connections and exchanges with cloud services, exchanges via web-based services, or exchanges of files via file transfer protocols, network protocols (e.g., IPv4, IPv6), email, or other organization-to-organization communications. Organizations consider the risk related to new or increased threats that may be introduced when systems exchange information with other systems that may have different security and privacy requirements and controls. This includes systems within the same organization and systems that are external to the organization. An example of an NCJA (public) is a county school board. An example of an NCJA (private) is a local bank.

Authorizing officials determine the risk associated with system information exchange and the controls needed for appropriate risk mitigation. The types of agreements selected are based on factors such as the impact level of the information being exchanged, the relationship between the organizations exchanging information (e.g., government to government, government to business, business to business, government or business to service provider, government or business to individual), or the level of access to the organizational system by users of the other system. If systems that exchange information have the same authorizing official, organizations need not develop agreements. Instead, the interface characteristics between the systems (e.g., how the information is being exchanged, how the information is protected) are described in the respective

security and privacy plans. If the systems that exchange information have different authorizing officials within the same organization, the organizations can develop agreements or provide the same information that would be provided in the appropriate agreement type from CA-3a in the respective security and privacy plans for the systems. Organizations may incorporate agreement information into formal contracts, especially for information exchanges established between federal agencies and nonfederal organizations (including service providers, contractors, system developers, and system integrators). Risk considerations include systems that share the same networks.

Related Controls: AC-4, AC-20, CA-6, IA-3, IR-4, PL-2, RA-3, SA-9, SC-7, SI-12.

CA-5 PLAN OF ACTION AND MILESTONES

[Priority 4]

Control:

- a. Develop a plan of action and milestones for the system to document the planned remediation actions of the organization to correct weaknesses or deficiencies noted during the assessment of the controls and to reduce or eliminate known vulnerabilities in the system; and
- b. Update existing plan of action and milestones at least every six (6) months or when new information is available based on the findings from control assessments, independent audits or reviews, and continuous monitoring activities.

Discussion: Plans of action and milestones are useful for any type of organization to track planned remedial actions. Related Controls: CA-2, CA-7, RA-7, SI-2, SI-12.

CA-6 AUTHORIZATION

[Priority 3]

Control:

- a. Assign a senior official as the responsible official for the system;
- b. Assign the CSO, SIB Chief, or IA Official as the authorizing official for common controls available for inheritance by organizational systems;
- c. Ensure that the authorizing official for the system, before commencing operations:
 1. Accepts the use of common controls inherited by the system; and
 2. Authorizes the system to operate;
- d. Ensure that the authorizing official for common controls authorizes the use of those controls for inheritance by organizational systems;
- e. Update the authorizations at least every three (3) years.

Discussion: Authorizations are official management decisions by senior officials to authorize operation of systems, authorize the use of common controls for inheritance by organizational systems, and explicitly accept the risk to organizational operations and assets, individuals, other organizations, and the Nation based on the implementation of agreed-upon controls. Authorizing

officials provide budgetary oversight for organizational systems and common controls or assume responsibility for the mission and business functions supported by those systems or common controls. Authorizing officials are both responsible and accountable for security and privacy risks associated with the operation and use of organizational systems.

Nonfederal organizations may have similar processes to authorize systems and senior officials that assume the authorization role and associated responsibilities.

Authorizing officials issue ongoing authorizations of systems based on evidence produced from implemented continuous monitoring programs. Robust continuous monitoring programs reduce the need for separate reauthorization processes. Through the employment of comprehensive continuous monitoring processes, the information contained in authorization packages (i.e., security and privacy plans, assessment reports, and plans of action and milestones) is updated on an ongoing basis. This provides authorizing officials, common control providers, and system owners with an up-to-date status of the security and privacy posture of their systems, controls, and operating environments. To reduce the cost of reauthorization, authorizing officials can leverage the results of continuous monitoring processes to the maximum extent possible as the basis for rendering reauthorization decisions.

Related Controls: CA-2, CA-3, CA-7, RA-3, SA-10, SI-12.

CA-7 CONTINUOUS MONITORING

[Priority 1]

Control:

Develop a system-level continuous monitoring strategy and implement continuous monitoring in accordance with the organization-level continuous monitoring strategy that includes:

a. Establishing the following system-level metrics to be monitored:

1. AC-2(g) Account Management
2. AC-17(1) Remote Access | Monitoring and Control
3. AT-4(a) Training Records
4. CM-3(f) Configuration Change Control
5. CM-6(d) Configuration Settings
6. CM-11(c) User-Installed Software
7. IR-5 Incident Monitoring
8. MA-2(b) Controlled Maintenance
9. MA-3(a) Maintenance Tool
10. MA-4(a) Nonlocal Maintenance
11. PE-3(d) Physical Access Control
12. PE-6 Monitoring Physical Access
13. PE-14(b) Environmental Controls

- 14. PE-16 Delivery and Removal
- 15. PS-7(e) External Personnel Security
- 16. SA-9(c) External System Services
- 17. SC-7(a) Boundary Protection
- 18. SC-7(24)(b) Boundary Protection | Personally Identifiable Information
- 19. SC-18(b) Mobile Code
- 20. SI-4 System Monitoring

- b. Establishing an ongoing frequency for monitoring and an ongoing frequency for assessment of control effectiveness;
- c. Ongoing control assessments in accordance with the continuous monitoring strategy;
- d. Ongoing monitoring of system and organization-defined metrics in accordance with the continuous monitoring strategy;
- e. Correlation and analysis of information generated by control assessments and monitoring;
- f. Response actions to address results of the analysis of control assessment and monitoring information; and
- g. Reporting the security and privacy status of the system to organizational personnel with information security, privacy responsibilities, and system/network administrators annually, when security events/incidents occur, and when requested.

Discussion: Continuous monitoring at the system level facilitates ongoing awareness of the system security and privacy posture to support organizational risk management decisions. The terms “continuous” and “ongoing” imply that organizations assess and monitor their controls and risks at a frequency sufficient to support risk-based decisions. Different types of controls may require different monitoring frequencies. The results of continuous monitoring generate risk response actions by organizations. When monitoring the effectiveness of multiple controls that have been grouped into capabilities, a root-cause analysis may be needed to determine the specific control that has failed. Continuous monitoring programs allow organizations to maintain the authorizations of systems and common controls in highly dynamic environments of operation with changing mission and business needs, threats, vulnerabilities, and technologies. Having access to security and privacy information on a continuing basis through reports and dashboards gives organizational officials the ability to make effective and timely risk management decisions, including ongoing authorization decisions.

Automation supports more frequent updates to hardware, software, and firmware inventories, authorization packages, and other system information. Effectiveness is further enhanced when continuous monitoring outputs are formatted to provide information that is specific, measurable, actionable, relevant, and timely. Continuous monitoring activities are scaled in accordance with the security categories of systems. Monitoring requirements, including the need for specific monitoring, may be referenced in other controls and control enhancements.

Related Controls: AC-2, AC-6, AC-17, AT-4, AU-6, CA-2, CA-5, CA-6, CM-3, CM-4, CM-6, CM-11, IA-5, IR-5, MA-2, MA-3, MA-4, PE-3, PE-6, PE-14, PE-16, PL-2, PS-7, RA-3, RA-5, RA-7, SA-8, SA-9, SA-11, SC-5, SC-7, SC-18, SI-3, SI-4, SI-12, SR-6.

Control Enhancements:

(1) CONTINUOUS MONITORING | INDEPENDENT ASSESSMENT

[Priority 1]

Control:

Employ independent assessors or assessment teams to monitor the controls in the system on an ongoing basis.

Discussion: Organizations maximize the value of control assessments by requiring that assessments be conducted by assessors with appropriate levels of independence. The level of required independence is based on organizational continuous monitoring strategies.

Assessor independence provides a degree of impartiality to the monitoring process. To achieve such impartiality, assessors do not create a mutual or conflicting interest with the organizations where the assessments are being conducted, assess their own work, act as management or employees of the organizations they are serving, or place themselves in advocacy positions for the organizations acquiring their services.

Related Controls: None.

(4) CONTINUOUS MONITORING | RISK MONITORING

[Priority 1]

Control:

Ensure risk monitoring is an integral part of the continuous monitoring strategy that includes the following:

- a. Effectiveness monitoring;
- b. Compliance monitoring; and
- c. Change monitoring.

Discussion: Risk monitoring is informed by the established organizational risk tolerance. Effectiveness monitoring determines the ongoing effectiveness of the implemented risk response measures. Compliance monitoring verifies that required risk response measures are implemented. It also verifies that security and privacy requirements are satisfied. Change monitoring identifies changes to organizational systems and environments of operation that may affect security and privacy risk.

Related Controls: None.

CA-9 INTERNAL SYSTEM CONNECTIONS

[Priority 3]

Control:

- a. Authorize internal connections of components with the capability to process, store, or transmit CJI to the system;
- b. Document, for each internal connection, the interface characteristics, security and privacy

requirements, and the nature of the information communicated;

- c. Terminate internal system connections when no longer required or authorized; and
- d. Review at least annually the continued need for each internal connection.

Discussion: Internal system connections are connections between organizational systems and separate constituent system components (i.e., connections between components that are part of the same system) including components used for system development. Intra-system connections include connections with mobile devices, notebook and desktop computers, tablets, printers, copiers, facsimile machines, scanners, sensors, and servers. Instead of authorizing each internal system connection individually, organizations can authorize internal connections for a class of system components with common characteristics and/or configurations, including printers, scanners, and copiers with a specified processing, transmission, and storage capability or smart phones and tablets with a specific baseline configuration. The continued need for an internal system connection is reviewed from the perspective of whether it provides support for organizational missions or business functions.

Related Controls: AC-3, AC-4, AC-18, AC-19, CM-2, IA-3, SC-7, SI-12.

CONFIGURATION MANAGEMENT (CM)

CM-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with configuration management responsibilities:
 1. Agency-level configuration management policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the configuration management policy and the associated configuration management controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the configuration management policy and procedures; and
- c. Review and update the current configuration management:
 1. Policy annually and following any hardware or software changes to systems which process, store, or transmit CJI; and
 2. Procedures annually and following any hardware or software changes to systems which process, store, or transmit CJI.

Discussion: Configuration management policy and procedures address the controls in the CM family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of configuration management policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission/business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to configuration management policy and procedures include, but are not limited to, assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SA-8, SI-12.

CM-2 BASELINE CONFIGURATION

[Existing] [Priority 1]

Control:

- a. Develop, document, and maintain under configuration control, a current baseline configuration of the system;
- b. Develop, document, and maintain a current and complete topological drawing depicting the interconnectivity of the agency network to criminal justice information systems and services; and
- c. Review and update the baseline configuration and topological drawing of the system:
 1. At least annually;
 2. When required due to security-relevant changes to the system and/or security incidents occur; and
 3. When system components are installed or upgraded.

Discussion: Baseline configurations for systems and system components include connectivity, operational, and communications aspects of systems. Baseline configurations are documented, formally reviewed, and agreed-upon specifications for systems or configuration items within those systems. Baseline configurations serve as a basis for future builds, releases, or changes to systems and include security and privacy control implementations, operational procedures, information about system components, network topology, and logical placement of components in the system architecture. Maintaining baseline configurations requires creating new baselines as organizational systems change over time. Baseline configurations of systems reflect the current enterprise architecture. See Appendix C for sample topological/network diagrams.

Related Controls: AC-19, AU-6, CA-9, CM-1, CM-3, CM-5, CM-6, CM-8, CM-9, CP-9, CP-10, MA-2, PL-8, SA-8, SA-10, SA-15, SC-18.

Control Enhancements:

(2) BASELINE CONFIGURATION | AUTOMATION SUPPORT FOR ACCURACY AND CURRENCY

[Priority 1]

Control:

Maintain the currency, completeness, accuracy, and availability of the baseline configuration of the system using automated mechanisms such as configuration management tools, hardware, software, firmware inventory tools, and network management tools.

Discussion: Automated mechanisms that help organizations maintain consistent baseline configurations for systems include configuration management tools, hardware, software, firmware inventory tools, and network management tools. Automated tools can be used at the

organization level, mission and business process level, or system level on workstations, servers, notebook computers, network components, or mobile devices. Tools can be used to track version numbers on operating systems, applications, types of software installed, and current patch levels. Automation support for accuracy and currency can be satisfied by the implementation of CM-8(2) for organizations that combine system component inventory and baseline configuration activities.

Related Controls: CM-7, IA-3, RA-5.

(3) BASELINE CONFIGURATION | RETENTION OF PREVIOUS CONFIGURATIONS

[Priority 1]

Control:

Retain at least one (1) previous version of baseline configurations of the system to support rollback.

Discussion: Retaining previous versions of baseline configurations to support rollback include hardware, software, firmware, configuration files, configuration records, and associated documentation.

Related Controls: None.

(7) BASELINE CONFIGURATION | CONFIGURE SYSTEMS AND COMPONENTS FOR HIGH-RISK AREAS

[Existing] [Priority 1]

Control:

- a. Issue devices (e.g., mobile devices) with CJISSECPOL compliant configurations to individuals traveling to locations that the organization deems to be of significant risk; and
- b. Apply the following controls to the systems or components when the individuals return from travel: examine the device for signs of physical tampering, purge and reimage disk drives and/or devices as required, and ensure all security controls are in place and functional.

Discussion: When it is known that systems or system components will be in high-risk areas external to the organization, additional controls may be implemented to counter the increased threat in such areas. For example, organizations can take actions for notebook computers used by individuals departing on and returning from travel. Actions include determining the locations that are of concern, defining the required configurations for the components, ensuring that components are configured as intended before travel is initiated, and applying controls to the components after travel is completed. Specially configured notebook computers include computers with sanitized hard drives, limited applications, and more stringent configuration settings. Controls applied to mobile devices upon return from travel include examining the mobile device for signs of physical tampering and purging and reimaging disk drives. Protecting information that resides on mobile devices is addressed in the MP (Media Protection) family.

Related Controls: MP-4, MP-5.

CM-3 CONFIGURATION CHANGE CONTROL

[Priority 2]

Control:

- a. Determine and document the types of changes to the system that are configuration-controlled;
- b. Review proposed configuration-controlled changes to the system and approve or disapprove such changes with explicit consideration for security and privacy impact analyses;
- c. Document configuration change decisions associated with the system;
- d. Implement approved configuration-controlled changes to the system;
- e. Retain records of configuration-controlled changes to the system for two (2) years;
- f. Monitor and review activities associated with configuration-controlled changes to the system; and
- g. Coordinate and provide oversight for configuration change control activities through personnel with configuration management responsibilities, a Configuration Control Board, or Change Advisory Board that convenes regularly or when hardware or software changes (i.e., updates, upgrades, replacements, etc.) to the information system are required.

Discussion: Configuration change control for organizational systems involves the systematic proposal, justification, implementation, testing, review, and disposition of system changes, including system upgrades and modifications. Configuration change control includes changes to baseline configurations, configuration items of systems, operational procedures, configuration settings for system components, remediate vulnerabilities, and unscheduled or unauthorized changes. Processes for managing configuration changes to systems include Configuration Control Boards or Change Advisory Boards that review and approve proposed changes. For changes that impact privacy risk, the senior agency official for privacy updates privacy impact assessments and system of records notices. For new systems or major upgrades, organizations consider including representatives from the development organizations on the Configuration Control Boards or Change Advisory Boards. Auditing of changes includes activities before and after changes are made to systems and the auditing activities required to implement such changes. See also SA-10.

Related Controls: CA-7, CM-2, CM-4, CM-5, CM-6, CM-9, CM-11, IA-3, MA-2, PE-16, SA-8, SA-10, SC-28, SI-2, SI-3, SI-4, SI-7, SI-10, SR-11.

Control Enhancements:

(2) CONFIGURATION CHANGE CONTROL | TESTING, VALIDATION, AND DOCUMENTATION OF CHANGES

[Priority 2]

Control:

Test, validate, and document changes to the system before finalizing the implementation of the changes.

Discussion: Changes to systems include modifications to hardware, software, or firmware

components and configuration settings defined in CM-6. Organizations ensure that testing does not interfere with system operations that support organizational mission and business functions. Individuals or groups conducting tests understand security and privacy policies and procedures, system security and privacy policies and procedures, and the health, safety, and environmental risks associated with specific facilities or processes. Operational systems may need to be taken offline, or replicated to the extent feasible, before testing can be conducted. If systems must be taken offline for testing, the tests are scheduled to occur during planned system outages whenever possible. If the testing cannot be conducted on operational systems, organizations employ compensating controls.

Related Controls: None.

(4) CONFIGURATION CHANGE CONTROL | SECURITY AND PRIVACY REPRESENTATIVES

[Priority 2]

Control:

Require organizational personnel with information security and privacy responsibilities to be members of the Configuration Control Board or Change Advisory Board.

Discussion: Information security and privacy representatives include system security officers, senior agency information security officers, senior agency officials for privacy, or system privacy officers. Representation by personnel with information security and privacy expertise is important because changes to system configurations can have unintended side effects, some of which may be security- or privacy-relevant. Detecting such changes early in the process can help avoid unintended, negative consequences that could ultimately affect the security and privacy posture of systems. The configuration change control element referred to in the second organization-defined parameter reflects the change control elements defined by organizations in CM-3g.

CM-4 IMPACT ANALYSES

[Priority 3]

Control:

Analyze changes to the system to determine potential security and privacy impacts prior to change implementation.

Discussion: Organizational personnel with security or privacy responsibilities conduct impact analyses. Individuals conducting impact analyses possess the necessary skills and technical expertise to analyze the changes to systems as well as the security or privacy ramifications. Impact analyses include reviewing security and privacy plans, policies, and procedures to understand control requirements; reviewing system design documentation and operational procedures to understand control implementation and how specific system changes might affect the controls; reviewing the impact of changes on organizational supply chain partners with stakeholders; and determining how potential changes to a system create new risks to the privacy of individuals and the ability of implemented controls to mitigate those risks. Impact analyses

also include risk assessments to understand the impact of the changes and determine if additional controls are required.

Related Controls: CA-7, CM-3, CM-8, CM-9, MA-2, RA-3, RA-5, SA-5, SA-8, SA-10, SI-2.

Control Enhancements:

(2) IMPACT ANALYSES | VERIFICATION OF CONTROLS

[Priority 3]

Control:

After-system changes, verify that the impacted controls are implemented correctly, operating as intended, and producing the desired outcome with regard to meeting the security and privacy requirements for the system.

Discussion: Implementation in this context refers to installing changed code in the operational system that may have an impact on security or privacy controls.

Related Controls: SA-11.

CM-5 ACCESS RESTRICTIONS FOR CHANGE

[Existing] [Priority 1]

Control:

Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.

Discussion: Changes to the hardware, software, or firmware components of systems or the operational procedures related to the system can potentially have significant effects on the security of the systems or individuals' privacy. Therefore, organizations permit only qualified and authorized individuals to access systems for purposes of initiating changes. Access restrictions include physical and logical access controls (see AC-3 and PE-3), software libraries, workflow automation, media libraries, abstract layers (i.e., changes implemented into external interfaces rather than directly into systems), and change windows (i.e., changes occur only during specified times).

Related Controls: AC-3, AC-5, AC-6, CM-9, PE-3, SC-28, SI-2, SI-10.

CM-6 CONFIGURATION SETTINGS

[Priority 1]

Control:

- a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using established best practices and guidelines such as Defense Information Systems Agency (DISA) Secure Technical Implementation Guidelines (STIGs), Center for Internet Security (CIS) Benchmarks, or Federal Information Processing Standards;

- b. Implement the configuration settings;
- c. Identify, document, and approve any deviations from established configuration settings for system components that store, process, or transmit CJI based on operational requirements; and
- d. Monitor and control changes to the configuration settings in accordance with organizational policies and procedures.

Discussion: Configuration settings are the parameters that can be changed in the hardware, software, or firmware components of the system that affect the security and privacy posture or functionality of the system. Information technology products for which configuration settings can be defined include mainframe computers, servers, workstations, operating systems, mobile devices, input/output devices, protocols, and applications. Parameters that impact the security posture of systems include registry settings; account, file, or directory permission settings; and settings for functions, protocols, ports, services, and remote connections. Privacy parameters are parameters impacting the privacy posture of systems, including the parameters required to satisfy other privacy controls. Privacy parameters include settings for access controls, data processing preferences, and processing and retention permissions. Organizations establish organization-wide configuration settings and subsequently derive specific configuration settings for systems. The established settings become part of the configuration baseline for the system.

Common secure configurations (also known as security configuration checklists, lockdown and hardening guides, and security reference guides) provide recognized, standardized, and established benchmarks that stipulate secure configuration settings for information technology products and platforms as well as instructions for configuring those products or platforms to meet operational requirements. Common secure configurations can be developed by a variety of organizations, including information technology product developers, manufacturers, vendors, federal agencies, consortia, academia, industry, and other organizations in the public and private sectors.

Implementation of a common secure configuration may be mandated at the organization level, mission and business process level, system level, or at a higher level, including by a regulatory agency. Common secure configurations include the United States Government Configuration Baseline [USGCB] and security technical implementation guides (STIGs), which affect the implementation of CM-6 and other controls such as AC-19 and CM-7. The Security Content Automation Protocol (SCAP) and the defined standards within the protocol provide an effective method to uniquely identify, track, and control configuration settings.

Related Controls: AC-3, AC-19, AU-2, AU-6, CA-9, CM-2, CM-3, CM-5, CM-7, CM-11, CP-7, CP-9, CP-10, IA-3, IA-5, PL-8, RA-5, SA-4, SA-5, SA-8, SA-9, SC-18, SC-28, SI-2, SI-4.

CM-7 LEAST FUNCTIONALITY

[Existing] [Priority 1]

Control:

- a. Configure the system to provide only essential capabilities to meet operational requirements; and

- b. Prohibit or restrict the use of specified functions, ports, protocols, software, and/or services which are not required.

Discussion: Systems provide a wide variety of functions and services. Some of the functions and services routinely provided by default may not be necessary to support essential organizational missions, functions, or operations. Additionally, it is sometimes convenient to provide multiple services from a single system component, but doing so increases risk over limiting the services provided by that single component. Where feasible, organizations limit component functionality to a single function per component. Organizations consider removing unused or unnecessary software and disabling unused or unnecessary physical and logical ports and protocols to prevent unauthorized connection of components, transfer of information, and tunneling. Organizations employ network scanning tools, intrusion detection and prevention systems, and end-point protection technologies, such as firewalls and host-based intrusion detection systems, to identify and prevent the use of prohibited functions, protocols, ports, and services. Least functionality can also be achieved as part of the fundamental design and development of the system (see SA- 8 and SC-2).

Related Controls: AC-3, AC-4, CM-2, CM-5, CM-6, CM-11, RA-5, SA-4, SA-5, SA-8, SA-9, SA-15, SC- 2, SC-7, SI-4.

Control Enhancements:

(1) LEAST FUNCTIONALITY | PERIODIC REVIEW

[Existing] [Priority 1]

Control:

- a. Review the system annually, as the system changes, or incidents occur to identify unnecessary and/or nonsecure functions, ports, protocols, software, and services; and
- b. Disable or remove functions, ports, protocols, software, and/or services within the system deemed to be unnecessary and/or unsecure.

Discussion: Organizations review functions, ports, protocols, and services provided by systems or system components to determine the functions and services that are candidates for elimination. Such reviews are especially important during transition periods from older technologies to newer technologies (e.g., transition from IPv4 to IPv6). These technology transitions may require implementing the older and newer technologies simultaneously during the transition period and returning to minimum essential functions, ports, protocols, and services at the earliest opportunity. Organizations can either decide the relative security of the function, port, protocol, and/or service or base the security decision on the assessment of other entities. Unsecure protocols include Bluetooth, FTP, and peer-to-peer networking.

Related Controls: AC-18.

(2) LEAST FUNCTIONALITY | PREVENT PROGRAM EXECUTION

[Priority 1]

Control:

Prevent program execution in accordance with rules of behavior and/or rules authorizing the

terms and conditions of software program usage.

Discussion: Prevention of program execution addresses organizational policies, rules of behavior, and/or access agreements that restrict software usage and the terms and conditions imposed by the developer or manufacturer, including software licensing and copyrights. Restrictions include prohibiting auto-execute features, restricting roles allowed to approve program execution, permitting or prohibiting specific software programs, or restricting the number of program instances executed at the same time.

Related Controls: CM-8, PL-4, PS-6.

(5) LEAST FUNCTIONALITY | AUTHORIZED SOFTWARE — ALLOW-BY-EXCEPTION

[Priority 1]

Control:

- a. Identify software programs authorized to execute on the system;
- b. Employ a deny-all, permit-by-exception policy to allow the execution of authorized software programs on the system; and
- c. Review and update the list of authorized software programs annually.

Discussion: Authorized software programs can be limited to specific versions or from a specific source. To facilitate a comprehensive authorized software process and increase the strength of protection for attacks that bypass application level authorized software, software programs may be decomposed into and monitored at different levels of detail. These levels include applications, application programming interfaces, application modules, scripts, system processes, system services, kernel functions, registries, drivers, and dynamic link libraries. The concept of permitting the execution of authorized software may also be applied to user actions, system ports and protocols, IP addresses/ranges, websites, and MAC addresses. Organizations consider verifying the integrity of authorized software programs using digital signatures, cryptographic checksums, or hash functions. Verification of authorized software can occur either prior to execution or at system startup. The identification of authorized URLs for websites is addressed in SC-7.

Related Controls: CM-2, CM-6, CM-8, CM-10, SA-10, SI-7.

CM-8 SYSTEM COMPONENT INVENTORY

[Priority 1]

Control:

- a. Develop and document an inventory of system components that:
 1. Accurately reflects the system;
 2. Includes all components within the system;
 3. Does not include duplicate accounting of components or components assigned to any other system;

4. Is at the level of granularity deemed necessary for tracking and reporting; and
5. Includes the following minimum information to achieve system component accountability: date of installation, model, serial number, manufacturer, supplier information, component type, software owner, software version number, software license information, and hardware and physical location; and

b. Review and update the system component inventory annually.

Discussion: System components are discrete, identifiable information technology assets that include hardware, software, and firmware. Organizations may choose to implement centralized system component inventories that include components from all organizational systems. In such situations, organizations ensure that the inventories include system-specific information required for component accountability. The information necessary for effective accountability of system components includes the system name, software owners, software version numbers, hardware inventory specifications, software license information, and for networked components, the machine names and network addresses across all implemented protocols (e.g., IPv4, IPv6). Inventory specifications include date of receipt, cost, model, serial number, manufacturer, supplier information, component type, and physical location.

Preventing duplicate accounting of system components addresses the lack of accountability that occurs when component ownership and system association is not known, especially in large or complex connected systems. Effective prevention of duplicate accounting of system components necessitates use of a unique identifier for each component. For software inventory, centrally managed software that is accessed via other systems is addressed as a component of the system on which it is installed and managed. Software installed on multiple organizational systems and managed at the system level is addressed for each individual system and may appear more than once in a centralized component inventory, necessitating a system association for each software instance in the centralized inventory to avoid duplicate accounting of components. Scanning systems implementing multiple network protocols (e.g., IPv4 and IPv6) can result in duplicate components being identified in different address spaces.

Related Controls: CM-2, CM-7, CM-9, CM-10, CM-11, CP-2, CP-9, MA-2, MA-6, SA-4, SA-5, SI-2.

Control Enhancements:

(1) SYSTEM COMPONENT INVENTORY | UPDATES DURING INSTALLATION AND REMOVAL

[Priority 1]

Control:

Update the inventory of system components as part of component installations, removals, and system updates.

Discussion: Organizations can improve the accuracy, completeness, and consistency of system component inventories if the inventories are updated as part of component installations or removals or during general system updates. If inventories are not updated at these key times, there is a greater likelihood that the information will not be appropriately captured and

documented. System updates include hardware, software, and firmware components.

Related Controls: None.

(3) SYSTEM COMPONENT INVENTORY | AUTOMATED UNAUTHORIZED COMPONENT DETECTION

[Priority 1]

Control:

- a. Detect the presence of unauthorized hardware, software, and firmware components within the system using automated mechanisms continuously or at least weekly; and
- b. Take the following actions when unauthorized components are detected: disable or isolate the unauthorized components and notify organizational personnel with security responsibilities.

Discussion: Automated unauthorized component detection is applied in addition to the monitoring for unauthorized remote connections and mobile devices. Monitoring for unauthorized system components may be accomplished on an ongoing basis or by the periodic scanning of systems for that purpose. Automated mechanisms may also be used to prevent the connection of unauthorized components. Automated mechanisms can be implemented in systems or in separate system components. When acquiring and implementing automated mechanisms, organizations consider whether such mechanisms depend on the ability of the system component to support an agent or supplicant in order to be detected since some types of components do not have or cannot support agents (e.g., IoT devices, sensors). Isolation can be achieved, for example, by placing unauthorized system components in separate domains or subnets or quarantining such components. This type of component isolation is commonly referred to as “sandboxing.”

Related Controls: AC-19, CA-7, RA-5, SC-3, SC-39, SI-3, SI-4, SI-7.

CM-9 CONFIGURATION MANAGEMENT PLAN

[Priority 2]

Control:

Develop, document, and implement a configuration management plan for the system that:

- a. Addresses roles, responsibilities, and configuration management processes and procedures;
- b. Establishes a process for identifying configuration items throughout the system development life cycle and for managing the configuration of the configuration items;
- c. Defines the configuration items for the system and places the configuration items under configuration management;
- d. Is reviewed and approved by organizational personnel with information security responsibilities and organizational personnel with configuration management responsibilities; and
- e. Protects the configuration management plan from unauthorized disclosure and modification.

Discussion: Configuration management activities occur throughout the system development life cycle. As such, there are developmental configuration management activities (e.g., the control of code and software libraries) and operational configuration management activities (e.g., control of installed components and how the components are configured). Configuration management plans satisfy the requirements in configuration management policies while being tailored to individual systems. Configuration management plans define processes and procedures for how configuration management is used to support system development life cycle activities.

Configuration management plans are generated during the development and acquisition stage of the system development life cycle. The plans describe how to advance changes through change management processes; update configuration settings and baselines; maintain component inventories; control development, test, and operational environments; and develop, release, and update key documents.

Organizations can employ templates to help ensure the consistent and timely development and implementation of configuration management plans. Templates can represent a configuration management plan for the organization with subsets of the plan implemented on a system by system basis. Configuration management approval processes include the designation of key stakeholders responsible for reviewing and approving proposed changes to systems, and personnel who conduct security and privacy impact analyses prior to the implementation of changes to the systems. Configuration items are the system components, such as the hardware, software, firmware, and documentation to be configuration managed. As systems continue through the system development life cycle, new configuration items may be identified, and some existing configuration items may no longer need to be under configuration control.

Related Controls: CM-2, CM-3, CM-4, CM-5, CM-8, PL-2, SA-10, SI-12.

CM-10 SOFTWARE USAGE RESTRICTIONS

[Priority 3]

Control:

- a. Use software and associated documentation in accordance with contract agreements and copyright laws;
- b. Track the use of software and associated documentation protected by quantity licenses to control copying and distribution; and
- c. Control and document the use of peer-to-peer file sharing technology to ensure that this capability is not used for the unauthorized distribution, display, performance, or reproduction of copyrighted work.

Discussion: Software license tracking can be accomplished by manual or automated methods, depending on organizational needs. Examples of contract agreements include software license agreements and non-disclosure agreements.

Related Controls: AC-17, AU-6, CM-7, CM-8, SC-7.

CM-11 USER-INSTALLED SOFTWARE

[Priority 2]

Control:

- a. Establish agency-level policies governing the installation of software by users;
- b. Enforce software installation policies through automated methods; and
- c. Monitor policy compliance through automated methods at least weekly.

Discussion: If provided the necessary privileges, users can install software in organizational systems. To maintain control over the software installed, organizations identify permitted and prohibited actions regarding software installation. Permitted software installations include updates and security patches to existing software and downloading new applications from organization-approved “app stores.” Prohibited software installations include software with unknown or suspect pedigrees or software that organizations consider potentially malicious. Policies selected for governing user-installed software are organization-developed or provided by some external entity. Policy enforcement methods can include procedural methods and automated methods.

Related Controls: AC-3, AU-6, CM-2, CM-3, CM-5, CM-6, CM-7, CM-8, PL-4, SI-4, SI-7.

CM-12 INFORMATION LOCATION

[Priority 2]

Control:

- a. Identify and document the location of CJI and the specific system components on which the information is processed, stored, or transmitted;
- b. Identify and document the users who have access to CJI and system components where CJI is processed, stored, or transmitted; and
- c. Document changes to the location (i.e., system or system components) where CJI is processed, stored, or transmitted.

Discussion: Information location addresses the need to understand where information is being processed and stored. Information location includes identifying where specific information types and information reside in system components and how information is being processed so that information flow can be understood, and adequate protection and policy management provided for such information and system components. The security category of the information is also a factor in determining the controls necessary to protect the information and the system component where the information resides. The location of the information and system components is also a factor in the architecture and design of the system (see SA-4, SA-8).

Related Controls: AC-2, AC-3, AC-4, AC-6, CM-8, RA-2, SA-4, SA-8, SC-4, SC-28, SI-4, SI-7.

Control Enhancements:

(1) INFORMATION LOCATION | AUTOMATED TOOLS TO SUPPORT INFORMATION LOCATION

[Priority 2]

Control:

Use automated tools to identify CJI on software and hardware system components to ensure controls are in place to protect organizational information and individual privacy.

Discussion: The use of automated tools helps to increase the effectiveness and efficiency of the information location capability implemented within the system. Automation also helps organizations manage the data produced during information location activities and share such information across the organization. The output of automated information location tools can be used to guide and inform system architecture and design decisions.

Related Controls: None.

Figure 5 – A Local Police Department’s Configuration Management Controls

A local police department decided to update their CAD system, and in doing so tracked all changes made to their infrastructure in a configuration management journal, updated their network topology documents to include all new components in their architecture, then marked all documentation as FOUO and stored them securely.

CONTINGENCY PLANNING (CP)

CP-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with contingency planning responsibilities:
 1. Agency-level contingency planning policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the contingency planning policy and the associated contingency planning controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the contingency planning policy and procedures; and
- c. Review and update the current contingency planning:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI, or training simulations or exercises; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI, or training simulations or exercises.

Discussion: Contingency planning policy and procedures address the controls in the CP family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of contingency planning policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to contingency planning policy and procedures include assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies,

standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

CP-2 CONTINGENCY PLAN

[Priority 2]

Control:

- a. Develop a contingency plan for the system that:
 1. Identifies essential mission and business functions and associated contingency requirements;
 2. Provides recovery objectives, restoration priorities, and metrics;
 3. Addresses contingency roles, responsibilities, assigned individuals with contact information;
 4. Addresses maintaining essential mission and business functions despite a system disruption, compromise, or failure;
 5. Addresses eventual, full system restoration without deterioration of the controls originally planned and implemented;
 6. Addresses the sharing of contingency information; and
 7. Is reviewed and approved by agency head or their designee;
- b. Distribute copies of the contingency plan to organizational personnel with contingency planning or incident response duties;
- c. Coordinate contingency planning activities with incident handling activities;
- d. Review the contingency plan for the system annually;
- e. Update the contingency plan to address changes to the organization, system, or environment of operation and problems encountered during contingency plan implementation, execution, or testing;
- f. Communicate contingency plan changes to organizational personnel with contingency planning or incident response duties;
- g. Incorporate lessons learned from contingency plan testing, training, or actual contingency activities into contingency testing and training; and
- h. Protect the contingency plan from unauthorized disclosure and modification.

Discussion: Contingency planning for systems is part of an overall program for achieving continuity of operations for organizational mission and business functions. Contingency planning addresses system restoration and implementation of alternative mission or business processes when systems are compromised or breached. Contingency planning is considered throughout the system development life cycle and is a fundamental part of the system design. Systems can be designed for redundancy, to provide backup capabilities, and for resilience.

Contingency plans reflect the degree of restoration required for organizational systems since not all systems need to fully recover to achieve the level of continuity of operations desired. System recovery objectives reflect applicable laws, executive orders, directives, regulations, policies, standards, guidelines, organizational risk tolerance, and system impact level.

Actions addressed in contingency plans include orderly system degradation, system shutdown, fallback to a manual mode, alternate information flows, and operating in modes reserved for when systems are under attack. By coordinating contingency planning with incident handling activities, organizations ensure that the necessary planning activities are in place and activated in the event of an incident. Incident response planning is part of contingency planning for organizations and is addressed in the IR (Incident Response) family.

Related Controls: CP-3, CP-4, CP-6, CP-7, CP-8, CP-9, CP-10, IR-4, IR-6, IR-8, MA-6, MP-2, MP-4, MP-5, PL-2, SA-15, SC-7, SC-23, SI-12.

Control Enhancements:

(1) CONTINGENCY PLAN | COORDINATE WITH RELATED PLANS

[Priority 2]

Control:

Coordinate contingency plan development with organizational elements responsible for related plans.

Discussion: Plans that are related to contingency plans include Business Continuity Plans, Disaster Recovery Plans, Critical Infrastructure Plans, Continuity of Operations Plans, Crisis Communications Plans, Insider Threat Implementation Plans, Data Breach Response Plans, Cyber Incident Response Plans, Breach Response Plans, and Occupant Emergency Plans.

Related Controls: None.

(3) CONTINGENCY PLAN | RESUME MISSION AND BUSINESS FUNCTIONS

[Priority 2]

Control:

Plan for the resumption of essential mission and business functions within twenty-four (24) hours of contingency plan activation.

Discussion: Organizations may choose to conduct contingency planning activities to resume mission and business functions as part of business continuity planning or as part of business impact analyses. Organizations prioritize the resumption of mission and business functions. The time period for resuming mission and business functions may be dependent on the severity and extent of the disruptions to the system and its supporting infrastructure.

Related Controls: None.

(8) CONTINGENCY PLAN | IDENTIFY CRITICAL ASSETS

[Priority 2]

Control:

Identify critical system assets supporting essential mission and business functions.

Discussion: Organizations may choose to identify critical assets as part of criticality analysis, business continuity planning, or business impact analyses. Organizations identify critical system assets so that additional controls can be employed (beyond the controls routinely implemented) to help ensure that organizational mission and business functions can continue to be conducted during contingency operations. The identification of critical information assets also facilitates the prioritization of organizational resources. Critical system assets include technical and operational aspects. Technical aspects include system components, information technology services, information technology products, and mechanisms. Operational aspects include procedures (i.e., manually executed operations) and personnel (i.e., individuals operating technical controls and/or executing manual procedures). Organizational program protection plans can assist in identifying critical assets.

Related Controls: CM-8, RA-9.

CP-3 CONTINGENCY TRAINING

[Priority 3]

Control:

- a. Provide contingency training to system users consistent with assigned roles and responsibilities:
 1. Within thirty (30) days of assuming a contingency role or responsibility;
 2. When required by system changes; and
 3. Annually thereafter; and
- b. Review and update contingency training content annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI, or training simulations or exercises.

Discussion: Contingency training provided by organizations is linked to the assigned roles and responsibilities of organizational personnel to ensure that the appropriate content and level of detail is included in such training. For example, some individuals may only need to know when and where to report for duty during contingency operations and if normal duties are affected; system administrators may require additional training on how to establish systems at alternate processing and storage sites; and organizational officials may receive more specific training on how to conduct mission-essential functions in designated off-site locations and how to establish communications with other governmental entities for purposes of coordination on contingency-related activities. Training for contingency roles or responsibilities reflects the specific continuity requirements in the contingency plan. Events that may precipitate an update to contingency training content include, but are not limited to, contingency plan testing or an actual contingency (lessons learned), assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. At the discretion of the organization, participation in a contingency plan test or exercise, including

lessons learned sessions subsequent to the test or exercise, may satisfy contingency plan training requirements.

Related Controls: AT-2, AT-3, AT-4, CP-2, CP-4, CP-8, IR-2, IR-4.

CP-4 CONTINGENCY PLAN TESTING

[Priority 3]

Control:

- a. Test the contingency plan for the system annually using the following tests to determine the effectiveness of the plan and the readiness to execute the plan: checklists, walk-through and tabletop exercises, simulations (parallel or full interrupt), or comprehensive exercises.
- b. Review the contingency plan test results; and
- c. Initiate corrective actions, if needed.

Discussion: Methods for testing contingency plans to determine the effectiveness of the plans and identify potential weaknesses include checklists, walk-through and tabletop exercises, simulations (parallel or full interrupt), and comprehensive exercises. Organizations conduct testing based on the requirements in contingency plans and include a determination of the effects on organizational operations, assets, and individuals due to contingency operations. Organizations have flexibility and discretion in the breadth, depth, and timelines of corrective actions.

Related Controls: AT-3, CP-2, CP-3, CP-8, CP-9, IR-3, IR-4, PL-2, SR-2.

Control Enhancements:

(1) CONTINGENCY PLAN TESTING | COORDINATE WITH RELATED PLANS

[Priority 3]

Control:

Coordinate contingency plan testing with organizational elements responsible for related plans.

Discussion: Plans related to contingency planning for organizational systems include Business Continuity Plans, Disaster Recovery Plans, Continuity of Operations Plans, Crisis Communications Plans, Critical Infrastructure Plans, Cyber Incident Response Plans, and Occupant Emergency Plans. Coordination of contingency plan testing does not require organizations to create organizational elements to handle related plans or to align such elements with specific plans. However, it does require that if such organizational elements are responsible for related plans, organizations coordinate with those elements.

Related Controls: IR-8.

CP-6 ALTERNATE STORAGE SITE

[Priority 2]

Control:

- a. Establish an alternate storage site, including necessary agreements to permit the storage and retrieval of system backup information; and
- b. Ensure that the alternate storage site provides controls equivalent to that of the primary site.

Discussion: Alternate storage sites are geographically distinct from primary storage sites and maintain duplicate copies of information and data if the primary storage site is not available. Similarly, alternate processing sites provide processing capability if the primary processing site is not available. Geographically distributed architectures that support contingency requirements may be considered alternate storage sites. Items covered by alternate storage site agreements include environmental conditions at the alternate sites, access rules for systems and facilities, physical and environmental protection requirements, and coordination of delivery and retrieval of backup media. Alternate storage sites reflect the requirements in contingency plans so that organizations can maintain essential mission and business functions despite compromise, failure, or disruption in organizational systems.

Related Controls: CP-2, CP-7, CP-8, CP-9, CP-10, MP-4, MP-5, PE-3.

Control Enhancements:

(1) ALTERNATE STORAGE SITE | SEPARATION FROM PRIMARY SITE

[Priority 2]

Control:

Identify an alternate storage site that is sufficiently separated from the primary storage site to reduce susceptibility to the same threats.

Discussion: Threats that affect alternate storage sites are defined in organizational risk assessments and include natural disasters, structural failures, hostile attacks, and errors of omission or commission. Organizations determine what is considered a sufficient degree of separation between primary and alternate storage sites based on the types of threats that are of concern. For threats such as hostile attacks, the degree of separation between sites is less relevant.

Related Controls: RA-3.

(3) ALTERNATE STORAGE SITE | ACCESSIBILITY

[Priority 2]

Control:

Identify potential accessibility problems to the alternate storage site in the event of an area-wide disruption or disaster and outline explicit mitigation actions.

Discussion: Area-wide disruptions refer to those types of disruptions that are broad in geographic scope with such determinations made by organizations based on organizational

assessments of risk. Explicit mitigation actions include duplicating backup information at other alternate storage sites if access problems occur at originally designated alternate sites or planning for physical access to retrieve backup information if electronic accessibility to the alternate site is disrupted.

Related Controls: RA-3.

CP-7 ALTERNATE PROCESSING SITE

[Priority 2]

Control:

- a. Establish an alternate processing site, including necessary agreements to permit the transfer and resumption of operations for essential mission and business functions within the time period defined in the system contingency plan(s) when the primary processing capabilities are unavailable;
- b. Make available at the alternate processing site, the equipment and supplies required to transfer and resume operations or put contracts in place to support delivery to the site within the organization-defined time period for transfer and resumption; and
- c. Provide controls at the alternate processing site that are equivalent to those at the primary site.

Discussion: Alternate processing sites are geographically distinct from primary processing sites and provide processing capability if the primary processing site is not available. The alternate processing capability may be addressed using a physical processing site or other alternatives, such as failover to a cloud-based service provider or other internally or externally provided processing service. Geographically distributed architectures that support contingency requirements may also be considered alternate processing sites. Controls that are covered by alternate processing site agreements include the environmental conditions at alternate sites, access rules, physical and environmental protection requirements, and the coordination for the transfer and assignment of personnel. Requirements are allocated to alternate processing sites that reflect the requirements in contingency plans to maintain essential mission and business functions despite disruption, compromise, or failure in organizational systems.

Related Controls: CP-2, CP-6, CP-8, CP-9, CP-10, MA-6, PE-3, PE-11, PE-12, PE-17.

Control Enhancements:

(1) ALTERNATE PROCESSING SITE | SEPARATION FROM PRIMARY SITE

[Priority 2]

Control:

Identify an alternate processing site that is sufficiently separated from the primary processing site to reduce susceptibility to the same threats.

Discussion: Threats that affect alternate processing sites are defined in organizational assessments of risk and include natural disasters, structural failures, hostile attacks, and errors of omission or commission. Organizations determine what is considered a sufficient degree of separation between primary and alternate processing sites based on the types of threats that are of concern. For threats such as hostile attacks, the degree of separation between sites is less relevant.

Related Controls: RA-3.

(2) ALTERNATE PROCESSING SITE | ACCESSIBILITY

[Priority 2]

Control:

Identify potential accessibility problems to alternate processing sites in the event of an area-wide disruption or disaster and outlines explicit mitigation actions.

Discussion: Area-wide disruptions refer to those types of disruptions that are broad in geographic scope with such determinations made by organizations based on organizational assessments of risk.

Related Controls: RA-3.

(3) ALTERNATE PROCESSING SITE | PRIORITY OF SERVICE

[Priority 2]

Control:

Develop alternate processing site agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives).

Discussion: Priority of service agreements refer to negotiated agreements with service providers that ensure that organizations receive priority treatment consistent with their availability requirements and the availability of information resources for logical alternate processing and/or at the physical alternate processing site. Organizations establish recovery time objectives as part of contingency planning.

Related Controls: None.

CP-8 TELECOMMUNICATIONS SERVICES

[Priority 2]

Control:

Establish alternate telecommunications services, including necessary agreements to permit the resumption of system operations for essential mission and business functions within the time period as defined in the system contingency plan(s) when the primary telecommunications capabilities are unavailable at either the primary or alternate processing or storage sites.

Discussion: Telecommunications services (for data and voice) for primary and alternate processing and storage sites are in scope for CP-8. Alternate telecommunications services reflect

the continuity requirements in contingency plans to maintain essential mission and business functions despite the loss of primary telecommunications services. Organizations may specify different time periods for primary or alternate sites. Alternate telecommunications services include additional organizational or commercial ground-based circuits or lines, network-based approaches to telecommunications, or the use of satellites. Organizations consider factors such as availability, quality of service, and access when entering into alternate telecommunications agreements.

Related Controls: CP-2, CP-6, CP-7, SC-7.

Control Enhancements:

(1) TELECOMMUNICATIONS SERVICES | PRIORITY OF SERVICE PROVISIONS

[Priority 2]

Control:

- a. Develop primary and alternate telecommunications service agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives); and
- b. Request Telecommunications Service Priority for all telecommunications services used for national security emergency preparedness if the primary and/or alternate telecommunications services are provided by a common carrier.

Discussion: Organizations consider the potential mission or business impact in situations where telecommunications service providers are servicing other organizations with similar priority of service provisions. Telecommunications Service Priority (TSP) is a Federal Communications Commission (FCC) program that directs telecommunications service providers (e.g., wireline and wireless phone companies) to give preferential treatment to users enrolled in the program when they need to add new lines or have their lines restored following a disruption of service, regardless of the cause. The FCC sets the rules and policies for the TSP program, and the Department of Homeland Security manages the TSP program. The TSP program is always in effect and not contingent on a major disaster or attack taking place. Federal sponsorship is required to enroll in the TSP program.

Related Controls: None.

(2) TELECOMMUNICATIONS SERVICES | SINGLE POINTS OF FAILURE

[Priority 2]

Control:

Obtain alternate telecommunications services to reduce the likelihood of sharing a single point of failure with primary telecommunications services.

Discussion: In certain circumstances, telecommunications service providers or services may share the same physical lines, which increases the vulnerability of a single failure point. It is

important to have provider transparency for the actual physical transmission capability for telecommunication services.

Related Controls: None.

CP-9 SYSTEM BACKUP

[Priority 2]

Control:

- a. Conduct backups of user-level information contained in operational systems for essential business functions as required by the contingency plans;
- b. Conduct backups of system-level information contained in the system as required by the contingency plans;
- c. Conduct backups of system documentation, including security- and privacy-related documentation as required by the contingency plans; and
- d. Protect the confidentiality, integrity, and availability of backup information.

Discussion: System-level information includes system state information, operating system software, middleware, application software, and licenses. User-level information includes information other than system-level information. Mechanisms employed to protect the integrity of system backups include digital signatures and cryptographic hashes. Protection of system backup information while in transit is addressed by MP-5 and SC-8. System backups reflect the requirements in contingency plans as well as other organizational requirements for backing up information. Organizations may be subject to laws, executive orders, directives, regulations, or policies with requirements regarding specific categories of information (e.g., criminal justice information). Organizational personnel consult with the senior agency official for privacy and legal counsel regarding such requirements.

Related Controls: CP-2, CP-6, CP-10, MP-4, MP-5, SC-8, SC-12, SC-13, SI-4.

Control Enhancements:

(1) SYSTEM BACKUP | TESTING FOR RELIABILITY AND INTEGRITY

[Priority 2]

Control:

Test backup information as required by the contingency plans to verify media reliability and information integrity.

Discussion: Organizations need assurance that backup information can be reliably retrieved. Reliability pertains to the systems and system components where the backup information is stored, the operations used to retrieve the information, and the integrity of the information being retrieved. Independent and specialized tests can be used for each of the aspects of reliability. For example, decrypting and transporting (or transmitting) a random sample of backup files from the alternate storage or backup site and comparing the information to the same information at the primary processing site can provide such assurance.

Related Controls: CP-4.

(8) SYSTEM BACKUP | CRYPTOGRAPHIC PROTECTION

[Priority 2]

Control:

Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of CJI.

Discussion: The selection of cryptographic mechanisms is based on the need to protect the confidentiality and integrity of backup information. The strength of mechanisms selected is commensurate with the security category or classification of the information. Cryptographic protection applies to system backup information in storage at both primary and alternate locations. Organizations that implement cryptographic mechanisms to protect information at rest also consider cryptographic key management solutions.

Related Controls: SC-12, SC-13, SC-28.

CP-10 SYSTEM RECOVERY AND RECONSTITUTION

[Priority 2]

Control:

Provide for the recovery and reconstitution of the system to a known state within the timeframe as required by the contingency plans after a disruption, compromise, or failure.

Discussion: Recovery is executing contingency plan activities to restore organizational mission and business functions. Reconstitution takes place following recovery and includes activities for returning systems to fully operational states. Recovery and reconstitution operations reflect mission and business priorities; recovery point, recovery time, and reconstitution objectives; and organizational metrics consistent with contingency plan requirements. Reconstitution includes the deactivation of interim system capabilities that may have been needed during recovery operations. Reconstitution also includes assessments of fully restored system capabilities, reestablishment of continuous monitoring activities, system reauthorization (if required), and activities to prepare the system and organization for future disruptions, breaches, compromises, or failures. Recovery and reconstitution capabilities can include automated mechanisms and manual procedures. Organizations establish recovery time and recovery point objectives as part of contingency planning.

Related Controls: CP-2, CP-4, CP-6, CP-7, CP-9, IR-4, SA-8.

Control Enhancements:

(2) SYSTEM RECOVERY AND RECONSTITUTION | TRANSACTION RECOVERY

[Priority 2]

Control:

Implement transaction recovery for systems that are transaction-based.

Discussion: Transaction-based systems include database management systems and transaction processing systems. Mechanisms supporting transaction recovery include transaction rollback and transaction journaling.

Related Controls: None.

IDENTIFICATION AND AUTHENTICATION (IA)

Identification is a unique, auditable representation of an identity within an information system usually in the form of a simple character string for each individual user, machine, software component, or any other entity. Authentication refers to mechanisms or processes to verify the identity of a user, process, or device, as a prerequisite to allowing access to a system's resources.

IA-0 USE OF ORIGINATING AGENCY IDENTIFIERS IN TRANSACTIONS AND INFORMATION EXCHANGES

[Existing]

An FBI authorized originating agency identifier (ORI) shall be used in each transaction on CJIS systems in order to identify the sending agency and to ensure the proper level of access for each transaction. The original identifier between the requesting agency and the CSA/SIB/Channeler shall be the ORI, and other agency identifiers, such as user identification or personal identifier, an access device mnemonic, or the Internet Protocol (IP) address.

Agencies may act as a servicing agency and perform transactions on behalf of authorized agencies requesting the service. Servicing agencies performing inquiry transactions on behalf of another agency may do so using the requesting agency's ORI. Servicing agencies may also use their own ORI to perform inquiry transactions on behalf of a requesting agency if the means and procedures are in place to provide an audit trail for the current specified retention period. Because the agency performing the transaction may not necessarily be the same as the agency requesting the transaction, the CSA/SIB/Channeler shall ensure that the ORI for each transaction can be traced, via audit trail, to the specific agency which is requesting the transaction.

Audit trails can be used to identify the requesting agency if there is a reason to inquire into the details surrounding why an agency ran an inquiry on a subject. Agencies assigned a limited access ORI shall not use the full access ORI of another agency to conduct an inquiry transaction.

NOTE: This control will be included in AC-3 Access Enforcement when modernized.

IA-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to authorized personnel:
 1. Agency/Entity identification and authentication policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the identification and authentication policy and the associated identification and authentication controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the identification and authentication policy and procedures; and

- c. Review and update the current identification and authentication:
 - 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 - 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

DISCUSSION: Identification and authentication policy and procedures address the controls in the IA family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of identification and authentication policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to identification and authentication policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: AC-1, PS-8, SI-12.

Control Enhancements: None.

IA-2 IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS)

[Existing] [Priority 1]

Control:

Uniquely identify and authenticate organizational users and associate that unique identification with processes acting on behalf of those users.

DISCUSSION: Organizations can satisfy the identification and authentication requirements by complying with the requirements in [HSPD 12]. Organizational users include employees or individuals who organizations consider to have an equivalent status to employees (e.g., contractors and guest researchers). Unique identification and authentication of users applies to all accesses other than those that are explicitly identified in AC-14 and that occur through the authorized use of group authenticators without individual authentication. Since processes execute on behalf of groups and roles, organizations may require unique identification of individuals in group accounts or for detailed accountability of individual activity.

Organizations employ passwords, physical authenticators, or biometrics to authenticate user identities or, in the case of multi-factor authentication, some combination thereof. Access to organizational systems is defined as either local access or network access. Local access is any access to organizational systems by users or processes acting on behalf of users, where access is

obtained through direct connections without the use of networks. Network access is access to organizational systems by users (or processes acting on behalf of users) where access is obtained through network connections (i.e., nonlocal accesses). Remote access is a type of network access that involves communication through external networks. Internal networks include local area networks and wide area networks.

The use of encrypted virtual private networks for network connections between organization-controlled endpoints and non-organization-controlled endpoints may be treated as internal networks with respect to protecting the confidentiality and integrity of information traversing the network. Identification and authentication requirements for non-organizational users are described in IA-8.

Related Controls: AC-2, AC-3, AC-4, AC-14, AC-17, AC-18, AU-1, AU-6, IA-4, IA-5, IA-8, MA-4, MA-5, PE-2, PL-4, SA-4, SA-8.

Control Enhancements:

(1) IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS) | MULTI-FACTOR AUTHENTICATION TO PRIVILEGED ACCOUNTS

[Priority 1]

Control:

Implement multi-factor authentication for access to privileged accounts.

DISCUSSION: Multi-factor authentication requires the use of two or more different factors to achieve authentication. The authentication factors are defined as follows: something you know (e.g., a personal identification number [PIN]), something you have (e.g., a physical authenticator such as a cryptographic private key), or something you are (e.g., a biometric). Multi-factor authentication solutions that feature physical authenticators include hardware authenticators that provide time-based or challenge-response outputs and smart cards such as the U.S. Government Personal Identity Verification (PIV) card or the Department of Defense (DoD) Common Access Card (CAC). In addition to authenticating users at the system level (i.e., at logon), organizations may employ authentication mechanisms at the application level, at their discretion, to provide increased security. Regardless of the type of access (i.e., local, network, remote), privileged accounts are authenticated using multi-factor options appropriate for the level of risk. Organizations can add additional security measures, such as additional or more rigorous authentication mechanisms, for specific types of access.

Related Controls: AC-5, AC-6.

(2) IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS) | MULTI-FACTOR AUTHENTICATION TO NON-PRIVILEGED ACCOUNTS

[Priority 1]

Control:

Implement multi-factor authentication for access to non-privileged accounts.

DISCUSSION: Multi-factor authentication requires the use of two or more different factors to achieve authentication. The authentication factors are defined as follows: something you know

(e.g., a personal identification number [PIN]), something you have (e.g., a physical authenticator such as a cryptographic private key), or something you are (e.g., a biometric). Multi-factor authentication solutions that feature physical authenticators include hardware authenticators that provide time-based or challenge-response outputs and smart cards such as the U.S. Government Personal Identity Verification card or the DoD Common Access Card. In addition to authenticating users at the system level, organizations may also employ authentication mechanisms at the application level, at their discretion, to provide increased information security. Regardless of the type of access (i.e., local, network, remote), non-privileged accounts are authenticated using multi-factor options appropriate for the level of risk. Organizations can provide additional security measures, such as additional or more rigorous authentication mechanisms, for specific types of access.

Related Controls: AC-5.

(8) IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS) | ACCESS TO ACCOUNTS — REPLAY RESISTANT

[Priority 1]

Control:

Implement replay-resistant authentication mechanisms for access to privileged and non-privileged accounts.

DISCUSSION: Authentication processes resist replay attacks if it is impractical to achieve successful authentications by replaying previous authentication messages. Replay-resistant techniques include protocols that use nonces or challenges such as time synchronous or cryptographic authenticators.

Related Controls: None.

(12) IDENTIFICATION AND AUTHENTICATION (ORGANIZATIONAL USERS) | ACCEPTANCE OF PIV CREDENTIALS

[Priority 1]

Control:

Accept and electronically verify Personal Identity Verification-compliant credentials.

DISCUSSION: Acceptance of Personal Identity Verification (PIV)-compliant credentials applies to organizations implementing logical access control and physical access control systems. PIV-compliant credentials are those credentials issued by federal agencies that conform to FIPS Publication 201 and supporting guidance documents. The adequacy and reliability of PIV card issuers are authorized using [SP 800-79-2]. Acceptance of PIV-compliant credentials includes derived PIV credentials, the use of which is addressed in [SP 800-166]. The DOD Common Access Card (CAC) is an example of a PIV credential.

Related Controls: None.

IA-3 DEVICE IDENTIFICATION AND AUTHENTICATION

[Priority 2]

Control:

Uniquely identify and authenticate agency-managed devices before establishing network connections. In the instance of local connection, the device must be approved by the agency and the device must be identified and authenticated prior to connection to an agency asset.

DISCUSSION: Devices that require unique device-to-device identification and authentication are defined by type, device, or a combination of type and device. Organization-defined device types include devices that are not owned by the organization. Systems use shared known information (e.g., Media Access Control [MAC], Transmission Control Protocol/Internet Protocol [TCP/IP] addresses) for device identification or organizational authentication solutions (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.1x and Extensible Authentication Protocol [EAP], RADIUS server with EAP-Transport Layer Security [TLS] authentication, Kerberos) to identify and authenticate devices on local and wide area networks. Organizations determine the required strength of authentication mechanisms based on the security categories of systems and mission or business requirements. Because of the challenges of implementing device authentication on a large scale, organizations can restrict the application of the control to a limited number/type of devices based on mission or business needs.

Related Controls: AC-17, AC-18, AC-19, AU-6, CA-3, CA-9, IA-4, IA-5, IA-11, SI-4.

IA-4 IDENTIFIER MANAGEMENT

[Existing] [Priority 2]

Control:

Manage system identifiers by:

- a. Receiving authorization from organizational personnel with identifier management responsibilities to assign an individual, group, role, service, or device identifier;
- b. Selecting an identifier that identifies an individual, group, role, service, or device;
- c. Assigning the identifier to the intended individual, group, role, service, or device; and
- d. Preventing reuse of identifiers for one (1) year.

DISCUSSION: Common device identifiers include Media Access Control (MAC) addresses, Internet Protocol (IP) addresses, or device-unique token identifiers. The management of individual identifiers is not applicable to shared system accounts. Typically, individual identifiers are the usernames of the system accounts assigned to those individuals. In such instances, the account management activities of AC-2 use account names provided by IA-4. Identifier management also addresses individual identifiers not necessarily associated with system accounts. Preventing the reuse of identifiers implies preventing the assignment of previously used individual, group, role, service, or device identifiers to different individuals, groups, roles, services, or devices.

Related Controls: AC-5, IA-2, IA-3, IA-5, IA-8, IA-12, MA-4, PE-2, PE-3, PE-4, PL-4, PS-3, PS-4, PS-5.

Control Enhancements:

(4) IDENTIFIER MANAGEMENT | IDENTIFY USER STATUS

[Priority 2]

Control:

Manage individual identifiers by uniquely identifying each individual as agency or nonagency.

DISCUSSION: Characteristics that identify the status of individuals include contractors, foreign nationals, and non-organizational users. Identifying the status of individuals by these characteristics provides additional information about the people with whom organizational personnel are communicating. For example, it might be useful for a government employee to know that one of the individuals on an email message is a contractor.

Related Controls: None.

IA-5 AUTHENTICATOR MANAGEMENT

[Priority 1]

Control:

Manage system authenticators by:

- a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;
- b. Establishing initial authenticator content for any authenticators issued by the organization;
- c. Ensuring that authenticators have sufficient strength of mechanism for their intended use;
- d. Establishing and implementing administrative procedures for initial authenticator distribution, for lost or compromised or damaged authenticators, and for revoking authenticators;
- e. Changing default authenticators prior to first use;
- f. Changing or refreshing memorized secret authenticators annually or when there is evidence of authenticator compromise; changing or refreshing all other authenticator types as they expire or when there is evidence of authenticator compromise;
- g. Protecting authenticator content from unauthorized disclosure and modification;
- h. Requiring individuals to take, and having devices implement, specific controls to protect authenticators; and
- i. Changing authenticators for group or role accounts when membership to those accounts changes.
- j. AAL2 Specific Requirements

Control:

All credential service providers (CSPs) authenticating claimants at Authenticator Assurance Level 2 (AAL2) SHALL be assessed on the following criteria:

- (1) Authentication SHALL occur by the use of either a multi-factor authenticator or a combination of two single-factor authenticators.

SUPPLEMENTAL GUIDANCE: A multi-factor authenticator requires two factors to execute a single authentication event, such as a cryptographically-secure device with an integrated biometric sensor that is required to activate the device. Nine different authenticator types are recognized, representing something you know (a memorized secret), something you have (a physical authenticator), or combinations of physical authenticators with either memorized secrets or biometric modalities (something you are). Multi-factor (MF) authentication is required at AAL2. MF authentication at AAL2 may be performed using the following AAL2 permitted authenticator types: MF OTP Device, MF Crypto Software, or MF Crypto Device; or a memorized secret used in combination with the following permitted single-factor authenticators: Look-Up Secret, Out-of-Band authenticator, SF OTP Device, SF Crypto Software, or SF Crypto Device.

- (2) If the multi-factor authentication process uses a combination of two single-factor authenticators, then it SHALL include a Memorized Secret authenticator and a possession-based authenticator.

SUPPLEMENTAL GUIDANCE: Multifactor authentication requires the use of two different authentication factors. See IA-5 j (1) for permitted authenticator types at AAL2.

- (3) Cryptographic authenticators used at AAL2 SHALL use approved cryptography.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation. Since verifiers and cryptographic authenticators must use the same algorithms to successfully authenticate, assessment of the verifier also assesses the authenticators that may be used.

- (4) At least one authenticator used at AAL2 SHALL be replay resistant.

SUPPLEMENTAL GUIDANCE: Replay resistance is a characteristic of most, although not all, physical authenticators. A given output of the authenticator is required to be accepted for only one authentication transaction. For example, the output of a time-based OTP device or an out-of-band device is considered replay resistant if it can only be used for at most one authentication transaction during its validity period. If it can be used for more than one during this period, it is not replay resistant.

- (5) Communication between the claimant and verifier SHALL be via an authenticated protected channel.

SUPPLEMENTAL GUIDANCE: Communication between claimant or user and verifier or agency is required to be via an encrypted channel that authenticates the verifier to provide confidentiality of the authenticator output and resistance to Man-in-the-Middle (MitM) attacks. This is typically accomplished using the Transport Level Security (TLS) protocol. Mutual authentication of the communication channel is not required unless that is part of the process of authenticating the claimant. Accordingly, the verifier is only responsible the use of an appropriately secure communications protocol.

- (6) Verifiers operated by federal government agencies at AAL2 SHALL be validated to meet the requirements of FIPS 140 Level 1.

SUPPLEMENTAL GUIDANCE: Verifiers operated by or on behalf of government agencies are required to be validated to meet FIPS 140 requirements. The FIPS 140 requirements generally apply to cryptographic modules (both hardware and software).

- (7) Authenticators procured by federal government agencies SHALL be validated to meet the requirements of FIPS 140 Level 1.

SUPPLEMENTAL GUIDANCE: The FIPS 140 requirements generally apply to cryptographic modules (both hardware and software). While authenticators are not directly the responsibility of the CSP (particularly in the case of bring- your-own authenticators), the CSP is still responsible for ensuring that a sufficiently strong and FIPS 140 validated authenticator is being used. Binding of CSP-supplied authenticators that are known to meet validation criteria is sufficient.

- (8) If a device such as a smartphone is used in the authentication process, then the unlocking of that device (typically done using a PIN or biometric) SHALL NOT be considered one of the authentication factors.

SUPPLEMENTAL GUIDANCE: This requirement applies to multi-factor authenticators resident on a smartphone or similar device; single-factor authenticators on such devices would only provide a single (physical) authentication factor. Unlocking of a device such as a smartphone may be done for any number of reasons unrelated to authentication, and such devices are normally in an unlocked state for a period of time thereafter. Human action such as entry of a memorized secret or presentation of a biometric factor needs to be provided that is directly associated with the authentication event. Generally, it is not possible for a verifier to know that the device had been locked or if the unlock process met the requirements for the relevant authenticator type.

- (9) If a biometric factor is used in authentication at AAL2, then the performance requirements stated in IA-5 m Biometric Requirements SHALL be met.

SUPPLEMENTAL GUIDANCE: Detailed conformance criteria applicable to the use of biometrics are contained in section IA-5 m Biometric Requirements. Since verification of biometric factors is not deterministic due to measurement errors in collection of the

biometric information, evaluation of performance, and, most importantly, false accept rate, is important to ensure security of the authentication process.

(10) Reauthentication of the subscriber SHALL be repeated at least once per 12 hours during an extended usage session.

SUPPLEMENTAL GUIDANCE: Reauthentication is required to mitigate the risks associated with an authenticated endpoint that has been abandoned by the subscriber or has been misappropriated by an attacker while authenticated. At AAL2, providing a memorized secret or biometric factor is sufficient for reauthentication prior to the expiration time.

(11) Reauthentication of the subscriber SHALL be repeated following any period of inactivity lasting 30 minutes or longer.

SUPPLEMENTAL GUIDANCE: Reauthentication is required to mitigate the risks associated with an authenticated endpoint that has been abandoned by the subscriber or has been misappropriated by an attacker while authenticated. At AAL2, providing a memorized secret or biometric factor is sufficient for reauthentication prior to the expiration time.

(12) The CSP SHALL employ appropriately tailored security controls from the moderate baseline of security controls defined in the CJISSECPOL.

The CSP SHALL ensure that the minimum assurance-related controls for moderate-impact systems are satisfied.

SUPPLEMENTAL GUIDANCE: NIST SP 800-53 provides a comprehensive catalog of controls, three security control baselines (low, moderate, and high impact), and guidance for tailoring the appropriate baseline to specific needs and risk environments for federal information systems. These controls are the operational, technical, and management safeguards to maintain the integrity, confidentiality, and security of federal information systems and are intended to be used in conjunction with the NIST risk management framework outlined in SP 800-37 and SP 800-63-3 section 5 Digital Identity Risk Management. NIST SP 800-53 presents security control baselines determined by the security categorization of the information system (low, moderate or high) from NIST FIPS 199 Standards for Security Categorization of Federal Information and Information Systems. For IAL2, the moderate baseline controls (see <https://nvd.nist.gov/800-53/Rev4/impact/moderate>) may be considered the starting point for the selection, enhancement, and tailoring of the security controls presented. Guidance on tailoring the control baselines to best meet the organization's risk environment, systems and operations is presented in SP 800-53 section 3.2. Tailoring Baseline Security Controls.

While SP 800-53 and other NIST Special Publications in the SP-800-XXX series apply to federal agencies for the implementation of the Federal Information Security Management Act (FISMA), non-federal entities providing services for federal information services also are subject to FISMA and should similarly use SP 800-53 and associated publications for appropriate controls. Non-federal entities may be subject to and conformant with other

applicable controls systems and processes for information system security (e.g., FEDRAMP, ISO/IEC 27001). SP-63A allows the application of equivalent controls from such standards and processes to meet conformance with this criterion.

- (13) The CSP SHALL comply with records retention policies in accordance with applicable laws and regulations.

SUPPLEMENTAL GUIDANCE: It is recommended that CSPs document any specific retention policies they are subject to, in accordance with applicable laws, regulations, or policies, including any National Archives and Records Administration (NARA) records retention schedules that may apply.

The CSP is responsible for the proper handling, protection, and retention or disposal of any sensitive data it collects, even after it ceases to provide identity proofing and enrollment services. A CSP may document its policies and procedures for the management of the data it collects in a data handling plan or other document.

- (14) If the CSP opts to retain records in the absence of any mandatory requirements, then the CSP SHALL conduct a risk management process, including assessments of privacy and security risks to determine how long records should be retained and SHALL inform subscribers of that retention policy.

SUPPLEMENTAL GUIDANCE: This is a conditional requirement and depends on the basis for CSP records retention. Absent clear jurisdictional requirements, risk management processes, including privacy and security risk assessment, need to be performed for records retention decisions. The records retention duration is required to be derived from a risk-based decision process.

k. Privacy requirements that apply to all CSPs, verifiers, and RPs.

- (1) The CSP SHALL employ appropriately tailored privacy controls from the CJISSECPOL.

SUPPLEMENTAL GUIDANCE: This requirement establishes overall privacy posture of the CSP.

- (2) If the CSP processes attributes for purposes other than identity proofing, authentication, or attribute assertions (collectively “identity service”), related fraud mitigation, or to comply with law or legal process, then the CSP SHALL implement measures to maintain predictability and manageability commensurate with the associated privacy risk.

SUPPLEMENTAL GUIDANCE: Predictability and manageability measures include providing clear notice, obtaining subscriber consent, and enabling selective use and disclosure of attributes. Predictability is meant to build trust and provide accountability and

requires full understanding (and disclosure) of how the attribute information will be used. Manageability also builds trust by demonstrating a CSPs ability to control attribute information throughout processing – collection, maintenance, retention.

1. General requirements applicable to AAL2 authentication process.

- (1) CSPs SHALL provide subscriber instructions on how to appropriately protect a physical authenticator against theft or loss.

SUPPLEMENTAL GUIDANCE: Instruction should address aspects of protecting the specific type of authenticator being used.

- (2) The CSP SHALL provide a mechanism to revoke or suspend the authenticator immediately upon notification from subscriber that loss or theft of the authenticator is suspected.

SUPPLEMENTAL GUIDANCE: The CSP needs to have a documented procedure to allow subscribers to report lost or stolen physical authenticators, and to revoke or suspend such authenticators promptly when reported. Subscribers need to be instructed (see GEN-1) the procedure for reporting loss or theft.

- (3) If required by the authenticator type descriptions in IA-5(1), then the verifier SHALL implement controls to protect against online guessing attacks.

SUPPLEMENTAL GUIDANCE: Throttling or rate limiting is key to resistance against online guessing attacks. This is generally required for memorized secrets or when the authenticator output of a look-up secret, OOB, or OTP authenticator may have less than 64 bits of entropy.

- (4) If required by the authenticator type descriptions in IA-5(1) and the description of a given authenticator does not specify otherwise, then the verifier SHALL limit consecutive failed authentication attempts on a single account to no more than 100.

SUPPLEMENTAL GUIDANCE: Throttling or rate limiting is key to resistance against online guessing attacks. It is important that it be implemented in a non-abrupt manner as described in the specification so that it is not usable as a denial-of-service mechanism by an attacker. Additional techniques MAY be used to reduce the likelihood that an attacker will lock the legitimate claimant out as a result of rate limiting. These include:

- Requiring the claimant to complete a CAPTCHA before attempting authentication.
- Requiring the claimant to wait following a failed attempt for a period of time that increases as the account approaches its maximum allowance for consecutive failed attempts (e.g., 30 seconds up to an hour).

- Accepting only authentication requests that come from a white list of IP addresses from which the subscriber has been successfully authenticated before. Leveraging other risk-based or adaptive authentication techniques to identify user behavior that falls within, or out of, typical norms. These might, for example, include use of IP address, geolocation, timing of request patterns, or browser metadata.

(5) If signed attestations are used, then they SHALL be signed using a digital signature that provides at least the minimum security strength specified in the latest revision of 112 bits as of the date of this publication.

SUPPLEMENTAL GUIDANCE: Attestations are sometimes provided by cryptographic authenticators to securely indicate their capabilities, e.g., that they are hardware-based or that they have characteristics such as two-factor capability. For the attestations to be useful, these signatures need to use algorithms and keys that are sufficiently strong.

(6) If the verifier and CSP are separate entities (as shown by the dotted line in Figure 6 Digital Identity Model), then communications between the verifier and CSP SHALL occur through a mutually-authenticated secure channel (such as a client-authenticated TLS connection).

SUPPLEMENTAL GUIDANCE: In cases where the verifier and CSP are separate, it is important that this not create additional security vulnerabilities as compared with an integrated verifier/CSP combination. This requirement ensures that there is not an opportunity to perform eavesdropping or active attacks on the channel between them.

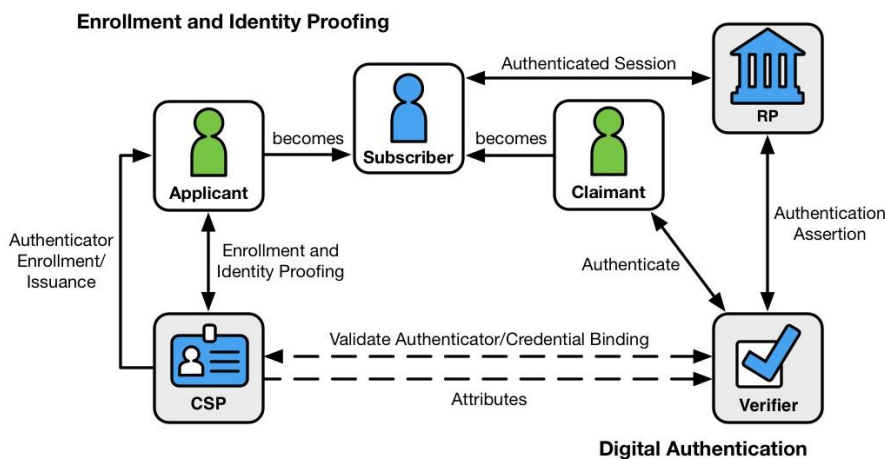


Figure 6 – Digital Identity Model

(7) If the CSP provides the subscriber with a means to report loss, theft, or damage to an authenticator using a backup or alternate authenticator, then that authenticator SHALL be either a memorized secret or a physical authenticator.

SUPPLEMENTAL GUIDANCE: It is important that the loss of control of an authenticator be quickly reported to the CSP. To balance between the need to easily and promptly report this and the risk of a fraudulent report, a backup authenticator, either a memorized

secret or physical authenticator, should be usable by the subscriber to make this report. Only a single, single-factor authenticator is required.

- (8) If the CSP chooses to verify an address of record (i.e., email, telephone, postal) and suspend authenticator(s) reported to have been compromised, then...The suspension SHALL be reversible if the subscriber successfully authenticates to the CSP using a valid (i.e., not suspended) authenticator and requests reactivation of an authenticator suspended in this manner.

SUPPLEMENTAL GUIDANCE: Reversibility of suspension is intended to minimize the impact of inadvertent loss reports from the subscriber and in some cases from an attacker who may be attempting to deny service to the subscriber.

- (9) If and when an authenticator expires, it SHALL NOT be usable for authentication.

SUPPLEMENTAL GUIDANCE: Expiration is used by some CSPs to limit the security exposure from an authenticator that is lost but the loss has not been detected/reported and revoked.

- (10) The CSP SHALL have a documented process to require subscribers to surrender or report the loss of any physical authenticator containing attribute certificates signed by the CSP as soon as practical after expiration or receipt of a renewed authenticator.

SUPPLEMENTAL GUIDANCE: The requirement for surrender or destruction of expired authenticators minimizes the possibility that authentication with an expired authenticator will be attempted. PKI-based authenticators that are collected or known to be destroyed also do not need to be included in certificate revocation lists.

- (11) CSPs SHALL revoke the binding of authenticators immediately upon notification when an online identity ceases to exist (e.g., subscriber's death, discovery of a fraudulent subscriber), when requested by the subscriber, or when the CSP determines that the subscriber no longer meets its eligibility requirements.

SUPPLEMENTAL GUIDANCE: Prompt revocation ensures that unauthorized parties are not able to use the authenticator to make unauthorized access to the subscriber account. Revocation at subscriber request can affect only a single authenticator; the other classes of revocation generally affect all authenticators associated with the subscriber's account.

- (12) The CSP SHALL have a documented process to require subscribers to surrender or report the loss of any physical authenticator containing certified attributes signed by the CSP within five (5) days after revocation or termination takes place.

SUPPLEMENTAL GUIDANCE: This requirement blocks the use of the authenticator's certified attributes in offline situations between revocation/termination and expiration

of the certification. Prompt revocation ensures that unauthorized parties are not able to use the authenticator to make unauthorized access to the subscriber account. Collection or destruction also minimizes the dependence on (and growth of) certificate revocation lists, which are not always 100% effective in accomplishing revocation, particularly in offline situations.

m. Biometric Requirements

- (1) Biometrics SHALL be used only as part of multi-factor authentication with a physical authenticator (something you have).

SUPPLEMENTAL GUIDANCE: For a variety of reasons listed here, a biometric factor is not considered to be an authenticator by itself. The risks associated with biometric factors are largely mitigated by binding the biometric with a specific physical authenticator.

- The biometric False Match Rate (FMR) does not provide confidence in the authentication of the subscriber by itself. In addition, FMR does not account for spoofing attacks.
- Biometric comparison is probabilistic, whereas the other authentication factors are deterministic.
- Biometric template protection schemes provide a method for revoking biometric credentials that is comparable to other authentication factors (e.g., PKI certificates and passwords). However, the availability of such solutions is limited, and standards for testing these methods are under development.
- Biometric characteristics do not constitute secrets. They can be obtained online or by taking a picture of someone with a camera phone (e.g., facial images) with or without their knowledge, lifted from objects someone touches (e.g., latent fingerprints), or captured with high resolution images (e.g., iris patterns). While presentation attack detection (PAD) technologies (e.g., liveness detection) can mitigate the risk of these types of attacks, additional trust in the sensor or biometric processing is required to ensure that PAD is operating in accordance with the needs of the CSP and the subscriber.

- (2) An authenticated protected channel between sensor (or an endpoint containing a sensor that resists sensor replacement) and verifier SHALL be established.

SUPPLEMENTAL GUIDANCE: This requirement ensures that biometric data that flows across the network to the verifier is protected from disclosure and that an attacker cannot substitute a “skimmer” or other fraudulent replacement for the biometric sensor. If the biometric factor is verified directly on a multi-factor authenticator and the sensor

is tightly integrated with it, that local connection does not require an authenticated protected channel.

- (3) The sensor or endpoint SHALL be authenticated prior to capturing the biometric sample from the claimant.

SUPPLEMENTAL GUIDANCE: This requirement ensures that the biometric data being verified is obtained from the expected sensor rather than from a device that may be spoofing biometric information. This is generally not required when the biometric factor is verified in an endpoint that is tightly integrated with the sensor in a manner that resists sensor replacement.

- (4) The biometric system SHALL operate with an FMR [ISO/IEC 2382-37] of 1 in 1000 or better. This FMR SHALL be achieved under conditions of a conformant attack (i.e., zero-effort impostor attempt) as defined in [ISO/IEC 30107-1].

SUPPLEMENTAL GUIDANCE: Since biometric comparison is an approximate match, an operating point threshold is chosen by the verifier that balances false matches and false non-matches. To operate adequately as a verifier, a 1 in 1000 or better false match rate is required.

- (5) The biometric system SHALL allow no more than 5 consecutive failed authentication attempts or 10 consecutive failed attempts if PAD demonstrating at least 90% resistance to presentation attacks is implemented.

SUPPLEMENTAL GUIDANCE: With a false accept rate of as much as 1 in 1000 zero-effort attempts, the ability to make a large number of biometric authentication attempts would result in an unacceptably high probability of mis-authentication. This limit is comparable to that provided by several commercial products (mobile devices) currently on the market.

- (6) Once the limit on authentication failures has been reached, the biometric authenticator SHALL either:
- i. Impose a delay of at least 30 seconds before the next attempt, increasing exponentially with each successive attempt, or
 - ii. disable the biometric user authentication and offer another factor (e.g., a different biometric modality or a PIN/Passcode if it is not already a required factor) if such an alternative method is already available.

SUPPLEMENTAL GUIDANCE: Following a number of consecutive biometric match failures that exceeds the limit in IA-5 m (5), subsequent attempts need to be either aggressively delayed (e.g., 1 minute before the following failed attempt, 2 minutes before the second following attempt) or another authentication or biometric modality associated with the same physical authenticator needs to be used.

- (7) The verifier SHALL make a determination of sensor and endpoint performance, integrity, and authenticity.

SUPPLEMENTAL GUIDANCE: The verifier needs to have a basis for determining that biometric verification meets the necessary performance requirements. This may be accomplished by authenticating the sensor or endpoint, by a certification by an approved accreditation authority, or by runtime interrogation of a signed attestation.

- (8) If biometric comparison is performed centrally, then use of the biometric as an authentication factor SHALL be limited to one or more specific devices that are identified using approved cryptography.

SUPPLEMENTAL GUIDANCE: The ability to use a biometric factor on an arbitrary device greatly increases the value of breached biometric data. For this reason, the use of the biometric factor is limited to specific devices for each subscriber. A separate key is required since the main authentication key is only unlocked upon successful comparison of the biometric factor.

- (9) If biometric comparison is performed centrally, then a separate key SHALL be used for identifying the device.

SUPPLEMENTAL GUIDANCE: Since the main authentication key has not yet been unlocked, a separate key is required for identifying the specific device(s) that the biometric may be used with.

- (10) If biometric comparison is performed centrally, then biometric revocation, referred to as biometric template protection in ISO/IEC 24745, SHALL be implemented.

SUPPLEMENTAL GUIDANCE: Central databases of biometric templates are an attractive target for attackers. The ability to securely revoke biometric factors is required in response to that threat.

- (11) If biometric comparison is performed centrally, all transmission of biometrics SHALL be over the authenticated protected channel.

SUPPLEMENTAL GUIDANCE: Because of the replay potential of biometric data, biometric information needs to be distributed in a manner that minimizes the opportunity for attackers to intercept the data either by eavesdropping on MitM attacks.

- (12) Biometric samples and any biometric data derived from the biometric sample such as a probe produced through signal processing SHALL be zeroized immediately after any training or research data has been derived.

SUPPLEMENTAL GUIDANCE: If the biometric factor is used for any supplemental purpose, it is important that it not be a mechanism for breach of subscribers' biometric data.

n. Authenticator binding refers to the establishment of an association between a specific authenticator and a subscriber's account, enabling the authenticator to be used — possibly in conjunction with other authenticators — to authenticate for that account.

- (1) Authenticators SHALL be bound to subscriber accounts by either issuance by the CSP as part of enrollment or associating a subscriber-provided authenticator that is acceptable to the CSP.

SUPPLEMENTAL GUIDANCE: In the past, many physical authenticators were provided by the CSP. More recently, there has been a trend toward BYO authenticators, which can be both cost-effective for CSPs and convenient for the subscriber. This requirement ensures that such BYO authenticators are subject to approval by the CSP, primarily to ensure that they meet security requirements.

- (2) Throughout the digital identity lifecycle, CSPs SHALL maintain a record of all authenticators that are or have been associated with each identity.

SUPPLEMENTAL GUIDANCE: In order to authenticate subscribers successfully, the CSP needs to maintain a record of authenticators bound to each subscriber's account. In addition, a record of authenticators formerly bound to each account needs to be kept for forensic purposes.

- (3) The CSP or verifier SHALL maintain the information required for throttling authentication attempts.

SUPPLEMENTAL GUIDANCE: In order to successfully support the throttling of authentication attempts (see requirement IA-5.1 (3)), the CSP needs to maintain information on the number of consecutive failed authentication attempts.

- (4) The CSP SHALL also verify the type of user-provided authenticator so verifiers can determine compliance with requirements at each AAL.

SUPPLEMENTAL GUIDANCE: In order to determine compliance with AAL-specific requirements, the CSP needs to reliably determine some authenticator characteristics, such as whether the authenticator is hardware-based, whether it is a single-factor or multi-factor authenticator, and performance characteristics of associated biometric sensors. Mechanisms to do this include attestation certificates from the manufacturer and examination of the authenticator (particularly at account issuance). In the absence of this information, the CSP needs to assume that the authenticator is the weakest type that is consistent with the authentication protocol being used.

- (5) The record created by the CSP SHALL contain the date and time the authenticator was bound to the account.

SUPPLEMENTAL GUIDANCE: For forensic purposes it is useful to have a record of the period of time each authenticator is bound to the subscriber's account.

- (6) When any new authenticator is bound to a subscriber account, the CSP SHALL ensure that the binding protocol and the protocol for provisioning the associated key(s) are done at AAL2.

SUPPLEMENTAL GUIDANCE: If the process of binding an authenticator is not strong enough, an authenticator that is fraudulently bound to the account could be used by an attacker to gain access to a subscriber's account. The authentication factor being bound to the account needs to be included in the authentication process for the session in which the authenticator is bound.

- (7) Protocols for key provisioning SHALL use authenticated protected channels or be performed in person to protect against MitM attacks.

SUPPLEMENTAL GUIDANCE: For the same reasons that MitM attacks are of concern during authentication, they could occur during provisioning, which could result in the binding of an attacker's key to the account rather than the subscriber's key.

- (8) Binding of multi-factor authenticators SHALL require multi-factor authentication (or equivalent) at identity proofing.

SUPPLEMENTAL GUIDANCE: In order to prevent a subscriber with only single-factor authentication from up-leveling to multi-factor, binding of a multi-factor authenticator requires that the subscriber be multi-factor authenticated at the time the new authenticator is bound

- (9) At enrollment, the CSP SHALL bind at least one, and SHOULD bind at least two, physical (something you have) authenticators to the subscriber's online identity, in addition to a memorized secret or one or more biometrics.

SUPPLEMENTAL GUIDANCE: Executive order 13681 requires the use of multi-factor authentication for the release of personal data. Therefore, it is important that the CSP associate sufficient authentication factors at enrollment to make this possible. While all identifying information is self-asserted at IAL1, preservation of online material or an online reputation makes it undesirable to lose control of an account due to the loss of an authenticator. The second authenticator makes it possible to securely recover from an authenticator loss. For this reason, a CSP SHOULD bind at least two physical authenticators to the subscriber's credential at IAL1 as well.

(10) At enrollment, authenticators at AAL2 and IAL2 SHALL be bound to the account.

SUPPLEMENTAL GUIDANCE: In order to support higher identity assurance, correspondingly high authenticator assurance levels are required to ensure the proper use of the identity.

(11) If enrollment and binding are being done remotely and cannot be completed in a single electronic transaction, then the applicant SHALL identify themselves in each new binding transaction by presenting a temporary secret which was either established during a prior transaction, or sent to the applicant's phone number, email address, or postal address of record.

SUPPLEMENTAL GUIDANCE: The issuance or binding of authenticators may occur well after the enrollment process, following adjudication and eligibility determinations. It is necessary to securely associate the applicant that appears for identity proofing with the person appearing for authenticator issuance/binding in order to avoid mis-issuance of authenticators. At this point it is not possible to fully authenticate the applicant, but the use of a temporary secret provides the necessary protection for this one-time transaction.

(12) If enrollment and binding are being done remotely and cannot be completed in a single electronic transaction, then long-term authenticator secrets are delivered to the applicant within a protected session.

SUPPLEMENTAL GUIDANCE: Long-term secrets need to be protected against disclosure while they are sent to the applicant. This applies primarily to symmetric keys, such as for OTP authenticators, that are sent to the applicant by the CSP. "Protected session" in this context refers to an authenticated protected.

(13) If enrollment and binding are being done in person and cannot be completed in a single physical encounter, the applicant SHALL identify themselves in person by either using a secret as described in IA-5 n (12) above, or through use of a biometric that was recorded during a prior encounter.

SUPPLEMENTAL GUIDANCE: The issuance or binding of authenticators may occur well after the enrollment process, following adjudication and eligibility determinations. It is necessary to securely associate the applicant that appears for identity proofing with the person appearing for authenticator issuance/binding in order to avoid mis-issuance of authenticators. At this point it is not possible to fully authenticate the applicant, but the use of a temporary secret provides the necessary protection for this one-time transaction.

(14) If enrollment and binding are being done in person and cannot be completed in a single physical encounter, temporary secrets SHALL NOT be reused.

SUPPLEMENTAL GUIDANCE: The issuance or binding of authenticators may occur well after the enrollment process, following adjudication and eligibility determinations. It is necessary to securely associate the applicant that appears for identity proofing with the person appearing for authenticator issuance/binding in order to avoid mis-issuance of authenticators. A new secret for this purpose is required for each subsequent encounter.

- (15) If enrollment and binding are being done in person and cannot be completed in a single physical encounter and the CSP issues long-term authenticator secrets during a physical transaction, they SHALL be loaded locally onto a physical device that is issued in person to the applicant or delivered in a manner that confirms the address of record.

SUPPLEMENTAL GUIDANCE: To avoid misappropriation of long-term authenticator secrets at enrollment, the CSP is required to load the secrets onto authenticators directly, or deliver them to the new subscriber in a manner that confirms the address of record, typically by sending a short-term secret to that address that the new subscriber uses to obtain the long-term secret.

- (16) Before adding a new authenticator to a subscriber's account, the CSP SHALL first require the subscriber to authenticate at AAL2 (or a higher AAL) at which the new authenticator will be used.

SUPPLEMENTAL GUIDANCE: In order to maintain the significance of AALs and prevent attackers from leveraging lower AAL authentication to gain access to higher AAL resources, subscribers binding additional authenticators need to do so at the maximum AAL at which they will be used.

- (17) If the subscriber's account has only one authentication factor bound to it, the CSP SHALL require the subscriber to authenticate at AAL1 in order to bind an additional authenticator of a different authentication factor.

SUPPLEMENTAL GUIDANCE: This is a special-case, one-time only exception to IA-5(n)(17) to allow a single-factor account not subject to identity proofing (IAL1) to be upgraded to a multi-factor account. This provides a mechanism for such accounts to increase their authentication security.

- (18) If a subscriber loses all authenticators of a factor necessary to complete multi-factor authentication and has been identity proofed at IAL2, that subscriber SHALL repeat the identity proofing process described in IA-12.

SUPPLEMENTAL GUIDANCE: Repeating the identity proofing process is an onerous requirement when a subscriber is no longer able to complete multi-factor authentication, but it is necessary to avoid the security problems typically present in "account recovery" situations. This is the primary reason that the binding of multiple authenticators is recommended, particularly in the case of physical authenticators. The entire identity

proofing process need not be repeated if the CSP has maintained enough records of the evidence presented to repeat the verification phase of identity proofing.

- (19) If a subscriber loses all authenticators of a factor necessary to complete multi-factor authentication and has been identity proofed at IAL2 or IAL3, the CSP SHALL require the claimant to authenticate using a n authenticator of the remaining factor, if any, to confirm binding to the existing identity.

SUPPLEMENTAL GUIDANCE: While use of an authenticator at a different factor is only a single authentication factor (and therefore only AAL1), authentication in conjunction with the repeated identity proofing process provides assurance that the claimant is who they claim to be.

- (20) If the CSP opts to allow binding of a new memorized secret with the use of two physical authenticators, then it requires entry of a confirmation code sent to an address of record.

SUPPLEMENTAL GUIDANCE: Loss of a memorized secret is different from the loss of a physical authenticator because it is not mitigated by the binding of multiple authenticators. This alternate method of associating a new memorized secret may be used by CSPs to avoid the need for repeating identity proofing (Refer to IA-12).

- (21) If the CSP opts to allow binding of a new memorized secret with the use of two physical authenticators, then the confirmation code SHALL consist of at least 6 random alphanumeric characters generated by an approved random bit generator [SP 800-90Ar1].

SUPPLEMENTAL GUIDANCE: The confirmation code is required to have sufficient entropy and to be generated in a manner that cannot be predicted by an attacker.

- (22) If the CSP opts to allow binding of a new memorized secret with the use of two physical authenticators, then the confirmation code SHALL be valid for a maximum of 7 days but MAY be made valid up to 21 days via an exception process to accommodate addresses outside the direct reach of the U.S. Postal Service. Confirmation codes sent by means other than physical mail SHALL be valid for a maximum of 5 minutes.

SUPPLEMENTAL GUIDANCE: The confirmation code has a limited lifetime to mitigate the risk of loss or misappropriation in transit.

o. Session Management: The following requirements apply to applications where a session is maintained between the subscriber and relying party to allow multiple interactions without repeating the authentication event each time.

Once an authentication event has taken place, it is often desirable to allow the subscriber to continue using the application across multiple subsequent interactions without requiring them

to repeat the authentication event. This requirement is particularly true for federation scenarios where the authentication event necessarily involves several components and parties coordinating across a network.

(1) Session Binding Requirements: A session occurs between the software that a subscriber is running — such as a browser, application, or operating system (i.e., the session subject) — and the RP or CSP that the subscriber is accessing (i.e., the session host).

- a. A session is maintained by a session secret which SHALL be shared between the subscriber's software and the service being accessed.

SUPPLEMENTAL GUIDANCE: This secret binds the two ends of the session, allowing the subscriber to continue using the service over time.

- b. The secret SHALL be presented directly by the subscriber's software or possession of the secret SHALL be proven using a cryptographic mechanism.

SUPPLEMENTAL GUIDANCE: The session secret is considered a short-term secret, so direct presentation of a shared secret is permitted, even at AAL2 or AAL3.

- c. The secret used for session binding SHALL be generated by the session host in direct response to an authentication event.

SUPPLEMENTAL GUIDANCE: The session secret needs to be directly associated with authentication so that it isn't inadvertently provided to the wrong session.

- d. A session SHALL NOT be considered at a higher AAL than the authentication event.

SUPPLEMENTAL GUIDANCE: Each session has an associated maximum AAL at which it can be used that is derived from the authentication AAL; this is associated with the session and its secret by the CSP/RP.

- e. Secrets used for session binding SHALL be generated by the session host during an interaction, typically immediately following authentication.

SUPPLEMENTAL GUIDANCE: It is the responsibility of the host (RP/CSP/Verifier) to generate session secrets, not the subscriber.

- f. Secrets used for session binding SHALL be generated by an approved random bit generator [SP 800-90Ar1].

SUPPLEMENTAL GUIDANCE: The use of a high-quality random bit generator is important to ensure that an attacker cannot guess the session secret.

- g. Secrets used for session binding SHALL contain at least 64 bits of entropy.

SUPPLEMENTAL GUIDANCE: The use of a high-quality random bit generator is important to ensure that an attacker cannot guess the session secret.

- h. Secrets used for session binding SHALL be erased or invalidated by the session subject when the subscriber logs out.

SUPPLEMENTAL GUIDANCE: At a minimum, the CSP/RP needs to ensure that the session secret can no longer be used following logout. If possible, the secret should be erased on the subscriber endpoint as well.

- i. Secrets used for session binding SHALL be sent to and received from the device using an authenticated protected channel.

SUPPLEMENTAL GUIDANCE: Session secrets, particularly when directly presented, need to be protected against eavesdropping and MitM attacks. This is typically accomplished using the Transport Level Security (TLS) protocol.

- j. Secrets used for session binding SHALL time out and not be accepted after the times specified in IA-5(j)(10) and IA-5(j)(11) as appropriate for the AAL.

SUPPLEMENTAL GUIDANCE: This requirement is in support of the reauthentication requirements in AAL2-*, AAL3-*, and REAUTH-*. The proper way to ensure that a session is logged out is to invalidate the session secrets associated with that session. A new session secret will need to be generated and associated with any session that is about to be established from the same endpoint.

- k. Secrets used for session binding SHALL NOT be available to insecure communications between the host and subscriber's endpoint.

SUPPLEMENTAL GUIDANCE: User endpoints such as browsers that support both secure and insecure communications typically have mechanisms to flag information (e.g., cookies) that are only available to secure sessions. These mechanisms are required to be used for session management secrets. See also IA-5 o (7).

- l. Authenticated sessions SHALL NOT fall back to an insecure transport, such as from https to http, following authentication.

SUPPLEMENTAL GUIDANCE: In some cases, endpoints supporting https provide, primarily for legacy purposes, the ability to connect via http as well. If not done properly, this can make the site vulnerable to a "downgrade attack" where a session switches from https to http. This must not happen for authenticated sessions. If session secrets are managed properly, this downgrade interferes with the continuity of the session.

- m. URLs or POST content SHALL contain a session identifier that SHALL be verified by the RP to ensure that actions taken outside the session do not affect the protected session.

SUPPLEMENTAL GUIDANCE: Unique session identifiers in the URL or POST content are used to ensure that sessions are not vulnerable to cross-site request forgery (CSRF). Note that the session identifier is separate and different from the session secret; under no circumstances should the session secret be included in a URL.

- n. Browser cookies SHALL be tagged to be accessible only on secure (HTTPS) sessions.

SUPPLEMENTAL GUIDANCE: Browser cookies have an optional “secure” flag to ensure that they are not accidentally transmitted over a non-secure channel. This flag must be set for session secrets.

- o. Browser cookies SHALL be accessible to the minimum practical set of hostnames and paths.

SUPPLEMENTAL GUIDANCE: Browser cookies have a scope parameter that limits the sites from to which the cookie can be sent; this should be specified as specifically as possible to limit access to the session secret as narrowly as practical.

- p. Expiration of browser cookies SHALL NOT be depended upon to enforce session timeouts.

SUPPLEMENTAL GUIDANCE: While browser cookies have an expiration time, enforcement of session timeouts must occur at the RP/CSP and not at the user endpoint. Cookie expiration may, however, be used to limit accumulation of cookies in the browser.

- q. The presence of an OAuth access token SHALL NOT be interpreted by the RP as presence of the subscriber, in the absence of other signals.

SUPPLEMENTAL GUIDANCE: Access tokens, used in federated identity systems, may be valid after the authentication session has ended and the subscriber has left.

(2) Reauthentication Requirements

- a. Continuity of authenticated sessions SHALL be based upon the possession of a session secret issued by the verifier at the time of authentication and optionally refreshed during the session.

SUPPLEMENTAL GUIDANCE: This is a reiteration of requirement IA-5 o (1).

- b. Session secrets SHALL be non-persistent, i.e., they SHALL NOT be retained across a restart of the associated application or a reboot of the host device.

SUPPLEMENTAL GUIDANCE: Session secrets are not to be maintained across a restart of the associated application or a reboot of the host device in order to minimize the likelihood that a misappropriated logged in device can be exploited.

- c. Periodic reauthentication of sessions (at least every 12 hours per session) SHALL be performed to confirm the continued presence of the subscriber at an authenticated session.

SUPPLEMENTAL GUIDANCE: In order to protect against a subscriber leaving a logged-in endpoint, timeouts are defined for session inactivity and overall session length. The timer for these timeouts is reset by a reauthentication transaction. Higher AALs have more stringent (shorter) reauthentication timeouts. Following expiration of the session timer, the subscriber is required to start a new session by authenticating.

- d. A session SHALL NOT be extended past the guidelines in IA-5 o (2) a – j based on presentation of the session secret alone.

SUPPLEMENTAL GUIDANCE: The existence and possession of a session secret does not consider whether the subscriber continued to be in control of the session endpoint. To mitigate this risk, the session secret is only valid for a limited period of time. While the session secret is “something you have”, it is not an authenticator.

- e. Prior to session expiration, the reauthentication time limit SHALL be extended by prompting the subscriber for the authentication factor(s) of a memorized secret or biometric.

SUPPLEMENTAL GUIDANCE: Before the session times out, the subscriber should be given an opportunity to reauthenticate to extend the session. The subscriber may be prompted when an idle timeout is about to expire, to allow them to cause activity and thereby avoid the need to reauthenticate.

Note: At AAL2, a memorized secret or biometric, and not a physical authenticator, is required because the session secret is something you have, and an additional authentication factor is required to continue the session.

- f. If federated authentication is being used, then since the CSP and RP often employ separate session management technologies, there SHALL NOT be any assumption of correlation between these sessions.

SUPPLEMENTAL GUIDANCE: When an RP session expires and the RP requires reauthentication, it is entirely possible that the session at the CSP has not expired and that a new assertion could be generated from this session at the CSP without reauthenticating the user.

- g. An RP requiring reauthentication through a federation protocol SHALL — if possible within the protocol — specify the maximum (see IA-5 j (10)) acceptable authentication age to the CSP.

SUPPLEMENTAL GUIDANCE: In some applications, RPs may require a “fresh” authentication to meet its authentication risk requirements. By specifying maximum age, the RP can proactively request the CSP to obtain a new authentication to meet that requirement.

- h. If federated authentication is being used and an RP has specific authentication age (see IA-5 j [10]) requirements that it has communicated to the CSP, then the CSP SHALL reauthenticate the subscriber if they have not been authenticated within that time period.

SUPPLEMENTAL GUIDANCE: When the RP communicates its authentication freshness requirements to the CSP, the CSP is expected to reauthenticate the subscriber to support a session that meets those requirements.

- i. If federated authentication is being used, the CSP SHALL communicate the authentication event time to the RP to allow the RP to decide if the assertion is sufficient for reauthentication and to determine the time for the next reauthentication event.

SUPPLEMENTAL GUIDANCE: When federation authentication is being used, the authentication assertion from the CSP needs to contain the authentication event time to allow the RP to request reauthentication at an appropriate interval if it has specific authentication age requirements.

DISCUSSION: Authenticators include passwords, cryptographic devices, biometrics, certificates, one-time password devices, and ID badges. Device authenticators include certificates and passwords. Initial authenticator content is the actual content of the authenticator (e.g., the initial password). In contrast, the requirements for authenticator content contain specific criteria or characteristics (e.g., minimum password length). Developers may deliver system components with factory default authentication credentials (i.e., passwords) to allow for initial installation and configuration. Default authentication credentials are often well known, easily discoverable, and present a significant risk. The requirement to protect individual authenticators may be implemented via control PL-4 or PS-6 for authenticators in the possession of individuals and by controls AC-3, AC-6, and SC-28 for authenticators stored in organizational

systems, including passwords stored in hashed or encrypted formats or files containing encrypted or hashed passwords accessible with administrator privileges.

Systems support authenticator management by organization-defined settings and restrictions for various authenticator characteristics (e.g., minimum password length, validation time window for time synchronous one-time tokens, and number of allowed rejections during the verification stage of biometric authentication). Actions can be taken to safeguard individual authenticators, including maintaining possession of authenticators, not sharing authenticators with others, and immediately reporting lost, stolen, or compromised authenticators. Authenticator management includes issuing and revoking authenticators for temporary access when no longer needed.

Related Controls: AC-3, AC-6, CM-6, IA-2, IA-4, IA-7, IA-8, MA-4, PE-2, PL-4, SC-12, SC-13.

Control Enhancements:

(1) AUTHENTICATOR MANAGEMENT | AUTHENTICATOR TYPES

[Priority 1]

Control:

(a) Memorized Secret Authenticators and Verifiers:

1. Maintain a list of commonly-used, expected, or compromised passwords via API or download from a third party. Update the list quarterly and when organizational passwords are suspected to have been compromised directly or indirectly. Compare current memorized secrets against the list quarterly;
2. Require immediate selection of a new password upon account recovery;
3. Allow user selection of long passwords and passphrases, including spaces and all printable characters;
4. Employ automated tools to assist the user in selecting strong password authenticators;
5. If chosen by the subscriber, memorized secrets SHALL be at least 8 characters in length.

SUPPLEMENTAL GUIDANCE: Memorized secret length is the most reliable metric determining strength against online and offline guessing attacks. The objective is primarily to defend against online attacks (with throttling of guesses) and to provide some protection against offline attacks, with the primary defense for such attacks being secure storage of the verifier.

6. If chosen by the CSP or verifier using an approved random number generator, memorized secrets SHALL be at least 6 characters in length.

SUPPLEMENTAL GUIDANCE: Memorized secret length is the most reliable metric determining strength against online and offline guessing attacks. The objective is primarily to defend against online attacks (with throttling of guesses) and to provide some protection against offline attacks, with the primary defense for such attacks being secure storage of the verifier.

7. Truncation of the secret SHALL NOT be performed.

SUPPLEMENTAL GUIDANCE: Memorized secrets that are longer than expected by the verifier might (but must not) be simply truncated to an acceptable length. This gives a false impression of security to the user if the verifier only checks a subset of the memorized secret.

8. Memorized secret verifiers SHALL NOT permit the subscriber to store a “hint” that is accessible to an unauthenticated claimant.

SUPPLEMENTAL GUIDANCE: The availability of memorized secret hints greatly weakens the strength of memorized secret authenticators.

9. Verifiers SHALL NOT prompt subscribers to use specific types of information (e.g., “What was the name of your first pet?”) when choosing memorized secrets.

SUPPLEMENTAL GUIDANCE: Prompts for specific information (often called Knowledge-based Authentication or Security Questions) encourage use of the same memorized secrets at multiple sites, which causes a vulnerability to “password stuffing” attacks. This guidance applies to account recovery situations as well as normal authentication.

10. When processing requests to establish and change memorized secrets, verifiers SHALL compare the prospective secrets against the list maintained as required by IA-5(1)(a)(1) that contains values known to be commonly used, expected, or compromised.

SUPPLEMENTAL GUIDANCE: The maintenance of a list of common memorized secrets that cannot be used by users protects provides protection against online attacks that might otherwise succeed before throttling mechanisms take effect to defend against these attacks. This is an alternative to the use of composition rules (requirements for particular character types, etc.) and can provide more customized protection against common memorized secrets. This list may include, but is not limited to:

- Passwords obtained from previous breach corpuses.
- Dictionary words.
- Repetitive or sequential characters (e.g., ‘aaaaaa’, ‘1234abcd’).
- Context-specific words, such as the name of the service, the username, and derivatives thereof.

11. If a chosen secret is found in the list, the CSP or verifier SHALL advise the subscriber that they need to select a different secret.

SUPPLEMENTAL GUIDANCE: The use of common memorized secrets greatly increases the vulnerability of the account to both online (guessing) and offline (cracking) attacks. This is an alternative to the use of composition rules (requirements for particular character

types, etc.) and can provide more customized protection against common memorized secrets.

12. If a chosen secret is found in the list, the CSP or verifier SHALL provide the reason for rejection.

SUPPLEMENTAL GUIDANCE: When a subscriber chooses a weak memorized secret, it is likely that they will choose another weak memorized secret that may or may not be on the blocklist. In addition to explaining to the user the reason for the rejection of their selection, it is helpful to provide coaching on better choices. Tools like password-strength meters are often useful in this situation.

13. If a chosen secret is found in the list, the CSP or verifier SHALL require the subscriber to choose a different value.

SUPPLEMENTAL GUIDANCE: When a subscriber chooses a weak memorized secret, the memorized secret change process is not complete until the subscriber has chosen a different value.

14. Verifiers SHALL implement a rate-limiting mechanism that effectively limits failed authentication attempts that can be made on the subscriber's account to no more than five.

SUPPLEMENTAL GUIDANCE: Rate limiting restricts the ability of an attacker to make many online guessing attacks on the memorized secret. Other requirements (e.g., minimum length of memorized secrets) depend on the existence of rate limiting, so effective rate limiting is an essential capability. Ideally, a rate limiting mechanism should restrict the attacker as much as possible without creating an opportunity for a denial-of-service attack against the subscriber.

15. Verifiers SHALL force a change of memorized secret if there is evidence of compromise of the authenticator.

SUPPLEMENTAL GUIDANCE: Although requiring routine periodic changes to memorized secrets is not recommended, it is important that verifiers have the capability to prompt memorized secrets on an emergency basis if there is evidence of a possible successful attack.

16. The verifier SHALL use approved encryption when requesting memorized secrets in order to provide resistance to eavesdropping and MitM attacks.

SUPPLEMENTAL GUIDANCE: As defined in Appendix A of the CJIS Security Policy, cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation.

17. The verifier SHALL use an authenticated protected channel when requesting memorized secrets in order to provide resistance to eavesdropping and MitM attacks.

SUPPLEMENTAL GUIDANCE: Communication between claimant and verifier is required to be via an encrypted channel that authenticates the verifier to provide confidentiality of the authenticator output and resistance to MitM attacks. This is typically accomplished using the Transport Level Security (TLS) protocol.

18. Verifiers SHALL store memorized secrets in a form that is resistant to offline attacks.

SUPPLEMENTAL GUIDANCE: Storage of memorized secret verifiers in a hashed form that is not readily reversed is a key protection against offline attacks. In no case should a verifier store memorized secrets in cleartext form. Criteria IA-5(1)(a)(20) – (22) provide more detail on how this is done.

19. Memorized secrets SHALL be salted and hashed using a suitable one-way key derivation function.

SUPPLEMENTAL GUIDANCE: Key derivation functions take a password, a salt, and a cost factor as inputs then generate a password hash. Their purpose is to make each password guessing trial by an attacker who has obtained a password hash file expensive and therefore the cost of a guessing attack high or prohibitive. Use of a key derivation with a salt, preferably with a time- and memory-hard key derivation function, provides the best protection against attackers that are able to obtain a copy of the verifier database.

20. The salt SHALL be at least 32 bits in length and be chosen arbitrarily to minimize salt value collisions among stored hashes.

SUPPLEMENTAL GUIDANCE: Salt values need to be large enough to make it impractical for an attacker to precompute hashed verifier values (so called rainbow tables). While rainbow tables are typically quite large, this requirement would increase their size by a factor of about 4.3 billion. If not chosen arbitrarily, the attacker might be able to anticipate the salt values that would be used, which would eliminate much of this advantage.

21. Both the salt value and the resulting hash SHALL be stored for each subscriber using a memorized secret authenticator

SUPPLEMENTAL GUIDANCE: In order to verify a memorized secret, it needs to be salted and hashed for comparison with the stored verifier (resulting hash). To do this, the salt value needs to be available, and since it is different for each user, needs to be stored with the verifier. It is impractical to verify a memorized secret if this is not done.

22. If an additional iteration of a key derivation function using a salt value known only to the verifier is performed, then this secret salt value SHALL be generated with an approved random bit generator and of sufficient length.

SUPPLEMENTAL GUIDANCE: An additional keyed hashing iteration using a key value that is secret and stored separately from the verifiers provides excellent protection against even attackers (“password crackers”) with substantial computing resources, provided the key is not also compromised. Accordingly, it is important that this salt, which is common to multiple users, be generated in a manner that is not vulnerable to compromise.

23. If an additional iteration of a key derivation function using a salt value known only to the verifier is performed, then this secret salt value SHALL provide at least the minimum-security strength.

SUPPLEMENTAL GUIDANCE: An additional keyed hashing iteration using a key value that is secret and stored separately from the verifiers provides excellent protection against even attackers (“password crackers”) with substantial computing resources, provided the key is not also compromised. Accordingly, it is important that this salt, which is common to multiple users, be of sufficient size to make cryptographic and brute-force attacks impractical.

Currently, the requirement is that the key be at least 112 bits in length.

24. If an additional iteration of a key derivation function using a salt value known only to the verifier is performed, then this secret salt value SHALL be stored separately from the memorized secrets.

SUPPLEMENTAL GUIDANCE: An additional keyed hashing iteration using a key value that is secret and stored separately from the verifiers provides excellent protection against even attackers (“password crackers”) with substantial computing resources, provided the key is not also compromised. Accordingly, it is important that this salt, which is common to multiple users, be stored separately so that it is unlikely to be compromised along with the verifier database. One way to do this is to perform this last hashing iteration on a physically separate processor, since it only requires a value to hash as input and provides the hashed value in response.

DISCUSSION: Password-based authentication applies to passwords regardless of whether they are used in single-factor or multi-factor authentication. Long passwords or passphrases are preferable over shorter passwords. Enforced composition rules provide marginal security benefits while decreasing usability. However, organizations may choose to establish certain rules for password generation (e.g., minimum character length for long passwords) under certain circumstances and can enforce this requirement in IA-5(1)(a)(5). Account recovery can occur, for example, in situations when a password is forgotten. Cryptographically protected passwords include salted one-way cryptographic hashes of passwords. The list of commonly used, compromised, or expected passwords includes passwords obtained from previous breach corpuses,

dictionary words, and repetitive or sequential characters. The list includes context-specific words, such as the name of the service, username, and derivatives thereof.

(b) Look-Up Secret Authenticators and Verifiers

1. CSPs creating look-up secret authenticators SHALL use an approved random bit generator to generate the list of secrets.

SUPPLEMENTAL GUIDANCE: The use of a high-quality random bit generator is important to ensure that an attacker cannot guess the look-up secret

2. Look-up secrets SHALL have at least 20 bits of entropy.

SUPPLEMENTAL GUIDANCE: Look-up secrets need to have enough entropy to ensure that brute-force guessing attacks do not succeed.

3. If look-up secrets are distributed online, then they SHALL be distributed over a secure channel in accordance with the post-enrollment binding requirements in IA-5(n)(16) through (22).

SUPPLEMENTAL GUIDANCE: Look-up secrets need to be distributed in a manner that minimizes the opportunity for attackers to intercept the secrets either by eavesdropping or MitM attacks.

4. Verifiers of look-up secrets SHALL prompt the claimant for the next secret from their authenticator or for a specific (e.g., numbered) secret.

SUPPLEMENTAL GUIDANCE: In most cases claimants will be prompted for the next unused memorized secret in a list but may be challenged to use a specific secret from a list.

5. A given secret from an authenticator SHALL be used successfully only once.

SUPPLEMENTAL GUIDANCE: Many threats, such as key logging, are enabled if the look-up secret can be used more than once.

6. If a look-up secret is derived from a grid (bingo) card, then each cell of the grid SHALL be used only once.

SUPPLEMENTAL GUIDANCE: Grid (bingo) cards are sometimes used to provide a rudimentary challenge-response authentication involving the claimant. However, an attacker such as a key logger that has persistent access to the endpoint can derive the contents of the grid, and potentially authenticate successfully, if grid entries are reused in subsequent authentication transactions.

Absent the ability to reuse grid squares, grid (bingo) cards will probably no longer be attractive as authenticators.

7. Verifiers SHALL store look-up secrets in a form that is resistant to offline attacks.

SUPPLEMENTAL GUIDANCE: Storage of look-up secret verifiers in a hashed form that is not readily reversed is a key protection against offline attacks. In no case should a verifier store look-up secrets in cleartext form.

8. If look-up secrets have at least 112 bits of entropy, then they SHALL be hashed with an approved one-way function

SUPPLEMENTAL GUIDANCE: Use of an approved one-way function effectively protects the look-up secrets from disclosure if the verifier is compromised. Salting of secrets with this amount of entropy is not required because it is not practical to mount brute-force or cryptographic attacks against secrets this large.

9. If look-up secrets have less than 112 bits of entropy, then they SHALL be salted and hashed using a suitable one-way key derivation function.

SUPPLEMENTAL GUIDANCE: Key derivation functions take a look-up secret, a salt, and a cost factor as inputs then generate a hash. Their purpose is to make each look-up secret guessing trial by an attacker who has obtained a look-up secret hash file expensive and therefore the cost of a guessing attack high or prohibitive. Use of a key derivation with a salt, preferably with a time- and memory-hard key derivation function, provides the best protection against attackers that are able to obtain a copy of the verifier database.

10. If look-up secrets have less than 112 bits of entropy, then the salt SHALL be at least 32 bits in length and be chosen arbitrarily to minimize salt value collisions among stored hashes.

SUPPLEMENTAL GUIDANCE: Salt values need to be large enough to make it impractical for an attacker to precompute hashed verifier values (so called rainbow tables). While rainbow tables are typically quite large, this requirement would increase their size by a factor of about 4.3 billion. If not chosen arbitrarily, the attacker might be able to anticipate the salt values that would be used, which would eliminate much of this advantage.

11. If look-up secrets have less than 112 bits of entropy, then both the salt value and the resulting hash SHALL be stored for each look-up secret.

SUPPLEMENTAL GUIDANCE: In order to verify a look-up secret, it needs to be salted and hashed for comparison with the stored verifier (resulting hash). To do this, the salt value needs to be available, and since it is different for each secret, needs to be stored with the verifier. It is impractical to verify a look-up secret if this is not done.

12. If look-up secrets that have less than 64 bits of entropy, then the verifier SHALL implement a rate-limiting mechanism that effectively limits the number of failed authentication attempts that can be made on the subscriber's account.

SUPPLEMENTAL GUIDANCE: Rate limiting restricts the ability of an attacker to make many online guessing attacks on the look-up secret. Other requirements (e.g., minimum length of look-up secrets) depend on the existence of rate limiting, so effective rate limiting is an essential capability. Ideally, a rate limiting mechanism should restrict the attacker as much as possible without creating an opportunity for a denial-of-service attack against the subscriber.

13. The verifier SHALL use approved encryption when requesting look-up secrets in order to provide resistance to eavesdropping and MitM attacks.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation.

14. The verifier SHALL use an authenticated protected channel when requesting look-up secrets in order to provide resistance to eavesdropping and MitM attacks.

SUPPLEMENTAL GUIDANCE: Communication between claimant and verifier is required to be via an encrypted channel that authenticates the verifier to provide confidentiality of the authenticator output and resistance to MitM attacks. This is typically accomplished using the Transport Level Security (TLS) protocol.

(c) Out-of-Band Authenticators and Verifiers

1. The out-of-band authenticator SHALL establish a separate channel with the verifier in order to retrieve the out-of-band secret or authentication request.

SUPPLEMENTAL GUIDANCE: A channel is considered to be out-of-band with respect to the primary communication channel (even if it terminates on the same device) provided the device does not leak information from one channel to the other without the authorization of the claimant.

2. Communication over the secondary channel SHALL be encrypted unless sent via the public switched telephone network (PSTN).

SUPPLEMENTAL GUIDANCE: The secondary channel requires protection to ensure that authentication secrets are not leaked to attackers. Legacy use of the PSTN as an OOB authentication medium is exempt from this requirement, although other requirements apply.

3. Methods that do not prove possession of a specific device, such as voice-over-IP (VoIP) or email, SHALL NOT be used for out-of-band authentication.

SUPPLEMENTAL GUIDANCE: Communication with VoIP phone numbers and email do not establish the possession of a specific device, so they are not suitable for use in out-of-band authentication which is used as a physical authenticator (something you have).

4. If PSTN is not being used for out-of-band communication, then the out-of-band authenticator SHALL uniquely authenticate itself by establishing an authenticated protected channel with the verifier.

SUPPLEMENTAL GUIDANCE: Communication between out-of-band device and verifier is required to be via an encrypted channel to provide confidentiality of the authenticator output and resistance to MitM attacks. This is typically accomplished using the Transport Level Security (TLS) protocol.

5. If PSTN is not being used for out-of-band communication, then the out-of-band authenticator SHALL communicate with the verifier using approved cryptography.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation.

6. If PSTN is not being used for out-of-band communication, then the key used to authenticate the out-of-band device SHALL be stored in suitably secure storage available to the authenticator application (e.g., keychain storage, TPM, TEE, secure element).

SUPPLEMENTAL GUIDANCE: The secret key associated with an out-of-band device or authenticator application is critical to the determination of “something you have” and needs to be well protected.

7. If the PSTN is used for out-of-band authentication and a secret is sent to the out-of-band device via the PSTN, then the out-of-band authenticator SHALL uniquely authenticate itself to a mobile telephone network using a SIM card or equivalent that uniquely identifies the device.

SUPPLEMENTAL GUIDANCE: Since the PSTN does not support the establishment of authenticated protected channels, the alternative method of authenticating the device via the PSTN is supported. Note that there are other specific requirements for use of the PSTN that also apply (see IA-5 (1) c (19) through (20)).

8. If the out-of-band authenticator sends an approval message over the secondary communication channel, it SHALL either accept transfer of a secret from the primary channel to be sent to the verifier via the secondary communications channel, or present a secret received via the secondary channel from the verifier and prompt the claimant to verify the consistency of that secret with the primary channel, prior to accepting a yes/no response from the claimant which it sends to the verifier.

SUPPLEMENTAL GUIDANCE: Most out-of-band verifiers operate by sending a secret over the secondary channel that the subscriber transfers to the primary channel (e.g., the capability to copy and paste from one app to another). Other methods are possible, however, specifically transferring from primary to secondary and user comparison of secrets sent to both channels (with approval being sent to the verifier over the secondary channel). It is good practice to display descriptive information relating to the authentication on the claimant's out-of-band device, to provide additional assurance that the transaction being approved by the subscriber is the correct one, and not from an attacker who exploits the subscriber's approval.

9. The verifier SHALL NOT store the identifying key itself, but SHALL use a verification method (e.g., an approved hash function or proof of possession of the identifying key) to uniquely identify the authenticator.

SUPPLEMENTAL GUIDANCE: In order for the out-of-band authenticator to be considered "something you have", it must be securely authenticated as a unique device or instance of a software-based authentication application. This is required to be done through proof of possession of a key by the authenticator, rather than presentation of the key itself. This provides verifier compromise resistance with respect to the authentication key.

10. Depending on the type of out-of-band authenticator, one of the following SHALL take place: transfer of a secret to the primary channel, transfer of a secret to the secondary channel, or verification of secrets by the claimant.

SUPPLEMENTAL GUIDANCE: Three different methods of associating the primary and secondary channel sessions are permitted. The intent of these methods is to establish approval for a specific authentication transaction, and to minimize the likelihood that an attacker with knowledge of when the subscriber authenticates can obtain approval for a rogue authentication.

11. If the out-of-band authenticator operates by transferring the secret to the primary channel, then the verifier SHALL transmit a random secret to the out-of-band authenticator and then wait for the secret to be returned on the primary communication channel.

SUPPLEMENTAL GUIDANCE: This is the most common form of out-of-band authentication where an authentication secret is transmitted to the out-of-band device and entered by the user for transmission on the primary channel.

12. If the out-of-band authenticator operates by transferring the secret to the secondary channel, then the verifier SHALL display a random authentication secret to the claimant via the primary channel and then wait for the secret to be returned on the secondary channel from the claimant's out-of-band authenticator.

SUPPLEMENTAL GUIDANCE: This is a less typical authentication flow but is also acceptable in that the secret securely associates possession and control of the out-of-band authenticator with the session being authenticated.

13. If the out-of-band authenticator operates by verification of secrets by the claimant, then the verifier SHALL display a random authentication secret to the claimant via the primary channel, send the same secret to the out-of-band authenticator via the secondary channel for presentation to the claimant, and then wait for an approval (or disapproval) message via the secondary channel.

SUPPLEMENTAL GUIDANCE: This is a somewhat more user-friendly authentication flow because it does not require the claimant to read and manually enter the authentication secret, but it carries the additional risk that the claimant will approve the authentication without actually comparing the secrets received from the independent channels. Approval is required to be obtained from the out-of-band authenticator rather than the primary channel because that at least establishes control of the authenticator.

14. The authentication SHALL be considered invalid if not completed within 10 minutes.

SUPPLEMENTAL GUIDANCE: Secrets used in out-of-band authentication are short-term secrets and need to have a definite lifetime. This requirement also relieves the verifier from the responsibility of long-term storage of the secrets.

15. Verifiers SHALL accept a given authentication secret only once during the validity period.

SUPPLEMENTAL GUIDANCE: In order to prevent an attacker who gains access to an authentication secret generated by the subscriber from using it, it is important that the secret only be valid for a single authentication. This requirement only applies when a secret is being transferred between the primary channel and the out-of-band authenticator.

16. The verifier SHALL generate random authentication secrets with at least 20 bits of entropy.

SUPPLEMENTAL GUIDANCE: Consistent with other short-term authentication secrets, 20 bits of entropy are required to provide resistance against brute force attacks. 6-digit numeric secrets (19.93 bits of entropy) are sufficiently close to 20 bits to be acceptable.

17. The verifier SHALL generate random authentication secrets using an approved random bit generator.

SUPPLEMENTAL GUIDANCE: The use of a high-quality random bit generator is important to ensure that an attacker cannot guess the out-of-band secret. Approved random bit generators are generally included in a FIPS 140-2 certified encryption module.

18. If the authentication secret has less than 64 bits of entropy, the verifier SHALL implement a rate-limiting mechanism that effectively limits the number of failed authentication attempts that can be made on the subscriber's account as described in IA-5 l (3) through (4).

SUPPLEMENTAL GUIDANCE: Rate limiting limits the opportunity for attackers to mount a brute-force attack on the out-of-band verifier. Since the out-of-band secret has a limited lifetime, it is sufficient to limit the number of attempts allowed during the (maximum) 10-minute lifetime of the secret.

19. If out-of-band verification is to be made using the PSTN, then the verifier SHALL verify that the pre-registered telephone number being used is associated with a specific physical device.

SUPPLEMENTAL GUIDANCE: Some telephone numbers, such as those that are associated with VoIP services, are not associated with a specific device and can receive calls and text messages without establishing possession and control of a specific device. Such telephone numbers are not suitable for OOB authentication. Services exist to distinguish telephone numbers that are associated with a device from those that aren't.

20. If out-of-band verification is to be made using the PSTN, then changing the pre-registered telephone number is considered to be the binding of a new authenticator and SHALL only occur as described in IA-5(n)(16) through (22).

SUPPLEMENTAL GUIDANCE: The binding of a new authenticator requires that the subscriber authenticate at the same or a higher AAL (currently AAL2) than that at which the authenticator will be used, and that a notification be sent to the subscriber. This is required to prevent attackers from changing the phone number of a PSTN-based out-of-band authenticator to one they control.

21. If PSTN is used for out-of-band authentication, then the CSP SHALL offer subscribers at least one alternate authenticator that is not RESTRICTED and can be used to authenticate at the required AAL.

SUPPLEMENTAL GUIDANCE: Use of the PSTN for out-of-band authentication involves additional risk, resulting in its being designated as a restricted authenticator. CSPs are required to provide subscribers with a meaningful alternative.

22. If PSTN is used for out-of-band authentication, then the CSP SHALL Provide meaningful notice to subscribers regarding the security risks of the RESTRICTED authenticator and availability of alternative(s) that are not RESTRICTED.

SUPPLEMENTAL GUIDANCE: Use of the PSTN for out-of-band authentication involves additional risk, resulting in its being designated as a restricted authenticator. CSPs are required to explain these risks to subscribers and offer more secure alternatives.

Currently, authenticators leveraging the public switched telephone network, including phone- and Short Message Service (SMS)-based one-time passwords (OTPs) are restricted. Other authenticator types may be added as additional threats emerge. Note that, among other requirements, even when using phone- and SMS-based OTPs, the agency also must verify that the OTP is being directed to a phone and not an IP address, such as with VoIP, as these accounts are not typically protected with multi-factor authentication.

23. If PSTN is used for out-of-band authentication, then the CSP SHALL address any additional risk to subscribers in its risk assessment.

SUPPLEMENTAL GUIDANCE: Use of the PSTN for out-of-band authentication involves additional risk, resulting in its being designated as a restricted authenticator. These risks need to be documented.

24. If PSTN is used for out-of-band authentication, then the CSP SHALL develop a migration plan for the possibility that the RESTRICTED authenticator is no longer acceptable at some point in the future and include this migration plan in its digital identity acceptance statement.

SUPPLEMENTAL GUIDANCE: Use of the PSTN for out-of-band authentication involves additional risk, resulting in its being designated as a restricted authenticator. A plan for eliminating them in the future needs to be documented.

(d) OTP Authenticators and Verifiers

1. The secret key and its algorithm SHALL provide at least the minimum security strength of 112 bits as of the date of this publication.

SUPPLEMENTAL GUIDANCE: The secret key used by an OTP authenticator needs to be sufficiently complex to resist online and offline attacks. An attacker may have the ability to observe the authenticator output at some point during its operation; it needs to be impractical for the secret key to be derived from a set of these observations.

2. The nonce SHALL be of sufficient length to ensure that it is unique for each operation of the device over its lifetime.

SUPPLEMENTAL GUIDANCE: If the nonce isn't long enough, the output of the authenticator will repeat, which represents an easily avoided vulnerability.

3. OTP authenticators — particularly software-based OTP generators — SHALL NOT facilitate the cloning of the secret key onto multiple devices.

SUPPLEMENTAL GUIDANCE: Like other physical authenticators, the use of OTP authenticators is premised upon the authenticator secret being present in a single authenticator so that it proves possession of a specific device. Mechanisms that would facilitate cloning the secret onto multiple devices include the ability to enroll more than one device producing the same OTP output and backup mechanisms, especially when software-based authenticators are used. Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

4. The authenticator output SHALL have at least 6 decimal digits (approximately 20 bits) of entropy.

SUPPLEMENTAL GUIDANCE: Consistent with other short-term authentication secrets, 20 bits of entropy are required to provide resistance against brute force attacks. 6-digit numeric secrets (19.93 bits of entropy) are sufficiently close to 20 bits to be acceptable.

5. If the nonce used to generate the authenticator output is based on a real-time clock, then the nonce SHALL be changed at least once every 2 minutes.

SUPPLEMENTAL GUIDANCE: The authenticator output needs to be changed often enough that there is reasonable assurance that it is in the possession of the claimant and that it is not susceptible to OTP-guessing attacks.

6. The OTP value associated with a given nonce SHALL be accepted only once.

SUPPLEMENTAL GUIDANCE: A fundamental premise of a “one-time” authenticator is that it can be used successfully only once during its validity period.

7. The symmetric keys used by authenticators are also present in the verifier and SHALL be strongly protected against compromise.

SUPPLEMENTAL GUIDANCE: Verifiers typically contain symmetric keys for all subscribers using OTP authenticators. This makes them a particularly rich target for attackers. While the protection of these keys is implementation-dependent and there is therefore no specific requirement for how the keys are protected, measures to prevent the exfiltration of the keys are needed. An example of such a measure is the storage of keys and generation of authenticator outputs in a separate device accessible only by the verifier.

8. If a single-factor OTP authenticator is being associated with a subscriber account, then the verifier or associated CSP SHALL use approved cryptography to either generate and exchange or to obtain the secrets required to duplicate the authenticator output.

SUPPLEMENTAL GUIDANCE: It is critical that authentication secrets be generated and transferred or negotiated securely. This includes the use of secure random number generators and protocols for transferring or negotiating (e.g., Diffie-Hellman) secret values.

Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation.

9. The verifier SHALL use approved encryption when collecting the OTP.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation.

10. The verifier SHALL use an authenticated protected channel when collecting the OTP.

SUPPLEMENTAL GUIDANCE: Communication between claimant and verifier is required to be via an encrypted channel that authenticates the verifier to provide confidentiality of the authenticator output and resistance to MitM attacks. This is typically accomplished using the Transport Level Security (TLS) protocol.

11. If a time-based OTP is used, it SHALL have a defined lifetime (recommended 30 seconds) that is determined by the expected clock drift — in either direction — of the authenticator over its lifetime, plus allowance for network delay and user entry of the OTP.

SUPPLEMENTAL GUIDANCE: The clocks on time-based authenticators are subject to drift because of cost and environmental factors such as temperature. Accordingly, verifiers need to accept authenticator outputs before and particularly after the intended validity period to allow use by authenticators that are not in synchronization.

12. Verifiers SHALL accept a given time-based OTP only once during the validity period.

SUPPLEMENTAL GUIDANCE: In order to prevent an attacker who gains access to an OTP authenticator output from using it, it is important that the secret only be valid for a single authentication.

13. If the authenticator output has less than 64 bits of entropy, the verifier SHALL implement a rate-limiting mechanism that effectively limits the number of failed authentication attempts that can be made on the subscriber's account as described in IA-5 l (3) through (4).

SUPPLEMENTAL GUIDANCE: OTPs whose output has less entropy are more vulnerable to online guessing attacks. To mitigate these attacks, rate limiting is required. Online guessing attacks are less of a concern for time-based OTP authenticators because of the limited validity window, but a limit on the number of guesses during a given validity period is effective in resisting automated attacks.

14. If the authenticator is multi-factor, then each use of the authenticator SHALL require the input of the additional factor.

SUPPLEMENTAL GUIDANCE: To ensure that a multi-factor authenticator cannot be stolen and used repeatedly following activation, a separate activation is required for each use of the authenticator. It is preferable for a multi-factor authenticator not to indicate that the wrong memorized secret or biometric were presented, but rather to produce an authenticator output that is invalid, although this is not required. This provides protection against guessing or presentation attacks on the authenticator itself.

15. If the authenticator is multi-factor and a memorized secret is used by the authenticator for activation, then that memorized secret SHALL be a randomly chosen numeric secret at least 6 decimal digits in length or other memorized secret meeting the requirements of IA-5 (1)(a).

SUPPLEMENTAL GUIDANCE: The requirement for memorized secrets used as activation factors is the same as that for memorized secrets used as distinct authenticators (see IA-5 (1)(a)).

16. If the authenticator is multi-factor, then use of a memorized secret for activation SHALL be rate limited as specified in IA-5 l (3) through (4).

SUPPLEMENTAL GUIDANCE: Rate limiting is required to provide protection against brute-force guessing attacks, particularly if the authenticator gives an indication when an incorrect secret is entered.

17. If the authenticator is multi-factor and is activated by a biometric factor, then that factor SHALL meet the requirements of IA-5 m, including limits on the number of consecutive authentication failures.

SUPPLEMENTAL GUIDANCE: General requirements for biometric activation factors include false accept rate criteria and the number of consecutive authentication failures that are allowed.

18. If the authenticator is multi-factor, then the unencrypted key and activation secret or biometric sample — and any biometric data derived from the biometric sample such as a probe produced through signal processing — SHALL be zeroized immediately after an OTP has been generated.

SUPPLEMENTAL GUIDANCE: It is important that the unencrypted key and associated data be zeroized to minimize the likelihood that it can be misappropriated by an attacker following a successful authentication. Each authentication requires a re-presentation of the activation factor (see IA-5(1)(d)14). Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

19. If the authenticator is multi-factor, the verifier or CSP SHALL establish, via the authenticator source, that the authenticator is a multi-factor device.

SUPPLEMENTAL GUIDANCE: From the standpoint of a verifier, a multi-factor OTP authenticator appears the same as a single-factor OTP authenticator. In order to establish that the authenticator meets the multi-factor requirements, the verifier or CSP can issue the authenticator, examine it in some way, or rely on an assertion from the manufacturer.

20. In the absence of a trusted statement that it is a multi-factor device, the verifier SHALL treat the authenticator as single-factor, in accordance with IA-5 (1) (d) (1) through (13).

SUPPLEMENTAL GUIDANCE: Authenticators of unknown provenance or that are not known by the CSP or verifier to meet all of the requirements for multi-factor OTP authenticators can be used, but only as single-factor authenticators.

(e) Cryptographic Authenticators and Verifiers (including single- and multi-factor cryptographic authenticators, both hardware- and software-based)

1. If the cryptographic authenticator is software based, the key SHALL be stored in suitably secure storage available to the authenticator application.

SUPPLEMENTAL GUIDANCE: Although dependent on the computing device on which the authenticator is operating, authenticator software needs to avail itself of the most secure storage available, considering issues like ability to extract the secret from the device and its potential to be included in backup data. Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

2. If the cryptographic authenticator is software based, the key SHALL be strongly protected against unauthorized disclosure by the use of access controls that limit access to the key to only those software components on the device requiring access.

SUPPLEMENTAL GUIDANCE: Although dependent on the computing device on which the authenticator is operating, authenticator software needs to store secret keys in a manner that limits access to keys to the maximum extent possible so that they cannot be accessed by other (possibly rogue) applications and/or users. Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

3. If the cryptographic authenticator is software based, it SHALL NOT facilitate the cloning of the secret key onto multiple devices.

SUPPLEMENTAL GUIDANCE: Like other physical authenticators, the use of cryptographic authenticators is premised upon the authenticator secret being present in a single authenticator so that it proves possession of a specific device. Mechanisms that would facilitate cloning the secret onto multiple devices include the ability to enroll more

than one device with the same key and backup mechanisms, especially when software-based authenticators are used. Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

4. If the authenticator is single-factor and hardware-based, secret keys unique to the device SHALL NOT be exportable (i.e., cannot be removed from the device).

SUPPLEMENTAL GUIDANCE: Cryptographic device authenticators are constructed so as not to allow the secret key to be obtained from the device. These devices are enrolled for authentication using the public cryptographic key, but the private key is never shared. This requirement addresses primarily functionality allowing the key to be exported; FIPS 140 requirements cover the resistance of the device to various forms of attack.

5. If the authenticator is hardware-based, the secret key and its algorithm SHALL provide at least the minimum-security length of 112 bits as of the date of this publication.

SUPPLEMENTAL GUIDANCE: The secret key used by a cryptographic authenticator needs to be sufficiently complex to resist online and offline attacks. An attacker may have the ability to observe the authenticator output at some point during its operation; it needs to be impractical for the secret key to be derived from a set of these observations. Since verifiers and cryptographic authenticators must use the same algorithms to successfully authenticate, assessment of the verifier also assesses the authenticators that may be used.

6. If the authenticator is hardware-based, the challenge nonce SHALL be at least 64 bits in length.

SUPPLEMENTAL GUIDANCE: This requirement applies to hardware-based cryptographic authenticators. The challenge nonce is required to be large enough that it will not be reused during the lifetime of the authenticator in order to provide replay protection. Since verifiers and cryptographic authenticators must use the same algorithms to successfully authenticate, assessment of the nonce generated by the verifier also assesses the authenticators that may be used.

7. If the authenticator is hardware-based, approved cryptography SHALL be used.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation. Since verifiers and cryptographic authenticators must use the same algorithms to successfully authenticate, assessment of the verifier also assesses the authenticators that may be used.

8. Cryptographic keys stored by the verifier SHALL be protected against modification.

SUPPLEMENTAL GUIDANCE: Protection against modification is required for all keys to ensure that an attacker can't substitute keys they control, which would permit them to

authenticate successfully. This protection could be provided by operating system access controls, or through integrity checks of the stored keys with separately stored hashes.

9. If symmetric keys are used, cryptographic keys stored by the verifier SHALL be protected against disclosure.

SUPPLEMENTAL GUIDANCE: Protection against disclosure is required for symmetric keys because their disclosure also would permit an attacker to authenticate successfully. This protection could be provided through operating system access controls.

10. The challenge nonce SHALL be at least 64 bits in length.

SUPPLEMENTAL GUIDANCE: This requirement applies to verifiers of cryptographic authentication. The challenge nonce is generated by the verifier and used by a cryptographic authenticator to compute the authenticator output. The challenge needs to be sufficiently long that it will not need to repeat during the lifetime of the authenticator, so the authenticator output, if available to an attacker, cannot be replayed.

11. The challenge nonce SHALL either be unique over the authenticator's lifetime or statistically unique (i.e., generated using an approved random bit generator).

SUPPLEMENTAL GUIDANCE: The challenge nonce is generated by the verifier used by a cryptographic authenticator to compute the authenticator output. The nonce cannot repeat during the lifetime of the authenticator, so the authenticator output, if available to an attacker, cannot be replayed. This can be accomplished by either deterministic means (e.g., an algorithm choosing values guaranteed not to repeat) or statistically (random values chosen from a range giving a very low probability that the same nonce will ever be seen twice).

12. The verification operation SHALL use approved cryptography.

SUPPLEMENTAL GUIDANCE: Cryptography is considered approved if it is specified or adopted in a FIPS or NIST recommendation. Since verifiers and cryptographic authenticators must use the same algorithms to successfully authenticate, assessment of the verifier also assesses the authenticators that may be used.

13. If a multi-factor cryptographic software authenticator is being used, then each authentication requires the presentation of the activation factor.

SUPPLEMENTAL GUIDANCE: The activation factor, either a memorized secret or a biometric, is required to be presented each time an authentication operation is requested by the authenticator to ensure that an activated authenticator cannot be used by an attacker.

14. If the authenticator is multi-factor, then any memorized secret used by the authenticator for activation SHALL be a randomly chosen numeric secret at least 6 decimal digits in length or other memorized secret meeting the requirements of IA-5 (1) (a).

SUPPLEMENTAL GUIDANCE: The requirement for memorized secrets used as activation factors is the same as that for memorized secrets used as distinct authenticators (see IA-5(1)a).

15. If the authenticator is multi-factor, then use of a memorized secret for activation SHALL be rate limited as specified in IA-5 l (3) through (4).

SUPPLEMENTAL GUIDANCE: Rate limiting is required to provide protection against brute-force guessing attacks, particularly if the authenticator gives an indication when an incorrect secret is entered.

16. If the authenticator is multi-factor and is activated by a biometric factor, then that factor SHALL meet the requirements of IA-5 m, including limits on the number of consecutive authentication failures.

SUPPLEMENTAL GUIDANCE: General requirements for biometric activation factors include false accept rate criteria and the number of consecutive authentication failures that are allowed.

17. If the authenticator is multi-factor, then the unencrypted key and activation secret or biometric sample — and any biometric data derived from the biometric sample such as a probe produced through signal processing — SHALL be zeroized immediately after an authentication transaction has taken place.

SUPPLEMENTAL GUIDANCE: It is important that the unencrypted key and associated data be zeroized to minimize the likelihood that it can be misappropriated by an attacker following a successful authentication. Verifiers are expected to make their best effort at determining that bring-your-own authenticators not issued by them meet this requirement and to have policies not allowing the use of non-compliant authenticators.

Related Controls: IA-6.

(2) AUTHENTICATOR MANAGEMENT | PUBLIC KEY BASED AUTHENTICATION

[Existing] [Priority 1]

Control:

(a) For public key-based authentication:

1. Enforce authorized access to the corresponding private key; and
2. Map the authenticated identity to the account of the individual or group; and

(b) When public key infrastructure (PKI) is used:

1. Validate certificates by constructing and verifying a certification path to an accepted trust anchor, including checking certificate status information; and
2. Implement a local cache of revocation data to support path discovery and validation.

DISCUSSION: Public key cryptography is a valid authentication mechanism for individuals, machines, and devices. For PKI solutions, status information for certification paths includes certificate revocation lists or certificate status protocol responses. For PIV cards, certificate validation involves the construction and verification of a certification path to the Common Policy Root trust anchor, which includes certificate policy processing. Implementing a local cache of revocation data to support path discovery and validation also supports system availability in situations where organizations are unable to access revocation information via the network. A local cache of revocation data is also known as a certificate revocation list. This list contains a list of revoked certificates and can be periodically downloaded to ensure certificates can still be checked for revocation when network access is not available or access to the Online Certificate Status Protocol (OCSP) server is not available. Without configuring a local cache of revocation data, there is the potential to allow access to users who are no longer authorized (users with revoked certificates).

Related Controls: IA-3, SC-17.

(6) AUTHENTICATOR MANAGEMENT | PROTECTION OF AUTHENTICATORS

[Existing] [Priority 1]

Control:

Protect authenticators commensurate with the security category of the information to which use of the authenticator permits access.

DISCUSSION: For systems that contain multiple security categories of information without reliable physical or logical separation between categories, authenticators used to grant access to the systems are protected commensurate with the highest security category of information on the systems. Security categories of information are determined as part of the security categorization process.

Related Controls: RA-2.

IA-6 AUTHENTICATION FEEDBACK

[Existing] [Priority 3] Control:

Obscure feedback of authentication information during the authentication process to protect the information from possible exploitation and use by unauthorized individuals.

DISCUSSION: Authentication feedback from systems does not provide information that would allow unauthorized individuals to compromise authentication mechanisms. For some types of systems, such as desktops or notebooks with relatively large monitors, the threat (referred to as shoulder surfing) may be significant. For other types of systems, such as mobile devices with small

displays, the threat may be less significant and is balanced against the increased likelihood of typographic input errors due to small keyboards. Thus, the means for obscuring authentication feedback is selected accordingly. Obscuring authentication feedback includes displaying asterisks when users type passwords into input devices or displaying feedback for a very limited time before obscuring it.

Related Controls: AC-3.

IA-7 CRYPTOGRAPHIC MODULE AUTHENTICATION

[Priority 2]

Control:

Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication.

DISCUSSION: Authentication mechanisms may be required within a cryptographic module to authenticate an operator accessing the module and to verify that the operator is authorized to assume the requested role and perform services within that role.

Related Controls: AC-3, IA-5, SA-4, SC-12, SC-13.

IA-8 IDENTIFICATION AND AUTHENTICATION (NON-ORGANIZATIONAL USERS)

[Priority 2]

Control:

Uniquely identify and authenticate non-organizational users or processes acting on behalf of non-organizational users.

DISCUSSION: Non-organizational users include system users other than organizational users explicitly covered by IA-2. Non-organizational users are uniquely identified and authenticated for accesses other than those explicitly identified and documented in AC-14. Identification and authentication of non-organizational users accessing federal systems may be required to protect federal, proprietary, or privacy-related information (with exceptions noted for national security systems). Organizations consider many factors—including security, privacy, scalability, and practicality—when balancing the need to ensure ease of use for access to federal information and systems with the need to protect and adequately mitigate risk.

Related Controls: AC-2, AC-6, AC-14, AC-17, AC-18, AU-6, IA-2, IA-4, IA-5, IA-11, MA-4, RA-3, SA-4, SC-8.

Control Enhancements:

(1) IDENTIFICATION AND AUTHENTICATION (NON-ORGANIZATIONAL USERS) | ACCEPTANCE OF PIV CREDENTIALS FROM OTHER AGENCIES

[Priority 2]

Control:

Accept and electronically verify Personal Identity Verification-compliant credentials from other federal, state, local, tribal, or territorial (SLTT) agencies.

DISCUSSION: Acceptance of Personal Identity Verification (PIV) credentials from other federal or SLTT agencies applies to both logical and physical access control systems. PIV credentials are those credentials issued by federal or SLTT agencies that conform to FIPS Publication 201 and supporting guidelines.

Related Controls: PE-3.

(2) IDENTIFICATION AND AUTHENTICATION (NON-ORGANIZATIONAL USERS) | ACCEPTANCE OF EXTERNAL AUTHENTICATORS

[Priority 2]

Control:

- (a) Accept only external authenticators that are NIST-compliant; and
- (b) Document and maintain a list of accepted external authenticators.

DISCUSSION: Acceptance of only NIST-compliant external authenticators applies to organizational systems that are accessible to the public (e.g., public-facing websites). External authenticators are issued by nonfederal government entities and are compliant with the CJISSECPOL. Approved external authenticators meet or exceed the minimum Federal Government-wide technical, security, privacy, and organizational maturity requirements. Meeting or exceeding Federal requirements allows Federal Government relying parties to trust external authenticators in connection with an authentication transaction at a specified authenticator assurance level.

Related Controls: None.

(4) IDENTIFICATION AND AUTHENTICATION (NON-ORGANIZATIONAL USERS) | USE OF DEFINED PROFILES

[Priority 2]

Control:

Conform to the following profiles for identity management: Security Assertion Markup Language (SAML) or OpenID Connect.

DISCUSSION: Organizations define profiles for identity management based on open identity management standards. To ensure that open identity management standards are viable, robust, reliable, sustainable, and interoperable as documented, the Federal Government assesses and scopes the standards and technology implementations against applicable laws, executive orders, directives, policies, regulations, standards, and guidelines.

Related Controls: None.

IA-11 RE-AUTHENTICATION

[Priority 2]

Control:

Require users to re-authenticate when: roles, authenticators, or credentials change, security categories of systems change, the execution of privileged functions occur, or every 12 hours.

DISCUSSION: In addition to the re-authentication requirements associated with device locks, organizations may require re-authentication of individuals in certain situations, including when roles, authenticators, or credentials change, when security categories of systems change, when the execution of privileged functions occurs, after a fixed time period, or periodically.

Related Controls: AC-3, AC-11, IA-2, IA-3, IA-4, IA-8.

IA-12 IDENTITY PROOFING

[Priority 2]

Control:

- a. Identity proof users that require accounts for logical access to systems based on appropriate identity assurance level requirements as specified in applicable standards and guidelines;
- b. Resolve user identities to a unique individual; and
- c. Collect, validate, and verify identity evidence.

DISCUSSION: Identity proofing is the process of collecting, validating, and verifying a user's identity information for the purposes of establishing credentials for accessing a system. Identity proofing is intended to mitigate threats to the registration of users and the establishment of their accounts. Organizations may be subject to laws, executive orders, directives, regulations, or policies that address the collection of identity evidence. Organizational personnel consult with the senior agency official for privacy and legal counsel regarding such requirements.

Related Controls: AC-5, IA-1, IA-2, IA-3, IA-4, IA-5, IA-6, IA-8.

Control Enhancements:

(2) IDENTITY PROOFING | IDENTITY EVIDENCE

[Priority 2]

Control:

Require evidence of individual identification be presented to the registration authority.

DISCUSSION: Identity evidence, such as documentary evidence or a combination of documents and biometrics, reduces the likelihood of individuals using fraudulent identification to establish an identity or at least increases the work factor of potential adversaries. The forms of acceptable evidence are consistent with the risks to the systems, roles, and privileges associated with the user's account.

Related Controls: None.

(3) IDENTITY PROOFING | IDENTITY EVIDENCE VALIDATION AND VERIFICATION

[Priority 2]

Control:

- a. Require that the presented identity evidence be validated and verified through agency-defined resolution, validation, and verification methods.

DISCUSSION: Validation and verification of identity evidence increases the assurance that accounts and identifiers are being established for the correct user and authenticators are being bound to that user. Validation refers to the process of confirming that the evidence is genuine and authentic, and the data contained in the evidence is correct, current, and related to an individual. Verification confirms and establishes a linkage between the claimed identity and the actual existence of the user presenting the evidence. Acceptable methods for validating and verifying identity evidence are consistent with the risks to the systems, roles, and privileges associated with the users account.

- b. Identity proofing SHALL NOT be performed to determine suitability or entitlement to gain access to services or benefits.

SUPPLEMENTAL GUIDANCE: The sole objective of identity proofing is to ensure the applicant is who they claim to be to a stated level of certitude.

- c.
 1. Collection of PII SHALL be limited to the minimum necessary to resolve to a unique identity in a given context.
 2. Collection of PII SHALL be limited to the minimum necessary to validate the existence of the claimed identity and associate the claimed identity with the applicant providing identity evidence for appropriate identity resolution, validation, and verification.

SUPPLEMENTAL GUIDANCE: The goal of identity resolution is to uniquely distinguish an individual within a given population or context. Effective identity resolution uses the smallest set of attributes necessary to resolve to a unique individual. It provides the CSP an important starting point in the overall identity proofing process, to include the initial detection of potential fraud, but in no way represents a complete and successful identity proofing transaction.

Collection of PII may include attributes are used to correlate identity evidence to authoritative sources and to provide RPs with attributes used to make authorization decisions. There may be many different sets that suffice as the minimum, so it is recommended that CSPs choose this set to balance privacy and the user's usability needs, as well as the likely attributes needed in future uses of the digital identity.

Examples of attributes that may be used for minimum identity attribute sets include:

- Name (first, last, middle) with combinations and variations,
- Address (#, Street, City, County, State, Zip code) with combinations and variations,
- Date of birth (DDMMYYYY) with combinations and variations,
- Email address,
- Phone number.

For population sets that are more defined than the general population (e.g., military veterans, Native Americans), these minimum attribute sets may be tailored to that specific community.

Additionally, it is recommended that CSPs document which alternative attributes it will accept in cases where an applicant cannot provide the minimum necessary attributes (e.g., applicant does not have a home address or phone number).

d. The CSP SHALL provide explicit notice to the applicant at the time of collection regarding the purpose for collecting and maintaining a record of the attributes necessary for identity proofing, including whether such attributes are voluntary or mandatory to complete the identity proofing process, and the consequences for not providing the attributes.

SUPPLEMENTAL GUIDANCE: Notice of proofing may contain at a minimum:

- Attribute information that is mandatory
- Attribute information that is voluntary
- What will be done with the information collected
- How the information will be protected
- Consequence of not providing mandatory attribute information (e.g., suspension/termination of the identity proofing process).

This notice may be delivered as an online screen (for remote identity proofing), a poster or printed notice at in-person proofing locations, or an oral notice delivered at the time of information collection.

e. If CSPs process attributes for purposes other than identity proofing, authentication, or attribute assertions (collectively “identity service”), related fraud mitigation, or to comply with law or legal process, then CSPs SHALL implement measures to maintain predictability and manageability commensurate with the privacy risk arising from the additional processing.

SUPPLEMENTAL GUIDANCE: Predictability and manageability measures include providing clear notice, obtaining subscriber consent, or enabling selective use or disclosure of attributes.

Predictability is meant to build trust and provide accountability and requires full understanding (and disclosure) of how the attribute information will be used. Manageability also builds trust by demonstrating a CSPs ability to control attribute information throughout processing – collection, maintenance, retention.

f. If the CSP employs consent as part of its measures to maintain predictability and manageability, ...then it SHALL NOT make consent for the additional processing a condition of the identity service.

SUPPLEMENTAL GUIDANCE: Consent involves collecting and recording an affirmative response from the applicant that they agree to the additional processing of their attributes. In order to make this consent meaningful, it is recommended that CSPs first disclose to its applicants which attributes are being collected and processed and why.

g. The CSP SHALL provide mechanisms for redress of applicant complaints or problems arising from the identity proofing.

These [redress] mechanisms SHALL be easy for applicants to find and use.

SUPPLEMENTAL GUIDANCE: The Privacy Act requires federal CSPs that maintain a system of records to follow procedures to enable applicants to access and, if incorrect, amend their records. Any Privacy Act Statement should include a reference to the applicable SORN(s), which provide the applicant with instructions on how to make a request for access or correction. It is recommended that non-federal CSPs have comparable procedures, including contact information for any third parties if they are the source of the information.

It is recommended that CSPs make the availability of any alternative methods for completing the identity proofing and enrollment processes clear to users (e.g., in person at a customer service center, if available) in the event an applicant is unable to properly complete the initial identity proofing and enrollment process requirements online.

Note: If the ID proofing process is not successful, it is recommended that CSPs inform the applicant of the procedures to address the issue but avoid informing the applicant of the specifics of why the registration failed.

To be effective, the use of a CSP's redress mechanism results in a timely correction of errors, resolution of the dispute or complaint, and the process should not be overly burdensome or complex.

It is recommended that the CSP document and publish, in a manner which is easy for Applicants to find and use, its mechanisms for redress of Applicant complaints or problems arising from the identity proofing processes.

h. The CSP SHALL assess the [redress] mechanisms for their efficacy in achieving resolution of complaints or problems.

SUPPLEMENTAL GUIDANCE: "Effective" in this requirement means that use of the redress mechanism will result in a timely correction of errors, resolution of the dispute or complaint, and the process shall not be overly burdensome or complex.

It is recommended that CSPs maintain a record or log of all cases – including outcomes – where applicants have sought redress for complaints or problems arising from the identity proofing and provide for the periodic review of these records.

i. The identity proofing and enrollment processes SHALL be performed according to an applicable written policy or *practice statement* that specifies the particular steps taken to verify identities.

SUPPLEMENTAL GUIDANCE: Having documented procedures is a prerequisite for transparency, accountability, quality control, auditability, and ease of interoperability among federated communities. The documentation, dissemination, review and update to identity and authentication processes is a core control under IA-1 Identification and Authentication Policy and Procedures.

j. The *practice statement* SHALL include control information detailing how the CSP handles proofing errors that result in an applicant not being successfully enrolled.

SUPPLEMENTAL GUIDANCE: “Proofing errors” in this context refer to circumstances that result in the inability or failure to complete the identity proofing and enrollment processes. Such circumstances may include:

- Applicant abandons the identity proofing and enrollment processes;
- Applicant fails to provide mandatory attribute information;
- Identity evidence of required strength is not provided;
- Identity evidence is rejected following inspection;
- Identity evidence and information do not correlate;
- Information from identity evidence is not validated by issuing or authoritative sources at the required strength;
- Identity evidence verification of binding to the applicant fails; and
- Applicant fails to confirm enrollment code within code validity period.

Depending on the circumstances above, it is recommended that the documentation include the number of retries allowed, proofing alternatives (e.g., in-person if remote fails), or fraud countermeasures when anomalies are detected. Additional controls for handling identity proofing errors include:

- Advising the applicant of identity proofing failure and recourse options; and,
- Recording the errors in enrollment records/audit logs, along with any mitigating actions.

k. The CSP SHALL maintain a record, including audit logs, of all steps taken to verify the identity of the applicant as long as the identity exists in the information system.

SUPPLEMENTAL GUIDANCE: Ideally, the CSP’s identity system includes the capability to securely record and log key security-related activities associated with the identity proofing process.

Examples of key steps that may be recorded in enrollment logs include:

- Identity information collected;

- Identity evidence provided;
- Identity evidence validated;
- Identity evidence validation source;
- Identity evidence binding verification method;
- Identity evidence verification result;
- Enrollment code confirmation result;
- enrollment result; and
- Authenticator enrollment binding

l. The CSP SHALL record the types of identity evidence presented in the proofing process.

SUPPLEMENTAL GUIDANCE: Ideally, the CSP's identity system includes the capability to securely record and log specific activities associated with the identity proofing process. For each piece of evidence collected or captured, the record should include:

1. Evidence type;
2. Determined strength;
3. Issuing source; and
4. Method of collection/capture*.

* Methods of collection and capture may include camera, flatbed scanner, bar code scanner.

m. The CSP SHALL conduct a risk management process, including assessments of privacy and security risks to determine:

1. Any steps that it will take to verify the identity of the applicant beyond any mandatory requirements specified herein;
2. The PII, including any biometrics, images, scans, or other copies of the identity evidence that the CSP will maintain as a record of identity proofing (Note: Specific federal requirements may apply); and
3. The schedule of retention for these records (Note: CSPs may be subject to specific retention policies in accordance with applicable laws, regulations, or policies, including any National Archives and Records Administration (NARA) records retention schedules that may apply).

SUPPLEMENTAL GUIDANCE: In accordance with its risk management processes, CSPs should conduct – and document the results of – privacy and security risk assessments. It is recommended that the scope of this assessment includes risks associated with:

- Any steps the CSP takes to verify applicant identities beyond what is required by the CJISSECPOL
- The CSP's collection, processing, and protection of PII, including any biometrics, images, scans, or other copies of the identity evidence that the CSP will maintain as a

- record of identity proofing;
- Retention and/or disposal of any records; and
- Adherence to any applicable federal requirements, laws, regulations or policies.

n. All PII collected as part of the enrollment process SHALL be protected to ensure confidentiality, integrity, and attribution of the information source.

SUPPLEMENTAL GUIDANCE: Unauthorized disclosure of PII can result in tangible and intangible harms to both the CSP as well as the subjects of the PII. After assessing the risks associated with collecting PII as part of its enrollment process, it is recommended that the CSP employ functional and technical mechanisms that adequately protect the confidentiality, integrity, and attribution of the PII under its control.

Such mechanisms may include:

- Limiting access to PII data;
- Privacy protecting policies;
- The use of encryption for data at rest and during transmission; and
- Integrity protection mechanisms such as hashes and record access logging.

o. "The entire proofing transaction, including transactions that involve a third party, SHALL occur over authenticated protected channels. "

SUPPLEMENTAL GUIDANCE: An encrypted communication channel uses approved cryptography where the connection initiator (client) has authenticated the recipient (server). Authenticated protected channels provide confidentiality and MitM attack protection and are frequently used in the user authentication process. Transport Layer Security* (TLS) is an example of an authenticated protected channel where the certificate presented by the recipient is verified by the initiator. Unless otherwise specified, authenticated protected channels do not require the server to authenticate the client.

Authentication of the server is often accomplished through a certificate chain leading to a trusted root rather than individually with each server.

*TLS version 1.2 or greater is recommended.

p. "If the CSP uses fraud mitigation measures, then the CSP SHALL conduct a privacy risk assessment for these mitigation measures. "

Such assessments SHALL include any privacy risk mitigations (e.g., risk acceptance or transfer, limited retention, use limitations, notice) or other technological mitigations (e.g., cryptography), and be documented per requirement IA-12(3) k – m above.

SUPPLEMENTAL GUIDANCE: This is a conditional requirement. CSPs may choose to obtain additional confidence in the identity proofing process beyond the requirements for IAL2 through additional fraud mitigation measures. Such measures may include:

- inspecting metadata information, such as by checking geolocation data associated with a mobile device used to send a photo or receive an SMS;
- examining the applicant's device characteristics;
- evaluating behavioral characteristics, such as typing mannerisms, gait, or voice characteristics; and
- checking against authoritative sources, such as the Death Master File.

Employing one or more of these fraud mitigation techniques may result in the collection of additional PII about an applicant. Additional PII increases the potential impact of the unauthorized disclosure of this data. As part of the privacy risk assessment on these additional fraud mitigation measures, it is recommended that CSPs consider, at a minimum, the additional data (PII) that is processed, the implications of retaining this additional PII, and ways the associated risks can be minimized without negating the effects of the additional measures.

These additional fraud mitigation measures are not intended to substitute or replace the mandatory requirements. CSPs employing these measures are still responsible for meeting all applicable requirements.

q. In the event a CSP ceases to conduct identity proofing and enrollment processes, then the CSP SHALL be responsible for fully disposing of or destroying any sensitive data including PII, or its protection from unauthorized access for the duration of retention.

SUPPLEMENTAL GUIDANCE: This is a conditional requirement for CSPs that cease to perform identity proofing and enrollment functions. The CSP is responsible for the proper handling, protection, and retention or disposal of any sensitive data it collects, even after it ceases to provide identity proofing and enrollment services. A CSP may document its policies and procedures to the management of the data it collects in a data handling plan or other document. Additionally, it is recommended that CSPs document any specific retention policies they are subject to, in accordance with applicable laws, regulations, or policies, including any National Archives and Records Administration (NARA) records retention schedules that may apply.

Specifically, it is recommended that the CSP defines and documents the practices it has in place for fully disposing of or destroying any sensitive data including PII, or its continued protection from unauthorized access for the duration of any period of retention.

r. Regardless of whether the CSP is a federal agency or non-federal entity, the following requirements apply to the federal agency offering or using the proofing service:

1. The agency SHALL consult with their Senior Agency Official for Privacy (SAOP) to conduct an analysis determining whether the collection of PII to conduct identity proofing triggers Privacy Act requirements.
2. The agency SHALL publish a System of Records Notice (SORN) to cover such collection, as applicable.
3. The agency SHALL consult with their SAOP to conduct an analysis determining whether the collection of PII to conduct identity proofing triggers E-Government Act of 2002 requirements.

4. The agency SHALL publish a Privacy Impact Assessment (PIA) to cover such collection, as applicable.

SUPPLEMENTAL GUIDANCE: This requirement applies to Federal agencies whether providing authentication services directly or through a commercial provider. This requirement directs Agencies to consult with their Senior Agency Official for Privacy (SAOP) and conduct an analysis to determine whether the collection of PII to issue or maintain authenticators triggers the requirements of the Privacy Act of 1974 or the requirements of the E-Government Act of 2002. Based on this consultation and analysis, the agency may need to publish a System of Records Notice (SORN) and/or a Privacy Impact Assessment (PIA) to cover such collections, as applicable. While this requirement specifically applies only to federal agencies, CSPs that provide services to federal agencies may be expected to provide information about their identity services in support of an Agency's privacy analysis and PIA.

- s. An enrollment code SHALL be comprised of one of the following:
 - 1 Minimally, a random six character alphanumeric or equivalent entropy. For example, a code generated using an approved random number generator or a serial number for a physical hardware authenticator; OR
 - 2 A machine-readable optical label, such as a QR Code, that contains data of similar or higher entropy as a random six character alphanumeric.

SUPPLEMENTAL GUIDANCE: The use of an enrollment code for address confirmation is a requirement for IAL2 remote identity proofing and enrollment. CSPs that perform in-person identity at IAL2 may voluntarily choose to use enrollment codes for such binding, but this is not required. Enrollment codes may also be used for in-person proofing and enrollment processes if an authenticator(s) is not registered to the subscribers' account at the time of in-person identity proofing and, therefore, the authenticator binding would need to occur at a later time. Enrollment codes may be used for authenticator binding to subscribers' accounts in such circumstances.

Enrollment code use for IAL2 remote identity proofing allows the CSP to confirm that the applicant controls a validated address of record. Authenticator binding may not be completed in the same session for in-person identity proofing. Enrollment codes may be used for binding an authenticator to subscribers' accounts at a later time in such circumstances. The requirements presented in this criterion apply to all enrollment codes that may be used by the CSP for any purpose.

Enrollment code use has the additional requirement for code validity periods. The validity period is determined by the type of address where the enrollment code is sent, as follows:

- 10 days, when sent to a postal address of record within the contiguous United States;
- 30 days, when sent to a postal address of record outside the contiguous United States;
- 10 minutes, when sent to a telephone of record (SMS or voice);
- 24 hours, when sent to an email address of record;
- 7 days if provided directly to the applicant during an in-person proofing session for authenticator binding at IAL2.

These validity periods are presented again in requirement IA-12 (5) g which presents the mandatory requirement for enrollment code confirmation for IAL2 remote identity proofing.

t. Training requirements for personnel validating evidence SHALL be based on the policies, guidelines, or requirements of the CSP or RP.

SUPPLEMENTAL GUIDANCE: The training requirement pertains to personnel performing the validation of identity evidence but does not specify training content. The CSP policies, guidelines, or requirements for validating identity evidence for identity proofing would be appropriate for the type of training intended by this requirement. Such content may include:

- the CSP's policy for types of evidence it collects and validates in order to meet the requirements of designated IALs;
- validation of security features for the types of identity evidence collected;
- detection of evidence alteration, falsification, or forgery for the types of identity evidence collected. Procedures for the validation of identity evidence information with issuing and authoritative sources.

This training may be accomplished through written training material, oral instruction, on-the-job training and mentoring, or other means. CSPs may perform some of the requirements for identity evidence validation through automated services and equipment. Therefore, personnel training would be based on the CSPs policies and procedures for the manual performance of evidence validation.

u. This criterion applies to CSPs that provide identity proofing and enrollment services to minors (under the age of 18):

If the CSP provides identity proofing and enrollment services to minors (under the age of 18), then...the CSP SHALL give special consideration to the legal restrictions of interacting with minors unable to meet the evidence requirements of identity proofing [to ensure compliance with the Children's Online Privacy Protection Act of 1998 (COPPA), and other laws, as applicable]. "

SUPPLEMENTAL GUIDANCE: In general, minors will not possess the types of evidence required to meet the CSP's minimum requirements for a given IAL. ICSPs that provide identity services to minors will need to determine and document the special considerations it applies to minors. Such special considerations may include the use of trusted referees and an expanded list of acceptable evidence types to include evidence a minor would likely possess, such as school IDs.

Requirements 'v' and 'w' apply to the collection of biometric characteristics for in-person (physical or supervised remote) identity proofing and are mandatory at IAL3. These criteria also apply to CSPs that optionally choose to collect biometric characteristics through in-person identity proofing and enrollment at IAL2.

v. The CSP SHALL have the operator view the biometric source (e.g., fingers, face) for presence of non-natural materials and perform such inspections as part of the proofing process.

SUPPLEMENTAL GUIDANCE: Applicants may try to defraud the identity proofing process by using fake fingers or by applying non-natural materials - such as latex, silicon, or glue - to their fingers, faces, or other sources of biometrics. It is recommended that identity proofing operators

be trained to recognize such practices and to examine all biometric sources used in the identity proofing for the presence of foreign materials.

It is recommended that the CSP documents and applies technologies and procedures which ensure that the proofing operator reviews the biometric source (e.g., fingers, face) for presence of non-natural materials and perform such inspections as part of the proofing process.

Requirements ‘v’ and ‘w’ apply to the collection of biometric characteristics for in-person (physical or supervised remote) identity proofing and are mandatory at IAL3. These criteria also apply to CSPs that collect biometric characteristics through in-person identity-proofing identity proofing and enrollment at IAL2.

w. The CSP SHALL collect biometrics in such a way that ensures that the biometric is collected from the applicant, and not another subject. All biometric performance requirements in IA-5 m (1) through (12) apply.

SUPPLEMENTAL GUIDANCE: Applicants may try to defraud the identity proofing process by having another person present themselves for biometric collection. The risk of this happening is increased if the identity proofing process is not completed in a single session and during supervised remote identity proofing processes.

Documenting the technologies and procedures the CSP employs to ensure that biometric samples are taken from the applicant him/herself and not another person facilitates the assessment against this requirement.

x. The CSP SHALL support in-person or remote identity proofing, or both.

SUPPLEMENTAL GUIDANCE: IAL2 allows for remote or in-person identity proofing. IAL2 supports a wide range of acceptable identity proofing techniques in order to increase user adoption, decrease false negatives (legitimate applicants that cannot successfully complete identity proofing), and detect to the best extent possible the presentation of fraudulent identities by a malicious applicant.

Remote proofing presents challenges to achieving the desired outcomes described above that can be overcome through the use specific processes and technologies. Potential processes and controls that CSPs may employ to mitigate risks associated with remote identity proofing at IAL2 include:

1. A remote operator is present during at least part of the identity proofing session and can provide positive confirmation that the requirements for IAL2 identity proofing are met. Employing real-time remote operators provides the capability for the identity proofing process to be completed in a single session and allows the remote operator to direct the applicant for proper presentation and examination of identity evidence and biometrics collection.
2. The CSP employs automated technologies and services (e.g., liveness detection, identity evidence verification and validation, and presentation attack detection, if applicable) which can ensure the requirements for IAL2 identity proofing are met and protect against spoofing attacks. This process also provides the capability for the identity proofing

process to be completed in a single session.

3. The CSP employs an off-line operator to evaluate the evidence and images collected during a previous identity proofing process. In this scenario, the identity proofing process requires more than one session with the applicant and is not completed until the operator provides a positive confirmation that all requirements for IAL2 identity proofing are met.

y. The CSP SHALL collect the following from the applicant:

1. One piece of SUPERIOR or STRONG evidence if the evidence's issuing source, during its identity proofing event, confirmed the claimed identity by collecting two or more forms of SUPERIOR or STRONG evidence and the CSP validates the evidence directly with the issuing source; OR
2. Two pieces of STRONG evidence; OR
3. One piece of STRONG evidence plus two pieces of FAIR evidence

SUPPLEMENTAL GUIDANCE: The goal of identity validation is to collect the most appropriate identity evidence (e.g., a passport or driver's license) from the applicant and determine its authenticity, validity, and accuracy. Identity validation is made up of three process steps: 1) collecting the appropriate identity evidence, 2) confirming the evidence is genuine and authentic, and 3) confirming the data contained on the identity evidence is valid, current, and related to a real-life subject.

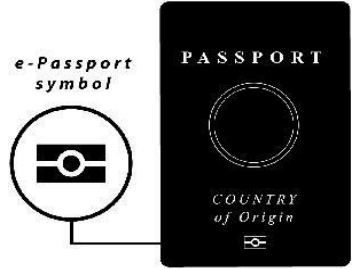
Figure 7 - Notional Strength of Evidence Types of this document presents notional strengths for types of evidence that may be presented for identity proofing purposes. Documenting the types and strengths of evidence the CSP collects for each proofing encounter demonstrates conformance for this requirement. (Also see IA-12 (3) 1.)

Examples of methods and how they can be used to capture identity evidence images or extract data for validation include:

- Cameras to capture images of identity evidence for the purposes of evidence validation;
- Document scanner to capture images of identity evidence for the purpose of evidence validation; and
- Bar-code scanner to capture and extract information from standardized barcodes embedded on identity evidence.

High resolution images of at least 300 ppi are necessary for proper evidence examination and validation.

Figure 7 – Notional Strengths of Evidence Types

Type of Evidence	Strength	Notes
US Passport	Superior	Includes US Passport cards
Foreign e-Passport	Superior	 The diagram illustrates a foreign e-Passport. On the left, a circular 'e-Passport symbol' contains a stylized 'e' and a passport icon. A line connects this symbol to the top of a dark passport cover on the right. The cover features the word 'PASSPORT' at the top, a circular emblem in the center, and 'COUNTRY of Origin' at the bottom.
Personal Identity Verification (PIV) card	Superior	
Common Access card (CAC)	Superior	
Personal Identity Verification Interoperable (PIV-I) card	Superior	
Transportation Worker Identification Credential (TWIC)	Superior	
Permanent Resident Card	Superior	Issued on or after May 11, 2010
Native American Enhanced Tribal Card	Superior	
REAL ID cards	Strong+	Includes REAL ID driver's licenses and ID cards. REAL ID cards have a star printed in the upper right-hand corner. Card and personal

Type of Evidence	Strength	Notes
		information must be validated with appropriate DMV or AAMVA.
Enhanced ID cards	Strong+	Includes Enhanced ID driver's licenses and ID cards. Must be validated with appropriate DMV or AAMVA.
U.S. Uniformed Services Privilege and Identification Card (U.S. Military ID)	Strong+	Includes Uniformed Services Dependent ID Cards. Must be validated with appropriate military issuing source.
Permanent Resident Card	Strong	Issued Prior to May 11, 2010
Native American Tribal Photo Identification Card	Strong	
Driver's License or ID card (REAL ID non-compliant)	Strong	
School ID card	Fair	Includes facial image photograph
Utility account statement	Fair	
Credit/debit card and account statement	Fair	
Financial institution account statement	Fair	
US Social Security Card	Weak	

Type of Evidence	Strength	Notes
Original or certified copy of a birth certificate issued by a state, county, municipal authority or outlying possession of the United States bearing an official seal	Weak	

z. The CSP SHALL validate each piece of evidence with a process that can achieve the same strength as the evidence presented (see ‘y’ above). For example, if two forms of STRONG identity evidence are presented, each piece of evidence will be validated at a strength of STRONG.

SUPPLEMENTAL GUIDANCE: The goal of identity validation is to collect the most appropriate identity evidence (e.g., a passport, driver’s license) from the applicant and determine its authenticity, validity, and accuracy. Identity validation is made up of three process steps: 1) collect the appropriate identity evidence, 2) confirm the evidence is genuine and authentic, and 3) confirm the data contained on the identity evidence is valid, current, and related to a real-life subject.

Evidence validation for authenticity involves examining the evidence for:

- Confirmation of required information completeness and format for the identity evidence type.
- Detection of evidence tampering or the creation of counterfeit or fraudulent evidence.
- Confirmation of security features. See Figure 8 - Types of Identity Evidence Security Features to this document for types of commonly used security features for identity evidence.

Most of the capabilities to confirm security features on identity evidence are dependent upon physically viewing the evidence directly, tactile feel of the evidence, and viewing the evidence under specialized lighting or through the use of specialized equipment (see Figure 8 - Types of Identity Evidence Security Features). Therefore, the validation of evidence that may be submitted remotely for remote identity proofing methods is particularly challenging. For this reason, CSPs opting to provide remote identity proofing may find it most effective to use automated evidence validation products and services. If automated evidence validation solutions are not used, CSPs may choose to apply similar procedures for IAL2 remote proofing as are required for IAL3 supervised remote proofing. These procedures provide that a trained operator can remotely supervise the evidence collection process, require the applicant to turn or tilt evidence or apply lighting to be able to confirm security features on evidence that is presented for the identity proofing encounter in a recorded video or webcast. Alternatively, a CSP may use an automated interface for the capture of identity evidence images that similarly can direct the applicant to turn, tilt or provide lighting on evidence presented for identity proofing purposes.

Figure 8 – Types of Identity Evidence Security Features

Security Feature (examination capability)	Description
Fine-line or Guilloche Pattern (visual)	Background pattern of continuous fine lines printed in wavy, overlapping pattern.
Ghost image (visual)	Half-tone reproduction of original image (e.g., facial image), may be printed behind printed data.
Overlapped data (visual)	Variable data (e.g., signature, seal, text) printed over another field such as facial image or seal.
Transparent image (visual)	See-through, window-like image feature (e.g., facial image) visible for both sides of the evidence.
Rainbow printing (visual)	Controlled color shifts of printed text in a continuous, linear fashion.
Holographic Images (visual, tilting)	Light field record of objects that will appear and change as view of evidence is tilted and turned. Most state-issued driver's licenses and IDs contain at least one holographic image.
Variable laser engraved images (visual, tilting)	Laser-engraved images at different angles so that image view changes with tilting angle of viewing evidence.
Iridescent Inks and Custom Foil Stamping (visual, tilting)	Custom designs and printing that will change color properties depending on the angle at which evidence is viewed.
Laser perforation (visual, light, tactile)	Perforated holes made by laser beam to form images. The images can be viewed under light source; image holes have tactile feel.
UV printing (visual, UV lighting)	A UV image or text that can only be viewed with special lighting. UV images may appear on the front or back of the evidence.
Microprinting (visual, magnifier)	Microtext of static or variable data that can be confirmed when viewed under a magnifier. Requires magnification of at least 10X to view.

Security Feature (examination capability)	Description
Laser embossing (tactile)	Use of laser to emboss image or text for tactile feel on only one side of the evidence.
Barcode (visual, barcode reader)	Machine readable, encoded data (typically personalized printed data) for 2-D barcode, readable with barcode reader.
UV printing (visual, UV lighting)	A UV image or text that can only be viewed with specialized lighting. UV images may appear on the front or back of a card.

The next step in identity evidence validation for authenticity and integrity is to verify the correctness of information from the identity evidence against the issuing source for the evidence or an authoritative source that has linkage to the issuing source. Results of these checks for authenticity and integrity should be recorded.

Figure 9 - Validating Identity Evidence lists strengths, ranging from unacceptable to superior, of identity validation performed by the CSP to validate the evidence presented for the current proofing session and the information contained therein.

Figure 9 – Validating Identity Evidence

Strength	Method(s) Performed by the CSP
Unacceptable	Evidence validation was not performed, or validation of the evidence failed.
Weak	All personal details from the evidence have been confirmed as valid by comparison with information held or published by an authoritative source.

Strength	Method(s) Performed by the CSP
Fair	- Attributes contained in the evidence have been confirmed as valid by comparison with information held or published by the issuing source or authoritative source(s), OR - The evidence has been confirmed as genuine using appropriate technologies, confirming the integrity of physical security features and that the evidence is not fraudulent or inappropriately modified, OR - The evidence has been confirmed as genuine by trained personnel, OR - The evidence has been confirmed as genuine by confirmation of the integrity of cryptographic security features.
Strong	- The evidence has been confirmed as genuine: - using appropriate technologies, confirming the integrity of physical security features and that the evidence is not fraudulent or inappropriately modified, OR - by trained personnel and appropriate technologies, confirming the integrity of the physical security features and that the evidence is not fraudulent or inappropriately modified, OR - by confirmation of the integrity of cryptographic security features. - All personal details and evidence details have been confirmed as valid by comparison with information held or published by the issuing source or authoritative source(s).
Superior	- The evidence has been confirmed as genuine by trained personnel and appropriate technologies including the integrity of any physical and cryptographic security features. - All personal details and evidence details from the evidence have been confirmed as valid by comparison with information held or published by the issuing source or authoritative source(s).

aa. The CSP SHALL verify identity evidence as follows:

At a minimum, the applicant's binding to identity evidence must be verified by a process that is able to achieve a strength of STRONG.

SUPPLEMENTAL GUIDANCE: The goal of identity verification is to confirm and establish a linkage between the validated evidence for the claimed identity and the real-life applicant presenting the evidence

The figure below shows IAL2 verification methods.

Figure 10 – Verification Methods and Strengths

Verification Strength	Verification Method	Description
Superior	Biometric Verification	Biometric comparison against biometric characteristics on the strongest piece(s) of evidence against live biometric capture for remote or in-person identity proofing. May be used for identity verification for FAIR, STRONG, and SUPERIOR strength.
Strong	In-Person Physical Verification	Physical comparison of applicant to facial-image photograph on strongest piece(s) of validated evidence. May be used for identity verification for FAIR and STRONG strength.
Strong	Remote Physical Verification	Physical comparison of applicant to facial-image photograph on strongest piece(s) of validated evidence. May be used for identity verification for FAIR and STRONG strength.

For IAL2 this linkage is achieved through a physical or biometric comparison of the facial image (i.e., photograph) on the strongest piece of evidence to the applicant or by a biometric comparison between information on the evidence and a biometric characteristic obtained from the applicant, most likely facial image.

Physical comparison is a comparison by a person (i.e., CSP-trained personnel) of the applicant to the photograph (i.e., facial image) on any of the strongest piece(s) of validated identity evidence collected. This comparison can be an in-person comparison for in-person identity proofing processes or may be conducted remotely for remote identity proofing. In both cases, the operator must perform a physical comparison of the applicant to the facial image photograph on the evidence. That is, the in-person proofing personnel will physically compare the facial image of the live applicant to the facial image photograph on the strongest piece of validated evidence. For remote physical comparison, the applicants' facial image may be captured by high resolution video or camera for physical comparison to the facial image photograph on the identity evidence.

For identity proofing verification, biometric comparison is an automated comparison of a biometric characteristic recorded on the strongest piece of identity evidence compared to the corresponding biometric characteristic of the applicant captured live during the identity proofing session.

Remote identity proofing requires the collection of both an image of the identity evidence and a live capture of the facial image of the applicant for physical or biometric comparison. The CSP must employ liveness and presentation attack detection capabilities to ensure that the applicant's facial image or other biometric characteristic used for comparison is "live" and not a spoofing or

presentation attack. Potential methods for remote identity proofing processes to mitigate such spoofing and presentation attacks are presented below.

- A remote operator is present during the identity proofing session (similar to supervised remote in-person proofing) and can conduct a real-time physical comparison between an image of the identity evidence and a live video of the applicant. In order to confirm the video stream is live and not pre-recorded, the Operator could direct the applicant to move their head in specific ways, or even ask the applicant a question. Once a positive confirmation is recorded from the operator, and all other requirements are met, the identity proofing can be completed in a single session.
- The CSP employs automated capabilities which are specifically designed to compare the image of the identity evidence with the applicant, and which also employ liveness detection technologies. Pending a positive confirmation from the automated comparison, and the satisfaction of all other requirements, the identity proofing can be completed in a single session.

The CSP employs liveness detection technology during the capture of the facial image, and an off-line operator performs the physical comparison of images captured during the identity proofing session. The identity proofing process requires more than one session with the applicant and is not completed until the operator provides a positive confirmation of the comparison and the other requirements are met.

bb. For IAL2 remote proofing: The collection of biometric characteristics for physical or biometric comparison of the applicant to the strongest piece of identity evidence provided to support the claimed identity performed remotely SHALL adhere to all requirements as specified in IA-5 m.

SUPPLEMENTAL GUIDANCE: See IA-5 m (1) – (12) for conformance criteria for the implementation and conformance assessment of requirements for the use of biometrics.

cc. Knowledge-based verification (KBV) SHALL NOT be used for in-person (physical or supervised remote) identity verification.

SUPPLEMENTAL GUIDANCE: identity verification is performed against the strongest piece of identity evidence submitted and validated. For IAL2 the strongest piece of evidence will always be either STRONG or SUPERIOR evidence. KBV (sometimes referred to as knowledge-based authentication) is only permitted as a verification method for evidence at the FAIR strength level; therefore, verification of FAIR evidence binding will never be required for IAL2.

dd. The CSP SHALL employ appropriately tailored security controls, to include control enhancements, from the moderate or high baseline of security controls defined in the CJISSECPOL.

The CSP SHALL ensure that the minimum assurance-related controls for moderate-impact systems are satisfied.

ee. Supervised Remote Identity Proofing: Supervised remote identity proofing is intended to provide controls for comparable levels of confidence and security to in-person IAL3 identity proofing for identity proofing processes that are performed remotely. Supervised remote identity proofing is optional for CSPs; that is, if a CSP chooses to use supervised remote identity proofing, then the following requirements, (1) through (8), would apply. It should be noted that the term “supervised remote identity proofing” has specialized meaning and is used only to refer to the specialized equipment and the following control requirements, (1) through (8). In addition to those requirements presented in this document, as well as the applicable identity validation and verification requirements, CSPs that provide supervised remote identity proofing services must demonstrate conformance with the requirements contained in this section. The following requirements for supervised remote proofing apply specifically to IAL3. If the equipment/facilities used for supervised remote proofing are used for IAL2 identity proofing, the following requirements, (1) through (8), for supervised remote proofing do not apply. In this case, the requirements for conventional remote identity proofing are applicable.

1. Supervised remote identity proofing and enrollment transactions SHALL meet the following requirements, in addition to the IAL3 validation and verification requirements specified in IA-12(3)s.

SUPPLEMENTAL GUIDANCE: Supervised remote identity proofing involves the use of a CSP-controlled station at a remote location that is connected to a trained operator at a central location. The goal of this arrangement is to permit identity proofing of individuals in remote locations where it is not practical for them to travel to the CSP for in-person identity proofing.

2. The CSP SHALL monitor the entire identity proofing session, from which the applicant SHALL NOT depart — for example, by a continuous high-resolution video transmission of the applicant.

SUPPLEMENTAL GUIDANCE: The integrity of supervised remote identity proofing depends upon the applicant being continuously present during the entire session. An applicant who steps away from an in-process session may do so to alter their biometric source or substitute a different person to complete the identity proofing process.

3. The CSP SHALL have a live operator participate remotely with the applicant for the entirety of the identity proofing session.

SUPPLEMENTAL GUIDANCE: Having a trained operator supervise and participate in a remote identity proofing session reduces the opportunity for an applicant to defraud the process. As described in Appendix A Terms and Definitions, the operator is a person who has received specific training on enrollment and identity proofing procedures and the detection of potential fraud by an applicant.

4. The CSP SHALL require all actions taken by the applicant during the identity proofing session to be clearly visible to the remote operator.

SUPPLEMENTAL GUIDANCE: The camera(s) a CSP employs to monitor the actions taken by a remote applicant during the identity proofing session should be positioned in such a way that the upper body, hands, and face of the applicant are always visible. Additionally, the components of the remote identity proofing station (including such things as keyboard, fingerprint capture device, signature pad, and scanner, as applicable) should be arranged such that all interactions with these devices is within the field of view.

5. The CSP SHALL require that all digital validation of evidence (e.g., via chip or wireless technologies) be performed by integrated scanners and sensors.

SUPPLEMENTAL GUIDANCE: Technologies exist that allow for the digital validation of identity evidence via electronic means (such as RFID to read the data off e-passports and chip readers for smartcards). The scanners and sensors employed to access these features should be integrated into the remote identity proofing stations in order to reduce the likelihood of being tampered with, removed, or replaced. To be integrated means the devices themselves are a component of the workstation (i.e., smartcard readers or fingerprint sensors built into a laptop) or the devices, and their connections, are secured in a protective case or locked box.

6. The CSP SHALL require operators to have undergone a training program to detect potential fraud and to properly perform a supervised remote proofing session.

SUPPLEMENTAL GUIDANCE: A comprehensive training program for supervised remote identity proofing operators may include some or all the following:

- Purpose and objectives of the identity proofing and enrollment process, as employed by the CSP;
- Supervised remote identity proofing process workflow;
- Identity evidence validation processes;
- Threats associated with the identity proofing process and how to detect potential fraud; and
- System and process troubleshooting and problem resolution.

7. The CSP SHALL employ physical tamper detection and resistance features appropriate for the environment in which it is located.

SUPPLEMENTAL GUIDANCE: For example, a kiosk located in a restricted area or one where it is monitored by a trusted individual requires less tamper detection than one that is located in a semi-public area such as a shopping mall concourse.

8. The CSP SHALL ensure that all communications occur over a mutually authenticated protected channel.

SUPPLEMENTAL GUIDANCE: Mutually authenticated protected channels employ approved cryptography to encrypt communications between the station/kiosk and server.

- ff. Trusted Referee: The use of trusted referees is optional for CSPs; that is, if a CSP chooses to use trusted referees for identity proofing and enrollment, then the following requirements, (1) through (3), would apply. The use of trusted referees is intended to assist in the identity proofing and enrollment for populations that are unable to meet IAL2 identity proofing requirements, or otherwise would be challenged to perform identity proofing and enrollment process requirements. Such populations may include, but are not limited to:

- disabled individuals;
- elderly individuals;
- homeless individuals,
- individuals with little or no access to online services or computing devices;
- unbanked and individuals with little or no credit history;
- victims of identity theft;
- children under 18; and
- immigrants.

In addition to those requirements presented in the General section of this document, as well as the applicable IAL requirements, CSPs that use trusted referees in their identity proofing services must demonstrate conformance with the requirements contained in this section.

1. If the CSP uses trusted referees, then...The CSP SHALL establish written policy and procedures as to how a trusted referee is determined and the lifecycle by which the trusted referee retains their status as a valid referee, to include any restrictions, as well as any revocation and suspension requirements.

SUPPLEMENTAL GUIDANCE: In instances where an individual cannot meet the identity evidence requirements specified in IA-12 (3) y – ee and IA-12 (5) b - i, the agency may use a trusted referee to assist in identity proofing the applicant. It is intended that CSPs using trusted referees for identity proofing and enrollment will document the procedures and controls in an applicable written policy or *practice statement* as described in IA-12 (3) h.

2. If the CSP uses trusted referees, then...The CSP SHALL proof the trusted referee at the same IAL as the applicant proofing.

SUPPLEMENTAL GUIDANCE: Trusted referees, who participate in the identity proofing process on behalf of an applicant need to be identity proofed themselves to the same level as that of the applicant. If CSPs allow the use of Trusted Referees, its documented policies should state this requirement.

3. If the CSP uses trusted referees, then...The CSP SHALL determine the minimum evidence required to bind the relationship between the trusted referee and the applicant.

SUPPLEMENTAL GUIDANCE: In addition to proofing a Trusted Referee to the same (or greater) IAL as that of the applicant, CSPs will need to determine its process for proving a legitimate relationship to the applicant. The CSP should consider and document the types of evidence (i.e., power of attorney) it will accept to “bind” the relationship between Trusted Referee and an applicant. This minimum evidence may vary based on IAL.

Related Controls: None.

(5) IDENTITY PROOFING | ADDRESS CONFIRMATION

[Priority 2]

Control:

- a. Require that a registration code or notice of proofing be delivered through an out-of-band channel to verify the user's address (physical or digital) of record.

DISCUSSION: To make it more difficult for adversaries to pose as legitimate users during the identity proofing process, organizations can use out-of-band methods to ensure that the individual associated with an address of record is the same individual that participated in the registration. Confirmation can take the form of a temporary enrollment code or a notice of proofing. The delivery address for these artifacts is obtained from records and not self-asserted by the user. The address can include a physical or digital address. A home address is an example of a physical address. Email addresses and telephone numbers are examples of digital addresses.

- b. The CSP SHALL confirm address of record.

SUPPLEMENTAL GUIDANCE: Valid records to confirm address are issuing source(s) or authoritative source(s). Ideally, the CSP will confirm an address of record through validation of the address contained on any supplied, valid piece of identity evidence. However, the CSP may confirm address of record by validating information supplied by the applicant that is not contained on any supplied piece of identity evidence. Postal addresses are preferred, however these guidelines support any type of address that can be validated against an issuing or authoritative source, whether physical or digital. Acceptable addresses of record include postal addresses, email addresses, and telephone numbers. The types of addresses of record a CSP accepts will determine, in part, the method it employs to validate them. For instance, postal addresses can be validated by confirming it against a piece of supplied, valid identity evidence. Email addresses may be confirmed by sending an email to the provided address.

- c. Valid records to confirm address SHALL be issuing source(s) or authoritative source(s).

Self-asserted address data that has not been confirmed in records SHALL NOT be used for confirmation.

SUPPLEMENTAL GUIDANCE: An address of record is a “validated and verified location (physical or digital) where an individual can receive communications using approved mechanisms.” IAL2 requires confirming an applicant’s address of record. This can be accomplished in two ways: 1) validation of the address contained on a valid piece of identity evidence, or 2) by employing a mechanism such as enrollment codes to validate an address not contained on a supplied piece of identity evidence.

Addresses that are supplied by an applicant, either verbally or on a non-valid piece of identity evidence, are not valid for confirming an applicant’s address of record.

- d. Note that IAL2-7 applies only to in-person proofing at IAL2.

If the CSP performs in-person proofing for IAL2 and provides an enrollment code directly to the subscriber for binding to an authenticator at a later time, then the enrollment code...SHALL be valid for a maximum of seven (7) days.

SUPPLEMENTAL GUIDANCE: Upon successful completion of the identity proofing process the CSP will typically register one or more authenticators to the subscribers’ account or may optionally choose to bind an authenticator(s) at a later time. If the CSP chooses to use an enrollment code provided directly to the applicant to authenticate for such later binding, the validity period for the enrollment code is a maximum of seven days (see IA-12 (3) s).

Note that conformance criteria IA-12 (5) ‘e’ through ‘i’ apply to remote identity proofing processes at IAL2.

- e. For remote identity proofing at IAL2:

The CSP SHALL send an enrollment code to a confirmed address of record for the applicant.

SUPPLEMENTAL GUIDANCE: Enrollment codes used for IAL2 remote identity proofing may be sent to any confirmed address of record – postal, mobile phone number for SMS, or email addresses.

- f. For remote identity proofing at IAL2:

The applicant SHALL present a valid enrollment code to complete the identity proofing process.

SUPPLEMENTAL GUIDANCE: Per IA-12 (5) e above, sending an enrollment code to a confirmed address of record, as captured during the identity proofing process, is required to complete the remote identity proofing process and provides additional confidence in the binding of that address to the applicant.

Valid enrollment codes mean that the correct enrollment code is submitted by the applicant within prescribed validity periods. Enrollment code validity periods depend on the type of address where the code is sent as shown in IA-12 (5) g below.

Information captured in the CSP's enrollment records or system logs facilitate assessment against this requirement. Ideally, this information would include details about the validity of the enrollment code (date and time applicant entered code; confirmation it was the correct code; and confirmation it was not expired).

- g. Note that the following enrollment code validity periods apply to enrollment codes sent to confirmed addresses of record for IAL2 remote in-person proofing only.

Enrollment codes shall have the following maximum validities:

1. 10 days, when sent to a postal address of record within the contiguous United States;
2. 30 days, when sent to a postal address of record outside the contiguous United States;
3. 10 minutes, when sent to a telephone of record (SMS or voice);
4. 24 hours, when sent to an email address of record.

SUPPLEMENTAL GUIDANCE: Enrollment codes sent to addresses of record are only valid for a limited amount of time, depending on the type of address of record to which they are sent. Applicants that present enrollment codes that are no longer valid (aka, expired) cannot use this code to complete their identity proofing process.

- h. If the enrollment code sent to the confirmed address of record as part of the remote identity proofing process at IAL2 is also intended to be an authentication factor, then...it SHALL be reset upon first use.

SUPPLEMENTAL GUIDANCE: Enrollment codes sent as an authentication factor for address confirmation may only be used once.

- i. If the CSP performs remote proofing at IAL2 and optionally sends notification of proofing in addition to sending the required enrollment code, then...The CSP SHALL ensure the enrollment code and notification of proofing are sent to different addresses of record.

SUPPLEMENTAL GUIDANCE: For example, if the CSP sends an enrollment code to a phone number validated in records, a proofing notification may be sent to the postal address validated in records or obtained from validated and verified evidence, such as a driver's license.

Related Controls: IA-12.

INCIDENT RESPONSE (IR)

IR-1 POLICY AND PROCEDURES

[Existing] [Priority 2]

Control:

- a. Develop, document, and disseminate to all personnel when their unescorted logical or physical access to any information system results in the ability, right, or privilege to view, modify, or make use of unencrypted CJI:
 1. Agency-level incident response policy that:
 - a. Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - b. Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the incident response policy and the associated incident response controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the incident response policy and procedures; and
- c. Review and update the current incident response:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Incident response policy and procedures address the controls in the IR family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of incident response policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to incident response policy and procedures include assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

IR-2 INCIDENT RESPONSE TRAINING

[Existing] [Priority 2]

Control:

- a. Provide incident response training to system users consistent with assigned roles and responsibilities:
 1. Prior to assuming an incident response role or responsibility or acquiring system access;
 2. When required by system changes; and
 3. Annually thereafter; and
- b. Review and update incident response training content annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Incident response training is associated with the assigned roles and responsibilities of organizational personnel to ensure that the appropriate content and level of detail are included in such training. For example, users may only need to know who to call or how to recognize an incident; system administrators may require additional training on how to handle incidents; and incident responders may receive more specific training on forensics, data collection techniques, reporting, system recovery, and system restoration. Incident response training includes user training in identifying and reporting suspicious activities from external and internal sources. Incident response training for users may be provided as part of AT-2 or AT-3.

Events that may precipitate an update to incident response training content include, but are not limited to, incident response plan testing or response to an actual incident (lessons learned), assessment or audit findings, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Related Controls: AT-2, AT-3, AT-4, CP-3, IR-3, IR-4, IR-8.

Control Enhancements:

(3) INCIDENT RESPONSE TRAINING | BREACH

[Priority 3]

Control:

Provide incident response training on how to identify and respond to a breach, including the organization's process for reporting a breach.

Discussion: For agencies, an incident that involves personally identifiable information is considered a breach. A breach results in the loss of control, compromise, unauthorized disclosure, unauthorized acquisition, or a similar occurrence where a person other than an authorized user accesses or potentially accesses personally identifiable information or an authorized user accesses or potentially accesses such information for other than authorized purposes. The incident response training emphasizes the obligation of individuals to report both confirmed and suspected breaches

involving information in any medium or form, including paper, oral, and electronic. Incident response training includes tabletop exercises that simulate a breach.

IR-3 INCIDENT RESPONSE TESTING

[Priority 3]

Control:

Test the effectiveness of the incident response capability for the system annually using the following tests: tabletop or walk-through exercises; simulations; or other agency-appropriate tests.

Discussion: Organizations test incident response capabilities to determine their effectiveness and identify potential weaknesses or deficiencies. Incident response testing includes the use of checklists, walk-through or tabletop exercises, and simulations (parallel or full interrupt). Incident response testing can include a determination of the effects on organizational operations and assets and individuals due to incident response. The use of qualitative and quantitative data aids in determining the effectiveness of incident response processes.

Related Controls: CP-3, CP-4, IR-2, IR-4, IR-8.

Control Enhancements:

(2) INCIDENT RESPONSE TESTING | COORDINATION WITH RELATED PLANS

[Priority 3]

Control:

Coordinate incident response testing with organizational elements responsible for related plans.

Discussion: Organizational plans related to incident response testing include business continuity plans, disaster recovery plans, continuity of operations plans, contingency plans, crisis communications plans, critical infrastructure plans, and occupant emergency plans.

IR-4 INCIDENT HANDLING

[Existing] [Priority 2]

Control:

- a. Implement an incident handling capability for incidents that is consistent with the incident response plan and includes preparation, detection and analysis, containment, eradication, and recovery;
- b. Coordinate incident handling activities with contingency planning activities;
- c. Incorporate lessons learned from ongoing incident handling activities into incident response procedures, training, and testing, and implement the resulting changes accordingly; and
- d. Ensure the rigor, intensity, scope, and results of incident handling activities are comparable and predictable across the organization.

Discussion: Organizations recognize that incident response capabilities are dependent on the capabilities of organizational systems and the mission and business processes being supported by those systems. Organizations consider incident response as part of the definition, design, and development of mission and business processes and systems. Incident-related information can be obtained from a variety of sources, including audit monitoring, physical access monitoring, and network monitoring; user or administrator reports; and reported supply chain events. An effective incident handling capability includes coordination among many organizational entities (e.g., mission or business owners, system owners, authorizing officials, human resources offices, physical security offices, personnel security offices, legal departments, risk executive [function], operations personnel, procurement offices). Suspected security incidents include the receipt of suspicious email communications that can contain malicious code. Suspected supply chain incidents include the insertion of counterfeit hardware or malicious code into organizational systems or system components. For agencies, an incident that involves personally identifiable information is considered a breach. A breach results in unauthorized disclosure, the loss of control, unauthorized acquisition, compromise, or a similar occurrence where a person other than an authorized user accesses or potentially accesses personally identifiable information or an authorized user accesses or potentially accesses such information for other than authorized purposes.

Related Controls: AC-19, AU-6, AU-7, CM-6, CP-2, CP-3, CP-4, IR-2, IR-3, IR-5, IR-6, IR-8, PE-6, PL-2, SA-8, SC-5, SC-7, SI-3, SI-4, SI-7.

Control Enhancements:

(1) INCIDENT HANDLING | AUTOMATED INCIDENT HANDLING PROCESSES

[Existing] [Priority 2]

Control:

Support the incident handling process using automated mechanisms (e.g., online incident management systems and tools that support the collection of live response data, full network packet capture, and forensic analysis).

Discussion: Automated mechanisms that support incident handling processes include online incident management systems and tools that support the collection of live response data, full network packet capture, and forensic analysis. Incident handling could be inherited from an upstream agency or could be part of a state-level process.

IR-5 INCIDENT MONITORING

[Existing] [Priority 2]

Control: Track and document incidents.

Discussion: Documenting incidents includes maintaining records about each incident, the status of the incident, and other pertinent information necessary for forensics as well as evaluating incident details, trends, and handling. Incident information can be obtained from a variety of sources, including network monitoring, incident reports, incident response teams, user complaints, supply

chain partners, audit monitoring, physical access monitoring, and user and administrator reports. IR-4 provides information on the types of incidents that are appropriate for monitoring.

Related Controls: AU-6, AU-7, IR-4, IR-6, IR-8, PE-6, SC-5, SC-7, SI-3, SI-4, SI-7.

IR-6 INCIDENT REPORTING

[Priority 2]

Control:

- a. Require personnel to report suspected incidents to the organizational incident response capability immediately but not to exceed one (1) hour after discovery; and
- b. Report incident information to organizational personnel with incident handling responsibilities, and the CSO, SIB Chief, or Interface Agency Official, and FBI CJIS ISO.

Discussion: The types of incidents reported, the content and timeliness of the reports, and the designated reporting authorities reflect applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Incident information can inform risk assessments, control effectiveness assessments, security requirements for acquisitions, and selection criteria for technology products.

Related Controls: CM-6, CP-2, IR-4, IR-5, IR-8.

Control Enhancements:

(1) INCIDENT REPORTING | AUTOMATED REPORTING

[Existing] [Priority 2]

Control:

Report incidents using automated mechanisms.

Discussion: The recipients of incident reports are specified in IR-6b. Automated reporting mechanisms include email, posting on websites (with automatic updates), and automated incident response tools and programs.

Related Controls: IR-7.

(3) INCIDENT REPORTING | SUPPLY CHAIN COORDINATION

[Priority 2]

Control:

Provide incident information to the provider of the product or service and other organizations involved in the supply chain or supply chain governance for systems or system components related to the incident.

Discussion: Organizations involved in supply chain activities include product developers, system integrators, manufacturers, packagers, assemblers, distributors, vendors, and resellers. Entities that provide supply chain governance include the Federal Acquisition Security Council (FASC). Supply chain incidents include compromises or breaches that involve information technology

products, system components, development processes or personnel, distribution processes, or warehousing facilities. Organizations determine the appropriate information to share and consider the value gained from informing external organizations about supply chain incidents, including the ability to improve processes or to identify the root cause of an incident.

Related Controls: SR-8.

IR-7 INCIDENT RESPONSE ASSISTANCE

[Existing] [Priority 3]

Control:

Provide an incident response support resource, integral to the organizational incident response capability, that offers advice and assistance to users of the system for the handling and reporting of incidents.

Discussion: Incident response support resources provided by organizations include help desks, assistance groups, automated ticketing systems to open and track incident response tickets, and access to forensics services or consumer redress services, when required.

Related Controls: AT-2, AT-3, IR-4, IR-6, IR-8, SA-9.

Control Enhancements:

(1) INCIDENT RESPONSE ASSISTANCE | AUTOMATION SUPPORT FOR AVAILABILITY OF INFORMATION AND SUPPORT

[Priority 3]

Control:

Increase the availability of incident response information and support using automated mechanisms described in the discussion.

Discussion: Automated mechanisms can provide a push or pull capability for users to obtain incident response assistance. For example, individuals may have access to a website to query the assistance capability, or the assistance capability can proactively send incident response information to users (general distribution or targeted) as part of increasing understanding of current response capabilities and support.

If the automated mechanisms include external assistance that will give unescorted physical or logical access to CJI, it is imperative to ensure that the appropriate controls/procedures (CJIS Security Addendum/Outsourcing Standard) are in place. Examples would include Cyber Incident Response Vendors (IT Security/Law Firms).

Related Controls: None.

IR-8 INCIDENT RESPONSE PLAN

[Existing] [Priority 2]

Control:

- a. Develop an incident response plan that:
 1. Provides the organization with a roadmap for implementing its incident response capability;
 2. Describes the structure and organization of the incident response capability;
 3. Provides a high-level approach for how the incident response capability fits into the overall organization;
 4. Meets the unique requirements of the organization, which relate to mission, size, structure, and functions;
 5. Defines reportable incidents;
 6. Provides metrics for measuring the incident response capability within the organization;
 7. Defines the resources and management support needed to effectively maintain and mature an incident response capability;
 8. Addresses the sharing of incident information;
 9. Is reviewed and approved by the organization's/agency's executive leadership annually; and
 10. Explicitly designates responsibility for incident response to organizational personnel with incident reporting responsibilities and CSO or CJIS WAN Official.
- b. Distribute copies of the incident response plan to organizational personnel with incident handling responsibilities;
- c. Update the incident response plan to address system and organizational changes or problems encountered during plan implementation, execution, or testing;
- d. Communicate incident response plan changes to organizational personnel with incident handling responsibilities; and
- e. Protect the incident response plan from unauthorized disclosure and modification.

Discussion: It is important that organizations develop and implement a coordinated approach to incident response. Organizational mission and business functions determine the structure of incident response capabilities. As part of the incident response capabilities, organizations consider the coordination and sharing of information with external organizations, including external service providers and other organizations involved in the supply chain. For incidents involving personally identifiable information (i.e., breaches), include a process to determine whether notice to oversight organizations or affected individuals is appropriate and provide that notice accordingly.

Related Controls: AC-2, CP-2, CP-4, IR-4, IR-7, PE-6, PL-2, SA-15, SI-12, SR-8.

Control Enhancements:

(1) INCIDENT RESPONSE PLAN | BREACHES

[Priority 2]

Control:

Include the following in the Incident Response Plan for breaches involving personally identifiable information:

- a. A process to determine if notice to individuals or other organizations, including oversight organizations, is needed;
- b. An assessment process to determine the extent of the harm, embarrassment, inconvenience, or unfairness to affected individuals and any mechanisms to mitigate such harms; and
- c. Identification of applicable privacy requirements.

Discussion: Organizations may be required by law, regulation, or policy to follow specific procedures relating to breaches, including notice to individuals, affected organizations, and oversight bodies; standards of harm; and mitigation or other specific requirements.

Figure 11 – Incident Response Process Initiated by an Incident in a Local Police Department

A state ISO received a notification from a local police department that suspicious network activity from a known botnet was detected on their network. The state ISO began the process of collecting all pertinent information about this incident, e.g., incident date/time, points-of-contact, systems affected, nature of the incident, actions taken, etc. and requested that the local police department confirm that their malware signatures were up to date. The state ISO contacted both the FBI CJIS ISO and state CSO to relay the preliminary details of this incident. The FBI CJIS ISO instructed the involved parties to continue their investigation and to submit an incident response form once all the information had been gathered. The FBI CJIS ISO contacted the lead for the FBI CSIRC to inform them that an incident response form was forthcoming. The state ISO gathered the remainder of the information from the local police department and submitted a completed incident response form to the FBI CJIS ISO who subsequently provided it to the FBI CSIRC. The FBI CSIRC notified the Department of Justice Computer Incident Response Team (DOJCIRT). The state ISO continued to monitor the situation, passing relevant details to the FBI CJIS ISO, ultimately determining that the botnet was eliminated from the local police department's infrastructure. Subsequent investigations determined that the botnet was restricted to the department's administrative infrastructure and thus no CJI was compromised.

MAINTENANCE (MA)

MA-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with system maintenance responsibilities:
 1. Agency-level maintenance policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the maintenance policy and the associated maintenance controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the maintenance policy and procedures; and
- c. Review and update the current maintenance:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Maintenance policy and procedures address the controls in the MA family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of maintenance policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to maintenance policy and procedures assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

MA-2 CONTROLLED MAINTENANCE

[Priority 3]

Control:

- a. Schedule, document, and review records of maintenance, repair, and replacement on system components in accordance with manufacturer or vendor specifications and/or organizational requirements;
- b. Approve and monitor all maintenance activities, whether performed on site or remotely and whether the system or system components are serviced on site or removed to another location;
- c. Require that organizational personnel with information security and privacy responsibilities explicitly approve the removal of the system or system components from organizational facilities for off-site maintenance, repair, or replacement;
- d. Sanitize equipment to remove information from associated media prior to removal from organizational facilities for off-site maintenance, repair, replacement, or destruction;
- e. Check all potentially impacted controls to verify that the controls are still functioning properly following maintenance, repair, or replacement actions; and
- f. Include the following information in organizational maintenance records:
 1. Component name
 2. Component serial number
 3. Date/time of maintenance
 4. Maintenance performed
 5. Name(s) of entity performing maintenance including escort if required.

Discussion: Controlling system maintenance addresses the information security aspects of the system maintenance program and applies to all types of maintenance to system components conducted by local or nonlocal entities. Maintenance includes peripherals such as scanners, copiers, and printers. Information necessary for creating effective maintenance records includes the date and time of maintenance, a description of the maintenance performed, names of the individuals or group performing the maintenance, name of the escort, and system components or equipment that are removed or replaced. Organizations consider supply chain-related risks associated with replacement components for systems.

Related Controls: CM-2, CM-3, CM-4, CM-5, CM-8, MA-4, MP-6, PE-16, SI-2, SR-3, SR-11.

MA-3 MAINTENANCE TOOLS

[Priority 4]

Control:

- a. Approve, control, and monitor the use of system maintenance tools; and
- b. Review previously approved system maintenance tools prior to each use.

Discussion: Approving, controlling, monitoring, and reviewing maintenance tools address security-related issues associated with maintenance tools that are not within system authorization boundaries and are used specifically for diagnostic and repair actions on organizational systems. Organizations have flexibility in determining roles for the approval of maintenance tools and how that approval is documented. A periodic review of maintenance tools facilitates the withdrawal of approval for outdated, unsupported, irrelevant, or no-longer-used tools. Maintenance tools can include hardware, software, and firmware items and may be pre-installed, brought in with maintenance personnel on media, cloud-based, or downloaded from a website. Such tools can be vehicles for transporting malicious code, either intentionally or unintentionally, into a facility and subsequently into systems. Maintenance tools can include hardware and software diagnostic test equipment and packet sniffers. The hardware and software components that support maintenance and are a part of the system (including the software implementing utilities such as “ping,” “ls,” “ipconfig,” or the hardware and software implementing the monitoring port of an Ethernet switch) are not addressed by maintenance tools.

Related Controls: MA-2, PE-16.

Control Enhancements:

(1) MAINTENANCE TOOLS | INSPECT TOOLS

[Priority 4]

Control:

Inspect the maintenance tools used by maintenance personnel for improper or unauthorized modifications.

Discussion: Maintenance tools can be directly brought into a facility by maintenance personnel or downloaded from a vendor’s website. If, upon inspection of the maintenance tools, organizations determine that the tools have been modified in an improper manner or the tools contain malicious code, the incident is handled consistent with organizational policies and procedures for incident handling.

Related Controls: SI-7.

(2) MAINTENANCE TOOLS | INSPECT MEDIA

[Priority 4]

Control:

Check media containing diagnostic and test programs for malicious code before the media are used in the system.

Discussion: If, upon inspection of media containing maintenance, diagnostic, and test programs, organizations determine that the media contains malicious code, the incident is handled consistent with organizational incident handling policies and procedures.

Related Controls: SI-3.

(3) MAINTENANCE TOOLS | PREVENT UNAUTHORIZED REMOVAL

[Priority 4]

Control:

Prevent the removal of maintenance equipment containing organizational information by:

- a. Verifying that there is no organizational information contained on the equipment;
- b. Sanitizing or destroying the equipment;
- c. Retaining the equipment within the facility; or
- d. Obtaining an exemption from organizational personnel with system maintenance responsibilities explicitly authorizing removal of the equipment from the facility.

Discussion: Organizational information includes all information owned by organizations and any information provided to organizations for which the organizations serve as information stewards.

Related Controls: MP-6.

MA-4 NONLOCAL MAINTENANCE

[Priority 3]

Control:

- a. Approve and monitor nonlocal maintenance and diagnostic activities;
- b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system;
- c. Employ strong authentication in the establishment of nonlocal maintenance and diagnostic sessions;
- d. Maintain records for nonlocal maintenance and diagnostic activities; and
- e. Terminate session and network connections when nonlocal maintenance is completed.

Discussion: Nonlocal maintenance and diagnostic activities are conducted by individuals who communicate through either an external or internal network. Local maintenance and diagnostic activities are carried out by individuals who are physically present at the system location and not communicating across a network connection. Authentication techniques used to establish nonlocal maintenance and diagnostic sessions reflect the network access requirements in IA-2. Strong authentication requires authenticators that are resistant to replay attacks and employ multi-factor authentication. Strong authenticators include PKI where certificates are stored on a token protected by a password, passphrase, or biometric. Enforcing requirements in MA-4 is accomplished, in part, by other controls.

Related Controls: AC-2, AC-3, AC-6, AC-17, AU-2, AU-3, IA-2, IA-4, IA-5, IA-8, MA-2, MA-5, PL-2, SC-7, SC-10.

MA-5 MAINTENANCE PERSONNEL

[Priority 3]

Control:

- a. Establish a process for maintenance personnel authorization and maintain a list of authorized maintenance organizations or personnel;

- b. Verify that non-escorted personnel performing maintenance on the system possess the required access authorizations; and
- c. Designate organizational personnel with required access authorizations and technical competence to supervise the maintenance activities of personnel who do not possess the required access authorizations.

Discussion: Maintenance personnel refers to individuals who perform hardware or software maintenance on organizational systems, while PE-2 addresses physical access for individuals whose maintenance duties place them within the physical protection perimeter of the systems. Technical competence of supervising individuals relates to the maintenance performed on the systems, while having required access authorizations refers to maintenance on and near the systems. Individuals not previously identified as authorized maintenance personnel—such as information technology manufacturers, vendors, systems integrators, and consultants—may require privileged access to organizational systems, such as when they are required to conduct maintenance activities with little or no notice. Based on organizational assessments of risk, organizations may issue temporary credentials to these individuals. Temporary credentials may be for one-time use or for very limited time periods.

Related Controls: AC-2, AC-3, AC-5, AC-6, IA-2, IA-8, MA-4, MP-2, PE-2, PE-3, PS-7, RA-3.

MA-6 TIMELY MAINTENANCE

[Priority 3]

Control:

Obtain maintenance support and/or spare parts for critical system components that process, store, or transmit CJI within agency-defined recovery time and recovery point objectives of failure.

Discussion: Organizations specify the system components that result in increased risk to organizational operations and assets, individuals, other organizations, or the Nation when the functionality provided by those components is not operational. Organizational actions to obtain maintenance support include having appropriate contracts in place.

Related Controls: CM-8, CP-2, CP-7, RA-7, SA-15, SR-2, SR-3.

MEDIA PROTECTION (MP)

Documented and implemented media protection policies and procedures ensure that access to digital and non-digital media in all forms is restricted to authorized individuals using authorized methods and processes.

MP-1 POLICY AND PROCEDURES

[Existing] [Priority 2]

Control:

- a. Develop, document, and disseminate to authorized individuals:
 1. Agency-level media protection policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among agency entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the media protection policy and the associated media protection controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the media protection policy and procedures; and
- c. Review and update the current media protection:
 1. Policy at least annually and following any security incidents involving digital and/or non-digital media; and
 2. Procedures at least annually and following any security incidents involving digital and/or non-digital media.

Discussion: Media protection policy and procedures address the controls in the MP family that are implemented within systems and agencies. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of media protection policy and procedures. Security and privacy program policies and procedures at the agency level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of agencies. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to media protection policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an agency policy or procedure.

Related Controls: PS-8, SI-12.

MP-2 MEDIA ACCESS

[Existing] [Priority 2]

Control:

Restrict access to digital and non-digital media to authorized individuals.

Discussion: System media includes digital and non-digital media. Digital media includes flash drives, diskettes, magnetic tapes, external or removable hard disk drives (e.g., solid state, magnetic), compact discs, and digital versatile discs. Non-digital media includes paper and microfilm. Denying access to hard copies of case file information stored in a locked filing cabinet is an example of restricting access to non-digital media. Limiting access to the design specifications stored on compact discs in the media library to individuals on the system development team is an example of restricting access to digital media.

Related Controls: AC-19, AU-9, CP-2, CP-9, CP-10, MA-5, MP-4, MP-6, PE-2, PE-3, SC-12, SC-13, SI-12.

MP-3 MEDIA MARKING

[Priority 3]

Control:

- a. Mark system media indicating the distribution limitations, handling caveats, and applicable security markings (if any) of the information; and
- b. Exempt digital and non-digital media containing CJI from marking if the media remain within physically secure locations or controlled areas.

Discussion: Security marking refers to the application or use of human-readable security attributes. Digital media includes diskettes, magnetic tapes, external or removable hard disk drives (e.g., solid state, magnetic), flash drives, compact discs, and digital versatile discs. Non-digital media includes paper and microfilm. Controlled unclassified information (CUI) is defined by the National Archives and Records Administration (NARA) along with the appropriate safeguarding and dissemination requirements for such information and is codified in [32 CFR 2002]. Security markings are generally not required for media that contains information determined by agencies to be in the public domain or to be publicly releasable. Some agencies may require markings for public information indicating that the information is publicly releasable. System media marking reflects applicable laws, executive orders, directives, policies, regulations, standards, and guidelines. Executive Order 13556 mandates the marking of CUI. The following categories of CJI fall under the designated NARA categories which direct marking as indicated.

1. Biometric Data—data derived from one or more intrinsic physical or behavioral traits of humans typically for the purpose of uniquely identifying individuals from within a population. Used to identify individuals, to include: fingerprints, palm prints, iris scans, and facial recognition data. NARA CUI Category: Sensitive Personally Identifiable Information. NARA Banner Marking: CUI

- ~~2. Identity History Data—textual data that corresponds with an individual’s biometric data, providing a history of criminal and/or civil events for the identified individual. NARA CUI Category: Criminal History Records Information. NARA Banner Marking: CUI//SP-CHRI~~
- ~~3. Biographic Data—information about individuals associated with a unique case, and not necessarily connected to identity data. Biographic data does not provide a history of an individual, only information related to a unique case. NARA CUI Category: General Law Enforcement. NARA Banner Marking: CUI~~
- ~~4. Property Data—information about vehicles and property associated with crime when accompanied by any personally identifiable information (PII). NARA CUI Category: General Law Enforcement. NARA Banner Marking: CUI~~
- ~~5. Case/Incident History—information about the history of criminal incidents. NARA CUI Category: General Law Enforcement. NARA Banner Marking: CUI~~

~~[32 CFR 2002] states the CUI Program prohibits using markings or practices not included in this part or the CUI Registry. If legacy markings remain on information, the legacy markings are void and no longer indicate that the information is protected or that it is or qualifies as CUI.~~

~~Related Controls: CP-9, MP-5, SI-12.~~

MP-4 MEDIA STORAGE

[Existing] [Priority 2]

Control:

- a. Physically control and securely store digital and non-digital media within physically secure locations or controlled areas and encrypt CJI on digital media when physical and personnel restrictions are not feasible; and
- b. Protect system media types defined in MP-4a until the media are destroyed or sanitized using approved equipment, techniques, and procedures.

Discussion: System media includes digital and non-digital media. Digital media includes flash drives, diskettes, magnetic tapes, external or removable hard disk drives (e.g., solid state, magnetic), compact discs, and digital versatile discs. Non-digital media includes paper and microfilm. Physically controlling stored media includes conducting inventories, ensuring procedures are in place to allow individuals to check out and return media to the library, and maintaining accountability for stored media. Secure storage includes a locked drawer, desk, or cabinet or a controlled media library. The type of media storage is commensurate with the security category or classification of the information on the media. Controlled areas are spaces that provide physical and procedural controls to meet the requirements established for protecting information and systems. Fewer controls may be needed for media that contains information determined to be in the public domain, publicly releasable, or have limited adverse impacts on agencies, operations, or individuals if accessed by other than authorized personnel. In these situations, physical access controls provide adequate protection.

Related Controls: AC-19, CP-2, CP-6, CP-9, CP-10, MP-2, MP-7, PE-3, PL-2, SC-12, SC-13, SC-28, SI-12.

MP-5 MEDIA TRANSPORT

[Existing] [Priority 2]

Control:

- a. Protect and control digital and non-digital media to help prevent compromise of the data during transport outside of the physically secure locations or controlled areas using encryption, as defined in SC-13 and SC-28 of this Policy. Physical media will be protected at the same level as the information would be protected in electronic form. Restrict the activities associated with transport of electronic and physical media to authorized personnel;
- b. Maintain accountability for system media during transport outside of the physically secure location or controlled areas;
- c. Document activities associated with the transport of system media; and
- d. Restrict the activities associated with the transport of system media to authorized personnel.

Discussion: System media includes digital and non-digital media. Digital media includes flash drives, diskettes, magnetic tapes, external or removable hard disk drives (e.g., solid state and magnetic), compact discs, and digital versatile discs. Non-digital media includes microfilm and paper. Controlled areas are spaces for which agencies provide physical or procedural controls to meet requirements established for protecting information and systems. Controls to protect media during transport include cryptography and locked containers. Cryptographic mechanisms can provide confidentiality and integrity protections depending on the mechanisms implemented. Activities associated with media transport include releasing media for transport, ensuring that media enters the appropriate transport processes, and the actual transport. Authorized transport and courier personnel may include individuals external to the agency. Maintaining accountability of media during transport includes restricting transport activities to authorized personnel and tracking and/or obtaining records of transport activities as the media moves through the transportation system to prevent and detect loss, destruction, or tampering. Agencies establish documentation requirements for activities associated with the transport of system media in accordance with agency assessments of risk. Agencies maintain the flexibility to define record-keeping methods for the different types of media transport as part of a system of transport-related records.

Related Controls: AC-7, AC-19, CP-2, CP-9, MP-3, MP-4, PE-16, PL-2, SC-12, SC-13, SC-28.

MP-6 MEDIA SANITIZATION

[Existing] [Priority 2]

Control:

- a. Sanitize or destroy digital media prior to disposal, release out of agency control, or release for reuse using overwrite technology at least three times or degaussing. Inoperable digital media will be destroyed. Non-digital media will be securely disposed of when no longer needed for investigative or security purposes, whichever is later. Non-digital media will be destroyed by crosscut shredding or incineration; and

- b. Employ sanitization mechanisms with the strength and integrity commensurate with the security category or classification of the information.

Discussion: Media sanitization applies to all digital and non-digital system media subject to disposal or reuse, whether or not the media is considered removable. Examples include digital media in scanners, copiers, printers, notebook computers, workstations, network components, mobile devices, and non-digital media (e.g., paper and microfilm). The sanitization process removes information from system media such that the information cannot be retrieved or reconstructed. Sanitization techniques—including clearing, purging, cryptographic erase, de-identification of personally identifiable information, and destruction—prevent the disclosure of information to unauthorized individuals when such media is reused or released for disposal. Agencies determine the appropriate sanitization methods, recognizing that destruction is sometimes necessary when other methods cannot be applied to media requiring sanitization.

Agencies use discretion on the employment of approved sanitization techniques and procedures for media that contains information deemed to be in the public domain or publicly releasable or information deemed to have no adverse impact on agencies or individuals if released for reuse or disposal. Sanitization of non-digital media includes destruction, removing a classified appendix from an otherwise unclassified document, or redacting selected sections or words from a document by obscuring the redacted sections or words in a manner equivalent in effectiveness to removing them from the document. NSA standards and policies control the sanitization process for media that contains classified information. NARA policies control the sanitization process for controlled unclassified information.

Related Controls: AC-3, AC-7, AU-11, MA-2, MA-3, MA-4, MA-5, SI-12, SR-11.

MP-7 MEDIA USE

[Existing] [Priority 2]

Control:

- a. Restrict the use of digital and non-digital media on agency owned systems that have been approved for use in the storage, processing, or transmission of criminal justice information by using technical, physical, or administrative controls (examples below); and
- b. Prohibit the use of personally owned digital media devices on all agency owned or controlled systems that store, process, or transmit criminal justice information; and
- c. Prohibit the use of digital media devices on all agency owned or controlled systems that store, process, or transmit criminal justice information when such devices have no identifiable owner.

Examples of technical controls: port disabling, access control lists (ACL), security groups, group policy objects (GPO), mobile device management (MDM).

Example of physical control: locked server cage, disconnect CD-ROM drive in PC, remove USB port.

Example of administrative controls: the agency's electronic media policy defining how flash drives are to be used within the agency rules of behavior.

Discussion: System media includes both digital and non-digital media. Digital media includes diskettes, magnetic tapes, flash drives, compact discs, digital versatile discs, and removable hard disk drives. Non-digital media includes paper and microfilm. Media use protections also apply to mobile devices with information storage capabilities. In contrast to MP-2, which restricts user access to media, MP-7 restricts the use of certain types of media on systems, for example, restricting or prohibiting the use of flash drives or external hard disk drives. Agencies use technical and nontechnical controls to restrict the use of system media. Agencies may restrict the use of portable storage devices, for example, by using physical cages on workstations to prohibit access to certain external ports or disabling or removing the ability to insert, read, or write to such devices. Agencies may also limit the use of portable storage devices to only approved devices, including devices provided by the agency, devices provided by other approved agencies, and devices that are not personally owned. Finally, agencies may restrict the use of portable storage devices based on the type of device, such as by prohibiting the use of writeable, portable storage devices and implementing this restriction by disabling or removing the capability to write to such devices. Requiring identifiable owners for storage devices reduces the risk of using such devices by allowing agencies to assign responsibility for addressing known vulnerabilities in the devices.

Related Controls: AC-19, AC-20, PL-4.

Figure 12 – A Local Police Department’s Media Management Policies

A local police department implemented a replacement CAD system that integrated to their state’s CSA and was authorized to process CJI. The police department contracted with an off-site media manager to store backups of their data in the contractor’s vaults, but the contractor was not authorized to process or store CJI. To ensure the confidentiality of the police department’s data while outside its perimeter, they encrypted all data going to the contractor with an encryption product that is FIPS 140-2 certified. The police department rotated and reused media through the contractor’s vaults periodically, and when it required destruction, the police department incinerated the media to irreversibly destroy any data on it.

PHYSICAL AND ENVIRONMENTAL PROTECTION (PE)

PE-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with physical and environmental protection responsibilities:
 1. Agency-level physical and environmental protection policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the physical and environmental protection policy and the associated physical and environmental protection controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the physical and environmental protection policy and procedures; and
- c. Review and update the current physical and environmental protection:
 1. Policy annually and following any physical, environmental, or security related incidents involving CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any physical, environmental, or security related incidents involving CJI or systems used to process, store, or transmit CJI.

Discussion: Physical and environmental protection policy and procedures address the controls in the PE family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of physical and environmental protection policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to physical and environmental protection policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Physical and environmental protection policy and procedures address the controls in the PE family that are implemented within organizations for protection of physically secure locations which are a facility, a criminal justice conveyance, or an area, a room, or a group of rooms, within a facility with both the physical and personnel security controls sufficient to protect CJI and associated information systems.

Related Controls: AT-3, PS-8, SI-12.

PE-2 PHYSICAL ACCESS AUTHORIZATIONS

[Existing] [Priority 2]

Control:

- a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides;
- b. Issue authorization credentials for facility access;
- c. Review the access list detailing authorized facility access by individuals annually and when personnel changes occur; and
- d. Remove individuals from the facility access list when access is no longer required.

Discussion: Physical access authorizations apply to employees and visitors. Individuals with permanent physical access authorization credentials are not considered visitors. Authorization credentials include ID badges, identification cards, and smart cards. Organizations determine the strength of authorization credentials needed consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Physical access authorizations may not be necessary to access certain areas within facilities that are designated as publicly accessible.

Related Controls: AT-3, AU-9, IA-4, MA-5, MP-2, PE-3, PE-4, PE-5, PE-8, PS-3, PS-4, PS-5, PS-6.

PE-3 PHYSICAL ACCESS CONTROL

[Existing] [Priority 2]

Control:

- a. Enforce physical access authorizations by:
 1. Verifying individual access authorizations before granting access to the facility; and
 2. Controlling ingress to the facility using agency-implemented procedures and controls;
- b. Maintain physical access audit logs for the physically secure location and agency-defined sensitive areas;
- c. Control access to areas within the facility designated as non-publicly accessible by implementing physical access devices including, but not limited to keys, locks, combinations, biometric readers, placards, and/or card readers;

- d. Escort visitors and control visitor activity in all physically secure locations;
- e. Secure keys, combinations, and other physical access devices;
- f. Inventory all agency-issued physical access devices annually; and
- g. Change combinations and keys when keys are lost, combinations are compromised, or when individuals possessing the keys or combinations are transferred or terminated.
- h. If the above conditions cannot be met refer to the requirements listed in PE-17.

Discussion: Physical access control applies to employees and visitors. Individuals with permanent physical access authorizations are not considered visitors. Physical access controls for publicly accessible areas may include physical access control logs/records, guards, or physical access devices and barriers to prevent movement from publicly accessible areas to non-public areas.

Organizations determine the types of guards needed, including professional security staff, system users, or administrative staff. Physical access devices include keys, locks, combinations, biometric readers, and card readers. Physical access control systems comply with applicable laws, executive orders, directives, policies, regulations, standards, and guidelines. Organizations have flexibility in the types of audit logs employed. Audit logs can be procedural, automated, or some combination thereof. Physical access points can include facility access points, interior access points to systems that require supplemental access controls, or both. Components of systems may be in areas designated as publicly accessible with organizations controlling access to the components.

Related Controls: AT-3, AU-2, AU-6, AU-9, CP-10, IA-3, IA-8, MA-5, MP-2, MP-4, PE-2, PE-4, PE-5, PE-8, PS-2, PS-3, PS-6, PS-7, RA-3, SC-28, SI-4, SR-3.

PE-4 ACCESS CONTROL FOR TRANSMISSION

[Existing] [Priority 2]

Control:

Control physical access to information system distribution and transmission lines and devices within organizational facilities using agency-implemented procedures and controls.

Discussion: Security controls applied to system distribution and transmission lines prevent accidental damage, disruption, and physical tampering. Such controls may also be necessary to prevent eavesdropping or modification of unencrypted transmissions. Security controls used to control physical access to system distribution and transmission lines include disconnected or locked spare jacks, locked wiring closets, protection of cabling by conduit or cable trays, and wiretapping sensors.

Related Controls: AT-3, IA-4, MP-2, MP-4, PE-2, PE-3, PE-5, PE-9, SC-7, SC-8.

PE-5 ACCESS CONTROL FOR OUTPUT DEVICES

[Existing] [Priority 3]

Control:

Control physical access to output from monitors, printers, scanners, audio devices, facsimile machines, and copiers to prevent unauthorized individuals from obtaining the output.

Discussion: Controlling physical access to output devices includes placing output devices in locked rooms or other secured areas with keypad or card reader access controls and allowing access to authorized individuals only, placing output devices in locations that can be monitored by personnel, installing monitor or screen filters, and using headphones. Examples of output devices include monitors, printers, scanners, audio devices, facsimile machines, and copiers.

Related Controls: PE-2, PE-3, PE-4.

PE-6 MONITORING PHYSICAL ACCESS

[Existing] [Priority 2]

Control:

- a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents;
- b. Review physical access logs quarterly and upon occurrence of any physical, environmental, or security-related incidents involving CJI or systems used to process, store, or transmit CJI; and
- c. Coordinate results of reviews and investigations with the organizational incident response capability.

Discussion: Physical access monitoring includes publicly accessible areas within organizational facilities. Examples of physical access monitoring include the employment of guards, video surveillance equipment (i.e., cameras), and sensor devices. Reviewing physical access logs can help identify suspicious activity, anomalous events, or potential threats. The reviews can be supported by audit logging controls, such as AU-2, if the access logs are part of an automated system. Organizational incident response capabilities include investigations of physical security incidents and responses to the incidents. Incidents include security violations or suspicious physical access activities. Suspicious physical access activities include accesses outside of normal work hours, repeated accesses to areas not normally accessed, accesses for unusual lengths of time, and out-of-sequence accesses.

Related Controls: AU-2, AU-6, AU-9, AU-12, CA-7, CP-10, IR-4, IR-8.

Control Enhancements:

(1) MONITORING PHYSICAL ACCESS | INTRUSION ALARMS AND SURVEILLANCE EQUIPMENT

[Existing] [Priority 2]

Control:

Monitor physical access to the facility where the system resides using physical intrusion alarms and surveillance equipment.

Discussion: Physical intrusion alarms can be employed to alert security personnel when unauthorized access to the facility is attempted. Alarm systems work in conjunction with physical barriers, physical access control systems, and security guards by triggering a response when these other forms of security have been compromised or breached. Physical intrusion alarms can include different types of sensor devices, such as motion sensors, contact sensors, and broken glass sensors. Surveillance equipment includes video cameras installed at strategic locations throughout the facility.

Related Controls: None.

PE-8 VISITOR ACCESS RECORDS

[Priority 4]

Control:

- a. Maintain visitor access records to the facility where the system resides for one (1) year;
- b. Review visitor access records quarterly; and
- c. Report anomalies in visitor access records to organizational personnel with physical and environmental protection responsibilities and organizational personnel with information security responsibilities.

Discussion: Visitor access records include the names and organizations of individuals visiting, visitor signatures, forms of identification, dates of access, entry and departure times, purpose of visits, and the names and organizations of individuals visited. Access record reviews determine if access authorizations are current and are still required to support organizational mission and business functions. Access records are not required for publicly accessible areas.

Related Controls: PE-2, PE-3, PE-6.

Control Enhancements:

(3) VISITOR ACCESS RECORDS | LIMIT PERSONALLY IDENTIFIABLE INFORMATION ELEMENTS

[Priority 4]

Control:

Limit personally identifiable information contained in visitor access records to the minimum PII necessary to achieve the purpose for which it is collected (see Section 4.3).

Note: Access to visitor access records is restricted to authorized agency personnel.

Discussion: Organizations may have requirements that specify the contents of visitor access records. Limiting personally identifiable information in visitor access records when such information is not needed for operational purposes helps reduce the level of privacy risk created by a system.

Related Controls: RA-3, SA-8.

PE-9 POWER EQUIPMENT AND CABLING

[Priority 2]

Control:

Protect power equipment and power cabling for the system from damage and destruction.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: Organizations determine the types of protection necessary for the power equipment and cabling employed at different locations that are both internal and external to organizational facilities and environments of operation. Types of power equipment and cabling include internal cabling and uninterruptable power sources in offices or data centers, generators and power cabling outside of buildings, and power sources for self-contained components such as satellites, vehicles, and other deployable systems.

Related Controls: PE-4.

PE-10 EMERGENCY SHUTOFF

[Priority 2]

Control:

- a. Provide the capability of shutting off power to all information systems in emergency situations;
- b. Place emergency shutoff switches or devices in easily accessible locations to facilitate access for authorized personnel; and
- c. Protect emergency power shutoff capability from unauthorized activation.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: Emergency power shutoff primarily applies to organizational facilities that contain concentrations of system resources, including data centers, mainframe computer rooms, server rooms, and areas with computer-controlled machinery.

Related Controls: PE-15.

PE-11 EMERGENCY POWER

[Priority 2]

Control:

Provide an uninterruptible power supply to facilitate an orderly shutdown of the information system or transition of the information system to an alternate power source in the event of a primary power source loss.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: An uninterruptible power supply (UPS) is an electrical system or mechanism that provides emergency power when there is a failure of the main power source. A UPS is typically used to protect computers, data centers, telecommunication equipment, or other electrical equipment where an unexpected power disruption could cause injuries, fatalities, serious mission or business disruption, or loss of data or information. A UPS differs from an emergency power system or backup generator in that the UPS provides near-instantaneous protection from unanticipated power interruptions from the main power source by providing energy stored in batteries, supercapacitors, or flywheels. The battery duration of a UPS is relatively short but provides sufficient time to start a standby power source, such as a backup generator, or properly shut down the system.

Related Controls: AT-3, CP-2, CP-7.

PE-12 EMERGENCY LIGHTING

[Priority 2]

Control:

Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: The provision of emergency lighting applies primarily to organizational facilities that contain concentrations of system resources, including data centers, server rooms, and mainframe computer rooms. Emergency lighting provisions for the system are described in the contingency plan for the organization. If emergency lighting for the system fails or cannot be provided, organizations consider alternate processing sites for power-related contingencies.

Related Controls: CP-2, CP-7.

PE-13 FIRE PROTECTION

[Priority 2]

Control:

Employ and maintain fire detection and suppression systems that are supported by an independent energy source.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: The provision of fire detection and suppression systems applies primarily to organizational facilities that contain concentrations of system resources, including data centers, server rooms, and mainframe computer rooms. Fire detection and suppression systems that may require an independent energy source include sprinkler systems and smoke detectors. An

independent energy source is an energy source, such as a microgrid, that is separate, or can be separated, from the energy sources providing power for the other parts of the facility.

Related Controls: AT-3.

Control Enhancements:

(1) FIRE PROTECTION | DETECTION SYSTEMS — AUTOMATIC ACTIVATION AND NOTIFICATION

[Priority 2]

Control:

Employ fire detection systems that activate automatically and notify organizational personnel with physical and environmental protection responsibilities and police, fire, or emergency medical personnel in the event of a fire.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: Organizations can identify personnel, roles, and emergency responders if individuals on the notification list need to have access authorizations or clearances (e.g., to enter to facilities where access is restricted due to the classification or impact level of information within the facility). Notification mechanisms may require independent energy sources to ensure that the notification capability is not adversely affected by the fire.

Related Controls: None.

PE-14 ENVIRONMENTAL CONTROLS

[Priority 2]

Control:

- a. Maintain adequate HVAC levels within the facility where the system resides at recommended system manufacturer levels; and
- b. Monitor environmental control levels continuously.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: The provision of environmental controls applies primarily to organizational facilities that contain concentrations of system resources (e.g., data centers, mainframe computer rooms, and server rooms). Insufficient environmental controls, especially in very harsh environments, can have a significant adverse impact on the availability of systems and system components that are needed to support organizational mission and business functions.

Related Controls: AT-3, CP-2.

PE-15 WATER DAMAGE PROTECTION

[Priority 2]

Control:

Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel.

Note: This control only applies to data centers as defined in Appendix A Terms and Definitions.

Discussion: The provision of water damage protection primarily applies to organizational facilities that contain concentrations of system resources, including data centers, server rooms, and mainframe computer rooms. Isolation valves can be employed in addition to or in lieu of master shutoff valves to shut off water supplies in specific areas of concern without affecting entire organizations.

Related Controls: AT-3, PE-10.

PE-16 DELIVERY AND REMOVAL

[Existing] [Priority 3]

Control:

- a. Authorize and control information system-related components entering and exiting the facility; and
- b. Maintain records of the system components.

Discussion: Enforcing authorizations for entry and exit of system components may require restricting access to delivery areas and isolating the areas from the system and media libraries.

Related Controls: CM-3, CM-8, MA-2, MA-3, MP-5, SR-2, SR-3, SR-6.

PE-17 ALTERNATE WORK SITE

[Existing] [Priority 3]

Control:

- a. Determine and document all alternate facilities or locations allowed for use by employees;
- b. Employ the following controls at alternate work sites:
 - 1. Limit access to the area during CJI processing times to only those personnel authorized by the agency to access or view CJI.
 - 2. Lock the area, room, or storage container when unattended.
 - 3. Position information system devices and documents containing CJI in such a way as to prevent unauthorized individuals from access and view.
 - 4. Follow the encryption requirements found in SC-13 and SC-28 for electronic storage (i.e., data in-transit and data at-rest) of CJI.
- c. Assess the effectiveness of controls at alternate work sites; and
- d. Provide a means for employees to communicate with information security and privacy

personnel in case of incidents.

Discussion: Alternate work sites include government facilities or the private residences of employees. While distinct from alternative processing sites, alternate work sites can provide readily available alternate locations during contingency operations. Organizations can define different sets of controls for specific alternate work sites or types of sites depending on the work-related activities conducted at the sites. Implementing and assessing the effectiveness of organization-defined controls and providing a means to communicate incidents at alternate work sites supports the contingency planning activities of organizations.

Related Controls: AC-17, AC-18, CP-7.

PLANNING (PL)

PL-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with planning responsibilities:
 1. Agency-level planning policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the planning policy and the associated planning controls;
- b. Designate organizational personnel with information security and privacy responsibilities to manage the development, documentation, and dissemination of the planning policy and procedures; and
- c. Review and update the current planning:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Planning policy and procedures for the controls in the PL family implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on their development. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission level or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission/business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to planning policy and procedures include, but are not limited to, assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

PL-2 SYSTEM SECURITY AND PRIVACY PLANS

[Priority 2]

Control:

- a. Develop security and privacy plans for the system that:
 1. Are consistent with the organization's enterprise architecture;
 2. Explicitly define the constituent system components;
 3. Describe the operational context of the system in terms of mission and business processes;
 4. Identify the individuals that fulfill system roles and responsibilities;
 5. Identify the information types processed, stored, and transmitted by the system;
 6. Provide the security categorization of the system, including supporting rationale;
 7. Describe any specific threats to the system that are of concern to the organization;
 8. Provide the results of a privacy risk assessment for systems processing personally identifiable information;
 9. Describe the operational environment for the system and any dependencies on or connections to other systems or system components;
 10. Provide an overview of the security and privacy requirements for the system;
 11. Identify any relevant control baselines or overlays, if applicable;
 12. Describe the controls in place or planned for meeting the security and privacy requirements, including a rationale for any tailoring decisions;
 13. Include risk determinations for security and privacy architecture and design decisions;
 14. Include security- and privacy-related activities affecting the system that require planning and coordination with organizational personnel with system security and privacy planning and plan implementation responsibilities; system developers; organizational personnel with information security and privacy responsibilities; and
 15. Are reviewed and approved by the authorizing official or designated representative prior to plan implementation.
- b. Distribute copies of the plans and communicate subsequent changes to the plans to organizational personnel with system security and privacy planning and plan implementation responsibilities; system developers; organizational personnel with information security and privacy responsibilities;
- c. Review the system security and privacy plans at least annually or when required due to system changes or modifications;
- d. Update the plans to address changes to the system and environment of operation or problems identified during plan implementation or control assessments; and
- e. Protect the plans from unauthorized disclosure and modification.

Discussion: System security and privacy plans are scoped to the system and system components within the defined authorization boundary and contain an overview of the security and privacy requirements for the system and the controls selected to satisfy the requirements. The plans describe the intended application of each selected control in the context of the system with a sufficient level of detail to correctly implement the control and to subsequently assess the effectiveness of the control. The control documentation describes how system-specific and hybrid controls are implemented and the plans and expectations regarding the functionality of the system. System security and privacy plans can also be used in the design and development of systems in support of life cycle-based security and privacy engineering processes. System security and privacy plans are living documents that are updated and adapted throughout the system development life cycle (e.g., during capability determination, analysis of alternatives, requests for proposal, and design reviews). The CJISSECPOL describes the different types of requirements that are relevant to organizations during the system development life cycle and the relationship between requirements and controls.

Organizations may develop a single, integrated security and privacy plan or maintain separate plans. Security and privacy plans relate security and privacy requirements to a set of controls and control enhancements. The plans describe how the controls and control enhancements meet the security and privacy requirements but do not provide detailed, technical descriptions of the design or implementation of the controls and control enhancements. Security and privacy plans contain sufficient information (including specifications of control parameter values for selection and assignment operations explicitly or by reference) to enable a design and implementation that is unambiguously compliant with the intent of the plans and subsequent determinations of risk to organizational operations and assets, individuals, other organizations, and the Nation if the plan is implemented.

Security and privacy plans need not be single documents. The plans can be a collection of various documents, including documents that already exist. Effective security and privacy plans make extensive use of references to policies, procedures, and additional documents, including design and implementation specifications where more detailed information can be obtained. The use of references helps reduce the documentation associated with security and privacy programs and maintains the security- and privacy-related information in other established management and operational areas, including enterprise architecture, system development life cycle, systems engineering, and acquisition. Security and privacy plans need not contain detailed contingency plan or incident response plan information but can instead provide—explicitly or by reference—sufficient information to define what needs to be accomplished by those plans.

Security- and privacy-related activities that may require coordination and planning with other individuals or groups within the organization include assessments, audits, inspections, hardware and software maintenance, acquisition and supply chain risk management, patch management, and contingency plan testing. Planning and coordination include emergency and nonemergency (i.e., planned or non-urgent unplanned) situations. The process defined by organizations to plan and coordinate security- and privacy-related activities can also be included in other documents, as appropriate.

Related Controls: AC-2, AC-6, AC-14, AC-17, AC-20, CA-2, CA-3, CA-7, CM-9, CP-2, CP-4, IR-4, IR-8, MA-4, MA-5, MP-4, MP-5, PL-8, PL-10, PL-11, RA-3, RA-9, SA-5, SA-22, SI-12, SR-2.

PL-4 RULES OF BEHAVIOR

[Priority 3]

Control:

- a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy;
- b. Receive a documented acknowledgment from such individuals, indicating that they have read, understand, and agree to abide by the rules of behavior, before authorizing access to information and the system;
- c. Review and update the rules of behavior at least annually; and
- d. Require individuals who have acknowledged a previous version of the rules of behavior to read and re-acknowledge annually, or when the rules are revised or updated.

Discussion: Rules of behavior represent a type of access agreement for organizational users. Other types of access agreements include nondisclosure agreements, conflict-of-interest agreements, and acceptable use agreements (see PS-6). Organizations consider rules of behavior based on individual user roles and responsibilities and differentiate between rules that apply to privileged users and rules that apply to general users. Establishing rules of behavior for some types of non-organizational users, including individuals who receive information from federal systems, is often not feasible given the large number of such users and the limited nature of their interactions with the systems. Rules of behavior for organizational and non-organizational users can also be established in AC-8. The related controls section provides a list of controls that are relevant to organizational rules of behavior. PL-4b, the documented acknowledgment portion of the control, may be satisfied by the literacy training and awareness and role-based training programs conducted by organizations if such training includes rules of behavior. Documented acknowledgements for rules of behavior include electronic or physical signatures and electronic agreement check boxes or radio buttons.

Related Controls: AC-2, AC-6, AC-8, AC-17, AC-18, AC-19, AC-20, AT-2, AT-3, CM-11, IA-2, IA-4, IA-5, MP-7, PS-6, PS-8, SA-5, SI-12.

Control Enhancements:

(1) RULES OF BEHAVIOR | SOCIAL MEDIA AND EXTERNAL SITE/APPLICATION USAGE RESTRICTIONS

[Priority 3]

Control:

Include in the rules of behavior, restrictions on:

- a. Use of social media, social networking sites, and external sites/applications;
- b. Posting organizational information on public websites; and
- c. Use of organization-provided identifiers (e.g., email addresses) and authentication secrets (e.g., passwords) for creating accounts on external sites/applications.

Discussion: Social media, social networking, and external site/application usage restrictions address rules of behavior related to the use of social media, social networking, and external sites when organizational personnel are using such sites for official duties or in the conduct of official business, when organizational information is involved in social media and social networking transactions, and when personnel access social media and networking sites from organizational systems. Organizations also address specific rules that prevent unauthorized entities from obtaining non-public organizational information from social media and networking sites either directly or through inference. Non-public information includes personally identifiable information and system account information.

Related Controls: AC-22.

PL-8 SECURITY AND PRIVACY ARCHITECTURES

[Priority 2]

Control:

- a. Develop security and privacy architectures for the system that:
 1. Describe the requirements and approach to be taken for protecting the confidentiality, integrity, and availability of organizational information;
 2. Describe the requirements and approach to be taken for processing personally identifiable information to minimize privacy risk to individuals;
 3. Describe how the architectures are integrated into and support the enterprise architecture; and
 4. Describe any assumptions about, and dependencies on, external systems and services;
- b. Review and update the architectures at least annually or when changes to the system or its environment occur to reflect changes in the enterprise architecture; and
- c. Reflect planned architecture changes in security and privacy plans, Concept of Operations (CONOPS), criticality analysis, organizational procedures, and procurements and acquisitions.

Discussion: The security and privacy architectures at the system level are consistent with the organization-wide security and privacy architectures are integral to and developed as part of the enterprise architecture. The architectures include an architectural description, the allocation of security and privacy functionality (including controls), security- and privacy-related information for external interfaces, information being exchanged across the interfaces, and the protection mechanisms associated with each interface. The architectures can also include other information, such as user roles and the access privileges assigned to each role; security and privacy requirements; types of information processed, stored, and transmitted by the system; supply chain risk management requirements; restoration priorities of information and system services; and other protection needs.

[SP 800-160-1] provides guidance on the use of security architectures as part of the system development life cycle process. Security and privacy architectures are reviewed and updated throughout the system development life cycle, from analysis of alternatives through review of the

proposed architecture in the RFP responses to the design reviews before and during implementation (e.g., during preliminary design reviews and critical design reviews).

In today's modern computing architectures, it is becoming less common for organizations to control all information resources. There may be key dependencies on external information services and service providers. Describing such dependencies in the security and privacy architectures is necessary for developing a comprehensive mission and business protection strategy. Establishing, developing, documenting, and maintaining under configuration control a baseline configuration for organizational systems is critical to implementing and maintaining effective architectures. The development of the architectures is coordinated with the senior agency information security officer and the senior agency official for privacy to ensure that the controls needed to support security and privacy requirements are identified and effectively implemented. In many circumstances, there may be no distinction between the security and privacy architecture for a system. In other circumstances, security objectives may be adequately satisfied, but privacy objectives may only be partially satisfied by the security requirements. In these cases, consideration of the privacy requirements needed to achieve satisfaction will result in a distinct privacy architecture. The documentation, however, may simply reflect the combined architectures.

Related Controls: CM-2, CM-6, PL-2, PL-7, PL-9, PM-5, RA-9, SA-3, SA-5, SA-8, SC-7.

PL-9 CENTRAL MANAGEMENT

[Priority 4]

Control:

The CJISSECPOL is centrally managed by the FBI CJIS ISO.

Discussion: Central management refers to organization-wide management and implementation of selected controls and processes. This includes planning, implementing, assessing, authorizing, and monitoring the organization-defined, centrally managed controls and processes. As the central management of controls is generally associated with the concept of common (inherited) controls, such management promotes and facilitates standardization of control implementations and management and the judicious use of organizational resources. Centrally managed controls and processes may also meet independence requirements for assessments in support of initial and ongoing authorizations to operate and as part of organizational continuous monitoring.

Automated tools (e.g., security information and event management tools or enterprise security monitoring and management tools) can improve the accuracy, consistency, and availability of information associated with centrally managed controls and processes. Automation can also provide data aggregation and data correlation capabilities; alerting mechanisms; and dashboards to support risk-based decision-making within the organization.

As part of the control selection processes, organizations determine the controls that may be suitable for central management based on resources and capabilities. It is not always possible to centrally manage every aspect of a control. In such cases, the control can be treated as a hybrid control with the control managed and implemented centrally or at the system level. The controls and control enhancements that are candidates for full or partial central management include but are not limited to: AC-2(1), AC-2(2), AC-2(3), AC-2(4), AC-4(all), AC-17(1), AC-17(2), AC-

17(3), AC-17(9), AC-18(1), AC-18(3), AC-18(4), AC-18(5), AC-19(4), AC-22, AC-23, AT-2(1), AT-2(2), AT-3(1), AT-3(2), AT-3(3), AT-4, AU-3, AU-6(1), AU-6(3), AU-6(5), AU-6(6), AU-6(9), AU-7(1), AU-7(2), AU-11, AU-13, AU-16, CA-2(1), CA-2(2), CA-2(3), CA-3(1), CA-3(2), CA-3(3), CA-7(1), CA-9, CM-2(2), CM-3(1), CM-3(4), CM-4, CM-6, CM-6(1), CM-7(2), CM-7(4), CM-7(5), CM-8(all), CM-9(1), CM-10, CM-11, CP-7(all), CP-8(all), SC-43, SI-2, SI-3, SI-4(all), SI-7, SI-8.

Related Controls: PL-8.

PL-10 BASELINE SELECTION

[Priority 3]

Control:

Select a control baseline for the system.

Discussion: Control baselines are predefined sets of controls specifically assembled to address the protection needs of a group, organization, or community of interest. Controls are chosen for baselines to either satisfy mandates imposed by laws, executive orders, directives, regulations, policies, standards, and guidelines or address threats common to all users of the baseline under the assumptions specific to the baseline. Baselines represent a starting point for the protection of individuals' privacy, information, and information systems with subsequent tailoring actions to manage risk in accordance with mission, business, or other constraints (see PL-11). Federal control baselines are provided in [SP 800-53B]. The selection of a control baseline is determined by the needs of stakeholders. Stakeholder needs consider mission and business requirements as well as mandates imposed by applicable laws, executive orders, directives, policies, regulations, standards, and guidelines. For example, the control baselines in [SP 800-53B] are based on the requirements from [FISMA] and [PRIVACT]. The requirements, along with the NIST standards and guidelines implementing the legislation, direct organizations to select one of the control baselines after the reviewing the information types and the information that is processed, stored, and transmitted on the system; analyzing the potential adverse impact of the loss or compromise of the information or system on the organization's operations and assets, individuals, other organizations, or the Nation; and considering the results from system and organizational risk assessments. [CNSSI 1253] provides guidance on control baselines for national security systems.

Related Controls: PL-2, PL-11, RA-2, RA-3, SA-8.

PL-11 BASELINE TAILORING

[Priority 3]

Control:

Tailor the selected control baseline by applying specified tailoring actions.

Discussion: The concept of tailoring allows organizations to specialize or customize a set of baseline controls by applying a defined set of tailoring actions. Tailoring actions facilitate such specialization and customization by allowing organizations to develop security and privacy plans

that reflect their specific mission and business functions, the environments where their systems operate, the threats and vulnerabilities that can affect their systems, and any other conditions or situations that can impact their mission or business success. Tailoring guidance is provided in [SP 800-53B]. Tailoring a control baseline is accomplished by identifying and designating common controls, applying scoping considerations, selecting compensating controls, assigning values to control parameters, supplementing the control baseline with additional controls as needed, and providing information for control implementation. The general tailoring actions in [SP 800-53B] can be supplemented with additional actions based on the needs of organizations. Tailoring actions can be applied to the baselines in [SP 800-53B] in accordance with the security and privacy requirements from [FISMA], [PRIVACT], and [OMB A-130]. Alternatively, other communities of interest adopting different control baselines can apply the tailoring actions in [SP 800-53B] to specialize or customize the controls that represent the specific needs and concerns of those entities.

Related Controls: PL-10, RA-2, RA-3, RA-9, SA-8.

PERSONNEL SECURITY (PS)

Having proper security measures against the insider threat is a critical component for the CJIS Security Policy. This section's security terms and requirements apply to all personnel who have unescorted access to unencrypted CJI. Regardless of the implementation model – physical data center, virtual cloud solution, or a hybrid model – unescorted access to unencrypted CJI must be determined by the criminal and noncriminal justice agency taking into consideration if those individuals have unescorted logical or physical access to any information system resulting in the ability, right, or privilege to view, modify, or make use of unencrypted CJI.

PS-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with personnel security responsibilities:
 1. Agency-level personnel security policy that:
 - a. Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - b. Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the personnel security policy and the associated personnel security controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the personnel security policy and procedures; and
- c. Review and update the current personnel security:
 1. Policy annually and following assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures annually and following assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Discussion: Personnel security policy and procedures for the controls in the PS family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on their development. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission level or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies reflecting the complex nature of organizations. Procedures can be established for security and privacy programs, for

mission/business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents.

Events that may precipitate an update to personnel security policy and procedures include, but are not limited to, assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

PS-2 POSITION RISK DESIGNATION

[Priority 2]

Control:

- a. Assign a risk designation to all organizational positions;
- b. Establish screening criteria for individuals filling those positions; and
- c. Review and update position risk designations as required.

Discussion: Position risk designations reflect Office of Personnel Management (OPM) policy and guidance. Proper position designation is the foundation of an effective and consistent suitability and personnel security program. The Position Designation System (PDS) assesses the duties and responsibilities of a position to determine the degree of potential damage to the efficiency or integrity of the service due to misconduct of an incumbent of a position and establishes the risk level of that position. The PDS assessment also determines if the duties and responsibilities of the position present the potential for position incumbents to bring about a material adverse effect on national security and the degree of that potential effect, which establishes the sensitivity level of a position. The results of the assessment determine what level of investigation is conducted for a position. Risk designations can guide and inform the types of authorizations that individuals receive when accessing organizational information and information systems.

Position screening criteria include explicit information security role appointment requirements. Parts 1400 and 731 of Title 5, Code of Federal Regulations, establish the requirements for federal organizations to evaluate relevant covered positions for a position sensitivity and position risk designation commensurate with the duties and responsibilities of those positions.

Related Controls: AC-5, AT-3, PE-2, PE-3, PL-2, PS-3, PS-6, SA-5, SI-12.

PS-3 PERSONNEL SCREENING

[Existing] [Priority 2]

Control:

- a. Screen individuals prior to authorizing access to the information system;
 1. To properly screen, state of residency and national fingerprint-based record checks shall be conducted. National fingerprint-based record checks must be conducted pursuant to an FBI approved authority such as a federal statute or a state statute approved pursuant to Public Law 92-544. If the person resides in a different state than that of the assigned agency, the agency shall also conduct state (of the agency) record checks. When appropriate, the screening shall be consistent with:
 - a. 5 CFR 731.106; and/or
 - b. Office of Personnel Management policy, regulations, and guidance; and/or
 - c. agency policy, regulations, and guidance.

Agencies authorized to bypass state repositories in compliance with federal law must only conduct a national fingerprint-based record check.

Agencies without approved statutory authority authorizing or requiring civil fingerprint-based record checks on personnel with access to CHRI are exempt from this requirement until such time as appropriate statutory authority is obtained to conduct such record checks.

2. All requests for access shall be made as specified by the CSO, Authorized Recipient Official, or IA Official. The CSO/designee, Authorized Recipient Official/designee, or IA Official is authorized to approve access to CJI. All CSO designees shall be from an authorized criminal justice agency. All Authorized Recipient designees shall be employed by the Authorized Recipient Agency.
3. If a criminal history record of any kind exists, access to CJI shall not be granted until the CSO/designee, Authorized Recipient Official/designee, or IA Official reviews the matter to determine if access is appropriate.
 - a. If a felony conviction of any kind exists, the agency shall deny access to CJI. However, the requesting agency may ask for a review by the CSO/designee, Authorized Recipient Official/designee, or IA Official in extenuating circumstances where the severity of the offense and the time that has passed would support a possible variance.
 - b. Applicants with a record of misdemeanor offense(s) may be granted access if the CSO/designee, Authorized Recipient Official/designee, or IA Official, determines the nature or severity of the misdemeanor offense(s) do not warrant disqualification. The requesting agency may request the CSO/designee, Authorized Recipient Official/designee, or IA Official review a denial of access determination.
 - c. If a criminal history record of any kind is found on a contractor, the CA shall be formally notified of continuing fitness determination and system access shall be delayed pending review of the criminal history record information. The CA shall in turn notify the contractor's security officer.
4. If the person appears to be a fugitive or has an arrest history without conviction, the CSO/designee, Authorized Recipient Official/designee, or IA Official shall review the

matter to determine if access to CJI is appropriate.

5. If the person already has access to CJI and is subsequently arrested and or convicted, continued access to CJI shall be determined by the CSO/designee, Authorized Recipient Official/designee, or IA Official. This does not implicitly grant hiring/firing authority with the CSA, Authorized Recipient, or IA Official only the authority to grant access to CJI. For offenses other than felonies, the CSO/designee, Authorized Recipient Official/designee, or IA Official has the latitude to delegate continued access determinations.
6. If the CSO/designee, Authorized Recipient Official/designee, or IA Official determines that access to CJI by the person would not be in the public interest, access shall be denied and the person's appointing authority shall be notified in writing of the access denial.
7. The criminal and non-criminal justice agency shall maintain a list of personnel who have been authorized unescorted access to unencrypted CJI and shall, upon request, provide a current copy of the access list to the CSO/designee, Authorized Recipient Official/designee, or IA Official.

and

- b. Recommend rescreening individuals in accordance with PS-3(a)(1) above. The authority authorized for the national fingerprint-based background check must also authorize the rescreening.

Discussion: Personnel screening and rescreening activities reflect applicable laws, executive orders, directives, regulations, policies, standards, guidelines, and specific criteria established for the risk designations of assigned positions. Examples of personnel screening include background investigations and agency checks.

Organizations may define different rescreening conditions and frequencies for personnel accessing systems based on types of information processed, stored, or transmitted by the systems.

Related Controls: AC-2, IA-4, MA-5, PE-2, PS-2, PS-6, PS-7.

PS-4 PERSONNEL TERMINATION

[Priority 2]

Control:

Upon termination of individual employment:

- a. Disable system access within twenty-four (24) hours;
- b. Terminate or revoke any authenticators and credentials associated with the individual;
- c. Conduct exit interviews that include a discussion of non-disclosure of CJI and PII;
- d. Retrieve all security-related organizational system-related property; and
- e. Retain access to organizational information and systems formerly controlled by

terminated individual.

Discussion: System property includes hardware authentication tokens, system administration technical manuals, keys, identification cards, and building passes. Exit interviews ensure that terminated individuals understand the security constraints imposed by being former employees and that proper accountability is achieved for system-related property. Security topics at exit interviews include reminding individuals of nondisclosure agreements and potential limitations on future employment. Exit interviews may not always be possible for some individuals, including in cases related to the unavailability of supervisors, illnesses, or job abandonment. Exit interviews are important for individuals with security clearances. The timely execution of termination actions is essential for individuals who have been terminated for cause. In certain situations, organizations consider disabling the system accounts of individuals who are being terminated prior to the individuals being notified.

Related Controls: AC-2, IA-4, PE-2, PS-6, PS-7.

PS-5 PERSONNEL TRANSFER

[Priority 3]

Control:

- a. Review and confirm ongoing operational need for current logical and physical access authorizations to systems and facilities when individuals are reassigned or transferred to other positions within the organization;
- b. Initiate appropriate actions such as closing and establishing accounts and changing system access authorizations within twenty-four (24) hours;
- c. Modify access authorization as needed to correspond with any changes in operational need due to reassignment or transfer; and
- d. Notify organizational personnel with information security responsibilities, organizational personnel with personnel security responsibilities, system/network administrators, and organizational personnel with account management responsibilities prior to personnel transfer.

Discussion: Personnel transfer applies when reassignments or transfers of individuals are permanent or of such extended duration as to make the actions warranted.

Organizations define actions appropriate for the types of reassignments or transfers, whether permanent or extended. Actions that may be required for personnel transfers or reassignments to other positions within organizations include returning old and issuing new keys, identification cards, and building passes; closing system accounts and establishing new accounts; changing system access authorizations (i.e., privileges); and providing for access to official records to which individuals had access at previous work locations and in previous system accounts.

Related Controls: AC-2, IA-4, PE-2, PS-4, PS-7.

PS-6 ACCESS AGREEMENTS

[Priority 4]

Control:

- a. Develop and document access agreements for organizational systems;
- b. Review and update the access agreements at least annually; and
- c. Verify that individuals requiring access to organizational information and systems:
 1. Sign appropriate access agreements prior to being granted access; and
 2. Re-sign access agreements to maintain access to organizational systems when access agreements have been updated or when signatories change.

Discussion: Access agreements include nondisclosure agreements, acceptable use agreements, rules of behavior, and conflict-of-interest agreements. Signed access agreements include an acknowledgement that individuals have read, understand, and agree to abide by the constraints associated with organizational systems to which access is authorized. Organizations can use electronic signatures to acknowledge access agreements unless specifically prohibited by organizational policy.

Related Controls: AC-17, PE-2, PL-4, PS-2, PS-3, PS-6, PS-7, PS-8, SI-12.

PS-7 EXTERNAL PERSONNEL SECURITY

[Existing] [Priority 2]

Control:

- a. Establish personnel security requirements, including security roles and responsibilities for external providers;
- b. Require external providers to comply with personnel security policies and procedures established by the organization;
- c. Document personnel security requirements;
- d. Require external providers to notify organizational personnel with information security responsibilities, organizational personnel with personnel security responsibilities, system/network administrators, or organizational personnel with account management responsibilities of any personnel transfers or terminations of external personnel who possess organizational credentials and/or badges, or who have system privileges within twenty-four (24) hours; and
- e. Monitor provider compliance with personnel security requirements.

Discussion: External provider refers to organizations other than the organization operating or acquiring the system. External providers include service bureaus, contractors, and other organizations that provide system development, information technology services, testing or assessment services, outsourced applications, and network/security management. Organizations explicitly include personnel security requirements in acquisition-related documents. External providers may have personnel working at organizational facilities with credentials, badges, or

system privileges issued by organizations. Notifications of external personnel changes ensure the appropriate termination of privileges and credentials. Organizations define the transfers and terminations deemed reportable by security-related characteristics that include functions, roles, and the nature of credentials or privileges associated with transferred or terminated individuals.

Related Controls: AT-2, AT-3, MA-5, PE-3, PS-2, PS-3, PS-4, PS-5, PS-6, SA-5, SA- 9.

PS-8 PERSONNEL SANCTIONS

[Priority 4]

Control:

- a. Employ a formal sanctions process for individuals failing to comply with established information security and privacy policies and procedures; and
- b. Notify organizational personnel with information security responsibilities, organizational personnel with personnel security responsibilities, system/network administrators, or organizational personnel with account management responsibilities within twenty-four (24) hours when a formal employee sanctions process is initiated, identifying the individual sanctioned and the reason for the sanction.

Discussion: Organizational sanctions reflect applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Sanctions processes are described in access agreements and can be included as part of general personnel policies for organizations and/or specified in security and privacy policies.

Organizations consult with the Office of the General Counsel regarding matters of employee sanctions.

Related Controls: All XX-1 Controls, PL-4, PS-6.

PS-9 POSITION DESCRIPTIONS

[Priority 4]

Control:

Incorporate security and privacy roles and responsibilities into organizational position descriptions.

Discussion: Specification of security and privacy roles in individual organizational position descriptions facilitates clarity in understanding the security or privacy responsibilities associated with the roles and the role-based security and privacy training requirements for the roles.

Figure 13 – Personnel Security Use Cases

Use Case 1 – A Local Police Department’s Personnel Security Controls

A local police department implemented a replacement CAD system that integrated to their state’s CSA and was authorized to process CJI. In addition to the physical and technical controls already in place, the police department implemented a variety of personnel security controls to reduce the insider threat. The police department used background screening consistent with the FBI CJIS Security Policy to vet those with unescorted access to areas in which CJI is processed, including the IT administrators employed by a contractor and all janitorial staff. The police department established sanctions against any vetted person found to be in violation of stated policies. The police department re-evaluated each person’s suitability for access to CJI every five years.

Use Case 2 – Infrastructure as a Service (IaaS) Cloud Service Implementation

This model provides the consumer the capability to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, including operating systems and applications.

When using the IaaS service model the consumer may have control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls), but does not manage or control the underlying cloud infrastructure – as defined in Appendix G.3.

A local agency implements an IaaS solution in a cloud environment leveraging an agency-implemented secure virtual private cloud which has been identified to meet the appropriate security controls in the CJIS Security Policy. The agency maintains sole access to the encryption keys. In this scenario, cloud service provider personnel have no logical or physical access to any information system resulting in the ability, right, or privilege to view, modify, or make use of unencrypted CJI; therefore, no fingerprint-based background checks are required to comply with the CJIS Security Policy. Refer to Appendix G.3 Cloud Computing for additional implementation guidance.

Use Case 3 – Platform as a Service (PaaS) Cloud Service Implementation

This model provides the consumer the capability to deploy consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider onto the cloud infrastructure.*

** This capability does not necessarily preclude the use of compatible programming languages, libraries, services, and tools from other sources.*

When using the PaaS service model, the consumer may have control over the deployed applications and possibly configuration settings for the application-hosting environment, but does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage – as defined in Appendix G.3.

An agency utilizes a cloud service solution involving PaaS to write code, build, or develop a new application. There is no CJI associated with the code development. Although cloud service personnel provide support for the platform (i.e., hardware and software maintenance), no CJI is being accessed during development. The cloud service provider has no unescorted logical or physical access to any information system and no ability, right, or privilege to view, modify, or make use of unencrypted CJI. Refer to Appendix G.3 Cloud Computing for additional implementation guidance.

Use Case 4 – Software as a Service (SaaS) Cloud Service Implementation

The SaaS service model is often referred to as “Software deployed as a hosted service and accessed over the Internet.”

The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface.

When using the SaaS service model, it should be understood that the consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings – as defined in Appendix G.3

An agency utilizes a SaaS provider for their records management system (RMS). The SaaS provider hosts their solution on a cloud provider. The SaaS provider directly supports the agency for updates and management of the solution and has unescorted logical or physical access to the information system resulting in the ability, right, or privilege to view, modify, or make use of unencrypted CJI within the RMS. The cloud provider, however, does not have unescorted logical or physical access to any information system resulting in the ability, right, or privilege to view, modify, or make use of unencrypted CJI as the SaaS provider maintains the encryption keys. In this scenario, the agency would be required to comply with Personnel Security (PS) for employees of the SaaS provider with access to CJI but would not need to do so for employees of the cloud provider used by the SaaS provider. Refer to Appendix G.3 Cloud Computing for additional implementation guidance.

RISK ASSESSMENT (RA)

RA-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with risk assessment responsibilities:
 1. Agency Level risk assessment policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the risk assessment policy and the associated risk assessment controls;
- b. Designate organizational personnel with information security and privacy responsibilities to manage the development, documentation, and dissemination of the risk assessment policy and procedures; and
- c. Review and update the current risk assessment:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Risk assessment policy and procedures address the controls in the RA family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of risk assessment policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies reflecting the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to risk assessment policy and procedures include assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

RA-2 SECURITY CATEGORIZATION

[Priority 2]

Control:

- a. Categorize the system and information it processes, stores, or transmits;
- b. Document the security categorization results, including supporting rationale, in the security plan for the system; and
- c. Verify that the authorizing official or authorizing official designated representative reviews and approves the security categorization decision.

Discussion: Security categories describe the potential adverse impacts or negative consequences to organizational operations, organizational assets, and individuals if organizational information and systems are compromised through a loss of confidentiality, integrity, or availability. Security categorization is also a type of asset loss characterization in systems security engineering processes that is carried out throughout the system development life cycle. Organizations can use privacy risk assessments or privacy impact assessments to better understand the potential adverse effects on individuals. The FBI CJIS Advisory Policy Board (APB) has assigned a security categorization of “moderate” for CJI and systems that process, store, and transmit CJI.

Organizations conduct the security categorization process as an organization-wide activity with the direct involvement of chief information officers, senior agency information security officers, senior agency officials for privacy, system owners, mission and business owners, and information owners or stewards.

Security categorization processes facilitate the development of inventories of information assets and, along with CM-8, mappings to specific system components where information is processed, stored, or transmitted. The security categorization process is revisited throughout the system development life cycle to ensure that the security categories remain accurate and relevant.

Related Controls: CM-8, MP-4, PL-2, PL-10, PL-11, RA-3, RA-5, RA-7, SA-8, SC-7, SI-12.

RA-3 RISK ASSESSMENT

[Priority 2]

Control:

- a. Conduct a risk assessment, including:
 1. Identifying threats to and vulnerabilities in the system;
 2. Determining the likelihood and magnitude of harm from unauthorized access, use, disclosure, disruption, modification, or destruction of the system, the information it processes, stores, or transmits, and any related information; and
 3. Determining the likelihood and impact of adverse effects on individuals arising from the processing of personally identifiable information;
- b. Integrate risk assessment results and risk management decisions from the organization and mission or business process perspectives with system-level risk assessments;

- c. Document risk assessment results in a risk assessment report;
- d. Review risk assessment results at least quarterly;
- e. Disseminate risk assessment results to organizational personnel with risk assessment responsibilities and organizational personnel with security and privacy responsibilities; and
- f. Update the risk assessment at least quarterly or when there are significant changes to the system, its environment of operation, or other conditions that may impact the security or privacy state of the system.

Discussion: Risk assessments consider threats, vulnerabilities, likelihood, and impact to organizational operations and assets, individuals, other organizations, and the Nation. Risk assessments also consider risk from external parties, including contractors who operate systems on behalf of the organization, individuals who access organizational systems, service providers, and outsourcing entities.

Organizations can conduct risk assessments at all three levels in the risk management hierarchy (i.e., organization level, mission/business process level, or information system level) and at any stage in the system development life cycle. Risk assessments can also be conducted at various steps in the Risk Management Framework, including preparation, categorization, control selection, control implementation, control assessment, authorization, and control monitoring. Risk assessment is an ongoing activity carried out throughout the system development life cycle.

Risk assessments can also address information related to the system, including system design, the intended use of the system, testing results, and supply chain-related information or artifacts. Risk assessments can play an important role in control selection processes, particularly during the application of tailoring guidance and in the earliest phases of capability determination.

Related Controls: CA-3, CA-6, CM-4, CP-6, CP-7, IA-8, MA-5, PE-3, PE-8, PL-2, PL-10, PL-11, RA-2, RA-5, RA-7, SA-8, SA-9, SI-12.

RA-5 VULNERABILITY MONITORING AND SCANNING

[Priority 1]

Control:

- a. Monitor and scan for vulnerabilities in the system and hosted applications at least monthly and when new vulnerabilities potentially affecting the system are identified and reported;
- b. Employ vulnerability monitoring tools and techniques that facilitate interoperability among tools and automate parts of the vulnerability management process by using standards for:
 - 1. Enumerating platforms, software flaws, and improper configurations;
 - 2. Formatting checklists and test procedures; and
 - 3. Measuring vulnerability impact;
- c. Analyze vulnerability scan reports and results from vulnerability monitoring;
- d. Remediate legitimate vulnerabilities within the number of days listed;
 - Critical—15 days

- High—30 days
 - Medium—60 days
 - Low—90 days; and
- e. Share information obtained from the vulnerability monitoring process and control assessments with organizational personnel with risk assessment, control assessment, and vulnerability scanning responsibilities to help eliminate similar vulnerabilities in other systems; and
 - f. Employ vulnerability monitoring tools that include the capability to readily update the vulnerabilities to be scanned.

Discussion: Security categorization of information and systems guides the frequency and comprehensiveness of vulnerability monitoring (including scans). Organizations determine the required vulnerability monitoring for system components, ensuring that the potential sources of vulnerabilities—such as infrastructure components (e.g., switches, routers, guards, sensors), networked printers, scanners, and copiers—are not overlooked. The capability to readily update vulnerability monitoring tools as new vulnerabilities are discovered and announced and as new scanning methods are developed helps to ensure that new vulnerabilities are not missed by employed vulnerability monitoring tools. The vulnerability monitoring tool update process helps to ensure that potential vulnerabilities in the system are identified and addressed as quickly as possible. Vulnerability monitoring and analyses for custom software may require additional approaches, such as static analysis, dynamic analysis, binary analysis, or a hybrid of the three approaches. Organizations can use these analysis approaches in source code reviews and in a variety of tools, including web-based application scanners, static analysis tools, and binary analyzers.

Vulnerability monitoring includes scanning for patch levels; scanning for functions, ports, protocols, and services that should not be accessible to users or devices; and scanning for flow control mechanisms that are improperly configured or operating incorrectly. Vulnerability monitoring may also include continuous vulnerability monitoring tools that use instrumentation to continuously analyze components. Instrumentation-based tools may improve accuracy and may be run throughout an organization without scanning. Vulnerability monitoring tools that facilitate interoperability include tools that are Security Content Automated Protocol (SCAP)-validated. Thus, organizations consider using scanning tools that express vulnerabilities in the Common Vulnerabilities and Exposures (CVE) naming convention and that employ the Open Vulnerability Assessment Language (OVAL) to determine the presence of vulnerabilities. Sources for vulnerability information include the Common Weakness Enumeration (CWE) listing and the National Vulnerability Database (NVD). Control assessments, such as red team exercises, provide additional sources of potential vulnerabilities for which to scan. Organizations also consider using scanning tools that express vulnerability impact by the Common Vulnerability Scoring System (CVSS).

Vulnerability monitoring includes a channel and process for receiving reports of security vulnerabilities from the public at-large. Vulnerability disclosure programs can be as simple as publishing a monitored email address or web form that can receive reports, including notification authorizing good-faith research and disclosure of security vulnerabilities. Organizations generally expect that such research is happening with or without their authorization and can use

public vulnerability disclosure channels to increase the likelihood that discovered vulnerabilities are reported directly to the organization for remediation.

Organizations may also employ the use of financial incentives (also known as “bug bounties”) to further encourage external security researchers to report discovered vulnerabilities. Bug bounty programs can be tailored to the organization’s needs. Bounties can be operated indefinitely or over a defined period of time and can be offered to the general public or to a curated group. Organizations may run public and private bounties simultaneously and could choose to offer partially credentialed access to certain participants in order to evaluate security vulnerabilities from privileged vantage points.

Related Controls: CA-2, CA-7, CM-2, CM-4, CM-6, CM-8, RA-2, RA-3, SA-11, SA-15, SI-2, SI-3, SI-4, SI-7, SR-11.

Control Enhancements:

(2) VULNERABILITY MONITORING AND SCANNING | UPDATE VULNERABILITIES TO BE SCANNED

[Priority 1]

Control:

Update the system vulnerabilities to be scanned within 24 hours prior to running a new scan or when new vulnerabilities are identified and reported.

Discussion: Due to the complexity of modern software, systems, and other factors, new vulnerabilities are discovered on a regular basis. It is important that newly discovered vulnerabilities are added to the list of vulnerabilities to be scanned to ensure that the organization can take steps to mitigate those vulnerabilities in a timely manner.

Related Controls: SI-5.

(5) VULNERABILITY MONITORING AND SCANNING | PRIVILEGED ACCESS

[Priority 1]

Control:

Implement privileged access authorization to information system components containing or processing CJI for vulnerability scanning activities requiring privileged access.

Discussion: In certain situations, the nature of the vulnerability scanning may be more intrusive, or the system component that is the subject of the scanning may contain classified or controlled unclassified information, such as personally identifiable information. Privileged access authorization to selected system components facilitates more thorough vulnerability scanning and protects the sensitive nature of such scanning.

Related Controls: None.

(11) VULNERABILITY MONITORING AND SCANNING | PUBLIC DISCLOSURE PROGRAM

[Priority 1]

Control:

Establish a public reporting channel for receiving reports of vulnerabilities in organizational systems and system components.

Discussion: The reporting channel is publicly discoverable and contains clear language authorizing good-faith research and the disclosure of vulnerabilities to the organization. The organization does not condition its authorization on an expectation of indefinite non-disclosure to the public by the reporting entity but may request a specific time period to properly remediate the vulnerability.

Related Controls: None.

RA-7 RISK RESPONSE

[Priority 2]

Control:

Respond to findings from security and privacy assessments, monitoring, and audits in accordance with organizational risk tolerance.

Discussion: Organizations have many options for responding to risk including mitigating risk by implementing new controls or strengthening existing controls, accepting risk with appropriate justification or rationale, sharing or transferring risk, or avoiding risk. The risk tolerance of the organization influences risk response decisions and actions. Risk response addresses the need to determine an appropriate response to risk before generating a plan of action and milestones entry. For example, the response may be to accept risk or reject risk, or it may be possible to mitigate the risk immediately so that a plan of action and milestones entry is not needed.

However, if the risk response is to mitigate the risk, and the mitigation cannot be completed immediately, a plan of action and milestones entry is generated.

Related Controls: CA-5, RA-2, RA-3, SR-2.

RA-9 CRITICALITY ANALYSIS

[Priority 2]

Control:

Identify critical system components and functions by performing a criticality analysis for information system components containing or processing CJI at the planning, design, development, testing, implementation, and maintenance stages of the system development life cycle.

Discussion: Not all system components, functions, or services necessarily require significant protections. For example, criticality analysis is a key tenet of supply chain risk management and informs the prioritization of protection activities. The identification of critical system components and functions considers applicable laws, executive orders, regulations, directives, policies, standards, system functionality requirements, system and component interfaces, and system and component dependencies. Systems engineers conduct a functional decomposition of

a system to identify mission-critical functions and components. The functional decomposition includes the identification of organizational missions supported by the system, decomposition into the specific functions to perform those missions, and traceability to the hardware, software, and firmware components that implement those functions, including when the functions are shared by many components within and external to the system.

The operational environment of a system or a system component may impact the criticality, including the connections to and dependencies on cyber-physical systems, devices, system-of-systems, and outsourced IT services. System components that allow unmediated access to critical system components or functions are considered critical due to the inherent vulnerabilities that such components create. Component and function criticality are assessed in terms of the impact of a component or function failure on the organizational missions that are supported by the system that contains the components and functions.

Criticality analysis is performed when an architecture or design is being developed, modified, or upgraded. If such analysis is performed early in the system development life cycle, organizations may be able to modify the system design to reduce the critical nature of these components and functions, such as by adding redundancy or alternate paths into the system design. Criticality analysis can also influence the protection measures required by development contractors. In addition to criticality analysis for systems, system components, and system services, criticality analysis of information is an important consideration. Such analysis is conducted as part of security categorization in RA-2.

Related Controls: CP-2, PL-2, PL-8, PL-11, RA-2, SA-8, SA-15, SR-5.

SYSTEM AND SERVICES ACQUISITION (SA)

SA-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to: organizational personnel with system and services acquisition responsibilities; organizational personnel with information security and privacy responsibilities; organizational personnel with supply chain risk management responsibilities:
 1. Agency/Entity information systems and services acquisition policy for systems used to process, store, or transmit CJI that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the system and services acquisition policy and the associated system and services acquisition controls;
- b. Designate organizational personnel with information security responsibilities and organizational personnel with system and services acquisition responsibilities to manage the development, documentation, and dissemination of the system and services acquisition policy and procedures; and
- c. Review and update the current system and services acquisition:
 1. Policy following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: System and services acquisition policy and procedures address the controls in the SA family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of system and services acquisition policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events

that may precipitate an update to system and services acquisition policy and procedures include assessment or audit findings, security incidents or breaches, or changes in laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SA-8, SI-12.

SA-2 ALLOCATION OF RESOURCES

[Priority 2]

Control:

- a. Determine the high-level information security and privacy requirements for the system or system service in mission and business process planning;
- b. Determine, document, and allocate the resources required to protect the system or system service as part of the organizational capital planning and investment control process.

Discussion: Resource allocation for information security and privacy includes funding for system and services acquisition, sustainment, and supply chain-related risks throughout the system development life cycle.

Related Controls: SA-9, SR-3, SR-5.

SA-3 SYSTEM DEVELOPMENT LIFE CYCLE

[Priority 2]

Control:

- a. Acquire, develop, and manage the system using an agency documented system development lifecycle process that incorporates information security and privacy considerations;
- b. Define and document information security and privacy roles and responsibilities throughout the system development life cycle;
- c. Identify individuals having information security and privacy roles and responsibilities; and
- d. Integrate the organizational information security and privacy risk management process into system development life cycle activities.

Discussion: A system development life cycle process provides the foundation for the successful development, implementation, and operation of organizational systems. The integration of security and privacy considerations early in the system development life cycle is a foundational principle of systems security engineering and privacy engineering. To apply the required controls within the system development life cycle requires a basic understanding of

information security and privacy, threats, vulnerabilities, adverse impacts, and risk to critical mission and business functions. The security engineering principles in SA-8 help individuals properly design, code, and test systems and system components. Organizations include qualified personnel (e.g., senior agency information security officers, senior agency officials for privacy, security and privacy architects, and security and privacy engineers) in system development life cycle processes to ensure that established security and privacy requirements are incorporated into organizational systems. Role-based security and privacy training programs can ensure that individuals with key security and privacy roles and responsibilities have the experience, skills, and expertise to conduct assigned system development life cycle activities.

The effective integration of security and privacy requirements into enterprise architecture also helps to ensure that important security and privacy considerations are addressed throughout the system life cycle and that those considerations are directly related to organizational mission and business processes. This process also facilitates the integration of the information security and privacy architectures into the enterprise architecture, consistent with the risk management strategy of the organization. Because the system development life cycle involves multiple organizations (e.g., external suppliers, developers, integrators, service providers), acquisition and supply chain risk management functions and controls play significant roles in the effective management of the system during the life cycle.

Related Controls: PL-8, SA-4, SA-5, SA-8, SA-11, SA-15, SA-22, SR-3, SR-5,

SA-4 ACQUISITION PROCESS

[Priority 2]

Control:

Include the following requirements, descriptions, and criteria, explicitly or by reference, using agency defined contract language in the acquisition contract for the system, system component, or system service:

- a. Security and privacy functional requirements;
- b. Strength of mechanism requirements;
- c. Security and privacy assurance requirements;
- d. Controls needed to satisfy the security and privacy requirements.
- e. Security and privacy documentation requirements;
- f. Requirements for protecting security and privacy documentation;
- g. Description of the system development environment and environment in which the system is intended to operate;
- h. Allocation of responsibility or identification of parties responsible for information security, privacy, and supply chain risk management; and
- i. Acceptance criteria.

Discussion: Security and privacy functional requirements are typically derived from the high-level security and privacy requirements described in SA-2. The derived requirements include security and privacy capabilities, functions, and mechanisms. Strength requirements associated with such capabilities, functions, and mechanisms include degree of correctness, completeness, resistance to tampering or bypass, and resistance to direct attack. Assurance requirements include development processes, procedures, and methodologies as well as the evidence from development and assessment activities that provide grounds for confidence that the required functionality is implemented and possesses the required strength of mechanism.

Controls can be viewed as descriptions of the safeguards and protection capabilities appropriate for achieving the particular security and privacy objectives of the organization and for reflecting the security and privacy requirements of stakeholders. Controls are selected and implemented in order to satisfy system requirements and include developer and organizational responsibilities.

Controls can include technical, administrative, and physical aspects. In some cases, the selection and implementation of a control may necessitate additional specification by the organization in the form of derived requirements or instantiated control parameter values. The derived requirements and control parameter values may be necessary to provide the appropriate level of implementation detail for controls within the system development life cycle.

Security and privacy documentation requirements address all stages of the system development life cycle. Documentation provides user and administrator guidance for the implementation and operation of controls. The level of detail required in such documentation is based on the security categorization or classification level of the system and the degree to which organizations depend on the capabilities, functions, or mechanisms to meet risk response expectations. Requirements can include mandated configuration settings that specify allowed functions, ports, protocols, and services. Acceptance criteria for systems, system components, and system services are defined in the same manner as the criteria for any organizational acquisition or procurement.

Related Controls: CM-6, CM-8, PS-7, SA-3, SA-5, SA-8, SA-11, SA-15, SR-3, SR-5.

Control Enhancements:

(1) ACQUISITION PROCESS | FUNCTIONAL PROPERTIES OF CONTROLS

[Priority 2]

Control:

Require the developer of the system, system component, or system service to provide a description of the functional properties of the controls to be implemented.

Discussion: Functional properties of security and privacy controls describe the functionality (i.e., security or privacy capability, functions, or mechanisms) visible at the interfaces of the controls and specifically exclude functionality and data structures internal to the operation of the controls.

Related Controls: None.

(2) ACQUISITION PROCESS | DESIGN AND IMPLEMENTATION INFORMATION FOR CONTROLS

[Priority 2]

Control:

Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: security-relevant external system interfaces; high-level design and a project plan that addresses sufficient detail to permit analysis and testing of the controls.

Discussion: Organizations may require different levels of detail in the documentation for the design and implementation of controls in organizational systems, system components, or system services based on mission and business requirements, requirements for resiliency and trustworthiness, and requirements for analysis and testing. Systems can be partitioned into multiple subsystems. Each subsystem within the system can contain one or more modules. The high-level design for the system is expressed in terms of subsystems and the interfaces between subsystems providing security-relevant functionality. The low-level design for the system is expressed in terms of modules and the interfaces between modules providing security-relevant functionality. Design and implementation documentation can include manufacturer, version, serial number, verification hash signature, software libraries used, date of purchase or download, and the vendor or download source. Source code and hardware schematics are referred to as the implementation representation of the system.

Related Controls: None.

(9) ACQUISITION PROCESS | FUNCTIONS, PORTS, PROTOCOLS, AND SERVICES IN USE

[Priority 2]

Control:

Require the developer of the system, system component, or system service to identify the functions, ports, protocols, and services intended for organizational use.

Discussion: The identification of functions, ports, protocols, and services early in the system development life cycle (e.g., during the initial requirements definition and design stages) allows organizations to influence the design of the system, system component, or system service. This early involvement in the system development life cycle helps organizations avoid or minimize the use of functions, ports, protocols, or services that pose unnecessarily high risks and understand the trade-offs involved in blocking specific ports, protocols, or services or requiring

system service providers to do so. Early identification of functions, ports, protocols, and services avoids costly retrofitting of controls after the system, component, or system service has been implemented. SA-9 describes the requirements for external system services. Organizations identify which functions, ports, protocols, and services are provided from external sources.

Related Controls: CM-7, SA-9.

(10) ACQUISITION PROCESS | USE OF APPROVED PIV PRODUCTS

[Priority 2]

Control:

Employ only information technology products on the FIPS 201-approved products list for Personal Identity Verification (PIV) capability implemented within organizational systems.

Discussion: Products on the FIPS 201-approved products list meet requirements for Personal Identity Verification (PIV) of Federal Employees and Contractors. PIV cards are used for multi-factor authentication in systems and organizations.

Related Controls: IA-2, IA-8.

SA-5 SYSTEM DOCUMENTATION

[Priority 3]

Control:

- a. Obtain or develop administrator documentation for the system, system component, or system service that describes:
 1. Secure configuration, installation, and operation of the system, component, or service;
 2. Effective use and maintenance of security and privacy functions and mechanisms; and
 3. Known vulnerabilities regarding configuration and use of administrative or privileged functions;
- b. Obtain or develop user documentation for the system, system component, or system service that describes:
 1. User-accessible security and privacy functions and mechanisms and how to effectively use those functions and mechanisms;
 2. Methods for user interaction, which enables individuals to use the system, component, or service in a more secure manner and protect individual privacy; and
 3. User responsibilities in maintaining the security of the system, component, or service and privacy of individuals;

- c. Document steps to obtain system, system component, or system service documentation when such documentation is either unavailable or nonexistent by contacting manufacturers, suppliers, or developers and conducting web-based searches in response; and
- d. Distribute documentation to organizational personnel with system and services responsibilities.

Discussion: System documentation helps personnel understand the implementation and operation of controls. Organizations consider establishing specific measures to determine the quality and completeness of the content provided. System documentation may be used to support the management of supply chain risk, incident response, and other functions. Personnel or roles that require documentation include system owners, system security officers, and system administrators. Attempts to obtain documentation include contacting manufacturers or suppliers and conducting web-based searches. The inability to obtain documentation may occur due to the age of the system or component or the lack of support from developers and contractors. When documentation cannot be obtained, organizations may need to recreate the documentation if it is essential to the implementation or operation of the controls. The protection provided for the documentation is commensurate with the security category or classification of the system.

Documentation that addresses system vulnerabilities may require an increased level of protection. Secure operation of the system includes initially starting the system and resuming secure system operation after a lapse in system operation.

Related Controls: CM-4, CM-6, CM-7, CM-8, PL-2, PL-4, PL-8, PS-2, SA-3, SA-4, SA-8, SA-9, SA-10, SA-11, SA-15, SI-12, SR-3.

SA-8 SECURITY AND PRIVACY ENGINEERING PRINCIPLES

[Priority 2]

Control:

Apply agency documented systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components.

Discussion: Systems security and privacy engineering principles are closely related to and implemented throughout the system development life cycle (see SA-3). Organizations can apply systems security and privacy engineering principles to new systems under development or to systems undergoing upgrades. For existing systems, organizations apply systems security and privacy engineering principles to system upgrades and modifications to the extent feasible, given the current state of hardware, software, and firmware components within those systems.

The application of systems security and privacy engineering principles helps organizations develop trustworthy, secure, and resilient systems and reduces the susceptibility to disruptions, hazards, threats, and the creation of privacy problems for individuals. Examples of system

security engineering principles include: developing layered protections; establishing security and privacy policies, architecture, and controls as the foundation for design and development; incorporating security and privacy requirements into the system development life cycle; delineating physical and logical security boundaries; ensuring that developers are trained on how to build secure software; tailoring controls to meet organizational needs; and performing threat modeling to identify use cases, threat agents, attack vectors and patterns, design patterns, and compensating controls needed to mitigate risk.

Organizations that apply systems security and privacy engineering concepts and principles can facilitate the development of trustworthy, secure systems, system components, and system services; reduce risk to acceptable levels; and make informed risk management decisions.

System security engineering principles can also be used to protect against certain supply chain risks, including incorporating tamper-resistant hardware into a design.

Related Controls: PL-8, RA-2, RA-3, RA-9, SA-3, SA-4, SA-15, SC-2, SC-39, SR-2, SR-3, SR- 5.

Control Enhancements:

(33) SECURITY AND PRIVACY ENGINEERING PRINCIPLES | MINIMIZATION

[Priority 2]

Control:

Implement the privacy principle of minimization using only the Personally Identifiable Information necessary to perform system engineering.

Discussion: The principle of minimization states that organizations should only process personally identifiable information that is directly relevant and necessary to accomplish an authorized purpose and should only maintain personally identifiable information for as long as is necessary to accomplish the purpose. Organizations have processes in place, consistent with applicable laws and policies, to implement the principle of minimization.

Related Controls: PE-8, SI-12.

SA-9 EXTERNAL SYSTEM SERVICES

[Existing] [Priority 2]

Control:

- a. Require that providers of external system services comply with organizational security and privacy requirements and employ system and services acquisition security controls in accordance with the CJISSECPOL including the following agreements when applicable:
 1. Management control agreement: A NCJA (government) designated to perform criminal justice functions for a CJA shall be eligible for access to the CJI. Access shall be permitted when such designation is authorized pursuant to executive order, statute, regulation, or interagency agreement. The NCJA shall sign and execute a management control agreement (MCA) with the CJA stipulating management control of the criminal justice function remains solely with the CJA. An example of an NCJA (government) is a city information technology (IT) department. An example of an MCA is provided in Appendix D.2.
 2. Security Addendum: The CJIS Security Addendum is a uniform addendum to an agreement between the government agency and a private contractor, approved by the Director of the FBI, acting for the U.S. Attorney General, as referenced in Title 28 CFR 20.33 (a)(7), which specifically authorizes access to CHRI, limits the use of the information to the purposes for which it is provided, ensures the security and confidentiality of the information is consistent with existing regulations and the CJISSECPOL, provides for sanctions, and contains such other provisions as the Attorney General may require.

Private contractors designated to:

- a. Perform criminal justice functions for a CJA; or
- b. Perform criminal justice dispatching functions or data processing/information services for a NCJA (government), shall be eligible for access to CJI pursuant to an agreement which specifically identifies the private contractor's purpose and scope of providing services for the administration of criminal justice. The agreement between the CJA/NCJA and the private contractor shall incorporate the CJISSECPOL.

Private contractors who perform criminal justice functions shall meet the same training and certification criteria required by governmental agencies performing a similar function and shall be subject to the same extent of audit review as are local user agencies. All private contractors who perform criminal justice functions shall acknowledge, via signing of the CJIS Security Addendum Certification page (page H- 8), compliance with all aspects of the CJIS Security Addendum. The CJIS Security Addendum is presented in Appendix H (pages H-6 and 7). Modifications to the CJIS Security Addendum shall be enacted only by the FBI.

3. Outsourcing Standards for Channeling: Channelers designated to request civil fingerprint-based background checks on behalf of a NCJA (public) or NCJA (private) for noncriminal justice functions shall be eligible for access to CJI. Access shall be permitted when such designation is authorized pursuant to federal law or state statute approved by the U.S. Attorney General. All Channelers accessing CJI shall be subject to the terms and conditions described in the Compact Council Security and

Management Control Outsourcing Standard. Each Channeler that directly accesses CJI shall also allow the FBI to conduct periodic penetration testing. Channelers leveraging CJI to perform civil functions on behalf of an Authorized Recipient shall meet the same training and certification criteria required by governmental agencies performing a similar function and shall be subject to the same extent of audit review as are local user agencies.

4. Outsourcing Standards for Non-Channeling: Contractors designated to perform noncriminal justice ancillary functions on behalf of a NCJA (public) or NCJA (private) for noncriminal justice functions shall be eligible for access to CJI. Access shall be permitted when such designation is authorized pursuant to federal law or state statute approved by the U.S. Attorney General. All contractors accessing CJI shall be subject to the terms and conditions described in the Compact Council Outsourcing Standard for Non-Channeling. Contractors leveraging CJI to perform civil functions on behalf of an Authorized Recipient shall meet the same training and certification criteria required by governmental agencies performing a similar function and shall be subject to the same extent of audit review as are local user agencies.
- b. Define and document organizational oversight and user roles and responsibilities with regard to external system services; and
 - c. Employ the following processes, methods, and techniques to monitor control compliance by external service providers on an ongoing basis:
 1. All agencies having access to CJI shall permit an inspection team to conduct an appropriate inquiry and audit of any alleged security violations.
 2. At a minimum, triennially audit all external service providers which have access to the information system in order to ensure compliance with applicable statutes, regulations, and policies.
 3. Have the authority to conduct unannounced security inspections and scheduled audits of external service providers facilities.
 4. Have the authority, on behalf of another CSA, to conduct a CJISSECPOL compliance audit of contractor facilities and provide the results to the requesting CSA. If a subsequent CSA requests an audit of the same contractor facility, the CSA may provide the results of the previous audit unless otherwise notified by the requesting CSA that a new audit be performed.

Note: Compliance audit requirements are outlined in the Security and Management Control Outsourcing Standard for Non-Channeling and Channeling related to outsourcing noncriminal justice administrative functions.

Discussion: External system services are provided by an external provider, and the organization has no direct control over the implementation of the required controls or the assessment of control effectiveness. Organizations establish relationships with external service providers in a variety of ways, including through business partnerships, contracts, interagency agreements, lines of business arrangements, licensing agreements, joint ventures, and supply chain exchanges. The responsibility for managing risks from the use of external system services remains with authorizing officials. For services external to organizations, a chain of trust requires that organizations establish and retain a certain level of confidence that each provider in the consumer-provider relationship provides adequate protection for the services rendered. The extent and nature of this chain of trust vary based on relationships between organizations and the external providers. Organizations document the basis for the trust relationships so that the relationships can be monitored. External system services documentation includes government, service providers, end user security roles and responsibilities, and service-level agreements.

Service-level agreements define the expectations of performance for implemented controls, describe measurable outcomes, and identify remedies and response requirements for identified instances of noncompliance.

Related Controls: AC-20, CA-3, CP-2, IR-4, IR-7, PL-10, PL-11, PS-7, SA-2, SA-4, SR-3, SR- 5.

Control Enhancements:

(2) EXTERNAL SYSTEM SERVICES | IDENTIFICATION OF FUNCTIONS, PORTS, PROTOCOLS, AND SERVICES

[Priority 2]

Control:

Require providers of the following external system services to identify the functions, ports, protocols, and other services required for the use of such services: any system with a local, network, or remote connection to an agency information system.

Discussion: Information from external service providers regarding the specific functions, ports, protocols, and services used in the provision of such services can be useful when the need arises to understand the trade-offs involved in restricting certain functions and services or blocking certain ports and protocols.

Related Controls: CM-6, CM-7.

SA-10 DEVELOPER CONFIGURATION MANAGEMENT

[Priority 2]

Control:

Require the developer of the system, system component, or system service to:

- a. Perform configuration management during system, component, or service during design, development, implementation, operation, and disposal;
- b. Document, manage, and control the integrity of changes to security configuration, network diagrams, and system components (hardware, software, firmware) by implementing access restrictions such as least privilege for changes;
- c. Implement only organization-approved changes to the system, component, or service;
- d. Document approved changes to the system, component, or service and the potential security and privacy impacts of such changes; and
- e. Track security flaws and flaw resolution within the system, component, or service and report findings to the individual(s) with information security responsibilities and an individual(s) with system and services acquisition responsibilities.

Discussion: Organizations consider the quality and completeness of configuration management activities conducted by developers as direct evidence of applying effective security controls.

Controls include protecting the master copies of material used to generate security-relevant portions of the system hardware, software, and firmware from unauthorized modification or destruction. Maintaining the integrity of changes to the system, system component, or system service requires strict configuration control throughout the system development life cycle to track authorized changes and prevent unauthorized changes.

The configuration items that are placed under configuration management include the formal model; the functional, high-level, and low-level design specifications; other design data; implementation documentation; source code and hardware schematics; the current running version of the object code; tools for comparing new versions of security-relevant hardware descriptions and source code with previous versions; and test fixtures and documentation.

Depending on the mission and business needs of organizations and the nature of the contractual relationships in place, developers may provide configuration management support during the operations and maintenance stage of the system development life cycle.

Related Controls: CM-2, CM-3, CM-4, CM-7, CM-9, SA-4, SA-5, SA-8, SA-15, SI-2, SR-3, SR-5, SR-6.

SA-11 DEVELOPER TESTING AND EVALUATION

[Priority 2]

Control:

Require the developer of the system, system component, or system service, at all post-design stages of the system development life cycle, to:

- a. Develop and implement a plan for ongoing security and privacy control assessments;

- b. Perform system and regression testing/evaluation at a level of comprehensive testing;
- c. Produce evidence of the execution of the assessment plan and the results of the testing and evaluation;
- d. Implement a verifiable flaw remediation process; and
- e. Correct flaws identified during testing and evaluation.

Discussion: Developmental testing and evaluation confirms that the required controls are implemented correctly, operating as intended, enforcing the desired security and privacy policies, and meeting established security and privacy requirements. Security properties of systems and the privacy of individuals may be affected by the interconnection of system components or changes to those components. The interconnections or changes—including upgrading or replacing applications, operating systems, and firmware—may adversely affect previously implemented controls. Ongoing assessment during development allows for additional types of testing and evaluation that developers can conduct to reduce or eliminate potential flaws.

Testing custom software applications may require approaches, such as manual code review, security architecture review, and penetration testing, as well as static analysis, dynamic analysis, binary analysis, or a hybrid of the three analysis approaches.

Developers can use the analysis approaches, along with security instrumentation and fuzzing, in a variety of tools and in source code reviews. The security and privacy assessment plans include the specific activities that developers plan to carry out, including the types of analyses, testing, evaluation, and reviews of software and firmware components; the degree of rigor to be applied; the frequency of the ongoing testing and evaluation; and the types of artifacts produced during those processes. The depth of testing and evaluation refers to the rigor and level of detail associated with the assessment process. The coverage of testing and evaluation refers to the scope (i.e., number and type) of the artifacts included in the assessment process. Contracts specify the acceptance criteria for security and privacy assessment plans, flaw remediation processes, and the evidence that the plans and processes have been diligently applied. Methods for reviewing and protecting assessment plans, evidence, and documentation are commensurate with the security category or classification level of the system. Contracts may specify protection requirements for documentation.

Related Controls: CA-2, CA-7, CM-4, SA-3, SA-4, SA-5, SA-8, SA-15, SI-2, SR-5, SR-6.

SA-15 DEVELOPMENT PROCESS, STANDARDS, AND TOOLS

[Priority 3]

Control:

- a. Require the developer of the system, system component, or system service to follow a documented development process that:
 - 1. Explicitly addresses security and privacy requirements;
 - 2. Identifies the standards and tools used in the development process;

3. Documents the specific tool options and tool configurations used in the development process; and
 4. Documents, manages, and ensures the integrity of changes to the process and/or tools used in development; and
- b. Review the development process, standards, tools, tool options, and tool configurations to determine if the process, standards, tools, tool options and tool configurations selected and employed can satisfy security and privacy requirements during design, development, implementation, operation, and disposal.

Discussion: Development tools include programming languages and computer-aided design systems. Reviews of development processes include the use of maturity models to determine the potential effectiveness of such processes. Maintaining the integrity of changes to tools and processes facilitates effective supply chain risk assessment and mitigation. Such integrity requires configuration control throughout the system development life cycle to track authorized changes and prevent unauthorized changes.

Related Controls: MA-6, SA-3, SA-4, SA-8, SA-10, SA-11, SR-3, SR-5, SR-6.

Control Enhancements:

(3) DEVELOPMENT PROCESS, STANDARDS, AND TOOLS |CRITICALITY ANALYSIS

[Priority 3]

Control:

Require the developer of the system, system component, or system service to perform a criticality analysis:

- a. At the following decision points in the system development life cycle: design, development, implementation, and operational; and
- b. At the following level of rigor: comprehensive testing.

Discussion: Criticality analysis performed by the developer provides input to the criticality analysis performed by organizations. Developer input is essential to organizational criticality analysis because organizations may not have access to detailed design documentation for system components that are developed as commercial off-the-shelf products. Such design documentation includes functional specifications, high-level designs, low-level designs, source code, and hardware schematics. Criticality analysis is important for organizational systems that are designated as high value assets. High value assets can be moderate- or high-impact systems due to heightened adversarial interest or potential adverse effects on the federal enterprise.

Developer input is especially important when organizations conduct supply chain criticality analyses.

Related Controls: RA-9.

SA-22 UNSUPPORTED SYSTEM COMPONENTS

[Priority 2]

Control:

- a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; or
- b. Provide the following options for alternative sources for continued support for unsupported components: original manufacturer support, or original contracted vendor support.

Discussion: Support for system components includes software patches, firmware updates, replacement parts, and maintenance contracts. An example of unsupported components includes when vendors no longer provide critical software patches or product updates, which can result in an opportunity for adversaries to exploit weaknesses in the installed components.

Exceptions to replacing unsupported system components include systems that provide critical mission or business capabilities where newer technologies are not available or where the systems are so isolated that installing replacement components is not an option.

Alternative sources for support address the need to provide continued support for system components that are no longer supported by the original manufacturers, developers, or vendors when such components remain essential to organizational mission and business functions. If necessary, organizations can establish in-house support by developing customized patches for critical software components or, alternatively, obtain the services of external providers who provide ongoing support for the designated unsupported components through contractual relationships. Such contractual relationships can include open-source software value-added vendors. The increased risk of using unsupported system components can be mitigated, for example, by prohibiting the connection of such components to public or uncontrolled networks or implementing other forms of isolation.

Related Controls: PL-2, SA-3.

SYSTEM AND COMMUNICATIONS PROTECTION (SC)

Examples of system and communications safeguards range from boundary and transmission protection to securing an agency's virtualized environment. In addition, applications, services, or information systems must have the capability to ensure system integrity through the detection and protection against unauthorized changes to software and information. This section details the requirements for protecting systems and communications infrastructures.

SC-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with system and communications protection responsibilities:
 1. Agency-level system and communications protection policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the system and communications protection policy and the associated system and communications protection controls;
- b. Designate organizational personnel with information security responsibilities to manage the development, documentation, and dissemination of the system and communications protection policy and procedures; and
- c. Review and update the current system and communications protection:
 1. Policy annually and following any changes and security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any changes and security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: System and communications protection policy and procedures address the controls in the SC family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of system and communications protection policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may

precipitate an update to system and communications protection policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SA-8, SI-12.

SC-2 SEPARATION OF SYSTEM AND USER FUNCTIONALITY

[Existing] [Priority 2]

Control:

Separate user functionality, including user interface services, from system management functionality.

Discussion: System management functionality includes functions that are necessary to administer databases, network components, workstations, or servers. These functions typically require privileged user access. The separation of user functions from system management functions is physical or logical. Organizations may separate system management functions from user functions by using different computers, instances of operating systems, central processing units, or network addresses; by employing virtualization techniques; or some combination of these or other methods. Separation of system management functions from user functions includes web administrative interfaces that employ separate authentication methods for users of any other system resources. Separation of system and user functions may include isolating administrative interfaces on different domains and with additional access controls. The separation of system and user functionality can be achieved by applying the systems security engineering design principles in SA-8.

Related Controls: AC-6, SA-4, SA-8, SC-7, SC-22, SC-39.

SC-4 INFORMATION IN SHARED SYSTEM RESOURCES

[Existing] [Priority 2]

Control:

Prevent unauthorized and unintended information transfer via shared system resources.

Discussion: Preventing unauthorized and unintended information transfer via shared system resources stops information produced by the actions of prior users or roles (or the actions of processes acting on behalf of prior users or roles) from being available to current users or roles (or current processes acting on behalf of current users or roles) that obtain access to shared system resources after those resources have been released back to the system. Information in shared system resources also applies to encrypted representations of information. In other contexts, control of information in shared system resources is referred to as object reuse and residual information protection. Information in shared system resources does not address information remanence, which refers to the residual representation of data that has been nominally deleted; covert channels (including storage and timing channels), where shared system resources are

manipulated to violate information flow restrictions; or components within systems for which there are only single users or roles.

Related Controls: AC-3, AC-4, SA-8.

SC-5 DENIAL-OF-SERVICE PROTECTION

[Priority 2]

Control:

- a. Protect against or limit the effects of the following types of denial-of-service events: distributed denial of service, DNS Denial of Service, etc.; and
- b. Employ the following controls to achieve the denial-of-service objective: boundary protection devices and intrusion detection or prevention devices.

Discussion: Denial-of-service events may occur due to a variety of internal and external causes, such as an attack by an adversary or a lack of planning to support organizational needs with respect to capacity and bandwidth. Such attacks can occur across a wide range of network protocols (e.g., IPv4, IPv6). A variety of technologies are available to limit or eliminate the origination and effects of denial-of-service events. For example, boundary protection devices can filter certain types of packets to protect system components on internal networks from being directly affected by or the source of denial-of-service attacks. Employing increased network capacity and bandwidth combined with service redundancy also reduces the susceptibility to denial-of-service events.

Related Controls: CP-2, IR-4, SC-7.

SC-7 BOUNDARY PROTECTION

[Existing] [Priority 1]

Control:

- a. Monitor and control communications at the external managed interfaces to the system and at key internal managed interfaces within the system;
- b. Implement subnetworks for publicly accessible system components that are physically or logically separated from internal organizational networks; and
- c. Connect to external networks or systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security and privacy architecture.

Discussion: Managed interfaces include gateways, routers, firewalls, guards, network-based malicious code analysis, virtualization systems, or encrypted tunnels implemented within a security architecture. Subnetworks that are physically or logically separated from internal networks are referred to as demilitarized zones or DMZs. Restricting or prohibiting interfaces within organizational systems includes restricting external web traffic to designated web servers within managed interfaces, prohibiting external traffic that appears to be spoofing internal addresses, and prohibiting internal traffic that appears to be spoofing external addresses. [SP 800-

189] provides additional information on source address validation techniques to prevent ingress and egress of traffic with spoofed addresses. Commercial telecommunications services are provided by network components and consolidated management systems shared by customers. These services may also include third party-provided access lines and other service elements. Such services may represent sources of increased risk despite contract security provisions. Boundary protection may be implemented as a common control for all or part of an organizational network such that the boundary to be protected is greater than a system-specific boundary (i.e., an authorization boundary).

Related Controls: AC-4, AC-17, AC-18, AC-19, AC-20, CA-3, CM-2, CM-4, CM-7, CM-10, CP-8, CP-10, IR-4, MA-4, PE-3, PL-8, SA-8, SC-5.

Control Enhancements:

(3) BOUNDARY PROTECTION | ACCESS POINTS

[Priority 1]

Control:

Limit the number of external network connections to the system.

Discussion: Limiting the number of external network connections facilitates monitoring of inbound and outbound communications traffic. A Trusted Internet Connection (TIC) initiative is an example of a federal guideline that requires limits on the number of external network connections. Limiting the number of external network connections to the system is important during transition periods from older to newer technologies (e.g., transitioning from IPv4 to IPv6 network protocols). Such transitions may require implementing the older and newer technologies simultaneously during the transition period and thus increase the number of access points to the system.

Related Controls: None.

(4) BOUNDARY PROTECTION | EXTERNAL TELECOMMUNICATIONS SERVICES

[Priority 1]

Control:

- a. Implement a managed interface for each external telecommunication service;
- b. Establish a traffic flow policy for each managed interface;
- c. Protect the confidentiality and integrity of the information being transmitted across each interface;
- d. Document each exception to the traffic flow policy with a supporting mission or business need and duration of that need;
- e. Review exceptions to the traffic flow policy annually, after any incident, and after any major changes impacting the information system, while removing exceptions that are no longer supported by an explicit mission or business need;
- f. Prevent unauthorized exchange of control plane traffic with external networks;

- g. Publish information to enable remote networks to detect unauthorized control plane traffic from internal networks; and
- h. Filter unauthorized control plane traffic from external networks.

Discussion: External telecommunications services can provide data and/or voice communications services. Examples of control plane traffic include Border Gateway Protocol (BGP) routing, Domain Name System (DNS), and management protocols. See [SP 800-189] for additional information on the use of the resource public key infrastructure (RPKI) to protect BGP routes and detect unauthorized BGP announcements.

Related Controls: AC-3, SC-8, SC-20, SC-21, SC-22.

(5) BOUNDARY PROTECTION | DENY BY DEFAULT — ALLOW BY EXCEPTION

[Priority 1]

Control:

Deny network communications traffic by default and allow network communications traffic by exception at boundary devices for information systems used to process, store, or transmit CJL.

Discussion: Denying by default and allowing by exception applies to inbound and outbound network communications traffic. A deny-all, permit-by-exception network communications traffic policy ensures that only those system connections that are essential and approved are allowed. Deny by default, allow by exception also applies to a system that is connected to an external system.

Related Controls: None.

(7) BOUNDARY PROTECTION | SPLIT TUNNELING FOR REMOTE DEVICES

[Priority 1]

Control:

Prevent split tunneling for remote devices connecting to organizational systems.

Discussion: Split tunneling is the process of allowing a remote user or device to establish a non-remote connection with a system and simultaneously communicate via some other connection to a resource in an external network. This method of network access enables a user to access remote devices and simultaneously, access uncontrolled networks. Split tunneling might be desirable by remote users to communicate with local system resources, such as printers or file servers. However, split tunneling can facilitate unauthorized external connections, making the system vulnerable to attack and to exfiltration of organizational information. Split tunneling can be prevented by disabling configuration settings that allow such capability in remote devices and by preventing those configuration settings from being configurable by users. Prevention can also be achieved by the detection of split tunneling (or of configuration settings that allow split tunneling) in the remote device, and by prohibiting the connection if the remote device is using split tunneling. A virtual private network (VPN) can be used to securely provision a split tunnel. A securely provisioned VPN includes locking connectivity to exclusive, managed, and named environments, or to a specific set of pre-approved addresses, without user control.

Related Controls: None.

(8) BOUNDARY PROTECTION | ROUTE TRAFFIC TO AUTHENTICATED PROXY SERVERS

[Existing] [Priority 1]

Control:

Route all internal communications traffic that may be proxied, except traffic specifically exempted by organizational personnel with information security responsibilities, to all untrusted networks through authenticated proxy servers at managed interfaces.

Discussion: External networks are networks outside of organizational control. A proxy server is a server (i.e., system or application) that acts as an intermediary for clients requesting system resources from non-organizational or other organizational servers. System resources that may be requested include files, connections, web pages, or services. Client requests established through a connection to a proxy server are assessed to manage complexity and provide additional protection by limiting direct connectivity. Web content filtering devices are one of the most common proxy servers that provide access to the Internet. Proxy servers can support the logging of Transmission Control Protocol sessions and the blocking of specific Uniform Resource Locators, Internet Protocol addresses, and domain names. Web proxies can be configured with organization-defined lists of authorized and unauthorized websites. Note that proxy servers may inhibit the use of virtual private networks (VPNs) and create the potential for “man-in-the-middle” attacks (depending on the implementation).

Related Controls: AC-3.

(24) BOUNDARY PROTECTION | PERSONALLY IDENTIFIABLE INFORMATION

[Priority 1]

Control:

For systems that process personally identifiable information:

- a. Apply the following processing rules to data elements of personally identifiable information: all applicable laws, executive orders, directives, regulations, policies, standards, and guidelines;
- b. Monitor for permitted processing at the external interfaces to the system and at key internal boundaries within the system;
- c. Document each processing exception; and
- d. Review and remove exceptions that are no longer supported.

Discussion: Managing the processing of personally identifiable information is an important aspect of protecting an individual’s privacy. Applying, monitoring for, and documenting exceptions to processing rules ensure that personally identifiable information is processed only in accordance with established privacy requirements.

Related Controls: None.

SC-8 TRANSMISSION CONFIDENTIALITY AND INTEGRITY

[Existing] [Priority 2]

Control:

Protect the confidentiality and integrity of transmitted information.

Metadata derived from unencrypted CJI shall be protected in the same manner as CJI and shall not be used for any advertising or other commercial purposes by any cloud service provider or other associated entity.

Discussion: Protecting the confidentiality and integrity of transmitted information applies to internal and external networks as well as any system components that can transmit information, including servers, notebook computers, desktop computers, mobile devices, printers, copiers, scanners, facsimile machines, and radios. Unprotected communication paths are exposed to the possibility of interception and modification. Protecting the confidentiality and integrity of information can be accomplished by physical or logical means. Physical protection can be achieved by using protected distribution systems. A protected distribution system is a wireline or fiber-optics telecommunications system that includes terminals and adequate electromagnetic, acoustical, electrical, and physical controls to permit its use for the unencrypted transmission of classified information. Logical protection can be achieved by employing encryption techniques. Organizations that rely on commercial providers who offer transmission services as commodity services rather than as fully dedicated services may find it difficult to obtain the necessary assurances regarding the implementation of needed controls for transmission confidentiality and integrity. In such situations, organizations determine what types of confidentiality or integrity services are available in standard, commercial telecommunications service packages. If it is not feasible to obtain the necessary controls and assurances of control effectiveness through appropriate contracting vehicles, organizations can implement appropriate compensating controls. The agency may permit limited use of metadata derived from unencrypted CJI when specifically approved by the agency and its “intended use” is detailed within the service agreement. Such authorized uses of metadata may include but are not limited to the following: spam and spyware filtering, data loss prevention, spillage reporting, transaction logs (events and content—similar to the AU controls), data usage/indexing metrics, and diagnostic/syslog data.

Related Controls: AC-17, AC-18, IA-3, IA-8, MA-4, PE-4, SA-4, SA-8, SC-7, SC-20, SC-23, SC-28.

Control Enhancements:

(1) TRANSMISSION CONFIDENTIALITY AND INTEGRITY | CRYPTOGRAPHIC PROTECTION

[Existing] [Priority 2]

Control:

Implement cryptographic mechanisms to prevent unauthorized disclosure and detect unauthorized changes or access to CJI during transmission.

Discussion: Encryption protects information from unauthorized disclosure and modification during transmission. Cryptographic mechanisms that protect the confidentiality and integrity of information during transmission include TLS and IPSec. Cryptographic mechanisms used to

protect information integrity include cryptographic hash functions that have applications in digital signatures, checksums, and message authentication codes.

Related Controls: SC-12, SC-13.

SC-10 NETWORK DISCONNECT

[Priority 3]

Control:

Terminate the network connection associated with a communications session at the end of the session or after one (1) hour of inactivity.

NOTE: In the interest of safety, devices that are: (1) part of a criminal justice conveyance; or (2) used to perform dispatch functions and located within a physically secure location; or (3) terminals designated solely for the purpose of receiving alert notifications (i.e., receive only terminals or ROT) and used within physically secure location facilities that remain staffed when in operation, are exempt from this requirement.

Discussion: Network disconnect applies to internal and external networks. Terminating network connections associated with specific communications sessions includes de-allocating TCP/IP address or port pairs at the operating system level and de-allocating the networking assignments at the application level if multiple application sessions are using a single operating system-level network connection. Periods of inactivity may be established by organizations and include time periods by type of network access or for specific network accesses.

Related Controls: AC-17, SC-23.

SC-12 CRYPTOGRAPHIC KEY ESTABLISHMENT AND MANAGEMENT

[Existing] [Priority 2]

Control:

Establish and manage cryptographic keys when cryptography is employed within the system in accordance with the following key management requirements: encryption key generation, distribution, storage, access, and destruction is controlled by the agency.

Discussion: Cryptographic key management and establishment can be performed using manual procedures or automated mechanisms with supporting manual procedures. Organizations define key management requirements in accordance with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines and specify appropriate options, parameters, and levels. Organizations manage trust stores to ensure that only approved trust anchors are part of such trust stores. This includes certificates with visibility external to organizational systems and certificates related to the internal operations of systems. [NIST CMVP] and [NIST CAVP] provide additional information on validated cryptographic modules and algorithms that can be used in cryptographic key management and establishment.

Related Controls: AC-17, AU-9, CM-3, IA-3, IA-7, SA-4, SA-8, SA-9, SC-8, SC-12, SC-13, SC-17, SC-20, SI-3, SI-7.

SC-13 CRYPTOGRAPHIC PROTECTION

[Existing] [Priority 2]

Control:

- a. Determine the use of encryption for CJI in-transit when outside a physically secure location; and
- b. Implement the following types of cryptography required for each specified cryptographic use: cryptographic modules which are Federal Information Processing Standard (FIPS) 140-3 certified, or a FIPS validated algorithm for symmetric key encryption and decryption (FIPS 197 [AES]), with a symmetric cipher key of at least 256-bit strength for CJI in-transit.

NOTE: Subsequent versions of approved cryptographic modules that are under current review for FIPS 140-3 compliancy can be used in the interim until certification is complete. FIPS 140-2 certificates will not be acceptable after September 21, 2026.

Discussion: Cryptography can be employed to support a variety of security solutions, including the protection of classified information and controlled unclassified information, the provision and implementation of digital signatures, and the enforcement of information separation when authorized individuals have the necessary clearances but lack the necessary formal access approvals. Cryptography can also be used to support random number and hash generation. Generally applicable cryptographic standards include FIPS-validated cryptography and NSA-approved cryptography. For example, organizations that need to protect classified information may specify the use of NSA-approved cryptography. Organizations that need to provision and implement digital signatures may specify the use of FIPS-validated cryptography. Cryptography is implemented in accordance with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Related Controls: AC-2, AC-3, AC-7, AC-17, AC-18, AC-19, AU-9, CM-11, CP-9, IA-3, IA-5, IA-7, MA-4, MP-2, MP-4, MP-5, SA-4, SA-8, SA-9, SC-8, SC-12, SC-20, SC-23, SC-28, SI-3, SI-7.

SC-15 COLLABORATIVE COMPUTING DEVICES AND APPLICATIONS

[Priority 2]

Control:

- a. Prohibit remote activation of collaborative computing devices and applications; and
- b. Provide an explicit indication of use to users physically present at the devices.

Discussion: Collaborative computing devices and applications include remote meeting devices and applications, networked white boards, cameras, and microphones. The explicit indication of use includes signals to users when collaborative computing devices and applications are activated.

Related Controls: AC-21.

SC-17 PUBLIC KEY INFRASTRUCTURE CERTIFICATES

[Existing] [Priority 2]

Control:

- a. Issue public key certificates under an agency-level certificate authority or obtain public key certificates from an approved service provider; and
- b. Include only approved trust anchors in trust stores or certificate stores managed by the organization.

Discussion: Public key infrastructure (PKI) certificates are certificates with visibility external to organizational systems and certificates related to the internal operations of systems, such as application-specific time services. In cryptographic systems with a hierarchical structure, a trust anchor is an authoritative source (i.e., a certificate authority) for which trust is assumed and not derived. A root certificate for a PKI system is an example of a trust anchor. A trust store or certificate store maintains a list of trusted root certificates.

Related Controls: IA-5, SC-12.

SC-18 MOBILE CODE

[Priority 3]

Control:

- a. Define acceptable and unacceptable mobile code and mobile code technologies; and
- b. Authorize, monitor, and control the use of mobile code within the system.

Discussion: Mobile code includes any program, application, or content that can be transmitted across a network (e.g., embedded in an email, document, or website) and executed on a remote system. Decisions regarding the use of mobile code within organizational systems are based on the potential for the code to cause damage to the systems if used maliciously. Mobile code technologies include Java applets, JavaScript, HTML5, WebGL, and VBScript. Usage restrictions and implementation guidelines apply to both the selection and use of mobile code installed on servers and mobile code downloaded and executed on individual workstations and devices, including notebook computers and smart phones. Mobile code policy and procedures address specific actions taken to prevent the development, acquisition, and introduction of unacceptable mobile code within organizational systems, including requiring mobile code to be digitally signed by a trusted source.

Related Controls: AU-2, AU-12, CM-2, CM-6, SI-3.

SC-20 SECURE NAME/ADDRESS RESOLUTION SERVICE (AUTHORITATIVE SOURCE)

[Priority 2]

Control:

- a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; and
- b. Provide the means to indicate the security status of child zones and (if the child supports secure resolution services) to enable verification of a chain of trust among parent and child domains, when operating as part of a distributed, hierarchical namespace.

Discussion: Providing authoritative source information enables external clients, including remote Internet clients, to obtain origin authentication and integrity verification assurances for the host/service name to network address resolution information obtained through the service. Systems that provide name and address resolution services include domain name system (DNS) servers. Additional artifacts include DNS Security Extensions (DNSSEC) digital signatures and cryptographic keys. Authoritative data includes DNS resource records. The means for indicating the security status of child zones include the use of delegation signer resource records in the DNS. Systems that use technologies other than the DNS to map between host and service names and network addresses provide other means to assure the authenticity and integrity of response data.

Related Controls: SC-8, SC-12, SC-13, SC-21, SC-22.

SC-21 SECURE NAME/ADDRESS RESOLUTION SERVICE (RECURSIVE OR CACHING RESOLVER)

[Priority 2]

Control:

Request and perform data origin authentication and data integrity verification on the name/address resolution responses the system receives from authoritative sources.

Discussion: Each client of name resolution services either performs this validation on its own or has authenticated channels to trusted validation providers. Systems that provide name and address resolution services for local clients include recursive resolving or caching domain name system (DNS) servers. DNS client resolvers either perform validation of DNSSEC signatures, or clients use authenticated channels to recursive resolvers that perform such validations. Systems that use technologies other than the DNS to map between host and service names and network addresses provide some other means to enable clients to verify the authenticity and integrity of response data.

Related Controls: SC-20, SC-22.

SC-22 ARCHITECTURE AND PROVISIONING FOR NAME/ADDRESS RESOLUTION SERVICE

[Priority 2]

Control:

Ensure the systems that collectively provide name/address resolution service for an organization are fault-tolerant and implement internal and external role separation.

Discussion: Systems that provide name and address resolution services include domain name system (DNS) servers. To eliminate single points of failure in systems and enhance redundancy, organizations employ at least two authoritative domain name system servers—one configured as the primary server and the other configured as the secondary server. Additionally, organizations typically deploy the servers in two geographically separated network subnetworks (i.e., not located in the same physical facility). For role separation, DNS servers with internal roles only process name and address resolution requests from within organizations (i.e., from internal clients). DNS servers with external roles only process name and address resolution information requests from clients external to organizations (i.e., on external networks, including the Internet). Organizations specify clients that can access authoritative DNS servers in certain roles (e.g., by address ranges and explicit lists).

Related Controls: SC-2, SC-20, SC-21.

SC-23 SESSION AUTHENTICITY

[Priority 2]

Control:

Protect the authenticity of communications sessions.

Discussion: Protecting session authenticity addresses communications protection at the session level, not at the packet level. Such protection establishes grounds for confidence at both ends of communications sessions in the ongoing identities of other parties and the validity of transmitted information. Authenticity protection includes protecting against “man-in-the-middle” attacks, session hijacking, and the insertion of false information into sessions.

Related Controls: SC-8, SC-10.

SC-28 PROTECTION OF INFORMATION AT REST

[Existing] [Priority 2]

Control:

Protect the confidentiality and integrity of the following information at rest: CJJ when outside physically secure locations using cryptographic modules which are certified FIPS 140-3 with a symmetric cipher key of at least 256-bit strength, or FIPS 197 with a symmetric cipher key of at least 256-bit strength.

Metadata derived from unencrypted CJI shall be protected in the same manner as CJI and shall not be used for any advertising or other commercial purposes by any cloud service provider or other associated entity.

The storage of CJI, regardless of encryption status, shall only be permitted in cloud environments (e.g., government or third-party/commercial datacenters, etc.) which reside within the physical boundaries of APB-member country (i.e., United States, U.S. territories, Indian Tribes, and Canada) and are under legal authority of an APB-member agency (i.e., United States–federal/state/territory, Indian Tribe, or the Royal Canadian Mounted Police).

Note: This restriction does not apply to exchanges of CJI with foreign government agencies under international exchange agreements (e.g., the Preventing and Combating Serious Crime agreements, fugitive extracts, and exchanges made for humanitarian and criminal investigatory purposes in particular circumstances).

Discussion: Information at rest refers to the state of information when it is not in process or in transit and is located on system components. Such components include internal or external hard disk drives, storage area network devices, or databases. However, the focus of protecting information at rest is not on the type of storage device or frequency of access but rather on the state of the information. Information at rest addresses the confidentiality and integrity of information and covers user information and system information. System-related information that requires protection includes configurations or rule sets for firewalls, intrusion detection and prevention systems, filtering routers, and authentication information. Organizations may employ different mechanisms to achieve confidentiality and integrity protections, including the use of cryptographic mechanisms and file share scanning. Integrity protection can be achieved, for example, by implementing write-once-read-many (WORM) technologies. When adequate protection of information at rest cannot otherwise be achieved, organizations may employ other controls, including frequent scanning to identify malicious code at rest and secure offline storage in lieu of online storage. The agency may permit limited use of metadata derived from unencrypted CJI when specifically approved by the agency and its “intended use” is detailed within the service agreement. Such authorized uses of metadata may include but are not limited to the following: spam and spyware filtering, data loss prevention, spillage reporting, transaction logs (events and content–similar to the AU controls), data usage/indexing metrics, and diagnostic/syslog data.

Related Controls: AC-3, AC-4, AC-6, AC-19, CA-7, CM-3, CM-5, CM-6, CP-9, MP-4, MP-5, PE-3, SC-8, SC-12, SC-13, SI-3, SI-7, SI-16.

Control Enhancements:

(1) PROTECTION OF INFORMATION AT REST | CRYPTOGRAPHIC PROTECTION

[Existing] [Priority 2]

Control:

Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on information systems and digital media outside physically secure locations: CJI.

Discussion: The selection of cryptographic mechanisms is based on the need to protect the confidentiality and integrity of organizational information. The strength of mechanism is commensurate with the security category or classification of the information. Organizations have the flexibility to encrypt information on system components or media or encrypt data structures, including files, records, or fields.

Related Controls: AC-19, SC-12, SC-13.

SC-39 PROCESS ISOLATION

[Existing] [Priority 2]

Control:

Maintain a separate execution domain for each executing system process.

Discussion: Systems can maintain separate execution domains for each executing process by assigning each process a separate address space. Each system process has a distinct address space so that communication between processes is performed in a manner controlled through the security functions, and one process cannot modify the executing code of another process. Maintaining separate execution domains for executing processes can be achieved, for example, by implementing separate address spaces. Process isolation technologies, including sandboxing or virtualization, logically separate software and firmware from other software, firmware, and data. Process isolation helps limit the access of potentially untrusted software to other system resources. The capability to maintain separate execution domains is available in commercial operating systems that employ multi-state processor technologies.

Related Controls: AC-3, AC-4, AC-6, SA-8, SC-2, SI-16.

Figure 14 – System and Communications Protection and Information Integrity Use Cases

Use Case 1 – A Local Police Department’s Information Systems & Communications Protections

A local police department implemented a replacement CAD system within a physically secure location that was authorized to process CJI using a FIPS 140-2 encrypted VPN tunnel over the Internet to the state’s CSA. In addition to the policies, physical and personnel controls already in place, the police department employed firewalls both at their border and at key points within their network, intrusion detection systems, a patch-management strategy that included automatic patch updates where possible, virus scanners, spam and spyware detection mechanisms that update signatures automatically, and subscribed to various security alert mailing lists and addressed vulnerabilities raised through the alerts as needed.

Use Case 2 – Faxing from a Single/Multi-function Device over a Traditional Telephone Line

A dispatcher from county A runs a NCIC query on an individual. The results are printed and then sent to an adjoining county using a single/multi-function device with facsimile capability. For faxing, the device is only connected to a traditional telephone line as is the device at the receiving county. Encryption of a document containing CJI is not required because the document travels over a traditional telephone line.

Use Case 3 – Faxing from a Multi-function Device over a Network

A dispatcher from city A runs a NCIC query on an individual. The results are printed and the dispatcher uses a multi-function copier to fax the file to a city in another state. The dispatcher enters the fax number of the receiver and sends the document. The document containing CJI is automatically converted to a digital file and routed to the receiver over the agency network and the Internet. Because the device uses a network and the Internet for transmitting documents containing CJI, encryption in transit using FIPS 140-2 certified 128 bit symmetric encryption is required.

SYSTEM AND INFORMATION INTEGRITY (SI)

SI-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to all organizational personnel with system and information integrity responsibilities and information system owners:
 1. Agency-level system and information integrity policy that:
 - (a) Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - (b) Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the system and information integrity policy and the associated system and information integrity controls;
- b. Designate organizational personnel with system and information integrity responsibilities to manage the development, documentation, and dissemination of the system and information integrity policy and procedures; and
- c. Review and update the current system and information integrity:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: System and information integrity policy and procedures address the controls in the SI family that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of system and information integrity policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to system and information integrity policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SA-8, SI-12.

SI-2 FLAW REMEDIATION

[Existing] [Priority 1]

Control:

- a. Identify, report, and correct system flaws;
- b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;
- c. Install security-relevant software and firmware updates within the number of days listed after the release of the updates;
 - Critical – 15 days
 - High – 30 days
 - Medium – 60 days
 - Low – 90 days; and
- d. Incorporate flaw remediation into the organizational configuration management process.

Discussion: The need to remediate system flaws applies to all types of software and firmware. Organizations identify systems affected by software flaws, including potential vulnerabilities resulting from those flaws, and report this information to designated organizational personnel with information security and privacy responsibilities. Security-relevant updates include patches, service packs, and malicious code signatures. Organizations also address flaws discovered during assessments, continuous monitoring, incident response activities, and system error handling. By incorporating flaw remediation into configuration management processes, required remediation actions can be tracked and verified.

Organization-defined time periods for updating security-relevant software and firmware may vary based on a variety of risk factors, including the security category of the system, the criticality of the update (i.e., severity of the vulnerability related to the discovered flaw), the organizational risk tolerance, the mission supported by the system, or the threat environment. Some types of flaw remediation may require more testing than other types. Organizations determine the type of testing needed for the specific type of flaw remediation activity under consideration and the types of changes that are to be configuration-managed. In some situations, organizations may determine that the testing of software or firmware updates is not necessary or practical, such as when implementing simple malicious code signature updates. In testing decisions, organizations consider whether security-relevant software or firmware updates are obtained from authorized sources with appropriate digital signatures.

Related Controls: CA-5, CM-3, CM-4, CM-5, CM-6, CM-8, MA-2, RA-5, SA-8, SA-10, SA-11, SI-3, SI- 5, SI-7, SI-11.

Control Enhancements:

(2) FLAW REMEDIATION | AUTOMATED FLAW REMEDIATION STATUS

[Priority 1]

Control:

Determine if system components have applicable security-relevant software and firmware updates installed using vulnerability scanning tools at least monthly or following any security incidents involving CJI or systems used to process, store, or transmit CJI.

Discussion: Automated mechanisms can track and determine the status of known flaws for system components.

Related Controls: CA-7, SI-4.

SI-3 MALICIOUS CODE PROTECTION

[Existing] [Priority 1]

Control:

- a. Implement signature-based malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;
- b. Automatically update malicious code protection mechanisms as new releases are available in accordance with organizational configuration management policy and procedures;
- c. Configure malicious code protection mechanisms to:
 1. Perform periodic scans of the system at least daily and real-time scans of files from external sources at network entry and exit points and on all servers and endpoint devices as the files are downloaded, opened, or executed in accordance with organizational policy; and
 2. Block or quarantine malicious code, take mitigating action(s), and when necessary, implement incident response procedures; and send alert to system/network administrators and/or organizational personnel with information security responsibilities in response to malicious code detection; and
- d. Address the receipt of false positives during malicious code detection and eradication and the resulting potential impact on the availability of the system.

Discussion: System entry and exit points include firewalls, remote access servers, workstations, electronic mail servers, web servers, proxy servers, notebook computers, and mobile devices. Malicious code includes viruses, worms, Trojan horses, and spyware. Malicious code can also be encoded in various formats contained within compressed or hidden files or hidden in files using techniques such as steganography. Malicious code can be inserted into systems in a variety of ways, including by electronic mail, the World Wide Web, and portable storage devices. Malicious code insertions occur through the exploitation of system vulnerabilities. A variety of technologies and methods exist to limit or eliminate the effects of malicious code.

Malicious code protection mechanisms include both signature- and nonsignature-based technologies. Nonsignature-based detection mechanisms include artificial intelligence techniques that use heuristics to detect, analyze, and describe the characteristics or behavior of malicious

code and to provide controls against such code for which signatures do not yet exist or for which existing signatures may not be effective. Malicious code for which active signatures do not yet exist or may be ineffective includes polymorphic malicious code (i.e., code that changes signatures when it replicates). Nonsignature-based mechanisms also include reputation-based technologies. In addition to the above technologies, pervasive configuration management, comprehensive software integrity controls, and anti-exploitation software may be effective in preventing the execution of unauthorized code. Malicious code may be present in commercial off-the-shelf software as well as custom-built software and could include logic bombs, backdoors, and other types of attacks that could affect organizational mission and business functions.

In situations where malicious code cannot be detected by detection methods or technologies, organizations rely on other types of controls, including secure coding practices, configuration management and control, trusted procurement processes, and monitoring practices to ensure that software does not perform functions other than the functions intended. Organizations may determine that, in response to the detection of malicious code, different actions may be warranted. For example, organizations can define actions in response to malicious code detection during periodic scans, the detection of malicious downloads, or the detection of maliciousness when attempting to open or execute files.

Related Controls: AC-4, AC-19, CM-3, CM-8, IR-4, MA-3, MA-4, RA-5, SC-7, SC-23, SC- 28, SI-2, SI-4, SI-7, SI-8.

SI-4 SYSTEM MONITORING

[Priority 1]

Control:

- a. Monitor the system to detect:
 1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives:
 - a. Intrusion detection and prevention
 - b. Malicious code protection
 - c. Vulnerability scanning
 - d. Audit record monitoring
 - e. Network monitoring
 - f. Firewall monitoring;and
 2. Unauthorized local, network, and remote connections;
- b. Identify unauthorized use of the system through the following techniques and methods: event logging (ref. Audit and Accountability (AU));
- c. Invoke internal monitoring capabilities or deploy monitoring devices:

1. Strategically within the system to collect organization-determined essential information; and
 2. At ad hoc locations within the system to track specific types of transactions of interest to the organization;
- d. Analyze detected events and anomalies;
 - e. Adjust the level of system monitoring activity when there is a change in risk to organizational operations and assets, individuals, other organizations, or the Nation;
 - f. Obtain legal opinion regarding system monitoring activities; and
 - g. Provide intrusion detection and prevention systems, malicious code protection software, scanning tools, audit record monitoring software, network monitoring, and firewall monitoring software logs to organizational personnel with information security responsibilities weekly.

Discussion: System monitoring includes external and internal monitoring. External monitoring includes the observation of events occurring at external interfaces to the system. Internal monitoring includes the observation of events occurring within the system. Organizations monitor systems by observing audit activities in real time or by observing other system aspects such as access patterns, characteristics of access, and other actions. The monitoring objectives guide and inform the determination of the events. System monitoring capabilities are achieved through a variety of tools and techniques, including intrusion detection and prevention systems, malicious code protection software, scanning tools, audit record monitoring software, and network monitoring software.

Depending on the security architecture, the distribution and configuration of monitoring devices may impact throughput at key internal and external boundaries as well as at other locations across a network due to the introduction of network throughput latency. If throughput management is needed, such devices are strategically located and deployed as part of an established organization-wide security architecture. Strategic locations for monitoring devices include selected perimeter locations and near key servers and server farms that support critical applications. Monitoring devices are typically employed at the managed interfaces associated with controls SC-7 and AC-17. The information collected is a function of the organizational monitoring objectives and the capability of systems to support such objectives. Specific types of transactions of interest include Hypertext Transfer Protocol (HTTP) traffic that bypasses HTTP proxies. System monitoring is an integral part of organizational continuous monitoring and incident response programs, and output from system monitoring serves as input to those programs. System monitoring requirements, including the need for specific types of system monitoring, may be referenced in other controls (e.g., AC-2g, AC-17(1), CM-3f, CM-6d, MA-3a, MA-4a, SC-5(3)(b), SC-7a, SC-18b). Adjustments to levels of system monitoring are based on law enforcement information, intelligence information, or other sources of information. The legality of system monitoring activities is based on applicable laws, executive orders, directives, regulations, policies, standards, and guidelines.

Related Controls: AC-2, AC-3, AC-4, AC-8, AC-17, AU-2, AU-6, AU-7, AU-9, AU-12, CA-7, CM-3, CM-6, CM-8, CM-11, IR-4, MA-3, MA-4, RA-5, SC-5, SC-7, SC-18, SI-3, SI-7, SR-10.

Control Enhancements:

(2) SYSTEM MONITORING | AUTOMATED TOOLS AND MECHANISMS FOR REAL-TIME ANALYSIS

[Existing] [Priority 1]

Control:

Employ automated tools and mechanisms to support near real-time analysis of events.

Discussion: Automated tools and mechanisms include host-based, network-based, transport-based, or storage-based event monitoring tools and mechanisms or security information and event management (SIEM) technologies that provide real-time analysis of alerts and notifications generated by organizational systems. Automated monitoring techniques can create unintended privacy risks because automated controls may connect to external or otherwise unrelated systems. The matching of records between these systems may create linkages with unintended consequences. Organizations assess and document these risks in their privacy impact assessment and make determinations that are in alignment with their privacy program plan.

Related Controls: None.

(4) SYSTEM MONITORING | INBOUND AND OUTBOUND COMMUNICATIONS TRAFFIC

[Existing] [Priority 1]

Control:

- a. Determine criteria for unusual or unauthorized activities or conditions for inbound and outbound communications traffic;
- b. Monitor inbound and outbound communications traffic continuously for unusual or unauthorized activities or conditions such as: the presence of malicious code or unauthorized use of legitimate code or credentials within organizational systems or propagating among system components, signaling to external systems, and the unauthorized exporting of information.

Discussion: Unusual or unauthorized activities or conditions related to system inbound and outbound communications traffic includes internal traffic that indicates the presence of malicious code or unauthorized use of legitimate code or credentials within organizational systems or propagating among system components, signaling to external systems, and the unauthorized exporting of information. Evidence of malicious code or unauthorized use of legitimate code or credentials is used to identify potentially compromised systems or system components.

Related Controls: None.

(5) SYSTEM MONITORING | SYSTEM-GENERATED ALERTS

[Existing] [Priority 1]

Control:

Alert organizational personnel with system monitoring responsibilities when the following system-generated indications of compromise or potential compromise occur: inappropriate or unusual activities with security or privacy implications.

Discussion: Alerts may be generated from a variety of sources, including audit records or inputs from malicious code protection mechanisms, intrusion detection or prevention mechanisms, or boundary protection devices such as firewalls, gateways, and routers. Alerts can be automated and may be transmitted telephonically, by electronic mail messages, or by text messaging. Organizational personnel on the alert notification list can include system administrators, mission or business owners, system owners, information owners/stewards, senior agency information security officers, senior agency officials for privacy, system security officers, or privacy officers. In contrast to alerts generated by the system, alerts generated by organizations in SI-4(12) focus on information sources external to the system, such as suspicious activity reports and reports on potential insider threats.

Related Controls: AU-4, AU-5, PE-6.

SI-5 SECURITY ALERTS, ADVISORIES, AND DIRECTIVES

[Existing] [Priority 2]

Control:

- a. Receive system security alerts, advisories, and directives from external source(s) (e.g., CISA, Multi-State Information Sharing & Analysis Center [MS-ISAC], U.S. Computer Emergency Readiness Team [USCERT], hardware/software providers, federal/state advisories, etc.) on an ongoing basis;
- b. Generate internal security alerts, advisories, and directives as deemed necessary;
- c. Disseminate security alerts, advisories, and directives to: organizational personnel implementing, operating, maintaining, and using the system; and
- d. Implement security directives in accordance with established time frames, or notify the issuing organization of the degree of noncompliance.

Discussion: The Cybersecurity and Infrastructure Security Agency (CISA) generates security alerts and advisories to maintain situational awareness throughout the federal government. Security directives are issued by OMB or other designated organizations with the responsibility and authority to issue such directives. Compliance with security directives is essential due to the critical nature of many of these directives and the potential (immediate) adverse effects on organizational operations and assets, individuals, other organizations, and the Nation should the directives not be implemented in a timely manner. External organizations include supply chain partners, external mission or business partners, external service providers, and other peer or supporting organizations.

Related Controls: RA-5, SI-2.

SI-7 SOFTWARE, FIRMWARE, AND INFORMATION INTEGRITY

[Priority 1]

Control:

- a. Employ integrity verification tools to detect unauthorized changes to software, firmware, and

information systems that contain or process CJI; and

- b. Take the following actions when unauthorized changes to the software, firmware, and information are detected: notify organizational personnel responsible for software, firmware, and/or information integrity and implement incident response procedures as appropriate.

Discussion: Unauthorized changes to software, firmware, and information can occur due to errors or malicious activity. Software includes operating systems (with key internal components, such as kernels or drivers), middleware, and applications. Firmware interfaces include Unified Extensible Firmware Interface (UEFI) and Basic Input/Output System (BIOS). Information includes personally identifiable information and metadata that contains security and privacy attributes associated with information. Integrity-checking mechanisms—including parity checks, cyclical redundancy checks, cryptographic hashes, and associated tools—can automatically monitor the integrity of systems and hosted applications.

Related Controls: AC-4, CM-3, CM-7, CM-8, MA-3, MA-4, RA-5, SA-8, SA-9, SA-10, SC-8, SC-12, SC-13, SC-28, SI-3, SR-3, SR-5, SR-6, SR-10, SR-11.

Control Enhancements:

(1) SOFTWARE, FIRMWARE, AND INFORMATION INTEGRITY | INTEGRITY CHECKS

[Priority 1]

Control:

Perform an integrity check of software, firmware, and information systems that contain or process CJI at agency-defined transitional states or security relevant events at least weekly or in an automated fashion.

Discussion: Security-relevant events include the identification of new threats to which organizational systems are susceptible and the installation of new hardware, software, or firmware. Transitional states include system startup, restart, shutdown, and abort.

Related Controls: None.

(7) SOFTWARE, FIRMWARE, AND INFORMATION INTEGRITY | INTEGRATION OF DETECTION AND RESPONSE

[Priority 1]

Control:

Incorporate the detection of the following unauthorized changes into the organizational incident response capability: unauthorized changes to established configuration setting or the unauthorized elevation of system privileges.

Discussion: Integrating detection and response helps to ensure that detected events are tracked, monitored, corrected, and available for historical purposes. Maintaining historical records is important for being able to identify and discern adversary actions over an extended time period and for possible legal actions. Security-relevant changes include unauthorized changes to established configuration settings or the unauthorized elevation of system privileges.

Related Controls: AU-2, AU-6, IR-4, IR-5, SI-4.

SI-8 SPAM PROTECTION

[Existing] [Priority 3]

Control:

- a. Employ spam protection mechanisms at system entry and exit points to detect and act on unsolicited messages; and
- b. Update spam protection mechanisms when new releases are available in accordance with organizational configuration management policy and procedures.

Discussion: System entry and exit points include firewalls, remote-access servers, electronic mail servers, web servers, proxy servers, workstations, notebook computers, and mobile devices. Spam can be transported by different means, including email, email attachments, and web accesses. Spam protection mechanisms include signature definitions.

Related Controls: SC-5, SC-7, SI-3, SI-4.

Control Enhancements:

(2) SPAM PROTECTION | AUTOMATIC UPDATES

[Priority 3]

Control:

Automatically update spam protection mechanisms at least daily.

Discussion: Using automated mechanisms to update spam protection mechanisms helps to ensure that updates occur on a regular basis and provide the latest content and protection capabilities.

Related Controls: None.

SI-10 INFORMATION INPUT VALIDATION

[Priority 1]

Control:

Check the validity of the following information inputs: all inputs to web/application servers, database servers, and any system or application input that might receive or process CJI.

Discussion: Checking the valid syntax and semantics of system inputs—including character set, length, numerical range, and acceptable values—verifies that inputs match specified definitions for format and content. For example, if the organization specifies that numerical values between 1-100 are the only acceptable inputs for a field in a given application, inputs of “387,” “abc,” or “%K%” are invalid inputs and are not accepted as input to the system. Valid inputs are likely to vary from field to field within a software application. Applications typically follow well-defined protocols that use structured messages (i.e., commands or queries) to communicate between software modules or system components. Structured messages can contain raw or unstructured data interspersed with metadata or control information. If software applications use attacker-

supplied inputs to construct structured messages without properly encoding such messages, then the attacker could insert malicious commands or special characters that can cause the data to be interpreted as control information or metadata. Consequently, the module or component that receives the corrupted output will perform the wrong operations or otherwise interpret the data incorrectly. Prescreening inputs prior to passing them to interpreters prevents the content from being unintentionally interpreted as commands. Input validation ensures accurate and correct inputs and prevents attacks such as cross-site scripting and a variety of injection attacks.

Related Controls: None.

SI-11 ERROR HANDLING

[Priority 3]

Control:

- a. Generate error messages that provide information necessary for corrective actions without revealing information that could be exploited; and
- b. Reveal error messages only to organizational personnel with information security responsibilities.

Discussion: Organizations consider the structure and content of error messages. The extent to which systems can handle error conditions is guided and informed by organizational policy and operational requirements. Exploitable information includes stack traces and implementation details; erroneous logon attempts with passwords mistakenly entered as the username; mission or business information that can be derived from, if not stated explicitly by, the information recorded; and personally identifiable information, such as account numbers, social security numbers, and credit card numbers. Error messages may also provide a covert channel for transmitting information.

Related Controls: AU-2, AU-3, SI-2.

SI-12 INFORMATION MANAGEMENT AND RETENTION

[Existing] [Priority 3]

Control:

Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.

Discussion: Information management and retention requirements cover the full life cycle of information, in some cases extending beyond system disposal. Information to be retained may also include policies, procedures, plans, reports, data output from control implementation, and other types of administrative information. The National Archives and Records Administration (NARA) provides federal policy and guidance on records retention and schedules. If organizations have a records management office, consider coordinating with records management personnel. Records produced from the output of implemented controls that may require management and retention include, but are not limited to: All XX-1, AC-6(9), AT-4,

AU-12, CA-2, CA-3, CA-5, CA-6, CA-7, CA-9, CM-2, CM-3, CM-4, CM-6, CM-8, CM-9, CM-12, CP-2, IR-6, IR-8, MA-2, MA-4, PE-2, PE-8, PE-16, PE-17, PL-2, PL-4, PL-8, PS-2, PS-6, PS-7, RA-2, RA-3, RA-5, SA-4, SA-5, SA-8, SA-10, SI-4, SR-2, SR-8.

Related Controls: AU-5, AU-11, CA-2, CA-3, CA-5, CA-6, CA-7, CA-9, CM-5, CM-9, CP-2, IR-8, MP-2, MP-3, MP-4, MP-6, PL-2, PL-4, PS-2, PS-6, RA-2, RA-3, SA-5, SA-8, SR-2.

Control Enhancements:

(1) INFORMATION MANAGEMENT AND RETENTION | LIMIT PERSONALLY IDENTIFIABLE INFORMATION ELEMENTS

[Existing] [Priority 3]

Control:

Limit personally identifiable information being processed in the information life cycle to the minimum PII necessary to achieve the purpose for which it is collected (see Section 4.3).

Discussion: Limiting the use of personally identifiable information throughout the information life cycle when the information is not needed for operational purposes helps to reduce the level of privacy risk created by a system. The information life cycle includes information creation, collection, use, processing, storage, maintenance, dissemination, disclosure, and disposition. Risk assessments as well as applicable laws, regulations, and policies can provide useful inputs to determining which elements of personally identifiable information may create risk.

Related Controls: None.

(2) INFORMATION MANAGEMENT AND RETENTION | MINIMIZE PERSONALLY IDENTIFIABLE INFORMATION IN TESTING, TRAINING, AND RESEARCH

[Priority 3]

Control:

Use the following techniques to minimize the use of personally identifiable information for research, testing, or training: data obfuscation, randomization, anonymization, or use of synthetic data.

Discussion: Organizations can minimize the risk to an individual's privacy by employing techniques such as de-identification or synthetic data. Limiting the use of personally identifiable information throughout the information life cycle when the information is not needed for research, testing, or training helps reduce the level of privacy risk created by a system. Risk assessments as well as applicable laws, regulations, and policies can provide useful inputs to determining the techniques to use and when to use them.

Related Controls: None.

(3) INFORMATION MANAGEMENT AND RETENTION | INFORMATION DISPOSAL

[Existing] [Priority 3]

Control:

Use the following techniques to dispose of, destroy, or erase information following the retention period: as defined in MP-6.

Discussion: Organizations can minimize both security and privacy risks by disposing of information when it is no longer needed. The disposal or destruction of information applies to originals as well as copies and archived records, including system logs that may contain personally identifiable information.

Related Controls: MP-6.

SI-16 MEMORY PROTECTION

[Priority 2]

Control:

Implement the following controls to protect the system memory from unauthorized code execution: data execution prevention and address space layout randomization.

Discussion: Some adversaries launch attacks with the intent of executing code in nonexecutable regions of memory or in memory locations that are prohibited. Controls employed to protect memory include data execution prevention and address space layout randomization. Data execution prevention controls can either be hardware-enforced or software-enforced with hardware enforcement providing the greater strength of mechanism.

Related Controls: SI-7.

SUPPLY CHAIN RISK MANAGEMENT (SR)

SR-1 POLICY AND PROCEDURES

[Priority 2]

Control:

- a. Develop, document, and disseminate to organizational personnel with supply chain risk management responsibilities:
 1. Agency-level supply chain risk management policy that:
 - a. Addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
 - b. Is consistent with applicable laws, executive orders, directives, regulations, policies, standards, and guidelines; and
 2. Procedures to facilitate the implementation of the supply chain risk management policy and the associated supply chain risk management controls;
- b. Designate organizational personnel with security responsibilities to manage the development, documentation, and dissemination of the supply chain risk management policy and procedures; and
- c. Review and update the current supply chain risk management:
 1. Policy annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI; and
 2. Procedures annually and following any security incidents involving unauthorized access to CJI or systems used to process, store, or transmit CJI.

Discussion: Supply chain risk management policy and procedures address the controls in the SR family as well as supply chain-related controls in other families that are implemented within systems and organizations. The risk management strategy is an important factor in establishing such policies and procedures. Policies and procedures contribute to security and privacy assurance. Therefore, it is important that security and privacy programs collaborate on the development of supply chain risk management policy and procedures. Security and privacy program policies and procedures at the organization level are preferable, in general, and may obviate the need for mission- or system-specific policies and procedures. The policy can be included as part of the general security and privacy policy or be represented by multiple policies that reflect the complex nature of organizations. Procedures can be established for security and privacy programs, for mission or business processes, and for systems, if needed. Procedures describe how the policies or controls are implemented and can be directed at the individual or role that is the object of the procedure. Procedures can be documented in system security and privacy plans or in one or more separate documents. Events that may precipitate an update to supply chain risk management policy and procedures include assessment or audit findings, security incidents or breaches, or changes in applicable laws, executive orders, directives, regulations, policies, standards, and guidelines. Simply restating controls does not constitute an organizational policy or procedure.

Related Controls: PS-8, SI-12.

Control Enhancements: None.

SR-2 SUPPLY CHAIN RISK MANAGEMENT PLAN

[Priority 3]

Control:

- a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or system services: systems used to process, store, or transmit CJI;
- b. Review and update the supply chain risk management plan annually or as required, to address threat, organizational or environmental changes; and
- c. Protect the supply chain risk management plan from unauthorized disclosure and modification.

Discussion: The dependence on products, systems, and services from external providers, as well as the nature of the relationships with those providers, present an increasing level of risk to an organization. Threat actions that may increase security or privacy risks include unauthorized production, the insertion or use of counterfeits, tampering, theft, insertion of malicious software and hardware, and poor manufacturing and development practices in the supply chain. Supply chain risks can be endemic or systemic within a system element or component, a system, an organization, a sector, or the Nation. Managing supply chain risk is a complex, multifaceted undertaking that requires a coordinated effort across an organization to build trust relationships and communicate with internal and external stakeholders.

Supply chain risk management (SCRM) activities include identifying and assessing risks, determining appropriate risk response actions, developing SCRM plans to document response actions, and monitoring performance against plans. The SCRM plan (at the system-level) is implementation specific, providing policy implementation, requirements, constraints and implications. It can either be stand-alone, or incorporated into system security and privacy plans. The SCRM plan addresses managing, implementation, and monitoring of SCRM controls and the development/sustainment of systems across the SDLC to support mission and business functions.

Because supply chains can differ significantly across and within organizations, SCRM plans are tailored to the individual program, organizational, and operational contexts. Tailored SCRM plans provide the basis for determining whether a technology, service, system component, or system is fit for purpose, and as such, the controls need to be tailored accordingly. Tailored SCRM plans help organizations focus their resources on the most critical mission and business functions based on mission and business requirements and their risk environment. Supply chain risk management plans include an expression of the supply chain risk tolerance for the organization, acceptable supply chain risk mitigation strategies or controls, a process for consistently evaluating and monitoring supply chain risk, approaches for implementing and communicating the plan, a description of and justification for supply chain risk mitigation

measures taken, and associated roles and responsibilities. Finally, supply chain risk management plans address requirements for developing trustworthy, secure, privacy-protective, and resilient system components and systems, including the application of the security design principles implemented as part of life cycle-based systems security engineering processes (see SA-8).

Related Controls: CA-2, CP-4, IR-4, MA-2, MA-6, PE-16, PL-2, RA-3, RA-7, SA-8, SI-4.

Control Enhancements:

(1) SUPPLY CHAIN RISK MANAGEMENT PLAN | ESTABLISH SCRM TEAM

[Priority 3]

Control:

Establish a supply chain risk management team consisting of individuals with security responsibilities and supply chain risk management responsibilities to lead and support the following SCRM activities: information technology, contracting, information security, privacy, mission or business, legal, supply chain and logistics, acquisition, business continuity, and other relevant functions.

Discussion: To implement supply chain risk management plans, organizations establish a coordinated, team-based approach to identify and assess supply chain risks and manage these risks by using programmatic and technical mitigation techniques. The team approach enables organizations to conduct an analysis of their supply chain, communicate with internal and external partners or stakeholders, and gain broad consensus regarding the appropriate resources for SCRM. The SCRM team consists of organizational personnel with diverse roles and responsibilities for leading and supporting SCRM activities, including risk executive, information technology, contracting, information security, privacy, mission or business, legal, supply chain and logistics, acquisition, business continuity, and other relevant functions. Members of the SCRM team are involved in various aspects of the SDLC and, collectively, have an awareness of and provide expertise in acquisition processes, legal practices, vulnerabilities, threats, and attack vectors, as well as an understanding of the technical aspects and dependencies of systems. The SCRM team can be an extension of the security and privacy risk management processes or be included as part of an organizational risk management team.

Related Controls: None.

SR-5 ACQUISITION STRATEGIES, TOOLS, AND METHODS

[Priority 2]

Control:

Employ the following acquisition strategies, contract tools, and procurement methods to protect against, identify, and mitigate supply chain risks: use preferred suppliers who can provide attestation or demonstration of compliance with state or federal standards.

Discussion: The use of the acquisition process provides an important vehicle to protect the supply chain. There are many useful tools and techniques available, including obscuring the end use of a system or system component, using blind or filtered buys, requiring tamper-evident packaging, or using trusted or controlled distribution. The results from a supply chain risk assessment can guide and inform the strategies, tools, and methods that are most applicable to the situation. Tools and techniques may provide protections against unauthorized production, theft, tampering, insertion of counterfeits, insertion of malicious software or backdoors, and poor development practices throughout the system development life cycle. Organizations also consider providing incentives for suppliers who implement controls, promote transparency into their processes and security and privacy practices, provide contract language that addresses the prohibition of tainted or counterfeit components, and restrict purchases from untrustworthy suppliers. Organizations consider providing training, education, and awareness programs for personnel regarding supply chain risk, available mitigation strategies, and when the programs should be employed. Methods for reviewing and protecting development plans, documentation, and evidence are commensurate with the security and privacy requirements of the organization. Contracts may specify documentation protection requirements.

Related Controls: AT-3, SA-2, SA-3, SA-4, SA-5, SA-8, SA-9, SA-10, SA-15, SR-6, SR-10, SR-11.

SR-8 NOTIFICATION AGREEMENTS

[Priority 3]

Control:

Establish agreements and procedures with entities involved in the supply chain for the system, system component, or system service for the notification of supply chain compromises to systems used to process, store, or transmit CJI.

Discussion: The establishment of agreements and procedures facilitates communications among supply chain entities. Early notification of compromises and potential compromises in the supply chain that can potentially adversely affect or have adversely affected organizational systems or system components is essential for organizations to effectively respond to such incidents. The results of assessments or audits may include open-source information that contributed to a decision or result and could be used to help the supply chain entity resolve a concern or improve its processes.

Related Controls: IR-4, IR-6, IR-8.

SR-10 INSPECTION OF SYSTEMS OR COMPONENTS

[Priority 3]

Control:

Inspect the following systems or system components upon initial procurement and periodically as needed to detect tampering: systems used to process, store, or transmit CJI.

Discussion: The inspection of systems or systems components for tamper resistance and detection addresses physical and logical tampering and is applied to systems and system components removed from organization-controlled areas. Indications of a need for inspection include changes in packaging, specifications, factory location, or entity in which the part is purchased, and when individuals return from travel to high-risk locations.

Related Controls: AT-3, SI-4, SI-7, SR-3, SR-5, SR-11.

SR-12 COMPONENT DISPOSAL

[Priority 3]

Control:

Dispose of CJI using the techniques and methods as described in Media Protection (MP).

Discussion: Data, documentation, tools, or system components can be disposed of at any time during the system development life cycle (not only in the disposal or retirement phase of the life cycle). For example, disposal can occur during research and development, design, prototyping, or operations/maintenance and include methods such as disk cleaning, removal of cryptographic keys, partial reuse of components.

Opportunities for compromise during disposal affect physical and logical data, including system documentation in paper-based or digital files; shipping and delivery documentation; memory sticks with software code; or complete routers or servers that include permanent media, which contain sensitive or proprietary information.

Additionally, proper disposal of system components helps to prevent such components from entering the gray market.

Related Controls: MP-6.

5.20 Policy Area 20: Mobile Devices

This policy area describes considerations and requirements for mobile devices including smartphones and tablets. Mobile devices are not limited to a single form factor or communications medium. The requirements in this section augment those in other areas of the Policy to address the gaps introduced by using mobile devices.

The agency shall: (i) establish usage restrictions and implementation guidance for mobile devices; and (ii) authorize, monitor, control wireless access to the information system. Wireless technologies, in the simplest sense, enable one or more devices to communicate without physical connections—without requiring network or peripheral cabling.

Appendix G provides reference material and additional information on mobile devices.

5.20.1 Wireless Communications Technologies

Examples of wireless communication technologies include, but are not limited to: 802.11, cellular, Bluetooth, satellite, microwave, and land mobile radio (LMR). Wireless technologies require at least the minimum security applied to wired technology and, based upon the specific technology or implementation, wireless technologies may require additional security controls as described below.

5.20.1.1 802.11 Wireless Protocols

Wired Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA) cryptographic algorithms, used by all pre-802.11i protocols, do not meet the requirements for FIPS 140-2 and shall not be used.

Agencies shall implement the following controls for all agency-managed wireless access points with access to an agency's network that processes unencrypted CJI:

1. Perform validation testing to ensure rogue APs (Access Points) do not exist in the 802.11 Wireless Local Area Network (WLAN) and to fully understand the wireless network security posture.
2. Maintain a complete inventory of all Access Points (APs) and 802.11 wireless devices.
3. Place APs in secured areas to prevent unauthorized physical access and user manipulation.
4. Test AP range boundaries to determine the precise extent of the wireless coverage and design the AP wireless coverage to limit the coverage area to only what is needed for operational purposes.
5. Enable user authentication and encryption mechanisms for the management interface of the AP.
6. Ensure that all APs have strong administrative passwords and ensure that all passwords are changed in accordance with Identification and Authentication (IA).
7. Ensure the reset function on APs is used only when needed and is only invoked by authorized personnel. Restore the APs to the latest security settings, when the reset functions are used, to ensure the factory default settings are not utilized.

8. Change the default service set identifier (SSID) in the APs. Disable the broadcast SSID feature so that the client SSID must match that of the AP. Validate that the SSID character string does not contain any agency identifiable information (division, department, street, etc.) or services.
9. Enable all security features of the wireless product, including the cryptographic authentication, firewall, and other available privacy features.
10. Ensure that encryption key sizes are at least 128-bits and the default shared keys are replaced by unique keys.
11. Ensure that the ad hoc mode has been disabled.
12. Disable all nonessential management protocols on the APs.
13. Ensure all management access and authentication occurs via FIPS compliant secure protocols (e.g., SFTP, HTTPS, SNMP over TLS, etc.). Disable non-FIPS compliant secure access to the management interface.
14. Enable logging (if supported) and review the logs on a recurring basis per local policy. At a minimum, logs shall be reviewed monthly.
15. Insulate, virtually (e.g., virtual local area network (VLAN) and ACLs) or physically (e.g., firewalls), the wireless network from the operational wired infrastructure. Limit access between wireless networks and the wired network to only operational needs.
16. When disposing of access points that will no longer be used by the agency, clear access point configuration to prevent disclosure of network configuration, keys, passwords, etc.

5.20.1.2 Cellular Devices

Cellular telephones, smartphones (i.e., Blackberry, iPhones, etc.), tablets, personal digital assistants (PDA), and “aircards” are examples of cellular handheld devices or devices that are capable of employing cellular technology. Additionally, cellular handheld devices typically include Bluetooth, infrared, and other wireless protocols capable of joining infrastructure networks or creating dynamic ad hoc networks.

Threats to cellular handheld devices stem mainly from their size, portability, and available wireless interfaces and associated services. Examples of threats to cellular handheld devices include:

1. Loss, theft, or disposal.
2. Unauthorized access.
3. Malware.
4. Spam.
5. Electronic eavesdropping.
6. Electronic tracking (threat to security of data and safety of the criminal justice professional).
7. Cloning (not as prevalent with later generation cellular technologies).
8. Server-resident data.

5.20.1.2.1 Cellular Service Abroad

Certain internal functions on cellular devices may be modified or compromised by the cellular carrier during international use as the devices are intended to have certain parameters configured by the cellular provider which is considered a “trusted” entity by the device.

When devices are authorized to access CJI outside the U.S., agencies shall perform an inspection to ensure that all controls are in place and functioning properly in accordance with the agency’s policies prior to and after deployment outside of the U.S.

5.20.1.2.2 Voice Transmissions Over Cellular Devices

Any cellular device used to transmit CJI via voice is exempt from the encryption and authentication requirements.

5.20.1.3 Bluetooth

Bluetooth is an open standard for short-range radio frequency (RF) communication. Bluetooth is used primarily to establish wireless personal area networks (WPAN). Bluetooth technology has been integrated into many types of business and consumer devices, including cell phones, laptops, automobiles, medical devices, printers, keyboards, mice, headsets, and biometric capture devices.

Bluetooth technology and associated devices are susceptible to general wireless networking threats (e.g., denial of service [DoS] attacks, eavesdropping, man-in-the-middle [MITM] attacks, message modification, and resource misappropriation) as well as specific Bluetooth-related attacks that target known vulnerabilities in Bluetooth implementations and specifications. Organizational security policy shall be used to dictate the use of Bluetooth and its associated devices based on the agency’s operational and business processes.

5.20.1.4 Mobile Hotspots

Many mobile devices include the capability to function as a Wi-Fi hotspot that allows other devices to connect through the device to the internet over the devices cellular network.

When an agency allows mobile devices that are approved to access or store CJI to function as a Wi-Fi hotspot connecting to the Internet, they shall be configured:

1. Enable encryption on the hotspot
2. Change the hotspot’s default SSID
 - a. Ensure the hotspot SSID does not identify the device make/model or agency ownership
3. Create a wireless network password (pre-shared key)
4. Enable the hotspot’s port filtering/blocking features if present
5. Only allow connections from agency-controlled devices

Note: Refer to the requirements in System and Communications Protection (SC) encryption for item #1. Refer to the requirements in Identification and Authentication (IA) for item #3.

OR

1. Have a MDM solution to provide the same security as identified in items 1 – 5 above.

5.20.2 Mobile Device Management (MDM)

Mobile Device Management (MDM) facilitates the implementation of sound security controls for mobile devices and allows for centralized oversight of configuration control, application usage, and device protection and recovery, if so desired by the agency.

Due to the potential for inconsistent network access or monitoring capability on mobile devices, methods used to monitor and manage the configuration of full-featured operating systems may not function properly on devices with limited-feature operating systems. MDM systems and applications coupled with device specific technical policy can provide a robust method for device configuration management if properly implemented.

Devices that have had any unauthorized changes made to them (including but not limited to being rooted or jailbroken) shall not be used to process, store, or transmit CJI data at any time. User agencies shall implement the following controls when directly accessing CJI from devices running a limited-feature operating system:

1. Ensure that CJI is only transferred between CJI authorized applications and storage areas of the device.
2. MDM with centralized administration configured and implemented to perform at least the following controls:
 - a. Remote locking of device
 - b. Remote wiping of device
 - c. Setting and locking device configuration
 - d. Detection of “rooted” and “jailbroken” devices
 - e. Enforcement of folder or disk level encryption
 - f. Application of mandatory policy settings on the device
 - g. Detection of unauthorized configurations
 - h. Detection of unauthorized software or applications
 - i. Ability to determine the location of agency-controlled devices
 - j. Prevention of unpatched devices from accessing CJI or CJI systems
 - k. Automatic device wiping after a specified number of failed access attempts

EXCEPTION: An MDM is not required when receiving CJI from an indirect access information system (i.e., the system provides no capability to conduct transactional activities on state and national repositories, applications or services). However, it is incumbent upon the authorized agency to ensure CJI is delivered to the appropriate requesting agency or individual. The CSO will make the final determination of whether access is considered indirect.

5.20.3 Wireless Device Risk Mitigations

Organizations shall, at a minimum, ensure that wireless devices:

1. Apply available critical patches and upgrades to the operating system as soon as they become available for the device and after necessary testing as described in System and Information Integrity (SI).
2. Are configured for local device authentication (see Section 5.20.7.1).

3. Use advanced authentication or CSO approved compensating controls as per Section 5.20.7.2.1.
4. Encrypt all CJI resident on the device.
5. Erase cached information, to include authenticators (see Identification and Authentication (IA)) in applications, when session is terminated.
6. Employ personal firewalls on full-featured operating system devices or run a Mobile Device Management (MDM) system that facilitates the ability to provide firewall services from the agency level.
7. Employ malicious code protection on full-featured operating system devices or run a MDM system that facilitates the ability to provide anti-malware services from the agency level.

5.20.4 System Integrity

Managing system integrity on limited function mobile operating systems may require methods and technologies significantly different from traditional full-featured operating systems. In many cases, the requirements of the System and Information Integrity control family cannot be met with a mobile device without the installation of a third-party MDM, application, or supporting service infrastructure.

5.20.4.1 Patching/Updates

Based on the varying connection methods for mobile devices, an always on connection cannot be guaranteed for patching and updating. Devices without always-on cellular connections may not be reachable for extended periods of time by the MDM or solution either to report status or initiate patching.

Agencies shall monitor mobile devices to ensure their patch and update state is current.

5.20.4.2 Malicious Code Protection

Appropriately configured MDM software is capable of checking the installed applications on the device and reporting the software inventory to a central management console in a manner analogous to traditional virus scan detection of unauthorized software and can provide a high degree of confidence that only known software or applications are installed on the device.

Agencies that allow smartphones and tablets to access CJI shall have a process to approve the use of specific software or applications on the devices. Any device natively capable of performing these functions without a MDM solution is acceptable under this section.

5.20.4.3 Personal Firewall

For the purpose of this policy, a personal firewall is an application that controls network traffic to and from a user device, permitting or denying communications based on policy. A personal firewall shall be employed on all mobile devices that have a full-feature operating system (i.e., laptops or tablets with Windows or Linux/Unix operating systems). At a minimum, the personal firewall shall perform the following activities:

1. Manage program access to the Internet.
2. Block unsolicited requests to connect to the user device.

3. Filter incoming traffic by IP address or protocol.
4. Filter incoming traffic by destination ports.
5. Maintain an IP traffic log.

Mobile devices with limited-feature operating systems (i.e., tablets, smartphones) may not support a personal firewall. However, these operating systems have a limited number of system services installed, carefully controlled network access, and to a certain extent, perform functions similar to a personal firewall on a device with a full-feature operating system. Appropriately configured MDM software is capable of controlling which applications are allowed on the device.

5.20.5 Incident Response

In addition to the requirements in Section 5.3 Incident Response, agencies shall develop additional or enhanced incident reporting and handling procedures to address mobile device operating scenarios. Rapid response to mobile device related incidents can significantly mitigate the risks associated with illicit data access either on the device itself or within online data resources associated with the device through an application or specialized interface.

Special reporting procedures for mobile devices shall apply in any of the following situations:

1. Loss of device control. For example:
 - a. Device known to be locked, minimal duration of loss
 - b. Device lock state unknown, minimal duration of loss
 - c. Device lock state unknown, extended duration of loss
 - d. Device known to be unlocked, more than momentary duration of loss
2. Total loss of device
3. Device compromise
4. Device loss or compromise outside the United States

5.20.6 Access Control

Multiple user accounts are not generally supported on limited-feature mobile operating systems. Access control (Section 5.5 Access Control) shall be accomplished by the application that accesses CJI.

5.20.7 Identification and Authentication

Due to the technical methods used for identification and authentication on many limited-feature mobile operating systems, achieving compliance may require many different components.

5.20.7.1 Local Device Authentication

When mobile devices are authorized for use in accessing CJI, local device authentication shall be used to unlock the device for use. The authenticator used shall meet the requirements in Identification and Authentication (IA).

5.20.7.2 Advanced Authentication

When accessing CJI from an authorized mobile device, advanced authentication shall be used by the authorized user unless the access to CJI is indirect as described in Identification and Authentication (IA). If access is indirect, then AA is not required.

5.20.7.2.1 Compensating Controls

CSO approved compensating controls to meet the AA requirement on agency-issued smartphones and tablets with limited-feature operating systems are permitted. Compensating controls are temporary control measures that are implemented in lieu of the required AA control measures when an agency cannot meet a requirement due to legitimate technical or business constraints. Before CSOs consider approval of compensating controls, Mobile Device Management (MDM) shall be implemented per Section 5.20.2. The compensating controls shall:

1. Meet the intent of the CJIS Security Policy AA requirement
2. Provide a similar level of protection or security as the original AA requirement
3. Not rely upon the existing requirements for AA as compensating controls
4. Expire upon the CSO approved date or when a compliant AA solution is implemented.

Additionally, compensating controls may rely upon other, non-AA, existing requirements as compensating controls and/or be combined with new controls to create compensating controls.

The compensating controls for AA are a combination of controls providing acceptable assurance only the authorized user is authenticating and not an impersonator or (in the case of agency-issued device used by multiple users) controls that reduce the risk of exposure if information is accessed by an unauthorized party.

The following minimum controls shall be implemented as part of the CSO approved compensating controls:

- Possession and registration of an agency issued smartphone or tablet as an indication it is the authorized user
- Use of device certificates per Section 5.20.7.3 Device Certificates
- Implemented CJIS Security Policy compliant standard authenticator protection on the secure location where CJI is stored

5.20.7.3 Device Certificates

Device certificates are often used to uniquely identify mobile devices using part of a public key pair on the device in the form of a public key certificate. While there is value to ensuring the device itself can authenticate to a system supplying CJI, and may provide a critical layer of device identification or authentication in a larger scheme, a device certificate alone placed on the device shall not be considered valid proof that the device is being operated by an authorized user.

When certificates or cryptographic keys used to authenticate a mobile device are used in lieu of compensating controls for advanced authentication, they shall be:

1. Protected against being extracted from the device
2. Configured for remote wipe on demand or self-deletion based on a number of unsuccessful login or access attempts
3. Configured to use a secure authenticator (i.e., password, PIN) to unlock the key for use

APPENDICES

This policy area contains the various appendices. The appendices are:

- Appendix A—Terms and Definitions
- Appendix B—Acronyms
- Appendix C—Network Topology Diagrams
- Appendix D—Sample Information Exchange Agreements
- Appendix E—Security Forums and Organizational Entities
- Appendix F—Sample Forms
- Appendix G—Best Practices
- Appendix H—Security Addendum
- Appendix I—References

APPENDIX A: TERMS AND DEFINITIONS

28 CFR Certification Indicator — True if the user has been trained and certified in the handling of criminal intelligence data in accordance with Code of Federal Regulations Title 28 (28 CFR) Part 23, false otherwise. Usage information: Assertion of this privilege requires the user to have been trained and certified in the handling of criminal intelligence data in accordance with Code of Federal Regulations Title 28 (28 CFR) Part 23. One way for a user to meet this requirement is by having taken and passed the online 28 CFR Part 23 training course and certification exam offered by the U.S. Department of Justice Bureau of Justice Assistance (BJA) via its Secured National Criminal Intelligence Resource Center (NCIRC) Web Site (<http://www.ncirc.gov/securedwebsite.cfm>). Alternatively, a user may meet this requirement by having taken and passed an equivalent offline 28 CFR Part 23 training course, offered by the Institute for Intergovernmental Research (IIR). (See <https://28cfr.iir.com/> for details.)

Access to Criminal Justice Information — The physical or logical (electronic) ability, right or privilege to view, modify or make use of Criminal Justice Information.

Administration of Criminal Justice — The detection, apprehension, detention, pretrial release, post-trial release, prosecution, adjudication, correctional supervision, or rehabilitation of accused persons or criminal offenders. It also includes criminal identification activities; the collection, storage, and dissemination of criminal history record information; and criminal justice employment. In addition, administration of criminal justice includes “crime prevention programs” to the extent access to criminal history record information is limited to law enforcement agencies for law enforcement programs (e.g., record checks of individuals who participate in Neighborhood Watch or “safe house” programs) and the result of such checks will not be disseminated outside the law enforcement agency.

Agency Controlled Mobile Device — A mobile device that is centrally managed by an agency for the purpose of securing the device for potential access to CJI. The device can be agency issued or BYOD (personally owned).

Agency Coordinator (AC) — A staff member of the Contracting Government Agency who manages the agreement between the Contractor and agency.

Agency Issued Mobile Device — A mobile device that is owned by an agency and issued to an individual for use. It is centrally managed by the agency for the purpose of securing the device for potential access to CJI. The device is not BYOD (personally owned).

Agency Liaison (AL) — Coordinator of activities between the criminal justice agency and the noncriminal justice agency when responsibility for a criminal justice system has been delegated by a criminal justice agency to a noncriminal justice agency, which has in turn entered into an agreement with a contractor. The agency liaison shall, inter alia, monitor compliance with system security requirements. In instances in which the noncriminal justice agency’s authority is directly from the CJIS systems agency, there is no requirement for the appointment of an agency liaison.

Asymmetric Encryption — A type of encryption that uses key pairs for encryption. One key is used to encrypt a message and another key to decrypt the message. Asymmetric encryption is also commonly known as public key encryption.

Authenticator Assurance Level — The authenticator assurance level as defined by NIST SP 800-63-3. There are three defined levels: AAL1 (low), AAL2 (medium), and AAL3 (high). Usage

information: IDPs should assert this attribute to indicate the strength of the authenticator used for the current authentication process; AAL1 (low), AAL2 (medium), and AAL3 (high). AAL2 and AAL3 require multifactor authentication; although the use of multiple factors does not automatically qualify for AAL2.

Authorized User/Personnel — An individual, or group of individuals, who have been appropriately vetted through a national fingerprint-based record check and have been granted access to CJI.

Authorized Recipient (AR) — (1) A criminal justice agency or federal agency authorized to receive CHRI pursuant to federal statute or executive order; (2) A nongovernmental entity authorized by federal statute or executive order to receive CHRI for noncriminal justice purposes; or (3) A government agency authorized by federal statute or executive order, or state statute which has been approved by the United States Attorney General to receive CHRI for noncriminal justice purposes.

Authorized Recipient Security Officer (ARSO) — the individual appointed by the AR to coordinate and oversee Information Security by ensuring that the Channeler is adhering to the CJISSECPOL and Outsourcing Standard, verifying the completion of annual Security Awareness Training, and communicating with the FBI CJIS Division on matters relating to Information Security.

Availability — The degree to which information, a system, subsystem, or equipment is operable and in a useable state; frequently represented as a proportion of time the element is in a functioning condition.

Basic Testing — A test methodology that assumes no knowledge of the internal structure and implementation detail of the assessment object. Also known as black box testing.

Comprehensive Testing — A test methodology that assumes explicit and substantial knowledge of the internal structure and implementation detail of the assessment object. Also known as white box testing.

Biographic Data — Information collected about individuals associated with a unique case, and not necessarily connected to identity data. Biographic Data does not provide a history of an individual, only information related to a unique case.

Biometric Data — When applied to CJI, it is used to identify individuals, and includes the following types: fingerprints, palm prints, DNA, iris, and facial recognition.

Case / Incident History — All relevant information gathered about an individual, organization, incident, or combination thereof, arranged so as to serve as an organized record to provide analytic value for a criminal justice organization. In regards to CJI, it is the information about the history of criminal incidents.

Certificate Authority (CA) Certificate — Digital certificates required for certificate-based authentication that are issued to tell the client computers and servers that it can trust other certificates that are issued by this CA.

Channeler — A FBI approved contractor, who has entered into an agreement with an Authorized Recipient(s), to receive noncriminal justice applicant fingerprint submissions and collect the associated fees. The Channeler ensures fingerprint submissions are properly and adequately

completed, electronically forwards fingerprint submissions to the FBI's CJIS Division for national noncriminal justice criminal history record check, and receives electronic record check results for dissemination to Authorized Recipients. A Channeler is essentially an "expediter" rather than a user of criminal history record check results.

Cloud Client — A machine or software application that accesses cloud services over a network connection, perhaps on behalf of a subscriber.

Cloud Computing — A distributed computing model that permits on-demand network access to a shared pool of configurable computing resources (i.e., networks, servers, storage, applications, and services), software, and information.

Cloud Provider — An organization that provides cloud computing services.

Cloud Subscriber — A person or organization that is a customer of a cloud computing service provider.

CJIS Advisory Policy Board (APB) — The governing organization within the FBI CJIS Advisory Process composed of representatives from criminal justice and national security agencies within the United States. The APB reviews policy, technical, and operational issues relative to CJIS Division programs and makes subsequent recommendations to the Director of the FBI.

CJIS Audit Unit (CAU) — The organization within the FBI CJIS Division responsible to perform audits of CSAs to verify compliance with the CJIS Security Policy.

CJIS Security Policy — The FBI CJIS Security Policy document as published by the FBI CJIS ISO; the document containing this glossary.

CJIS Systems Agency (CSA) — A duly authorized state, federal, international, tribal, or territorial criminal justice agency on the CJIS network providing statewide (or equivalent) service to its criminal justice users with respect to the CJI from various systems managed by the FBI CJIS Division. There shall be only one CSA per state or territory. In federal agencies, the CSA may be the interface or switch to other federal agencies connecting to the FBI CJIS systems.

CJIS Systems Agency Information Security Officer (CSA ISO) — The CSO appointed information security personnel responsible to coordinate information security efforts throughout the CSA's user community.

CJIS Systems Officer (CSO) — The individual located within the CJIS Systems Agency responsible for the administration of the CJIS network on behalf of the CJIS Systems Agency.

Compact Council — The entity created by the National Crime Prevention and Privacy Compact of 1998 that has the authority to promulgate rules and procedures governing the use of the III system for noncriminal justice purposes.

Compact Officers — The leadership of the Compact Council, oversees the infrastructure established by the National Crime Prevention and Privacy Compact Act of 1998, which is used by ratifying states to exchange criminal records for noncriminal justice purposes. Their primary responsibilities are to promulgate rules and procedures for the effective and appropriate use of the III system.

Compensating Controls — Compensating controls are temporary control measures implemented in lieu of the required control measures when an agency cannot meet the AA requirement due to legitimate technical or business constraints. The compensating controls must:

1. Meet the intent of the CJIS Security Policy AA requirement
2. Provide a similar level of protection or security as the original AA requirement
3. Not rely upon the existing requirements for AA as compensating controls

Additionally, compensating controls may rely upon other, non-AA, existing requirements as compensating controls and/or be combined with new controls to create compensating controls.

Computer Security Incident Response Capability (CSIRC) — A collection of personnel, systems, and processes that are used to efficiently and quickly manage a centralized response to any sort of computer security incident which may occur.

Confidentiality — The concept of ensuring that information is observable only to those who have been granted authorization to do so.

Contractor — A private business, agency or individual which has entered into an agreement for the administration of criminal justice or noncriminal justice functions with a Criminal Justice Agency or a Noncriminal Justice Agency. Also, a private business approved by the FBI CJIS Division to contract with Noncriminal Justice Agencies to perform noncriminal justice functions associated with civil fingerprint submission for hiring purposes.

Contracting Government Agency (CGA) — The government agency, whether a Criminal Justice Agency or a Noncriminal Justice Agency (public or private), which enters into an agreement with a private contractor.

Controlled Unclassified Information (CUI) — Information the Government creates or possesses, or that an entity creates or possesses for or on behalf of the Government, that a law, regulation, or Government-wide policy requires or permits an agency to handle using safeguarding or dissemination controls. (32 CFR Vol 6 Part 2002)

Counter Terrorism Data Self Search Home Privilege Indicator — True if the user has permission to search on behalf of himself/herself (NOT on behalf of the user's home agency) for counter-terrorism data and documents within the user's home system, network, or agency. False otherwise. Usage information: Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Crime Reports Data — The data collected through the Uniform Crime Reporting program and reported upon annually by the FBI CJIS division used to analyze the crime statistics for the United States.

Criminal History Data Self Search Home Privilege Indicator — True if the user has permission to search on behalf of himself/herself (NOT on behalf of the user's home agency) for criminal history data and documents within the user's home system, network, or agency. False otherwise. Usage information: Example data sources include National Crime Information Center (NCIC) Criminal History. User eligibility requirements are decided by federation members, but may include: Fingerprint (FP) based background, NCIC training, access to Law Enforcement criminal history data in home agency, member of agency with Law Enforcement Originating Agency (ORI) code. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Criminal History Record Information (CHRI) — Information collected by criminal justice agencies on individuals consisting of identifiable descriptions and notations of arrests, detentions,

indictments, informations, or other formal criminal charges, and any disposition arising therefrom, including acquittal, sentencing, correctional supervision, and release. The term does not include identification information such as fingerprint records if such information does not indicate the individual's involvement with the criminal justice system.

Criminal Intelligence Data Self Search Home Privilege Indicator — True if the user has permission to search on behalf of himself/herself (NOT on behalf of the user's home agency) for criminal intelligence data and documents within the user's home system, network, or agency. False otherwise. Usage information: Example data sources include Criminal Law Enforcement Reporting Information System (CLERIS), Regional Information Sharing Systems (RISS), Joint Regional Information Exchange System (JRIES), and Law Enforcement Intelligence Units (LEIU). User eligibility requirements are decided by federation members, but may include: 28 CFR (23) training, Fingerprint (FP) based background, access to intelligence data in home agency, member of agency with Law Enforcement Originating Agency (ORI) code. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Criminal Investigative Data Self Search Home Privilege Indicator — True if the user has permission to search on behalf of himself/herself (NOT on behalf of the user's home agency) for criminal investigative data and documents within the user's home system, network, or agency. False otherwise. Usage information: Example data sources include Criminal Law Enforcement Reporting Information System (CLERIS) and Contact and Information Management System (CIMS). User eligibility requirements are decided by federation members, but may include: Fingerprint (FP) based background, access to investigative data in home agency, member of agency with Law Enforcement Originating Agency (ORI) code. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Criminal Justice Agency (CJA) — The courts, a governmental agency, or any subunit of a governmental agency which performs the administration of criminal justice pursuant to a statute or executive order and which allocates a substantial part of its annual budget to the administration of criminal justice. State and federal Inspectors General Offices are included.

Criminal Justice Agency User Agreement — A terms-of-service agreement that must be signed prior to accessing CJI. This agreement is required by each CJA and spells out user's responsibilities, the forms and methods of acceptable use, penalties for their violation, disclaimers, and so on.

Criminal Justice Conveyance — A criminal justice conveyance is any enclosed mobile vehicle used for the purposes of criminal justice activities with the capability to comply, during operational periods, with the requirements of PE-3 Physical Access Control.

Criminal Justice Information (CJI) — Criminal Justice Information is the abstract term used to refer to all of the FBI CJIS provided data necessary for law enforcement agencies to perform their mission and enforce the laws, including but not limited to: biometric, identity history, person, organization, property (when accompanied by any personally identifiable information), and case/incident history data. In addition, CJI refers to the FBI CJIS-provided data necessary for civil agencies to perform their mission; including, but not limited to data used to make hiring decisions. The following type of data are exempt from the protection levels required for CJI: transaction control type numbers (e.g., ORI, NIC, UCN, etc.) when not accompanied by information that reveals CJI or PII.

Criminal Justice Information Services Division (FBI CJIS or CJIS) — The FBI division responsible for the collection, warehousing, and timely dissemination of relevant CJI to the FBI and to qualified law enforcement, criminal justice, civilian, academic, employment, and licensing agencies.

Data — See Information and CJI.

Data Center — A physical room, building, or facility housing Information Technology infrastructure for developing, running, and delivering applications and services associated with storing and managing data, as determined by the CSO, his/her designee, or Interface Agency Official.

Decryption — The inverse cryptographic operation used to convert encrypted information back into a plaintext (readable) format.

Degauss — Neutralize a magnetic field to erase information from a magnetic disk or other storage device. In the field of information technology, degauss has become synonymous with erasing information whether or not the medium is magnetic. In the event the device to be degaussed is not magnetic (e.g., solid state drive, USB storage device), steps other than magnetic degaussing may be required to render the information irretrievable from the device.

Department of Justice (DoJ) — The Department within the U.S. Government responsible to enforce the law and defend the interests of the United States according to the law, to ensure public safety against threats foreign and domestic, to provide federal leadership in preventing and controlling crime, to seek just punishment for those guilty of unlawful behavior, and to ensure fair and impartial administration of justice for all Americans.

Digital Media — Any form of electronic media designed to store data in a digital format. This includes, but is not limited to: memory device in laptops, computers, and mobile devices; and any removable, transportable electronic media, such as magnetic tape or disk, optical disk, flash drives, external hard drives, or digital memory card.

Digital Signature — A digital signature consists of three algorithms: (1) A key generation algorithm that selects a private key uniformly at random from a set of possible private keys. The algorithm outputs the private key and a corresponding public key. (2) A signing algorithm that, given a message and a private key, produces a signature. (3) A signature verifying algorithm that, given a message, public key, and a signature, either accepts or rejects the message's claim to authenticity. Two main properties are required. First, a signature generated from a fixed message and fixed private key should verify the authenticity of that message by using the corresponding public key. Secondly, it should be computationally infeasible to generate a valid signature for a party who does not possess the private key.

Direct Access — Having the authority to access systems managed by the FBI CJIS Division, whether by manual or automated methods, not requiring the assistance of, or intervention by, any other party or agency (28 CFR, Chapter 1, Part 20).

Display Name — The user's display name is a text string formatted for display purposes in applications and correspondence.

Dissemination — The transmission/distribution of CJI to Authorized Recipients within an agency.

Email Address Text — The electronic mailing address by which the user may be contacted.

Employer Name — The name of the organization that is the user's primary employer.

Employer ORI — A unique identifier assigned to the organization that is the user's primary employer. ORIs are generally assigned by the FBI; however, in some cases they may be assigned by other agencies.

Employer Organization General Category Code —The general category of the organization that is the user's primary employer.

Employer State Code — The state, commonwealth, province, or other such geopolitical subdivision of the country in which is located the primary business office of the organization that is the user's primary employer.

Encryption — A form of cryptology that applies a cryptographic operation to provide confidentiality of (sensitive) information.

Escort — Authorized personnel who accompany a visitor at all times while within a physically secure location to ensure the protection and integrity of the physically secure location and any Criminal Justice Information therein. The use of cameras or other electronic means used to monitor a physically secure location does not constitute an escort.

Facsimile (Fax) — Facsimile is: (a) a document received and printed on a single or multi-function stand-alone device, (b) a single or multi-function stand-alone device for the express purpose of transmitting and receiving documents from a like device over a standard telephone line, or (c) a facsimile server, application, service which implements email-like technology and transfers documents over a network.

Federal Bureau of Investigation (FBI) — The agency within the DOJ responsible to protect and defend the United States against terrorist and foreign intelligence threats, to uphold and enforce the criminal laws of the United States, and to provide leadership and criminal justice services to federal, state, municipal, and international agencies and partners.

FBI CJIS Information Security Officer (FBI CJIS ISO) — The FBI personnel responsible for the maintenance and dissemination of the FBI CJIS Security Policy; the liaison between the FBI and the CSA's ISOs and other relevant security points-of-contact (POCs); the provider of technical guidance as to the intent and implementation of technical policy issues; the POC for computer incident notification which also disseminates security alerts to the CSOs and ISOs.

Federal Information Security Management Act (FISMA) — The Federal Information Security Management Act of 2002, a US Federal law that established information security standards for the protection of economic and national security interests of the United States. It requires each federal agency to develop, document, and implement an agency-wide program to provide information security for the information and information systems that support the operations and assets of the agency, including those provided or managed by another agency, contractor, or other source.

Federation Assurance Level — There are 3 Federation Assurance Levels representing the assurance level of the user's federated assertion: FAL1 (low assurance), FAL2 (moderate assurance), and FAL3 (high assurance) based on NIST SP 800-63-3. Usage information: IDPs should assert this for all assertions sent to RPs correctly identifying the level of assurance of the assertion. The RP can also derive the FAL if not asserted.

Federation Id — The persistent, federation-unique identifier for the user, comprising a federation part, an optional trusted identity broker (TIB) part, an identity provider (IDP) part, and a local ID.

Focused Testing — A test methodology that assumes some knowledge of the internal structure and implementation detail of the assessment object. Also known as gray box testing.

For Official Use Only (FOUO) — A caveat applied to unclassified sensitive information that may be exempt from mandatory release to the public under the Freedom of Information Act (FOIA), 5 U.S.C 522. In general, information marked FOUO shall not be disclosed to anybody except Government (Federal, State, tribal, or local) employees or contractors with a need to know.

Full-feature Operating System — Full-feature operating systems are traditional operating systems used by a standard desktop computer (e.g., Microsoft Windows, Apple OSX/macOS, LINUX/UNIX, etc.). These operating systems are generally open to user control and configuration and therefore require configuration management to properly secure, or “harden”, these devices from malicious network based technical attacks (e.g., malware, spyware, hackers, etc.). These operating systems require traditional protection applications such as antivirus programs and personal firewalls.

General User — A user, but not a process, who is authorized to use an information system.

Given Name — The first name of the user.

Government Data Self Search Home Privilege Indicator — True if the user has permission to search on behalf of himself/herself (NOT on behalf of the user's home agency) for government data and documents within the user's home system, network, or agency. False otherwise. Usage information: Example data sources include Tax Data, Labor Data, Uniform Commercial Code (UCC) Filings, Property Records, Department of Motor Vehicles (DMV), Drivers License (DL), Boat Ownership, Corporate Records, and Protected Critical Infrastructure Information (PCII). User eligibility requirements are decided by federation members, but may include: Agency vetting, application training, state law, and Federal law. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Guest Operating System — An operating system that has emulated hardware presented to it by a host operating system. Also referred to as the virtual machine (VM).

Hashing — The process of applying a mathematical algorithm to data to produce an alphanumeric value (i.e., hash value) to be used as a representative of that data.

Hash Value — The term that refers to an alphanumeric value which represents the result of applying a cryptographic hash function to data.

Host Operating System — In the context of virtualization, the operating system that interfaces with the actual physical hardware and arbitrates between it and the guest operating systems. It is also referred to as a hypervisor.

Hybrid Encryption — A type of encryption where both asymmetric encryption and symmetric encryption keys are used creating what is referred to as cipher suites. In a hybrid solution, the asymmetric encryption keys are used for client/server certificate exchange to provide session integrity while the symmetric encryption keys are used for bulk data encryption to provide data confidentiality.

Hypervisor — See Host Operating System.

Identity Assurance Level — The maximum NIST identity assurance level as defined by NIST SP 800-63-3 for which the identity proofing process of this Identity Provider Organization qualifies. There are three defined levels: IAL1 (low), IAL2 (medium), and IAL3 (high). Usage information: IDPs should use this attribute to indicate the assurance level of their identity and attribute proofing processes.

Identity History Data — Textual data that corresponds with an individual's biometric data, providing a history of criminal and/or civil events for the identified individual.

Identity Provider Id — A unique name that clearly identifies the Identity Provider. To ensure uniqueness and meaningfulness, it is recommended that a resolvable fully qualified domain name be used, represented as a URL.

In-Band — The communication service channel (network connection, email, SMS text, phone call, etc.) used to obtain an authenticator is the same as the one used for login.

Indirect Access — Having the authority to access systems containing CJI without providing the user the ability to conduct transactional activities (the capability to query or update) on state and national systems (e.g., CJIS Systems Agency (CSA), State Identification Bureau (SIB), or national repositories).

Information — See data and CJI.

Information Exchange Agreement — An agreement that codifies the rules by which two parties engage in the sharing of information. These agreements typically include language which establishes some general duty-of-care over the other party's information, whether and how it can be further disseminated, penalties for violations, the laws governing the agreement (which establishes venue), procedures for the handling of shared information at the termination of the agreement, and so on. This document will ensure consistency with applicable federal laws, directives, policies, regulations, standards and guidance.

Information Security Officer (ISO) — Typically a member of an organization who has the responsibility to establish and maintain information security policy, assesses threats and vulnerabilities, performs risk and control assessments, oversees the governance of security operations, and establishes information security training and awareness programs. The ISO also usually interfaces with security operations to manage implementation details and with auditors to verify compliance to established policies.

Information System — A system of people, data, and processes, whether manual or automated, established for the purpose of managing information.

Integrated Automated Fingerprint Identification System (IAFIS) — The national fingerprint and criminal history system maintained by the FBI CJIS Division that provides the law enforcement community with automated fingerprint search capabilities, latent searching capability, electronic image storage, and electronic exchange of fingerprints and responses.

Integrity — The perceived consistency of expected outcomes, actions, values, and methods of an individual or organization. As it relates to data, it is the concept that data is preserved in a consistent and correct state for its intended use.

Intelligence Analyst Indicator — True if the user is an Intelligence Analyst (IA) for a government agency, false otherwise. Usage information: An Identity Provider (IdP) may assert that a user is

an IA if they work for a government agency, and/or company who works with the government, in order to provide information assessments about criminal or security threats.

Interconnection Security Agreement (ISA) — An agreement much like an Information Exchange Agreement as mentioned above, but concentrating more on formalizing the technical and security requirements pertaining to some sort of interface between the parties' information systems.

Interface Agency — A criminal justice agency (including law enforcement) or any federally authorized agency/entity, other than a CSA or State Identification Bureau (SIB), that leverages a direct connection to the FBI CJIS Division.

Internet Protocol (IP) — A protocol used for communicating data across a packet-switched internetwork using the Internet Protocol Suite, also referred to as TCP/IP. IP is the primary protocol in the Internet Layer of the Internet Protocol Suite and has the task of delivering distinguished protocol datagrams (packets) from the source host to the destination host solely based on their addresses.

Interstate Identification Index (III) — The CJIS service that manages automated submission and requests for CHRI that is warehoused subsequent to the submission of fingerprint information. Subsequent requests are directed to the originating State as needed.

Intrusion Detection — The process of monitoring the events occurring in an information system or network and analyzing them for signs of possible incidents.

Intrusion Detection System — Software which automates the intrusion detection process.

Intrusion Prevention — The process of monitoring events occurring in an information system or network and analyzing them for signs of possible incidents and attempting to stop detected possible incidents.

Intrusion Prevention System — Software which has all the capabilities of an intrusion detection system and can also attempt to stop possible incidents.

Jailbreak (Jailbroken) — The process of attaining privileged control (known as “root access”) of a device running the Apple iOS operating system that ultimately allows a user the ability to alter or replace system applications and settings, run specialized applications that require administrator-level permissions, or perform other operations that are otherwise not allowed.

Laptop Devices — Laptop devices are mobile devices with a full-featured operating system (e.g., Microsoft Windows, Apple OSX/macOS, LINUX/UNIX, etc.). Laptops are typically intended for transport via vehicle mount or portfolio-sized carry case, but not on the body. This definition does not include pocket/handheld devices (e.g., smartphones), or mobile devices that feature a limited-feature operating system (e.g., tablets).

Law Enforcement Enterprise Portal (LEEP) — A secure, Internet-based communications portal provided by the FBI CJIS Division for use by law enforcement, first responders, criminal justice professionals, and anti-terrorism and intelligence agencies around the globe. Its primary purpose is to provide a platform on which various law enforcement agencies can collaborate on FOUO matters.

Limited-feature Operating System — Limited-feature operating systems are designed specifically for the mobile environment where battery life and power efficiency are primary design

drivers (e.g., Apple iOS, Android, Windows Mobile, Blackberry OS, etc.). These operating systems permit limited user control, but are inherently more resistant than a full-feature operating system to certain types of network based technical attacks due to the limited-feature sets. Devices using these operating systems are required to be managed by a mobile device management solution.

Local Id — The unique local identifier associated with the user for internal purposes within the user's identity provider (IDP). This identifier is assigned and maintained by a federation member agency or partner organization and used for local authentication and identification. The identifier typically has local significance and is integrated into the existing legacy IT infrastructure and/or business processes of the federation member agency or partner organization.

Logical Access — The technical means (e.g., read, create, modify, delete a file, execute a program, or use an external connection) for an individual or other computer system to utilize CJI or CJIS applications.

Logical Partitioning — When the host operating system, or hypervisor, allows multiple guest operating systems to share the same physical resources.

Local Agency Security Officer (LASO) — The primary Information Security contact between a local law enforcement agency and the CSA under which this agency interfaces with the FBI CJIS Division. The LASO actively represents their agency in all matters pertaining to Information Security, disseminates Information Security alerts and other material to their constituents, maintains Information Security documentation (including system configuration data), assists with Information Security audits of hardware and procedures, and keeps the CSA informed as to any Information Security needs and problems.

Management Control Agreement (MCA) — An agreement between parties that wish to share or pool resources that codifies precisely who has administrative control over, versus overall management and legal responsibility for, assets covered under the agreement. An MCA must ensure the CJA's authority remains with regard to all aspects of Section 3.2.2. The MCA usually results in the CJA having ultimate authority over the CJI supporting infrastructure administered by the NCJA.

Metadata — Structured information that describes, explains, locates or otherwise makes it easier to retrieve, use or manage an information resource. Metadata is commonly referred to as data about data, information about information, or information describing the characteristics of data.

Mobile Device — Any portable device used to access CJI via a wireless connection (e.g., cellular, Wi-Fi, Bluetooth, etc.).

Mobile Device Management (MDM) — Centralized administration and control of mobile devices specifically including, but not limited to, cellular phones, smart phones, and tablets. Management typically includes the ability to configure device settings and prevent a user from changing them, remotely locating a device in the event of theft or loss, and remotely locking or wiping a device. Management can also include over-the-air distribution of applications and updating installed applications.

Mobile (Wi-Fi) Hotspot — A mobile (Wi-Fi) hotspot is a zone or area associated with a mobile device (e.g., smartphone, air card) allowing wireless connectivity to the Internet typically through a cellular connection.

National Crime Information Center (NCIC) — An information system which stores CJI which can be queried by appropriate Federal, state, and local law enforcement and other criminal justice agencies.

National Instant Criminal Background Check System (NICS) — A system mandated by the Brady Handgun Violence Prevention Act of 1993 that is used by Federal Firearms Licensees (FFLs) to instantly determine via telephone or other electronic means whether the transfer of a firearm would be in violation of Section 922 (g) or (n) of Title 18, United States Code, or state law, by evaluating the prospective buyer's criminal history.

National Institute of Standards and Technology (NIST) — Founded in 1901, NIST is a non-regulatory federal agency within the U.S. Department of Commerce whose mission is to promote U.S. innovation and industrial competitiveness by advancing measurement science, standards, and technology in ways that enhance economic and national security.

Noncriminal Justice Agency (NCJA) — A governmental agency, or any subunit thereof, that provides services primarily for purposes other than the administration of criminal justice. Examples of services include, but not limited to, employment suitability, licensing determinations, immigration and naturalization matters, and national security clearances.

NCJA (Government) — A Federal, state, local, or tribal governmental agency or any subunit thereof whose charter does not include the responsibility to administer criminal justice, but may have a need to process CJI. An example would be the central IT organization within a state government that administers equipment on behalf of a state law-enforcement agency.

NCJA (Private) — A private agency or subunit thereof whose charter does not include the responsibility to administer criminal justice, but may have a need to process CJI. An example would include a local bank.

NCJA (Public) — A public agency or sub-unit thereof whose charter does not include the responsibility to administer criminal justice, but may have a need to process CJI. An example would include a county school board which uses CHRI to assist in employee hiring decisions.

NCIC Certification Indicator — True if the user has a valid, active certification in the usage and handling of data in accordance with National Crime Information Center (NCIC) rules and regulations, false otherwise.

N-DEx Privilege Indicator — True if the user has privileges to access the Federal Bureau of Investigation (FBI) Law Enforcement National Data Exchange (N-DEx), false otherwise. Usage information: Assertion of this privilege requires the user to meet certain requirements which are not currently documented in the GFIPM Metadata Specification. Future versions of this spec will properly document these requirements. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Noncriminal Justice Purpose — The uses of criminal history records for purposes authorized by federal or state law other than purposes relating to the administration of criminal justice, including employment suitability, licensing determinations, immigration and naturalization matters, and national security clearances.

Non-digital Media — Non-digital media means a hard copy or physical representation of information, including, but not limited to, paper copies, printer ribbons, drums, microfilm, platens, and other forms of preserved or preservable information.

Office of Management and Budget (OMB) — The agency within the Executive Branch of the Federal government responsible to oversee the preparation of the federal budget, to assist in the supervision of other Executive Branch agencies, and to oversee and coordinate the Presidential Administration's procurement, financial management, information, and regulatory policies.

One-time Password — A disposable, single-use standard authenticator for access CJI. One-time passwords are: minimum of six (6) randomly generated characters, valid for a single session, and if not used, expire within a minimum of five (5) minutes after issuance.

Organizational Personnel with Security Responsibilities — Personnel with the responsibility to ensure the confidentiality, integrity, and availability of CJI and the implementation of technology in a manner compliant with the CJISSECPOL.

Out-of-Band — The communication service channel (network connection, email, SMS text, phone call, etc.) used to obtain an authenticator is separate from that used for login.

Outsourcing — The process of delegating in-house operations to a third-party. For instance, when the administration of criminal justice functions (network operations, dispatch functions, system administration operations, etc.) are performed for the criminal justice agency by a city or county information technology department or are contracted to be performed by a vendor.

Outsourcing Standard — National Crime Prevention and Privacy Compact Council's Outsourcing Standard. The Compact Council's uniform standards and processes for the interstate and Federal-State exchange of criminal history records for noncriminal justice purposes.

Partitioning — Managing guest operating system, or virtual machine, access to hardware so that each guest OS can access its own resources but cannot encroach on the other guest operating systems resources or any resources not allocated for virtualization use.

Password Verifier (Verifier) — An entity or process that verifies the claimant's identity by verifying the claimant's possession and control of one or two authenticators using an authentication protocol. To do this, the Verifier may also need to validate credentials that link the authenticator(s) to the subscriber's identifier and check their status.

PCII Certification Indicator — True if the user has a valid, active certification in the usage and handling of Protected Critical Infrastructure Information (PCII) data in accordance with US Department of Homeland Security (DHS) rules and regulations, false otherwise. Usage information: Information about PCII authorized user training is available at the following URL. <http://www.dhs.gov/receive-pcii-authorized-user-training>

Personal Firewall — An application which controls network traffic to and from a computer, permitting or denying communications based on a security policy.

Personally Identifiable Information (PII) — PII is information which can be used to distinguish or trace an individual's identity, such as name, social security number, or biometric records, alone or when combined with other personal or identifying information which is linked or linkable to a specific individual, such as date and place of birth, or mother's maiden name.

Physical Access — The physical ability, right or privilege to view, modify or make use of Criminal Justice Information (CJI) by means of physical presence within the proximity of computers and network devices (e.g., the ability to insert a boot disk or other device into the system, make a physical connection with electronic equipment, etc.).

Physical Media — Physical media refers to media in printed form. This definition includes, but is not limited to, printed documents, printed imagery, printed facsimile.

Physical Partitioning — When the host operating system, or hypervisor, assigns separate physical resources to each guest operating systems, or virtual machine.

Physically Secure Location — A facility, a criminal justice conveyance, or an area, a room, or a group of rooms, within a facility with both the physical and personnel security controls sufficient to protect CJI and associated information systems.

Pocket/Handheld Mobile Device — Pocket/Handheld mobile devices (e.g., smartphones) are intended to be carried in a pocket or holster attached to the body and feature an operating system with limited functionality (e.g., iOS, Android, BlackBerry, etc.). This definition does not include tablet and laptop devices.

Privileged User — A user that is authorized (and, therefore, trusted) to perform security-relevant functions that general users are not authorized to perform.

Property Data — Information about vehicles and property associated with a crime.

Public Safety Officer Indicator — True if the user is a public safety officer (PSO), false otherwise. Usage information: An IDP may assert that a user is a PSO if the user is authorized to act in a role pursuant to the safety and welfare of the public within a government jurisdiction. This may include firefighters, emergency medical technicians (EMTs), hazardous materials (HAZMAT) cleanup specialists, etc.

Rap Back — A NGI service that allows authorized agencies to receive notification of subsequent criminal activity reported to the FBI committed by persons of interest.

Receive-Only Terminal (ROT) — A device that is configured to accept a limited type of data but is technically prohibited from forming or transmitting data, browsing or navigating internal or external networks, or otherwise performing outside the scope of receive only (e.g., a printer, dumb terminal, etc.).

Repository Manager, or Chief Administrator — The designated manager of the agency having oversight responsibility for a CSA's fingerprint identification services. If both state fingerprint identification services and CJIS systems control are managed within the same state agency, the repository manager and CSO may be the same person.

Root (Rooting, Rooted) — The process of attaining privileged control (known as “root access”) of a device running the Android operating system that ultimately allows a user the ability to alter or replace system applications and settings, run specialized applications that require administrator-level permissions, or perform other operations that are otherwise not allowed.

Salting — The process of applying a non-secret value to data prior to applying a cryptographic process, such as hashing. This process changes the value to be hashed in a manner designed to ensure an attacker cannot reuse the results of computations for one instance.

Secondary Dissemination — The promulgation of CJI from a releasing agency to an authorized recipient agency when the recipient agency has not been previously identified in a formal information exchange agreement.

Security Addendum (SA) — A uniform addendum to an agreement between the government agency and a private contractor, approved by the Attorney General of the United States, which

specifically authorizes access to criminal history record information, limits the use of the information to the purposes for which it is provided, ensures the security and confidentiality of the information consistent with existing regulations and the CJIS Security Policy, provides for sanctions, and contains such other provisions as the Attorney General may require.

Sensitive But Unclassified (SBU) — Designation of information in the United States federal government that, though unclassified, often requires strict controls over its distribution. SBU is a broad category of information that includes material covered by such designations as For Official Use Only (FOUO), Law Enforcement Sensitive (LES), Sensitive Homeland Security Information, Security Sensitive Information (SSI), Critical Infrastructure Information (CII), etc. Some categories of SBU information have authority in statute or regulation (e.g., SSI, CII) while others, including FOUO, do not. As of May 9, 2008, the more appropriate terminology to use is Controlled Unclassified Information (CUI).

Server/Client Computer Certificate (device-based) — Digital certificates that are issued to servers or client computers or devices by a CA and used to prove device identity between server and/or client computer devices during the authentication process.

Service — The organized system of apparatus, appliances, personnel, etc., that supply some tangible benefit to the consumers of this service. In the context of CJI, this usually refers to one of the applications that can be used to process CJI.

Shredder — A device used for shredding documents, often as a security measure to prevent unapproved persons from reading them. Strip-cut shredders, also known as straight-cut or spaghetti-cut, slice the paper into long, thin strips but are not considered secure. Cross-cut shredders provide more security by cutting paper vertically and horizontally into confetti-like pieces.

Smartphone — See pocket/handheld mobile devices.

Social Engineering — The act of manipulating people into performing actions or divulging confidential information. While similar to a confidence trick or simple fraud, the term typically applies to trickery or deception for the purpose of information gathering, fraud, or computer system access; in most cases the attacker never comes face-to-face with the victim.

Software Patch — A piece of software designed to fix problems with, or update, a computer program or its supporting data. This includes fixing security vulnerabilities and other bugs and improving the usability or performance. Though meant to fix problems, poorly designed patches can sometimes introduce new problems. As such, patches should be installed in a test environment prior to being installed in a live, operational system. Patches often can be found in multiple locations but should be retrieved only from sources agreed upon through organizational policy.

State and Federal Agency User Agreement — A written agreement that each CSA or SIB Chief shall execute with the FBI CJIS Division stating their willingness to demonstrate conformance with the FBI CJIS Security Policy prior to the establishment of connectivity between organizations. This agreement includes the standards and sanctions governing use of CJIS systems, as well as verbiage to allow the FBI to periodically audit the CSA as well as to allow the FBI to penetration test its own network from the CSA's interfaces to it.

State Compact Officer — The representative of a state that is party to the National Crime Prevention and Privacy Compact, and is the chief administrator of the state's criminal history

record repository or a designee of the chief administrator who is a regular full-time employee of the repository.

State Identification Bureau (SIB) — The state agency with the responsibility for the state's fingerprint identification services.

State Identification Bureau (SIB) Chief — The SIB Chief is the designated manager of state's SIB. If both state fingerprint identification services and CJIS systems control are managed within the same state agency, the SIB Chief and CSO may be the same person.

State of Residency — A state of residency is the state in which an individual claims and can provide documented evidence as proof of being his/her permanent living domicile. CJIS Systems Officers have the latitude to determine what documentation constitutes acceptable proof of residency.

Sur Name — The last name or family name of the user.

Sworn Law Enforcement Officer Indicator — True if the user is a sworn law enforcement officer (SLEO), false otherwise. Usage information: An IDP may assert that a user is a SLEO if all of the following conditions are true.

1. The user is a full-time employee of a state-recognized law enforcement agency.
2. The user is authorized (has the authority) to make an arrest.
3. The user is certified by a State Certifying Authority (i.e., Peace Officer Standards and Training (POST)), or equivalent.

Alternatively, an IDP may assert that a user is a SLEO if the user is a full time employee of a state-recognized law enforcement agency, acting on behalf of a SLEO, in performance of the user's assigned duties. Legal values for this attribute are "true", "false", "1", and "0", where "1" indicates true and "0" indicates false.

Symmetric Encryption — A type of encryption where the same key is used to encrypt and decrypt a message. Symmetric encryption is also known as secret key encryption.

System — Refer to connections to the FBI's criminal justice information repositories and the equipment used to establish said connections. In the context of CJI, this usually refers to applications and all interconnecting infrastructure required to use those applications that process CJI.

Tablet Devices — Tablet devices are mobile devices with a limited-feature operating system (e.g., iOS, Android, Windows RT, etc.). Tablets typically consist of a touch screen without a permanently attached keyboard intended for transport via vehicle mount or portfolio-sized carry case but not on the body. This definition does not include pocket/handheld devices (e.g., smartphones) or mobile devices with full-featured operating systems (e.g., laptops).

Telephone Number — The telephone number for a telecommunication device by which the user may be contacted.

Terminal Agency Coordinator (TAC) — Serves as the point-of-contact at the local agency for matters relating to CJIS information access. A TAC administers CJIS systems programs within the local agency and oversees the agency's compliance with CJIS systems policies.

Unique Subject Id — A persistent unique identifier for the subject or user that identifies both the subject or user and their identity provider. The identity provider should be identified by a fully qualified domain name.

User Certificate (user-based) — Digital certificates that are unique and issued to individuals by a CA. Though not always required to do so, these specific certificates are often embedded on smart cards or other external devices as a means of distribution to specified users. This certificate is used when individuals need to prove their identity during the authentication process.

Virtual Escort — Authorized personnel who actively monitor a remote maintenance session on Criminal Justice Information (CJI)-processing systems. The escort must have the ability to end the session at any time deemed necessary to ensure the protection and integrity of CJI at all times.

Virtual Machine (VM) — See Guest Operating System

Virtualization — Refers to a methodology of dividing the resources of a computer (hardware and software) into multiple execution environments, by applying one or more concepts or technologies such as hardware and software partitioning, time-sharing, partial or complete machine simulation or emulation allowing multiple operating systems, or images, to run concurrently on the same hardware.

Voice over Internet Protocol (VoIP) — A set of software, hardware, and standards designed to make it possible to transmit voice over packet switched networks, either an internal Local Area Network, or across the Internet.

Wireless Access Point — A wireless access point is a device that logically connects a wireless client device to an organization's enterprise network which processes unencrypted CJI.

Wireless (Wi-Fi) Hotspot — A wireless (Wi-Fi) hotspot is a zone or area within a fixed location allowing wireless connectivity to the Internet typically through a wired connection. Hotspots are typically available in public areas such as airports, hotels and restaurants.

APPENDIX B: ACRONYMS

Acronym	Term
AA	Advanced Authentication
AC	Agency Coordinator
ACL	Access Control List
AES	Advanced Encryption Standard
AP	Access Point
APB	Advisory Policy Board
BD-ADDR	Bluetooth-Enabled Wireless Devices and Addresses
BYOD	Bring Your Own Device
CAD	Computer-Assisted Dispatch
CAU	CJIS Audit Unit
CFR	Code of Federal Regulations
CGA	Contracting Government Agency
CHRI	Criminal History Record Information
CISA	Cybersecurity & Infrastructure Security Agency
CJA	Criminal Justice Agency
CJI	Criminal Justice Information
CJIS	Criminal Justice Information Services
ConOps	Concept of Operations
CSA	CJIS Systems Agency
CSIRC	Computer Security Incident Response Capability
CSO	CJIS Systems Officer
CSP	Credential Service Provider

CUI	Controlled Unclassified Information
DAA	Designated Approving Authority
DoJ	Department of Justice
DoJCERT	DoJ Computer Emergency Response Team
FBI	Federal Bureau of Investigation
FIPS	Federal Information Processing Standards
FISMA	Federal Information Security Management Act
FOIA	Freedom of Information Act
FOUO	For Official Use Only
HIDS	Host-based Intrusion Detection System
HIPS	Host-based Intrusion Prevention System
HTTP	Hypertext Transfer Protocol
IAFIS	Integrated Automated Fingerprint Identification System
IDS	Intrusion Detection System
III	Interstate Identification Index
IP	Internet Protocol
IPS	Intrusion Prevention System
IPSEC	Internet Protocol Security
ISA	Interconnection Security Agreement
ISO	Information Security Officer
IT	Information Technology
LASO	Local Agency Security Officer
LEEP	Law Enforcement Enterprise Portal
LMR	Land Mobile Radio

MAC	Media Access Control
MCA	Management Control Agreement
MDM	Mobile Device Management
MITM	Man-in-the-Middle
MOU	Memorandum of Understanding
MS-ISAC	Multi-State Information Sharing & Analysis Center
NARA	National Archives and Records Administration
NCIC	National Crime Information Center
NCJA	Noncriminal Justice Agency
NICS	National Instant Criminal Background Check System
NIDS	Network-based Intrusion Detection System
NIPS	Network-based Intrusion Prevention System
NIST	National Institute of Standards and Technology
OMB	Office of Management and Budget
ORI	Originating Agency Identifier
OTP	One-time Password
PBX	Private Branch Exchange
PCSC	Preventing and Combating Serious Crime
PDA	Personal Digital Assistant
PII	Personally Identifiable Information
PIN	Personal Identification Number
PKI	Public Key Infrastructure
POC	Point-of-Contact
PSTN	Public Switched Telephone Network

QA	Quality Assurance
QoS	Quality of Service
RCMP	Royal Canadian Mounted Police
RF	Radio Frequency
RP	Relying Party
SA	Security Addendum
SCO	State Compact Officer
SIB	State Identification Bureau
SIG	Special Interest Group
SP	Special Publication
SPRC	Security Policy Resource Center
SSID	Service Set Identifier
TAC	Terminal Agency Coordinator
TSC	Threat Screening Center
TLS	Transport Layer Security
UCN	Universal Control Number
USCERT	U.S. Computer Emergency Readiness Team
VLAN	Virtual Local Area Network
VM	Virtual Machine
VoIP	Voice Over Internet Protocol
VPN	Virtual Private Network
WEP	Wired Equivalent Privacy
WLAN	Wireless Local Area Network
WPA	Wi-Fi Protected Access

APPENDIX C: NETWORK TOPOLOGY DIAGRAMS

Network diagrams, i.e., topological drawings, are an essential part of solid network security. Through graphical illustration, a comprehensive network diagram provides the “big picture” – enabling network managers to quickly ascertain the interconnecting nodes of a network for a multitude of purposes, including troubleshooting and optimization. Network diagrams are integral to demonstrating the manner in which each agency ensures criminal justice data is afforded appropriate technical security protections and is protected during transit and at rest.

The following diagrams, labeled Appendix C.1-A through C.1-D, are examples for agencies to utilize during the development, maintenance, and update stages of their own network diagrams. By using these example drawings as a guideline, agencies can form the foundation for ensuring compliance with CM-2 Baseline Configuration of the CJIS Security Policy.

The purpose for including the following diagrams in this Policy is to aid agencies in their understanding of diagram expectations and should not be construed as a mandated method for network topologies. It should also be noted that agencies are not required to use the identical icons depicted in the example diagrams and should not construe any depiction of a particular vendor product as an endorsement of that product by the FBI CJIS Division.

Appendix C.1-A is a conceptual overview of the various types of agencies that can be involved in handling of CJI, and illustrates several ways in which these interconnections might occur. This diagram is not intended to demonstrate the level of detail required for any given agency’s documentation, but it provides the reader with some additional context through which to digest the following diagrams. Take particular note of the types of network interfaces in use between agencies, in some cases dedicated circuits with encryption mechanisms, and in other cases VPNs over the Internet. This diagram attempts to show the level of diversity possible within the law enforcement community. These diagrams in no way constitute a standard for network engineering, but rather, for the expected quality of documentation.

The next three topology diagrams, C.1-B through C.1-D, depict conceptual agencies. For C.1-B through C.1-D, the details identifying specific “moving parts” in the diagrams by manufacturer and model are omitted, but it is expected that any agencies producing such documentation will provide diagrams with full manufacturer and model detail for each element of the diagram. Note that the quantities of clients should be documented in order to assist the auditor in understanding the scale of assets and information being protected.

Appendix C.1-B depicts a conceptual state law enforcement agency’s network topology and demonstrates a number of common technologies that are in use throughout the law enforcement community (some of which are compulsory per CJIS policy, and some of which are optional) including Mobile Broadband cards, VPNs, Firewalls, Intrusion Detection Devices, VLANs, and so forth. Note that although most state agencies will likely have highly-available configurations, the example diagram shown omits these complexities and only shows the “major moving parts” for clarity but please note the Policy requires the logical location of all components be shown. The level of detail depicted should provide the reader with a pattern to model future documentation from, but should not be taken as network engineering guidance.

Appendix C.1-C depicts a conceptual county law enforcement agency. A number of common technologies are presented merely to reflect the diversity in the community, including proprietary

Packet-over-RF infrastructures and advanced authentication techniques, and to demonstrate the fact that agencies can act as proxies for other agencies.

Appendix C.1-D depicts a conceptual municipal law enforcement agency, presumably a small one that lacks any precinct-to-patrol data communications. This represents one of the smallest designs that could be assembled that, assuming all other details are properly considered, would meet the criteria for CM-2 Baseline Configuration. This diagram helps to demonstrate the diversity in size that agencies handling criminal justice data exhibit.

Figure C-1-A Overview: Conceptual Connections Between Various Agencies

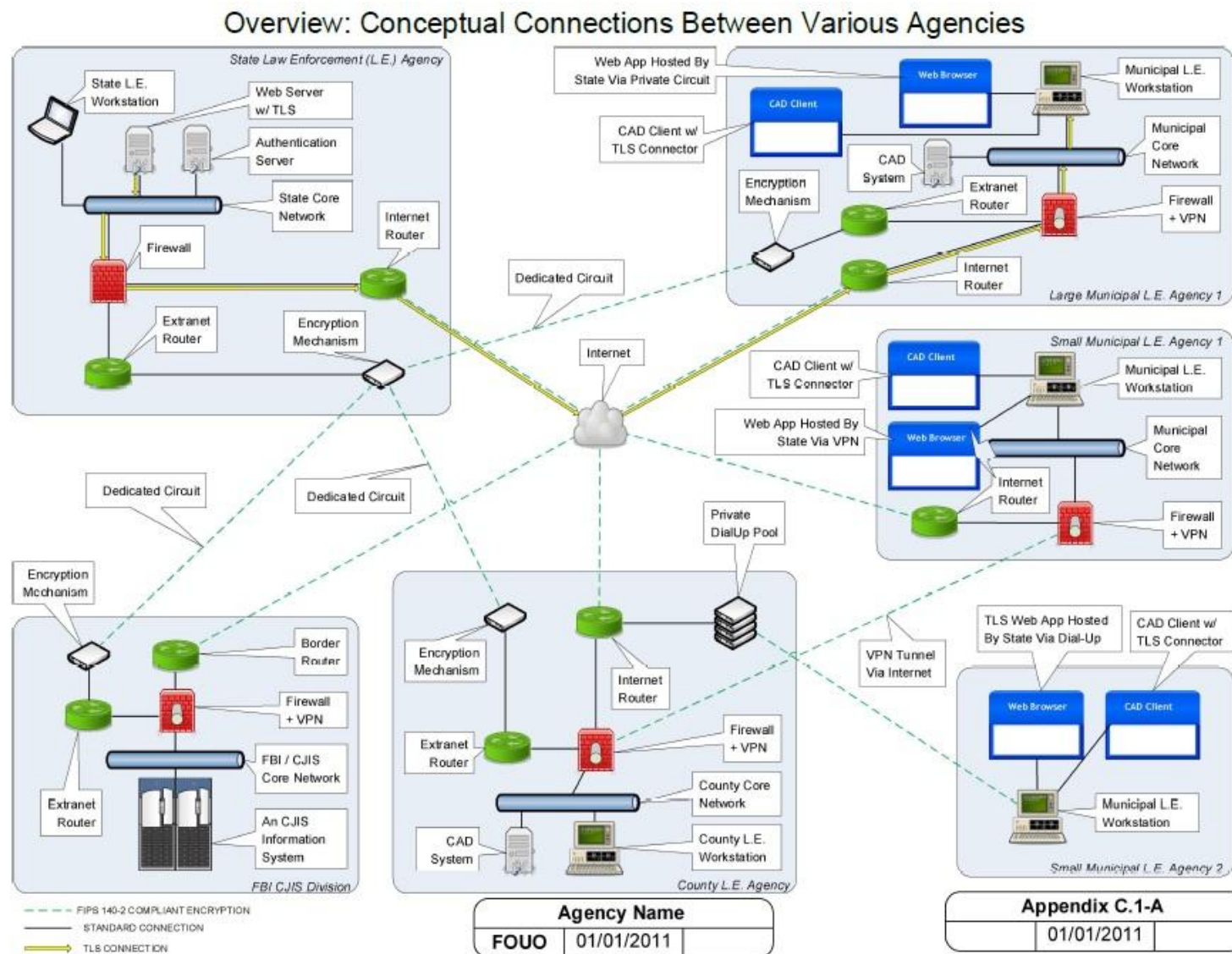


Figure C-1-B Conceptual Topology Diagram for a State Law Enforcement Agency

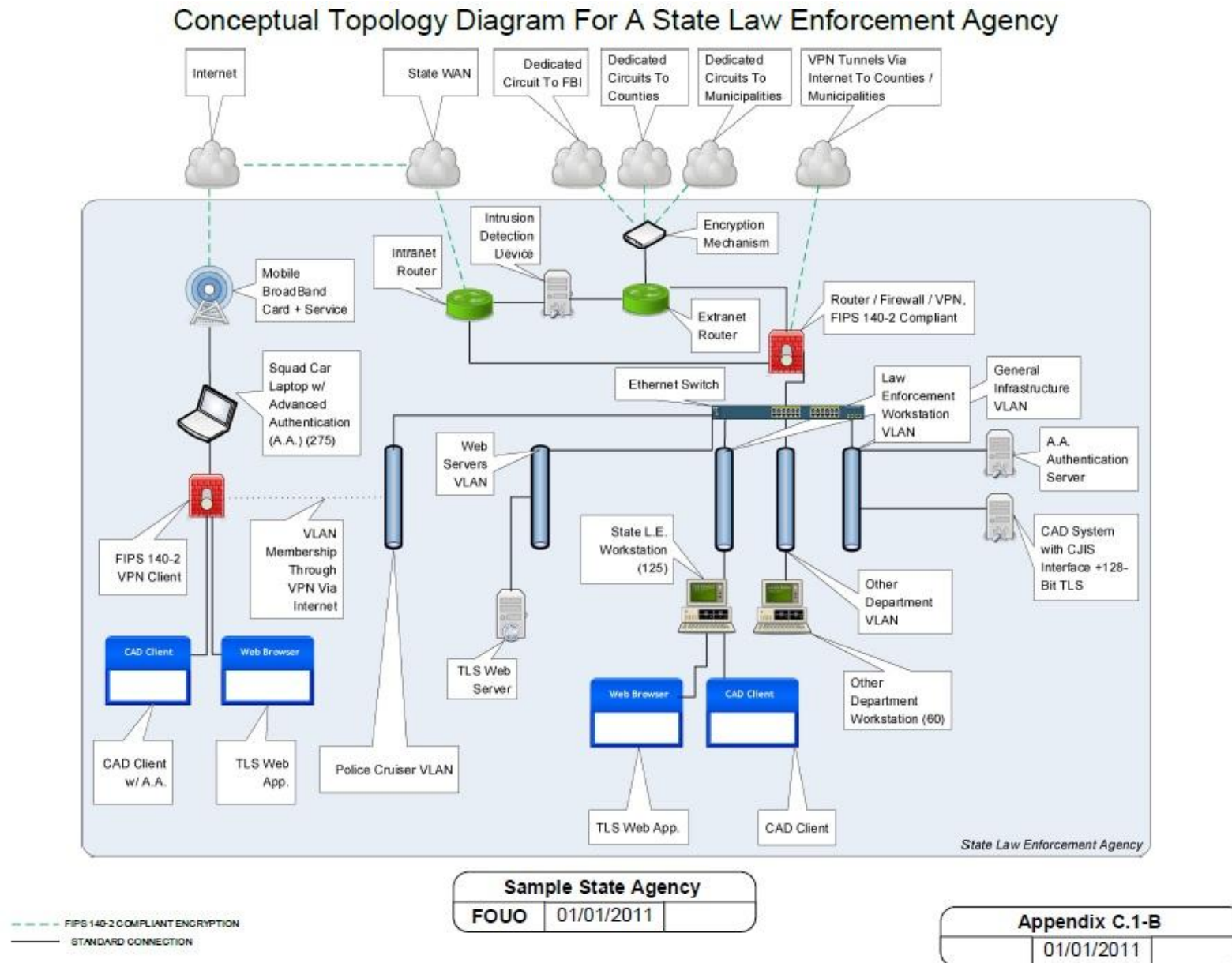


Figure C-1-C Conceptual Topology Diagram for a County Law Enforcement Agency

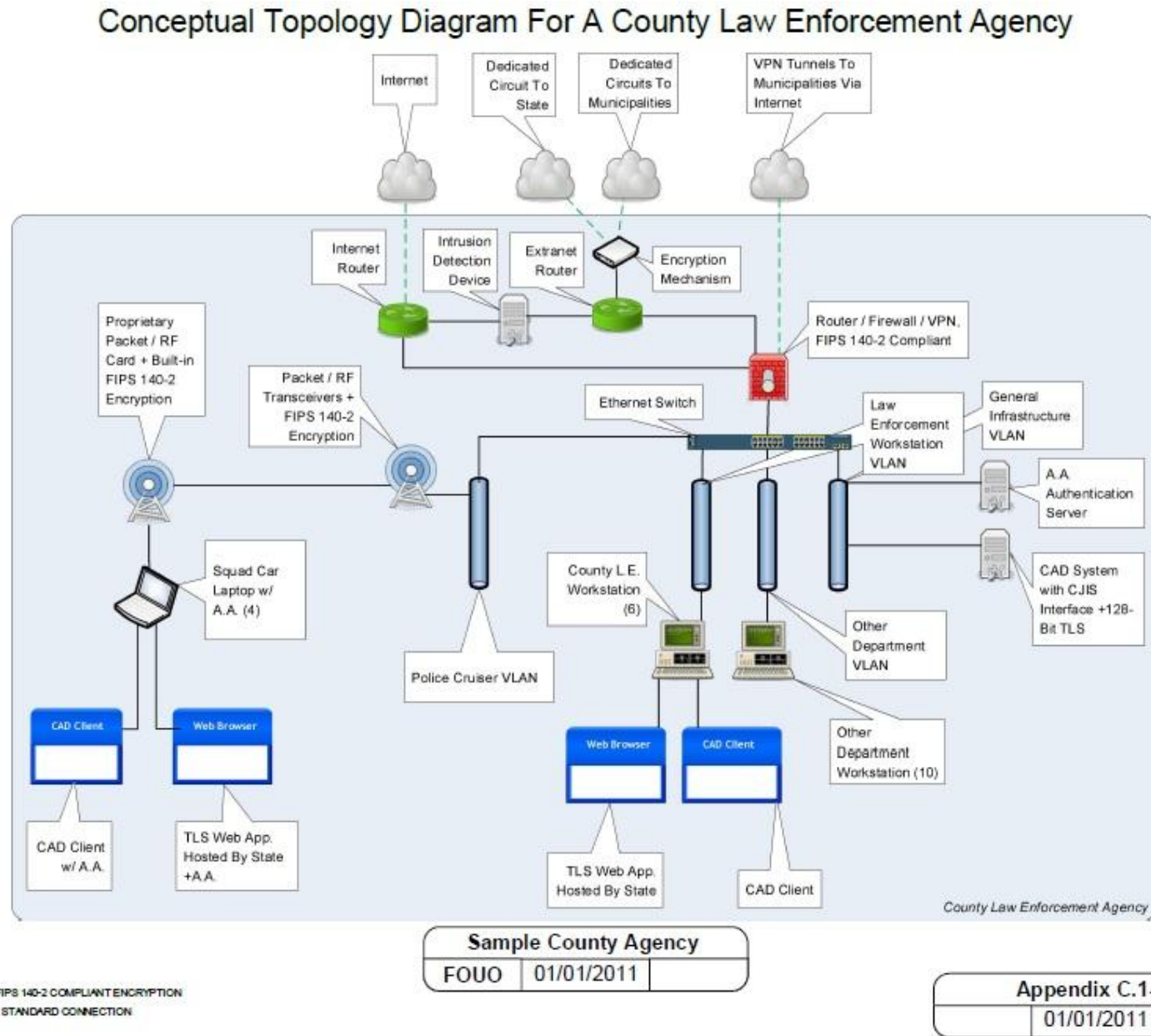
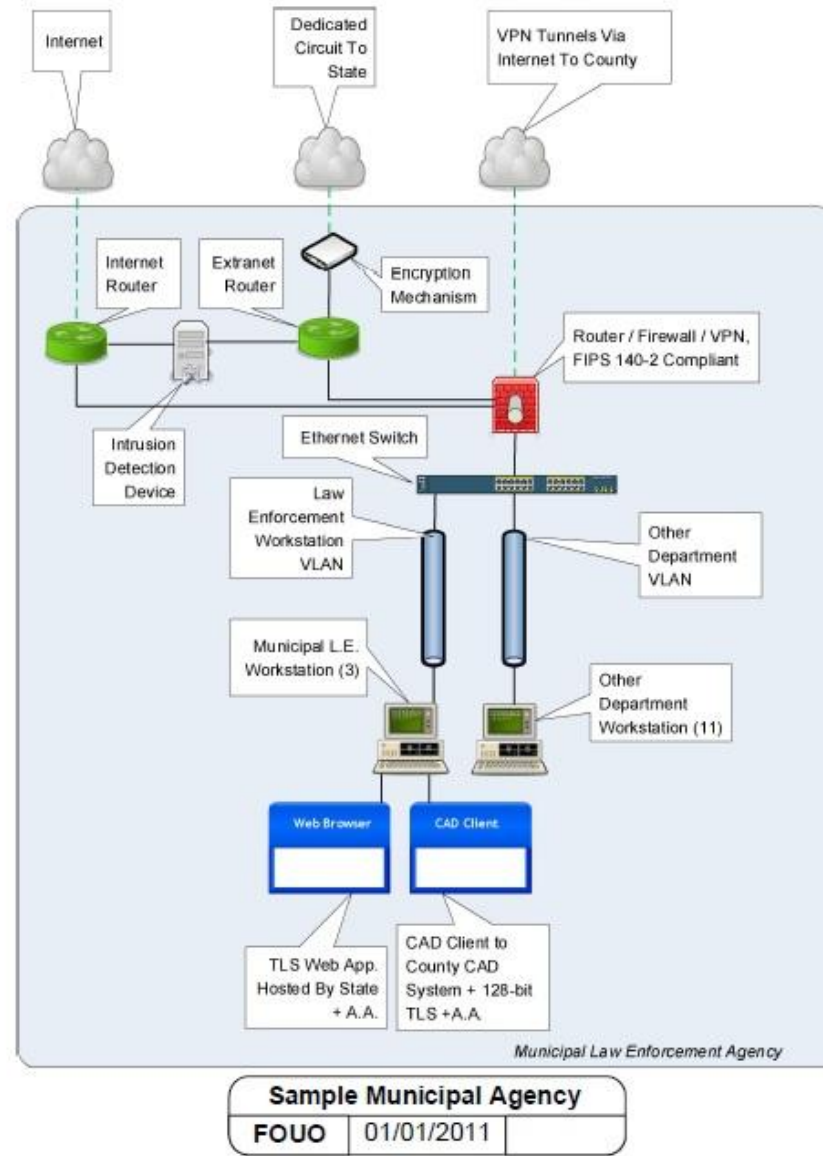


Figure C-1-D Conceptual Topology Diagram for a Municipal Law Enforcement Agency

Conceptual Topology Diagram For A Municipal Law Enforcement Agency



APPENDIX D: SAMPLE INFORMATION EXCHANGE AGREEMENTS

This appendix contains sample information exchange agreements.

D.1 CJIS User Agreement

CRIMINAL JUSTICE INFORMATION SERVICES (CJIS) DIVISION'S SYSTEM AND PROGRAM USER AGREEMENT

The FBI's CJIS Division provides state-of-the-art systems, programs, and services to the federal, state, local, tribal, regional, territorial, and foreign criminal and noncriminal justice communities. Through a shared management concept, the CJIS Division's systems, programs, and services are administered and maintained by the FBI's CJIS Division and managed in cooperation with the CJIS Advisory Policy Board (APB) and the Council established by the National Crime Prevention and Privacy Compact Act of 1998 (Compact). The CJIS Division's System and Program User Agreement (hereafter referenced as the User Agreement) covers the following: Law Enforcement Enterprise Portal (LEEP); National Crime Information Center (NCIC); National Instant Criminal Background Check System (NICS); National Data Exchange (N-DEx) System; Next Generation Identification (NGI) System; and the Uniform Crime Reporting (UCR) Program.

This User Agreement is intended to identify the categories of users, as defined by the *CJIS Security Policy (CJISSECPOL)*, and outlines the roles and responsibilities associated with participating in the CJIS Division's systems, programs, and services available to the above-listed criminal and noncriminal justice communities. This User Agreement identifies additional documentation as a valuable resource available to those individuals responsible for reviewing and signing the User Agreement.

PART 1-USER DEFINITIONS

A **CJIS Systems Agency (CSA)** is a duly authorized state, federal, international, tribal, or territorial criminal justice agency on the CJIS network providing statewide (or equivalent) service to its criminal justice users with respect to the criminal justice information (CJI) from various systems managed by the FBI's CJIS Division. There shall be only one CSA per state or territory. In federal agencies, the CSA may be the interface or switch to other federal agencies connecting to FBI CJIS Division's systems.

A **CJIS Systems Officer (CSO)** is an individual located within the CSA responsible for the administration of the CJIS network for the CSA.

An **Interface Agency (IA)** is a criminal justice agency (including law enforcement) or any federally authorized agency/entity, other than a CSA or State Identification Bureau (SIB), that leverages a direct connection to the FBI CJIS Division.

An **IA official** is an employee of the IA responsible for planning necessary hardware, software, funding, and training for the IA's authorized access to the CJIS Division's systems. The IA official shall not be a contract employee.

A **SIB** is the state (to include United States (U.S.) territories) agency with the responsibility for the state's fingerprint identification services.

An **SIB Chief** is the designated manager of the state's SIB. If both state fingerprint identification services and CJIS systems control are managed within the same state agency, the SIB Chief and CSO may be the same person.

A **state Compact officer** is the representative of a state that is party to the Compact and is the chief administrator of the state's criminal history record repository or a designee of the chief administrator who is a regular full-time employee of the repository.

The **FBI Compact officer** is appointed by the Director of the FBI and is responsible for administering the Compact within the Department of Justice and other federal agencies.

PART 2-CJIS DIVISION'S SYSTEMS, SERVICES, PROGRAMS AND RESPONSIBILITIES

The FBI's CJIS Division makes available access to the following systems and major programs (Note: Place a checkmark in the bulleted box to indicate use of specific service(s) by the agency reviewing and signing this User Agreement (applicable documents identified in Part 3)):

- ☐ LEEP
- ☐ NCIC
- ☐ NICS
- ☐ N-DEx
- ☐ NGI
- ☐ UCR

The FBI CJIS Division supports a multitude of services within the above-listed systems and programs (e.g., Interstate Identification Index (III) within NGI, NCIC offline search, UCR's Law Enforcement Suicide Data Collection, and LEEP's virtual command centers).

The FBI's CJIS Division provides the following to its authorized users, as applicable:

1. Operational, technical, and investigative assistance.
2. Secure telecommunication methods, such as wide area network (WAN) and virtual private network (VPN) connections.
3. Timely information on the CJIS Division's systems, programs, and services, by means of procedures and operating manuals, policy and implementation guides, code manuals, technical and operational updates, web pages, presentations, brochures, various newsletters, CJIS Information Letters, frequently asked questions, and other relevant documents.

4. On-site and virtual training, assistance, and system and program reference materials. The FBI's CJIS Division may prioritize on-site training opportunities to CSAs, SIBs, or IAs with open compliance findings, FBI priority initiatives, or other time-sensitive matters impacting the criminal and noncriminal justice communities.
5. Ongoing assistance through meetings and briefings to discuss operational and policy issues.
6. The CJIS Advisory Process through which authorized users have input as to the policies and procedures governing the operation of CJIS systems, programs, and services.
7. The Compact Office through education and assistance concerning the noncriminal justice use of the III of the NGI System, including facilitating engagement with the Compact Council, as needed.
8. Formal audits to ensure compliance with applicable statutes, regulations, and policies.

PART 3-CJIS DIVISION'S SYSTEMS AND PROGRAM DOCUMENTS AND AUTHORITIES

The following documents and authorities are incorporated by reference and made part of this User Agreement regarding agency access and use; this list is not all inclusive. Other policies, regulations, and manuals will be developed, published, and/or amended as CJIS Division systems, programs, and services evolve. Please contact the appropriate program office and visit the FBI CJIS Division's website for additional information: www.fbi.gov/cjis.

Security and Information Exchange Documents

CJISSECPOL

The CJISSECPOL provides criminal justice and noncriminal justice agencies with the minimum set of security requirements for access to FBI CJIS Division systems and information and to protect and safeguard CJI.

Information Exchange Package Documentation (IEPD)

An IEPD is a specification for a data exchange and defines a particular data exchange for a system.

Enterprise Documents

CJIS Information Letters

These letters provide users of CJIS Division's systems and programs with policy and procedural updates.

CSO Reference Guide

The purpose of this guide is to provide an overview of the roles and responsibilities of the CSO position.

LEEP Documents

LEEP Procedures and Operations Manual

This manual addresses the roles, responsibilities, and commitments between Identity Providers, Service Providers, and the FBI.

NCIC Documents

NCIC Operating Manual

This manual provides guidelines on the proper operation and use of the NCIC System, including the standards, procedures, formats, rules, and criteria.

NCIC Technical and Operational Updates (TOUs)

TOUs are documents that describe technical and operational changes to the NCIC System.

NICS Documents

NICS Policy Reference Guide

This manual serves as a training guide for NICS Point-of-Contact states who are processing name-based background checks through NICS before transferring a firearm to a prospective transferee.

Disposition of Firearms (DoF) User Guide

This manual serves as a training guide for law enforcement and criminal justice agencies disposing of (returning) firearms.

NICS Resource Entry and Maintaining Entries into the NICS Indices User Guide

This manual serves as a guide to sufficient source documentation to support a NICS Indices entry upon appeal or audit.

NICS Indices Reference Guide for Contributors

This manual serves as a guide for entering information into the NICS Indices.

N-DEx Documents

N-DEx Policy and Operating Manual

This manual applies to all entities with access to, or who operate in support of, N-DEx System, services, and information.

NGI Documents

III/National Fingerprint File (NFF) Operational and Technical Manual

This manual describes the III and NFF programs, including technical and operational details; procedures for entering, maintaining, and updating III and NFF records; and requirements for criminal and noncriminal fingerprint submissions.

III TOUs

TOUs are documents that describe technical and operational changes to the III System.

Electronic Biometric Transmission Specification (EBTS)

The purpose of the EBTS is to provide standards for electronically encoding and transmitting biometric images, identification, and arrest data which extends the American National Standards Institute/National Institute of Standards and Technology and Information Technology Laboratory standard. These standards ensure data is formatted for acceptance into NGI.

Compact Council's Security and Management Control Outsourcing Standards

These documents provide noncriminal justice agencies with information on the required procedures, responsibilities, and controls to maintain adequate security and integrity of criminal history record information (CHRI) while under the control or management of an outsourced, third-party contractor.

NGI Interstate Photo System (IPS) Policy and Reference Guide

This guide is made available to authorized law enforcement users and describes the policy and technical requirements for authorized use of the NGI IPS, as well as the best practices for NGI IPS users. It describes the types of NGI IPS enrollment and search transactions accepted, responses returned, training required, and additional resources available for NGI IPS users.

NGI Rap Back Service Criminal Justice Policy and Implementation Guide

This guide informs the SIBs and federal agencies of the policy and technical requirements for participation in the criminal justice Rap Back Service.

NGI Rap Back Service Noncriminal Justice Policy and Implementation Guide

This guide provides policy, technical, legal, and privacy requirements to the SIBs, federal submitting agencies, and authorized contractors who choose to participate in the FBI's NGI Noncriminal Justice Rap Back Service.

NGI Rap Back Service Noncriminal Justice Outsourcing Policy and Implementation Guide

This guide provides direction on the required policies and operations for FBI-approved channelers and nonchannelers who assist an authorized recipient with their noncriminal justice participation in the FBI's NGI Rap Back Service.

NGI Repository for Individuals of Special Concern (RISC) Policy and Implementation Guide

This guide is for the use of CSAs, vendors, and authorized criminal law enforcement agency interested in the mobile fingerprint identification search of the NGI RISC. It defines the types of search transactions that are accepted, the types of responses returned to contributors, and the system description.

NGI Iris Service Policy and Implementation Guide

This guide includes the policy and technical requirements for authorized criminal justice agency users to utilize the NGI Iris Service. It defines the types of NGI Iris Service enrollments and search transactions that are accepted, the responses returned, and references to additional technical resources.

UCR Documents

NIBRS User Manual

This manual is used to assist law enforcement agencies in reporting their crime statistics via NIBRS. This manual addresses NIBRS policies, including but not limited to, the types of offenses reported via NIBRS and guidelines for an agency to become certified to submit NIBRS data to the FBI.

NIBRS Technical Specification

This manual provides information necessary to create proper UCR NIBRS flat file submissions and is to be used in conjunction with the *NIBRS User Manual*.

Hate Crime Data Collection Guidelines and Training Manual

This manual is intended to assist law enforcement agencies in establishing an updated hate crime training program to allow their personnel to collect and submit hate crime data to the FBI's UCR Program.

Cargo Theft User Manual

This manual identifies policy and provides information regarding the types of offenses that constitute a cargo theft incident, how to identify a cargo theft, and guidelines for reporting cargo theft.

National Use-of-Force Data Collection Flat File and Bulk Load Technical Specification

This manual provides the record layout for report segments and related data elements. The bulk load feature enables report submissions outside of the web application interface.

Authorities

Title 28, United States Code (U.S.C.), Section 534

Among other things, 28 U.S.C. § 534 authorizes the United States Attorney General to collect, preserve, and exchange identification, criminal identification, crime, and other records.

Title 28, Code of Federal Regulations (C.F.R.), Part 20 and the Appendix

The purpose of these regulations is to ensure that CHRI, wherever it appears, is collected, stored, and disseminated in a manner to ensure the accuracy, completeness, currency, integrity, and security of such information and to protect individual privacy. These regulations apply to authorized federal, state, local, tribal, foreign, and international criminal justice agencies to the extent that they access the various criminal justice information systems managed by the CJIS Division or use or disseminate the records and information residing in those systems.

28 C.F.R., Part 25

The purpose of this subpart is to establish policy and procedures implementing the Brady Handgun Violence Prevention Act.

28 C.F.R. § 50.12

This section is intended to ensure that all relevant criminal record information is made available to provide for the public safety and further to protect the interest of the prospective employee/licensee who may be affected by the information or lack of information in an identification record.

Title 34, U.S.C. § 40311-40316

The Compact Act provides the legal framework for the noncriminal justice use of the III system and to facilitate complete decentralization of criminal history records. Established pursuant to the Compact, the Compact Council is to promulgate rules and procedures governing the use of the III system for noncriminal justice purposes.

PART 4- RESPONSIBILITIES OF THE CSA/IA/SIB

The purpose of a designated CSA/IA/SIB (also referenced as Signatory) is to unify responsibility for the administration and oversight of the users of the CJIS Division's systems, programs, and services, and ensure adherence to established procedures and policies within each Signatory federal, state, local, tribal, regional, territorial, and foreign agency and by each user. The CSA/IA/SIB may impose more stringent protection measures than outlined in this User Agreement. Such decisions shall be documented and kept current.

In addition, and as applicable, the CSO/IA official acknowledges state and FBI Compact officer responsibilities to ensure that Compact statutory provisions and rules, procedures, and standards established by the Compact Council are complied with in their respective area of responsibility.

Furthermore, this User Agreement applies to SIBs who have access to the CJIS Division's systems separate and apart from their state's CSA. In addition, and as applicable, the SIB Chief acknowledges the state Compact officers' responsibilities to ensure that Compact statutory provisions and rules, procedures, and standards are complied with in their respective area of responsibility.

To ensure continued access as set forth above, the Signatory agrees to adhere to all applicable policies of the CJIS Division including, but not limited to, the following:

1. As applicable, the Signatory agencies will provide fingerprints that meet submission criteria for all qualifying arrests. In addition, federal, state, local, tribal, regional, territorial, and foreign agencies will make their records available for interstate exchange for criminal justice and other authorized purposes unless restricted by federal, state, local, tribal, regional, territorial, and foreign laws, and, where applicable, continue to move toward participation in the III and, upon ratification of the Compact, the NFF. [Article I(4) of the National Crime Prevention and Privacy Compact Act (Title 34, United States Code, section 40316) defines criminal history records as (A) . . . information collected by criminal justice agencies on individuals consisting of identifiable descriptions and notations of arrests, detentions, indictments, or other formal criminal charges, and any disposition arising therefrom, including acquittal, sentencing, correctional supervision, or release; and (B) does not include identification information such as fingerprint records if such information does not indicate involvement of the individual with the criminal justice system.]
2. Appropriate and reasonable quality assurance procedures, e.g., hit confirmation, audits for record timeliness, and validation for all CJIS Division systems, programs, and services in which they participate, e.g., biometrics, must be in place to ensure that only complete, accurate, and valid information is maintained in the CJIS Division's systems.
3. Biannual file synchronization of information entered into the III by participating agencies.
4. Each Signatory is responsible for the minimum-security measures as defined by the *CJISSECPOL*.
5. Each CSA/IA/SIB is responsible for completing a triennial audit of all agencies with access to the CJIS Division's systems through the system(s) of the CSA/IA/SIB. In addition, each authorized CSA/IA/SIB user is subject to a triennial audit by the CJIS Division.
6. Each CSA/IA/SIB shall be responsible for training requirements, including compliance with operator training mandates of the CJIS Division's systems, programs, and services, as outlined by the above-referenced systems and program documents.
7. Each CSA/IA/SIB shall be responsible for maintaining the integrity of their information systems in accordance with established policies to ensure only authorized terminal access; only authorized transaction submission; and proper protection, handling, and dissemination of CJI. Each CSA/IA/SIB shall be responsible for security incident reporting as required by the *CJISSECPOL*.

8. In the event that any state/local/tribal agency for which the designated CSA/IA/SIB is responsible receives a request pursuant to a freedom of information law, the civil or criminal discovery process, or other judicial, legislative, or administrative process, to disclose FBI information concerning the subject matter of this agreement, the designated CSA/IA/SIB will ensure that the state/local/tribal agency immediately notifies the FBI of any such request in order to allow the FBI sufficient time to seek to protect its equities through appropriate channels, if necessary. For purposes of this paragraph, FBI information includes, but is not limited to, e-mails or other correspondence with the FBI and the information referenced in Part 2, paragraphs 3 through 5 of this agreement. This provision survives the termination of this agreement.

PART 5-GENERAL PROVISIONS

Funding:

Unless otherwise agreed in writing, each signatory to this User Agreement (hereafter referenced as party) shall bear its own costs in relation to this User Agreement. Expenditures will be subject to budgetary processes and availability of funds pursuant to applicable laws and regulations. The parties expressly acknowledge that this User Agreement in no way implies that Congress will appropriate funds for such expenditures.

Termination and Other Topics

1. All activities of the parties under this User Agreement will be carried out in accordance with the above-described provisions.
2. This User Agreement may be amended or terminated by the mutual written consent of the parties' authorized representatives.
3. The parties acknowledge that this User Agreement does not alter applicable law governing any claim for civil liability arising out of any activity conducted pursuant to this User Agreement or otherwise relating to this User Agreement.
4. This User Agreement is not intended, and should not be construed, to create any right or benefit, substantive or procedural, enforceable at law or otherwise by any third party against the parties, the United States, or the officers, employees, agents, or other associated personnel thereof. No assignment of rights, duties, or obligations of this User Agreement shall be made by any party without the express written approval of a duly authorized representative of all other parties.
5. Each party agrees to notify the other party in the event of receipt of a civil claim arising from this User Agreement. The parties agree to cooperate fully with one another in the event of any investigation arising from alleged negligence or misconduct arising

from acts or omissions related to this User Agreement. Nothing in this paragraph prevents any party from conducting an independent administrative review of any incident giving rise to a claim.

6. Either party may terminate this User Agreement upon 30-days, written notification to the other party. Such notice will be the subject of immediate consultation by the parties to decide upon the appropriate course of action. In the event of such termination, the following rules apply:

- a. The parties will continue participation, financial or otherwise, up to the effective date of termination.
- b. Each party will pay the costs it incurs as a result of termination.
- c. All information and rights therein received under the provisions of this User Agreement, prior to the termination, will be retained by the parties subject to the provisions of this User Agreement.

PART 6-ACKNOWLEDGMENT AND CERTIFICATION

I hereby certify that I am a U.S. Citizen and acknowledge the duties and responsibilities as set out in this User Agreement. I acknowledge these duties and responsibilities have been developed and approved by the users of the CJIS Division's systems to ensure the reliability, confidentiality, completeness, and accuracy of all information contained in, or obtained by means of, the CJIS Division's systems. I further acknowledge that failure to comply with these duties and responsibilities may result in the imposition of non-monetary sanctions against the offending signatory agency. The APB or the Compact Council may approve sanctions to include the termination of CJIS services.

I hereby certify that I am familiar with all applicable documents that are made part of this User Agreement and all applicable federal and state laws and regulations relevant to the receipt and dissemination of information provided through the CJIS Division's systems, programs, and services.

Should the designated official vacate their position, the FBI designated federal officer must be notified via E-mail at AGMU@leo.gov. The name and telephone number of the acting CSO, IA official, or SIB chief and when known, the name and telephone number of the new CSO, IA official, or SIB chief must be provided, and a new User Agreement must be executed by the appropriate designated official. In addition, if the use and participation in the acknowledged systems and/or programs in Part 2 of this User Agreement change, a new User Agreement must be executed. In addition, this User Agreement will be reviewed every three years by the FBI CJIS Division.

This User Agreement is a formal expression of the purpose and intent of the parties and is effective when signed. It may be amended by the deletion or modification of any provision contained therein, or by the addition of new provisions, after written concurrence of the parties. The "Acknowledgment and Certification" is being executed by the appropriate authorized official in a representative capacity. Accordingly, this User Agreement will remain in effect after the authorized official vacates his/her position or until it is affirmatively amended or rescinded in writing. This User Agreement does not confer, grant, or authorize any rights, privileges, or obligations to any third party.

PART 7-SIGNATORY AUTHORITY

You are signing this CJIS Division's System and Program User Agreement as one of the following officials: CSO, IA Official, or SIB Chief.

CSO and CSA SIGNATORIES

CJIS Systems Officer
(Name)
(Title)
(Agency Name)
Date: _____

CONCURRENCE OF CSA HEAD:

CSA Head
Date: _____

INTERFACE AGENCY SIGNATORY

Interface Agency Official
(Name)
(Title)
(Agency Name)
Date: _____

CONCURRENCE OF INTERFACE AGENCY HEAD:

Interface Agency Head
Date: _____

STATE IDENTIFICATION BUREAU SIGNATORY

State Identification Bureau Chief
(Name)
(Title)
(Agency Name)

Date: _____

CONCURRENCE OF STATE IDENTIFICATION BUREAU AGENCY HEAD:

State Identification Bureau Agency Head

Date: _____

FBI CJIS DIVISION SIGNATORY

Name of Assistant Director
Assistant Director
FBI CJIS Division

Date: _____

Revised: 08/25/2023

D.2 Management Control Agreement

Management Control Agreement

Pursuant to the CJIS Security Policy, it is agreed that with respect to administration of that portion of computer systems and network infrastructure interfacing directly or indirectly with the state network (Network Name) for the interstate exchange of criminal history/criminal justice information, the (Criminal Justice Agency) shall have the authority, via managed control, to set, maintain, and enforce:

- (1) Priorities.
- (2) Standards for the selection, supervision, and termination of personnel access to Criminal Justice Information (CJI).
- (3) Policy governing operation of justice systems, computers, access devices, circuits, hubs, routers, firewalls, and any other components, including encryption, that comprise and support a telecommunications network and related criminal justice systems to include but not limited to criminal history record/criminal justice information, insofar as the equipment is used to process or transmit criminal justice systems information guaranteeing the priority, integrity, and availability of service needed by the criminal justice community.
- (4) Restriction of unauthorized personnel from access or use of equipment accessing the State network.
- (5) Compliance with all rules and regulations of the (Criminal Justice Agency) Policies and CJIS Security Policy in the operation of all information received.

“...management control of the criminal justice function remains solely with the Criminal Justice Agency.” Section 5.1.1.4

This agreement covers the overall supervision of all (Criminal Justice Agency) systems, applications, equipment, systems design, programming, and operational procedures associated with the development, implementation, and maintenance of any (Criminal Justice Agency) system to include NCIC Programs that may be subsequently designed and/or implemented within the (Criminal Justice Agency).

John Smith, CIO
Any State Department of Administration

Date

Joan Brown, CIO
(Criminal Justice Agency)

Date

D.3 Noncriminal Justice Agency Agreement & Memorandum of Understanding

MEMORANDUM OF UNDERSTANDING

BETWEEN

THE FEDERAL BUREAU OF INVESTIGATION

AND

(Insert Name of Requesting Organization)

FOR

THE ESTABLISHMENT AND ACCOMMODATION OF
THIRD-PARTY CONNECTIVITY TO THE
CRIMINAL JUSTICE INFORMATION SERVICES DIVISION'S WIDE AREA NETWORK

1. **PURPOSE:** This Memorandum of Understanding (MOU) between the Federal Bureau of Investigation (FBI) and **(insert requesting organization's name)**, hereinafter referred to as the "parties," memorializes each party's responsibilities with regard to establishing connectivity to records services accessible via the Wide Area Network (WAN) of the FBI's Criminal Justice Information Services (CJIS) Division.

2. **BACKGROUND:** The requesting organization, **(insert requesting organization's name)**, being approved for access to systems of records accessible via the CJIS WAN, desires connectivity to the CJIS WAN or via a secure Virtual Private Network (VPN) Connection (Internet) to the CJIS WAN. The CJIS Division has created a framework for accommodating such requests based on the type of connection.

In preparing for such non-CJIS-funded connectivity to the CJIS WAN, the parties plan to acquire, configure, and place needed communications equipment at suitable sites and to make electronic connections to the appropriate systems of records via the CJIS WAN.

To ensure that there is a clear understanding between the parties regarding their respective roles in this process, this MOU memorializes each party's responsibilities regarding the development, operation, and maintenance of third-party connectivity to the CJIS WAN. Unless otherwise contained in an associated contract, the enclosed terms apply. If there is a conflict between terms and provisions contained in both the contract and this MOU, the contract will prevail.

3. **AUTHORITY:** The FBI is entering into this MOU under the authority provided by Title 28, United States Code (U.S.C.), Section 534; 42 U.S.C. § 14616; and/or Title 28, Code of Federal Regulations, Part 906.

4. **SCOPE:**

a. The CJIS Division agrees to:

- i. Provide the requesting organization with a "CJIS WAN Third-Party Connectivity Package" that will detail connectivity requirements and options compatible with the CJIS Division's WAN architecture upon receipt of a signed nondisclosure statement.
- ii. Configure the requesting organization's connection termination equipment suite at Clarksburg, West Virginia, and prepare it for deployment or shipment under the CJIS WAN option. In the Secure VPN arrangement only, the third party will develop, configure, manage, and maintain its network connectivity to its preferred service provider.
- iii. Work with the requesting organization to install the connection termination equipment suite and verify connectivity.
- iv. Perform installation and/or routine maintenance on the requesting organization's third-party dedicated CJIS WAN connection termination equipment after coordinating with the requesting organization's designated point of contact (POC) and during a time when the CJIS Division's technical personnel are near the requesting organization's site.
- v. Perform periodic monitoring and troubleshooting of the requesting organization's CJIS WAN connection termination equipment. Software patches will be maintained on the dedicated CJIS WAN connected network equipment only. Under the Secure VPN option, no availability or data throughput rates will be guaranteed.
- vi. Provide 24 hours a day, 7 days a week uninterrupted monitoring from the CJIS Division's Network Operations Center.
- vii. Provide information regarding potential hardware end-of-life replacement cycles to the requesting organization for its budgeting purposes.
- viii. Maintain third-party dedicated CJIS WAN connection termination equipment as if in the CJIS Division's operational environment.
- ix. Update the appropriate software on the requesting organization's dedicated connection termination equipment connected to the CJIS WAN (i.e., Cisco

- Internetwork Operating System, SafeNet frame relay encryptor firmware, etc.) pursuant to the requesting organization's authorized maintenance contracts.
- x. Provide a POC and telephone number for MOU-related issues.

b. The **(insert requesting organization's name)** agrees to:

- i. Coordinate requests for third-party connectivity to the CJIS WAN or the Secure VPN with the CJIS Division's POC.
- ii. Purchase hardware and software that are compatible with the CJIS WAN.
- iii. Pay for the telecommunications infrastructure that supports its connection to the CJIS WAN or Secure VPN.
- iv. Maintain telecommunication infrastructure in support of Secure VPN connectivity.
- v. Provide any/all hardware and software replacements and upgrades as mutually agreed to by the parties.
- vi. Pay for all telecommunication requirements related to its connectivity.
- vii. Provide required information for dedicated service relating to Data Link Connection Identifiers, Circuit Identifier, Permanent Virtual Circuit Identifiers, Local Exchange Carrier Identifier, POC, location, etc., as determined by the parties.
- viii. Transport the CJIS WAN connection termination equipment suite to the CJIS Division for configuration and preparation for deployment under the dedicated service option.
- ix. Provide registered Internet Protocol information to be used by the requesting organization's system to the CJIS Division.
- x. Provide the CJIS Division with six months advance notice or stated amount of time for testing activities (i.e., disaster recovery exercises).
- xi. Provide the CJIS Division with applicable equipment maintenance contract numbers and level of service verifications needed to perform software upgrades on connection termination equipment.
- xii. Provide the CJIS Division with applicable software upgrade and patch images (or information allowing the CJIS Division to access such images).
- xiii. Transport only official, authorized traffic over the Secure VPN.
- xiv. Provide a POC and telephone number for MOU-related issues.

5. **FUNDING:** There are no reimbursable expenses associated with this level of support. Each party will fund its own activities unless otherwise agreed to in writing. This MOU is not an obligation or commitment of funds, nor a basis for transfer of funds, but rather is a basic statement of understanding between the parties hereto of the nature of the relationship for the connectivity efforts. Unless otherwise agreed to in writing, each party shall bear its own costs in relation to this MOU. Expenditures by each party will be subject to its budgetary processes and to the availability of funds and resources pursuant to applicable laws, regulations, and policies. The parties expressly acknowledge that the above language in no way implies that Congress will appropriate funds for such expenditures.

6. SETTLEMENT OF DISPUTES: Disagreements between the parties arising under or relating to this MOU will be resolved only by consultation between the parties and will not be referred to any other person or entity for settlement.

7. SECURITY: It is the intent of the parties that the actions carried out under this MOU will be conducted at the unclassified level. No classified information will be provided or generated under this MOU.

8. AMENDMENT, TERMINATION, ENTRY INTO FORCE, AND DURATION:

- a. All activities of the parties under this MOU will be carried out in accordance with the above – described provisions.
- b. This MOU may be amended or terminated by the mutual written consent of the parties' authorized representatives.
- c. Either party may terminate this MOU upon 30-days written notification to the other party. Such notice will be the subject of immediate consultation by the parties to decide upon the appropriate course of action. In the event of such termination, the following rules apply:
 - i. The parties will continue participation, financial or otherwise, up to the effective date of the termination.
 - ii. Each party will pay the costs it incurs as a result of the termination.
 - iii. All information and rights therein received under the provisions of this MOU prior to the termination will be retained by the parties, subject to the provisions of this MOU.

9. FORCE AND EFFECT: This MOU, which consists of nine numbered sections, will enter into effect upon signature of the parties and will remain in effect until terminated. The parties should review the contents of this MOU annually to determine whether there is a need for the deletion, addition, or amendment of any provision. This MOU is not intended, and should not be construed, to create any right or benefit, substantive or procedural, enforceable at law or otherwise by any third party against the parties, their parent agencies, the United States, or the officers, employees, agents, or other associated personnel thereof.

The foregoing represents the understandings reached between the parties.

FOR THE FEDERAL BUREAU OF INVESTIGATION

[Name]

Date

Assistant Director

Criminal Justice Information Services Division

FOR THE (insert requesting organization name)

Date

D.4 Interagency Connection Agreement

CRIMINAL JUSTICE INFORMATION SERVICES (CJIS)

Wide Area Network (WAN) USER AGREEMENT

BY INTERIM REMOTE LATENT USERS

The responsibility of the FBI CJIS Division is to provide state-of-the-art identification and information services to the local, state, federal, and international criminal justice communities, as well as the civil community for licensing and employment purposes. The data provided by the information systems administered and maintained by the FBI CJIS Division are routed to and managed in cooperation with the designated interface agency official. This information includes, but is not limited to, the Interstate Identification Index (III), National Crime Information Center (NCIC), Uniform Crime Reporting (UCR)/National Incident-Based Reporting System (NIBRS), and the Integrated Automated Fingerprint Identification System (IAFIS) programs.

In order to fulfill this responsibility, the FBI CJIS Division provides the following services to its users:

- Operational, technical, and investigative assistance;
- Telecommunications lines to local, state, federal and authorized interfaces;
- Legal and legislative review of matters pertaining to IAFIS, CJIS WAN and other related services;
- Timely information on all aspects of IAFIS, CJIS WAN, and other related programs by means of technical and operational updates, various newsletters, and other relative documents;
- Shared management through the CJIS Advisory Process and the Compact Council;

- Training assistance and up-to-date materials provided to each designated agency official, and;

- Audit.

The concept behind a designated interface agency official is to unify responsibility for system user discipline and ensure adherence to system procedures and policies within each interface agency. These individuals are ultimately responsible for planning necessary hardware, software, funding, training, and the administration of policy and procedures including security and integrity for complete access to CJIS related systems and CJIS WAN related data services by authorized agencies.

The following documents and procedures are incorporated by reference and made part of this agreement:

- *CJIS Security Policy*;

- *Title 28, Code of Federal Regulations, Part 20*;

- Computer Incident Response Capability (CIRC);

- Applicable federal and state laws and regulations.

To ensure continued access as set forth above, the designated interface agency agrees to adhere to all CJIS policies, including, but not limited to, the following:

1. The signatory criminal agency will provide fingerprints for all qualifying arrests either via electronic submission or fingerprint card that meet submission criteria. In addition, the agency will make their records available for interstate exchange for criminal justice and other authorized purposes.

2. The signatory civil agency with legislative authority will provide all qualifying fingerprints via electronic submission or fingerprint card that meet submission criteria.
3. Appropriate and reasonable quality assurance procedures must be in place to ensure that only complete, accurate, and valid information is maintained in the system.
4. Security – Each agency is responsible for appropriate security measures as applicable to physical security of terminals and telecommunications lines; Interim Distributed Imaging System (IDIS) equipment shall remain stand-alone devices and be used only for authorized purposes; personnel security to meet background screening requirements; technical security to protect against unauthorized use; data security, dissemination, and logging for audit purposes; and actual security of criminal history records. Additionally, each agency must establish an information security structure that provides for an Information Security Officer (ISO) or a security point of contact.
5. Audit – Each agency shall be responsible for complying with the appropriate audit requirements.
6. Training – Each agency shall be responsible for training requirements, including compliance with training mandates.
7. Integrity of the system shall be in accordance with FBI CJIS Division and interface agency policies. Computer incident reporting shall be implemented.

Until states are able to provide remote latent connectivity to their respective latent communities via a state WAN connection, the CJIS Division may provide direct connectivity to IAFIS via a dial-up connection or through the Combined DNA Index System (CODIS) and/or National Integrated Ballistics Information Network (NIBIN) connections. When a state implements a latent management system and is able to provide intrastate connectivity and subsequent forwarding to IAFIS, this agreement may be terminated. Such termination notice will be provided in writing by either the FBI or the state CJIS Systems Agency.

It is the responsibility of the local remote latent user to develop or acquire an IAFIS compatible workstation. These workstations may use the software provided by the FBI or develop their own software, provided it is IAFIS compliant.

The CJIS Division will provide the approved modem and encryptors required for each dial-up connection to IAFIS. The CJIS Communication Technologies Unit will configure and test the encryptors before they are provided to the user. Users requesting remote latent

connectivity through an existing CODIS and/or NIBIN connection must receive verification from the FBI that there are a sufficient number of Ethernet ports on the router to accommodate the request.

If at any time search limits are imposed by the CJIS Division, these individual agency connections will be counted toward the total state allotment.

ACKNOWLEDGMENT AND CERTIFICATION

As a CJIS WAN interface agency official serving in the CJIS system, I hereby acknowledge the duties and responsibilities as set out in this agreement. I acknowledge that these duties and responsibilities have been developed and approved by CJIS system users in order to ensure the reliability, confidentiality, completeness, and accuracy of all information contained in or obtained by means of the CJIS system. I further acknowledge that a failure to comply with these duties and responsibilities may subject our agency to various sanctions adopted by the CJIS Advisory Policy Board and approved by the Director of the FBI. These sanctions may include the termination of CJIS service.

As the designated CJIS WAN interface agency official serving in the CJIS system, I hereby certify that I am familiar with the contents of the *Title 28, Code of Federal Regulations, Part 20; CJIS Security Policy; Computer Incident Response Capability*; and applicable federal or state laws and regulations applied to IAFIS and CJIS WAN Programs for the dissemination of criminal history records for criminal and noncriminal justice purposes.

*

Signature

Print or Type

CJIS WAN Agency Official

Date

CONCURRENCE OF FEDERAL/REGULATORY AGENCY HEAD OR STATE CJIS SYSTEMS OFFICER (CSO):

*

Signature

Print or Type

*

Title

Date

State CSO

FBI CJIS DIVISION:

Signature – [Name]

Assistant Director _____

Title

Date

* If there is a change in the CJIS WAN interface agency official, the FBI Designated Federal Employee must be notified in writing 30 days prior to the change.

5/27/2004 UA modification reflects change in CTO title to CSO.

APPENDIX E: SECURITY FORUMS AND ORGANIZATIONAL ENTITIES

Online Security Forums / Organizational Entities
AntiOnline
Black Hat
CIO.com
CSO Online
CyberSpeak Podcast
FBI Criminal Justice Information Services Division (CJIS)
Forrester Security Forum
Forum of Incident Response and Security Teams (FIRST)
Information Security Forum (ISF)
Information Systems Audit and Control Association (ISACA)
Information Systems Security Association (ISSA)
Infosyssec
International Organization for Standardization (ISO)
International Information Systems Security Certification Consortium, Inc. (ISC)
Metasploit
Microsoft Developer Network (MSDN) Information Security
National Institute of Standards and Technology (NIST)
Open Web Application Security Project (OWASP)
SANS (SysAdmin, Audit, Network, Security) Institute
SC Magazine
Schneier.com
Security Focus
The Register
US Computer Emergency Response Team (CERT)
US DoJ Computer Crime and Intellectual Property Section (CCIPS)

APPENDIX F: SAMPLE FORMS

This appendix contains sample forms.

F.1 Security Incident Response Form

FBI CJIS DIVISION INFORMATION SECURITY OFFICER (ISO) SECURITY INCIDENT REPORTING FORM

NAME OF PERSON REPORTING THE INCIDENT: _____

DATE OF REPORT: _____ (mm/dd/yyyy)

DATE OF INCIDENT: _____ (mm/dd/yyyy)

POINT(S) OF CONTACT (Include Phone/Extension/Email): _____

LOCATION(S) OF INCIDENT: _____

INCIDENT DESCRIPTION: _____

SYSTEM(S) AFFECTED: _____

SYSTEM(S) AFFECTED (e.g., CAD, RMS, file server, etc.): _____

METHOD OF DETECTION: _____

ACTIONS TAKEN/RESOLUTION: _____

Copies To:

John C. Weatherly

(FBI CJIS Division ISO)

1000 Custer Hollow Road

Clarksburg, WV 26306-0102

iso@fbi.gov

APPENDIX G: BEST PRACTICES

This appendix contains best practices.

G.1 Virtualization

Virtualization

This appendix documents security considerations for implementing and operating virtual environments that process, store, or transmit Criminal Justice Information.

The FBI CJIS ISO has fielded several inquiries from various states requesting guidance on implementing virtual environments within their data centers. With the proliferation of virtual environments across industry in general there is a realistic expectation that FBI CJIS Auditors will encounter virtual environments during the upcoming year. Criminal Justice Agencies (CJAs) and Noncriminal Justice Agencies (NCJAs) alike need to understand and appreciate the foundation of security protection measures required for virtual environments.

From Microsoft's Introduction to Windows Server 2008

<http://www.microsoft.com/windowsserver2008/en/us/hyperv.aspx>:

"Server virtualization, also known as hardware virtualization, is a hot topic in the IT world because of the potential for serious economic benefits. Server virtualization enables multiple operating systems to run on a single physical machine as virtual machines (VMs). With server virtualization, you can consolidate workloads across multiple underutilized server machines onto a smaller number of machines. Fewer physical machines can lead to reduced costs through lower hardware, energy, and management overhead, plus the creation of a more dynamic IT infrastructure."

From a trade publication, kernelthread.com

<http://www.kernelthread.com/publications/virtualization/>:

"Virtualization is a framework or methodology of dividing the resources of a computer into multiple execution environments, by applying one or more concepts or technologies such as hardware and software partitioning, time-sharing, partial or complete machine simulation, emulation, quality of service, and many others."

From an Open Source Software developer

<http://www.kallasoft.com/pc-hardware-virtualization-basics/>:

"Virtualization refers to virtualizing hardware in software, allowing multiple operating systems, or images, to run concurrently on the same hardware. There are two main types of virtualization software:

- *"Type-1 Hypervisor, which runs 'bare-metal' (on top of the hardware)*
- *"Type-2 Hypervisor which requires a separate application to run within an operating system*

"Type1 hypervisors usually offer the best in efficiency, while Type-2 hypervisors allow for greater support of hardware that can be provided by the operating system. For the developer, power user, and small business IT professionals, virtualization offers the same basic idea of collapsing multiple physical boxes into one. For

instance, a small business can run a web server and an Exchange server without the need for two boxes. Developers and power users can use the ability to contain different development environments without the need to modify their main operating system. Big businesses can also benefit from virtualization by allowing software maintenance to be run and tested on a separate image on hardware without having to take down the main production system.”

Industry leaders and niche developers are bringing more products to market every day. The following article excerpts, all posted during September 2008, on www.virtualization.com are examples of industry offerings.

“Microsoft and Novell partnered together for joint virtualization solution. Microsoft and Novell are announcing the availability of a joint virtualization solution optimized for customers running mixed-source environments. The joint offering includes SUSE Linux Enterprise Server configured and tested as an optimized guest operating system running on Windows Server 2008 Hyper-V, and is fully supported by both companies’ channel partners. The offering provides customers with the first complete, fully supported and optimized virtualization solution to span Windows and Linux environments.”

“Sun Microsystems today announced the availability of Sun xVM Server software and Sun xVM Ops Center 2.0, key components in its strategy. Sun also announced the addition of comprehensive services and support for Sun xVM Server software and xVM Ops Center 2.0 to its virtualization suite of services. Additionally, Sun launched xVMserver.org, a new open source community, where developers can download the first source code bundle for Sun xVM Server software and contribute to the direction and development of the product.”

“NetEx, specialist in high-speed data transport over TCP, today announced Virtual HyperIP bandwidth optimization solutions for VMware environments that deliver a threefold to tenfold increase in data replication performance. Virtual HyperIP is a software-based Data Transport Optimizer that operates on the VMware ESX server and boosts the performance of storage replication applications from vendors such as EMC, NetApp, Symantec, IBM, Data Domain, and FalconStor. Virtual HyperIP mitigates TCP performance issues that are common when moving data over wide-area network (WAN) connections because of bandwidth restrictions, latency due to distance and/or router hop counts, packet loss and network errors. Like the company’s award-winning appliance-based HyperIP, Virtual HyperIP eliminates these issues with an innovative software design developed specifically to accelerate traffic over an IP based network.”

From several sources, particularly:

<http://www.windowsecurity.com/articles/security-virtualization.html>

<http://csrc.nist.gov/publications/drafts/6--64rev2/draft-sp800-64-Revision2.pdf>

Virtualization provides several benefits:

- Make better use of under-utilized servers by consolidating to fewer machines saving on hardware, environmental costs, management, and administration of the server infrastructure.

- Legacy applications unable to run on newer hardware and/or operating systems can be loaded into a virtual environment – replicating the legacy environment.
- Provides for isolated portions of a server where trusted and untrusted applications can be ran simultaneously – enabling hot standbys for failover.
- Enables existing operating systems to run on shared memory multiprocessors.
- System migration, backup, and recovery are easier and more manageable.

Virtualization also introduces several vulnerabilities:

- Host Dependent.
- If the host machine has a problem then all the VMs could potentially terminate.
- Compromise of the host makes it possible to take down the client servers hosted on the primary host machine.
- If the virtual network is compromised then the client is also compromised.
- Client share and host share can be exploited on both instances. Potentially this can lead to files being copied to the share that fill up the drive.

These vulnerabilities can be mitigated by the following factors:

- Apply “least privilege” technique to reduce the attack surface area of the virtual environment and access to the physical environment.
- Configuration and patch management of the virtual machine and host, i.e., Keep operating systems and application patches up to date on both virtual machines and hosts.
- Install the minimum applications needed on host machines.
- Practice isolation from host and virtual machine.
- Install and keep updated antivirus on virtual machines and the host.
- Segregation of administrative duties for host and versions.
- Audit logging as well as exporting and storing the logs outside the virtual environment.
- Encrypting network traffic between the virtual machine and host IDS and IPS monitoring.
- Firewall each virtual machine from each other and ensure that only allowed protocols will transact.

G.2 Voice over Internet Protocol

Voice over Internet Protocol (VoIP)

Attribution:

The following information has been extracted from NIST Special Publication 800-58, Security Considerations for Voice over IP Systems.

Definitions:

Voice over Internet Protocol (VoIP) – A set of software, hardware, and standards designed to make it possible to transmit voice over packet switched networks, either an internal Local Area Network, or across the Internet.

Internet Protocol (IP) – A protocol used for communicating data across a packet-switched internetwork using the Internet Protocol Suite, also referred to as TCP/IP. IP is the primary protocol in the Internet Layer of the Internet Protocol Suite and has the task of delivering distinguished protocol datagrams (packets) from the source host to the destination host solely based on their addresses.

Summary:

Voice over Internet Protocol (VoIP) has been embraced by organizations globally as an addition to, or replacement for, public switched telephone network (PSTN) and private branch exchange (PBX) telephone systems. The immediate benefits are alluring since the typical cost to operate VoIP is less than traditional telephone services and VoIP can be installed in-line with an organization's existing Internet Protocol services. Unfortunately, installing a VoIP network is not a simple "plug-and-play" procedure. There are myriad security concerns, cost issues with new networking hardware requirements, and overarching quality of service (QoS) factors that have to be considered carefully.

What are some of the advantages of VoIP?

- a. Cost – a VoIP system is usually cheaper to operate than an equivalent office telephone system with a Private Branch Exchange and conventional telephone service.
- b. Integration with other services – innovative services are emerging that allow customers to combine web access with telephone features through a single PC or terminal. For example, a sales representative could discuss products with a customer using the

company's web site. In addition, the VoIP system may be integrated with video across the Internet, providing a teleconferencing facility.

What are some of the disadvantages of VoIP?

- a. Startup cost – although VoIP can be expected to save money in the long run, the initial installation can be complex and expensive. In addition, a single standard has not yet emerged for many aspects of VoIP, so an organization must plan to support more than one standard, or expect to make relatively frequent changes as the VoIP field develops.
- b. Security – the flexibility of VoIP comes at a price: added complexity in securing voice and data. Because VoIP systems are connected to the data network, and share many of the same hardware and software components, there are more ways for intruders to attack a VoIP system than a conventional voice telephone system or PBX.

VoIP Risks, Threats, and Vulnerabilities

This section details some of the potential threats and vulnerabilities in a VoIP environment, including vulnerabilities of both VoIP phones and switches. Threat discussion is included because the varieties of threats faced by an organization determine the priorities in securing its communications equipment. Not all threats are present in all organizations. A commercial firm may be concerned primarily with toll fraud, while a government agency may need to prevent disclosure of sensitive information because of privacy or national security concerns. Information security risks can be broadly categorized into the following three types: confidentiality, integrity, and availability, (which can be remembered with the mnemonic "CIA"). Additional risks relevant to switches are fraud and risk of physical damage to the switch, physical network, or telephone extensions.

Packet networks depend for their successful operation on a large number of configurable parameters: IP and MAC (physical) addresses of voice terminals, addresses of routers and firewalls, and VoIP specific software such as Call Managers and other programs used to place and route calls. Many of these network parameters are established dynamically every time a network component is restarted, or when a VoIP telephone is restarted or added to the network. Because there are so many places in a network with dynamically configurable parameters, intruders have a wide array of potentially vulnerable points to attack.

Vulnerabilities described in this section are generic and may not apply to all systems, but investigations by NIST and other organizations have found these vulnerabilities in a number of VoIP systems. In addition, this list is not exhaustive; systems may have security weaknesses that are not included in the list. For each potential vulnerability, a recommendation is included to eliminate or reduce the risk of compromise.

Confidentiality and Privacy

Confidentiality refers to the need to keep information secure and private. For home computer users, this category includes confidential memoranda, financial information, and security information such as passwords. In a telecommunications switch, eavesdropping on conversations is an obvious concern, but the confidentiality of other information on the switch must be protected to defend against toll fraud, voice and data interception, and

denial of service attacks. Network IP addresses, operating system type, telephone extension to IP address mappings, and communication protocols are all examples of information that, while not critical as individual pieces of data, can make an attacker's job easier.

With conventional telephones, eavesdropping usually requires either physical access to tap a line, or penetration of a switch. Attempting physical access increases the intruder's risk of being discovered, and conventional PBXs have fewer points of access than VoIP systems. With VoIP, opportunities for eavesdroppers increase dramatically, because of the many nodes in a packet network.

Switch Default Password Vulnerability

It is common for switches to have a default login/password set, e.g., admin/admin, or root /root. This vulnerability also allows for wiretapping conversations on the network with port mirroring or bridging. An attacker with access to the switch administrative interface can mirror all packets on one port to another, allowing the indirect and unnoticeable interception of all communications. Failing to change default passwords is one of the most common errors made by inexperienced users.

REMEDIATION: If possible, remote access to the graphical user interface should be disabled to prevent the interception of plaintext administration sessions. Some devices provide the option of a direct USB connection in addition to remote access through a web browser interface. Disabling port mirroring on the switch should also be considered.

Classical Wiretap Vulnerability

Attaching a packet capture tool or protocol analyzer to the VoIP network segment makes it easy to intercept voice traffic.

REMEDIATION: A good physical security policy for the deployment environment is a general first step to maintaining confidentiality. Disabling the hubs on IP Phones as well as developing an alarm system for notifying the administrator when an IP Phone has been disconnected will allow for the possible detection of this kind of attack.

ARP Cache Poisoning and ARP Floods

Because many systems have little authentication, an intruder may be able to log onto a computer on the VoIP network segment, and then send ARP commands corrupting ARP caches on sender(s) of desired traffic, then activate IP. An ARP flood attack on the switch could render the network vulnerable to conversation eavesdropping. Broadcasting ARP replies blind is sufficient to corrupt many ARP caches. Corrupting the ARP cache makes it possible to re-route traffic to intercept voice and data traffic.

REMEDIATION: Use authentication mechanisms wherever possible and limit physical access to the VoIP network segment.

Web Server interfaces

Both VoIP switches and voice terminals are likely to have a web server interface for remote or local administration. An attacker may be able to sniff plaintext HTTP packets to gain

confidential information. This would require access to the local network on which the server resides.

REMEDIATION: If possible, do not use an HTTP server. If it is necessary to use a web server for remote administration, use the more secure HTTPS (HTTP over SSL or TLS) protocol.

IP Phone Netmask Vulnerability

A similar effect of the ARP Cache Vulnerability can be achieved by assigning a subnet mask and router address to the phone crafted to cause most or all of the packets it transmits to be sent to an attacker's MAC address. Again, standard IP forwarding makes the intrusion all but undetectable.

REMEDIATION: A firewall filtering mechanism can reduce the probability of this attack. Remote access to IP phones is a severe risk.

Extension to IP Address Mapping Vulnerability

Discovering the IP address corresponding to any extension requires only calling that extension and getting an answer. A protocol analyzer or packet capture tool attached to the hub on the dialing instrument will see packets directly from the target instrument once the call is answered. Knowing the IP address of a particular extension is not a compromise in itself, but makes it easier to accomplish other attacks. For example, if the attacker is able to sniff packets on the local network used by the switch, it will be easy to pick out packets sent and received by a target phone. Without knowledge of the IP address of the target phone, the attacker's job may be much more difficult to accomplish and require much longer, possibly resulting in the attack being discovered.

REMEDIATION: Disabling the hub on the IP Phone will prevent this kind of attack. However, it is a rather simple task to turn the hub back on.

Integrity Issues

Integrity of information means that information remains unaltered by unauthorized users. For example, most users want to ensure that bank account numbers cannot be changed by anyone else, or that passwords are changed only by the user or an authorized security administrator. Telecommunication switches must protect the integrity of their system data and configuration. Because of the richness of feature sets available on switches, an attacker who can compromise the system configuration can accomplish nearly any other goal. For example, an ordinary extension could be re-assigned into a pool of phones that supervisors can listen in on or record conversations for quality control purposes. Damaging or deleting information about the IP network used by a VoIP switch results in an immediate denial of service.

The security system itself provides the capabilities for system abuse and misuse. That is, compromise of the security system not only allows system abuse but also allows the elimination of all traceability and the insertion of trapdoors for intruders to use on their next visit. For this reason, the security system must be carefully protected. Integrity threats include any in which system functions or data may be corrupted, either accidentally or as

a result of malicious actions. Misuse may involve legitimate users (i.e., insiders performing unauthorized operations) or intruders.

A legitimate user may perform an incorrect, or unauthorized, operations function (e.g., by mistake or out of malice) and may cause deleterious modification, destruction, deletion, or disclosure of switch software and data. This threat may be caused by several factors including the possibility that the level of access permission granted to the user is higher than what the user needs to remain functional.

Intrusion – An intruder may masquerade as a legitimate user and access an operations port of the switch. There are a number of serious intrusion threats. For example, the intruder may use the permission level of the legitimate user and perform damaging operations functions such as:

- Disclosing confidential data
- Causing service deterioration by modifying the switch software
- Crashing the switch
- Removing all traces of the intrusion (e.g., modifying the security log) so that it may not be readily detected

Insecure state – At certain times the switch may be vulnerable due to the fact that it is not in a secure state. For example:

- After a system restart, the old security features may have been reset to insecure settings, and new features may not yet be activated. (For example, all old passwords may have reverted to the default system-password, even though new passwords are not yet assigned.) The same may happen at the time of a disaster recovery.
- At the time of installation the switch may be vulnerable until the default security features have been replaced.

DHCP Server Insertion Attack

It is often possible to change the configuration of a target phone by exploiting the DHCP response race when the IP phone boots. As soon as the IP phone requests a DHCP response, a rogue DHCP server can initiate a response with data fields containing false information.

This attack allows for possible man in the middle attacks on the IP-media gateway, and IP Phones. Many methods exist with the potential to reboot the phone remotely, e.g., “social engineering”, ping flood, MAC spoofing (probably SNMP hooks, etc.).

REMEDIATION: If possible, use static IP addresses for the IP Phones. This will remove the necessity of using a DHCP server. Further, using a state based intrusion detection system can filter out DHCP server packets from IP Phone ports, allowing this traffic only from the legitimate server.

TFTP Server Insertion Attack

It is possible to change the configuration of a target phone by exploiting the TFTP response race when the IP phone is resetting. A rogue TFTP server can supply spurious information before the legitimate server is able to respond to a request. This attack allows an attacker to change the configuration of an IP Phone.

REMEDIATION: Using a state based intrusion detection system can filter out DHCP server packets from IP Phone ports, allowing such traffic only from the legitimate server. Organizations looking to deploy VoIP systems should look for IP Phone instruments that can download signed binary files.

Availability and Denial of Service

Availability refers to the notion that information and services be available for use when needed. Availability is the most obvious risk for a switch. Attacks exploiting vulnerabilities in the switch software or protocols may lead to deterioration or even denial of service or functionality of the switch. For example: if unauthorized access can be established to any branch of the communication channel (such as a CCS link or a TCP/IP link), it may be possible to flood the link with bogus messages causing severe deterioration (possibly denial) of service. A voice over IP system may have additional vulnerabilities with Internet connections. Because intrusion detection systems fail to intercept a significant percentage of Internet based attacks, attackers may be able to bring down VoIP systems by exploiting weaknesses in Internet protocols and services.

Any network may be vulnerable to denial of service attacks, simply by overloading the capacity of the system. With VoIP the problem may be especially severe, because of its sensitivity to packet loss or delay.

CPU Resource Consumption Attack without any account information.

An attacker with remote terminal access to the server may be able to force a system restart (shutdown all/restart all) by providing the maximum number of characters for the login and password buffers multiple times in succession. Additionally, IP Phones may reboot as a result of this attack.

In addition to producing a system outage, the restart may not restore uncommitted changes or, in some cases, may restore default passwords, which would introduce intrusion vulnerabilities.

REMEDIATION: The deployment of a firewall disallowing connections from unnecessary or unknown network entities is the first step to overcoming this problem. However, there is still the opportunity for an attacker to spoof his MAC and IP address, circumventing the firewall protection.

Default Password Vulnerability

It is common for switches to have a default login/password set, e.g., admin/admin, or root /root. Similarly, VoIP telephones often have default keypad sequences that can be used to unlock and modify network information.

This vulnerability would allow an attacker to control the topology of the network remotely, allowing for not only complete denial of service to the network, but also a port mirroring attack to the attacker's location, giving the ability to intercept any other conversations taking place over the same switch. Further, the switch may have a web server interface, providing an attacker with the ability to disrupt the network without advance knowledge of switch operations and commands. In most systems, telephones download their configuration data on startup using TFTP or similar protocols. The configuration specifies the IP addresses for Call Manager nodes, so an attacker could substitute another IP address pointing to a call manager that would allow eavesdropping or traffic analysis.

REMEDIATION: Changing the default password is crucial. Moreover, the graphical user interface should be disabled to prevent the interception of plaintext administration sessions.

Exploitable software flaws

Like other types of software, VoIP systems have been found to have vulnerabilities due to buffer overflows and improper packet header handling. These flaws typically occur because the software is not validating critical information properly. For example, a short integer may be used as a table index without checking whether the parameter passed to the function exceeds 32,767, resulting in invalid memory accesses or crashing of the system.

Exploitable software flaws typically result in two types of vulnerabilities: denial of service or revelation of critical system parameters. Denial of service can often be implemented remotely, by passing packets with specially constructed headers that cause the software to fail. In some cases the system can be crashed, producing a memory dump in which an intruder can find IP addresses of critical system nodes, passwords, or other security-relevant information. In addition, buffer overflows that allow the introduction of malicious code have been found in VoIP software, as in other applications.

REMEDIATION: These problems require action from the software vendor, and distribution of patches to administrators. Intruders monitor announcements of vulnerabilities, knowing that many organizations require days or weeks to update their software. Regular checking for software updates and patches is essential to reducing these vulnerabilities. Automated patch handling can assist in reducing the window of opportunity for intruders to exploit known software vulnerabilities.

Account Lockout Vulnerability

An attacker will be able to provide several incorrect login attempts at the telnet prompt until the account becomes locked out. (This problem is common to most password-protected systems, because it prevents attackers from repeating login attempts until the correct password is found by trying all possible combinations.)

The account is unable to connect to the machine for the set lockout time.

REMEDIATION: If remote access is not available, this problem can be solved with physical access control.

NIST Recommendations.

Because of the integration of voice and data in a single network, establishing a secure VoIP and data network is a complex process that requires greater effort than that required for data-only networks. In particular, start with these general guidelines, recognizing that practical considerations, such as cost or legal requirements, may require adjustments for the organization:

9. Develop appropriate network architecture.

- Separate voice and data on logically different networks if feasible. Different subnets with separate RFC 1918 address blocks should be used for voice and data traffic, with separate DHCP servers for each, to ease the incorporation of intrusion detection and VoIP firewall protection at the voice gateway, which interfaces with the PSTN, disallow H.323, SIP, or other VoIP protocols from the data network. Use strong authentication and access control on the voice gateway system, as with any other critical network component. Strong authentication of clients towards a gateway often presents difficulties, particularly in key management. Here, access control mechanisms and policy enforcement may help.
- A mechanism to allow VoIP traffic through firewalls is required. There are a variety of protocol dependent and independent solutions, including application level gateways (ALGs) for VoIP protocols, Session Border Controllers, or other standards-based solutions when they mature.
- Stateful packet filters can track the state of connections, denying packets that are not part of a properly originated call. (This may not be practical when multimedia protocol inherent security or lower layer security is applied, e.g., H.235 Annex D for integrity provision or TLS to protect SIP signaling).
- Use Ipsec or Secure Shell (SSH) for all remote management and auditing access. If practical, avoid using remote management at all and do IP PBX access from a physically secure system.
- If performance is a problem, use encryption at the router or other gateway, not the individual endpoints, to provide for Ipsec tunneling. Since some VoIP endpoints are not computationally powerful enough to perform encryption, placing this burden at a central point ensures all VoIP traffic emanating from the enterprise network has been encrypted. Newer IP phones are able to provide Advanced Encryption System (AES) encryption at reasonable cost. Note that Federal

Information Processing Standard (FIPS) 140-2, Security Requirements for Cryptographic Modules, is applicable to all Federal agencies that use cryptographic-based security systems to protect sensitive information in computer and telecommunication systems (including voice systems) as defined in Section 5131 of the Information Technology Management Reform Act of 1996, Public Law 104-106.

10. Ensure that the organization has examined and can acceptably manage and mitigate the risks to their information, system operations, and continuity of essential operations when deploying VoIP systems.

VoIP can provide more flexible service at lower cost, but there are significant tradeoffs that must be considered. VoIP systems can be expected to be more vulnerable than conventional telephone systems, in part because they are tied in to the data network, resulting in additional security weaknesses and avenues of attack (see VoIP Risks, Threats, and Vulnerabilities section for more detailed discussion of vulnerabilities of VoIP and their relation to data network vulnerabilities).

Confidentiality and privacy may be at greater risk in VoIP systems unless strong controls are implemented and maintained. An additional concern is the relative instability of VoIP technology compared with established telephony systems. Today, VoIP systems are still maturing and dominant standards have not emerged. This instability is compounded by VoIP's reliance on packet networks as a transport medium. The public switched telephone network is ultra-reliable. Internet service is generally much less reliable, and VoIP cannot function without Internet connections, except in the case of large corporate or other users who may operate a private network. Essential telephone services, unless carefully planned, deployed, and maintained, will be at greater risk if based on VoIP.

11. Special consideration should be given to E-911 emergency services communications, because E-911 automatic location service is not available with VoIP in some cases.

Unlike traditional telephone connections, which are tied to a physical location, VoIP's packet switched technology allows a particular number to be anywhere. This is convenient for users, because calls can be automatically forwarded to their locations. But the tradeoff is that this flexibility severely complicates the provision of E-911 service, which normally provides the caller's location to the 911 dispatch office. Although most VoIP vendors have workable solutions for E-911 service, government regulators and vendors are still working out standards and procedures for 911 services in a VoIP environment. Agencies must carefully evaluate E-911 issues in planning for VoIP deployment.

12. Agencies should be aware that physical controls are especially important in a VoIP environment and deploy them accordingly.

Unless the VoIP network is encrypted, anyone with physical access to the office LAN could potentially connect network monitoring tools and tap into telephone conversations. Although conventional telephone lines can also be monitored when physical access is obtained, in most offices there are many more points to connect with a LAN without arousing suspicion. Even if encryption is used, physical access to VoIP servers and gateways may allow an attacker to do traffic analysis (i.e., determine which parties are

communicating). Agencies therefore should ensure that adequate physical security is in place to restrict access to VoIP network components. Physical security measures, including barriers, locks, access control systems, and guards, are the first line of defense. Agencies must make sure that the proper physical countermeasures are in place to mitigate some of the biggest risks such as insertion of sniffers or other network monitoring devices. Otherwise, practically speaking this means that installation of a sniffer could result in not just data but all voice communications being intercepted.

13. VoIP-ready firewalls and other appropriate protection mechanisms should be employed.

Agencies must enable, use, and routinely test the security features that are included in VoIP systems.

Because of the inherent vulnerabilities (e.g., susceptibility to packet sniffing) when operating telephony across a packet network, VoIP systems incorporate an array of security features and protocols. Organization security policy should ensure that these features are used. In particular, firewalls designed for VoIP protocols are an essential component of a secure VoIP system.

14. If practical, “softphone” systems, which implement VoIP using an ordinary PC with a headset and special software, should not be used where security or privacy are a concern.

Worms, viruses, and other malicious software are extraordinarily common on PCs connected to the internet, and very difficult to defend against. Well-known vulnerabilities in web browsers make it possible for attackers to download malicious software without a user’s knowledge, even if the user does nothing more than visit a compromised web site. Malicious software attached to email messages can also be installed without the user’s knowledge, in some cases even if the user does not open the attachment. These vulnerabilities result in unacceptably high risks in the use of “softphones”, for most applications. In addition, because PCs are necessarily on the data network, using a softphone system conflicts with the need to separate voice and data networks to the greatest extent practical.

15. If mobile units are to be integrated with the VoIP system, use products implementing Wi-Fi Protected Access (WPA), rather than 802.11 Wired Equivalent Privacy (WEP).

The security features of 802.11 WEP provide little or no protection because WEP can be cracked with publicly available software. The more recent Wi-Fi Protected Access (WPA), a snapshot of the ongoing 802.11i standard, offers significant improvements in security, and can aid the integration of wireless technology with VoIP. NIST strongly recommends that the WPA (or WEP if WPA is unavailable) security features be used as part of an overall defense-in-depth strategy. Despite their weaknesses, the 802.11 security mechanisms can provide a degree of protection against unauthorized disclosure, unauthorized network access, or other active probing attacks. However, the Federal Information Processing Standard (FIPS) 140-2, Security Requirements for Cryptographic Modules, is mandatory and binding for Federal agencies that have determined that certain information must be protected via cryptographic means. As currently defined, neither WEP nor WPA meets the FIPS 140-2 standard. In these cases, it will be necessary to employ higher level cryptographic protocols and applications such as secure shell (SSH), Transport Level Security (TLS) or Internet Protocol Security (Ipsec) with FIPS 140-2 validated

cryptographic modules and associated algorithms to protect information, regardless of whether the nonvalidated data link security protocols are used.

16. Carefully review statutory requirements regarding privacy and record retention with competent legal advisors.

Although legal issues regarding VoIP are beyond the scope of this document, readers should be aware that laws and rulings governing interception or monitoring of VoIP lines, and retention of call records, may be different from those for conventional telephone systems. Agencies should review these issues with their legal advisors. See Section 2.5 for more on these issues.

G.3 Cloud Computing

Cloud Computing

Purpose:

This paper is provided to define and describe cloud computing, discuss CJIS Security Policy (CSP) compliance, detail security and privacy, and provide general recommendations.

Attribution:

- NIST SP 800-144, Guidelines on Security and Privacy in Public Cloud Computing (Dec. 2011)
- NIST SP 800-145, the NIST Definition of Cloud Computing (Sept. 2011)
- NIST SP 800-146, Cloud Computing Synopsis and Recommendations (May 2011)
- CJIS Security Policy, Version 5.0

Definitions and Terms:

Cloud computing – A distributed computing model that permits on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services), software, and information.

Cloud subscriber – A person or organization that is a customer of a cloud

Cloud client – A machine or software application that accesses a cloud over a network connection, perhaps on behalf of a subscriber

Cloud provider – An organization that provides cloud services

Summary:

With many law enforcement agencies looking for ways to attain greater efficiency while grappling with reduced budgets, the idea of cloud computing to maintain data and applications is a viable business solution. But the unique security and legal characteristics of law enforcement agencies means any migration to cloud services may be challenging. Anytime the security of information and transactions must be maintained, as it must be with access to the FBI's CJIS systems and the protection of Criminal Justice Information (CJI), security and policy compliance concerns are bound to arise.

Cloud computing has become a popular and sometimes contentious topic of discussion for both the private and public sectors. This is in part because of the difficulty in describing cloud computing in general terms, because it is not a single kind of system. The “cloud” spans a spectrum of underlying technologies, configuration possibilities, service and deployment models. Cloud computing offers the ability to conveniently rent access to fully featured applications, software development and deployment environments, and computing infrastructure assets - such as network-accessible data storage and processing from a cloud service provider.

One of the benefits of cloud computing is the ability to outsource many of the technical functions agencies may not want to perform for various reasons. Ultimately, the move to cloud computing is a business and security risk decision in which the following relevant factors are given proper consideration:

- readiness of existing applications for cloud deployment
- transition costs
- life-cycle costs
- maturity of service orientation in existing infrastructure
- security and privacy requirements – federal, state, and local

Achieving CJIS Security Policy Compliance:

The question that is often asked is, “Can an Agency be compliant with the CJIS Security Policy and also cloud compute?”

Because the CJIS Security Policy is device and architecture independent (per CSP Section 2.2), the answer is yes, and this can be accomplished— assuming the vendor of the cloud technology is able to meet the existing requirements of the CJIS Security Policy.

There are security challenges that must be addressed if CJI is to be sent into or through, stored within, or accessed from the cloud.

Admittedly, the existing CJIS Security Policy requirements may be difficult for some cloud-computing vendors due to the sheer numbers and the geographic disbursement of their personnel; however, the requirements aren’t new to vendors serving the criminal justice community and many vendors have been successfully meeting the Policy requirements for years. Even so, they are the minimum security requirements which will provide an acceptable level of assurance that law enforcement and personally identifiable information (PII) will be protected when shared with other law enforcement agencies across the nation.

General CJIS Security Policy Applicability Questions

Before tackling these challenges, the cloud subscriber should first be aware of what security and legal requirements they are subject to prior to entering into any agreement with a cloud provider. Asking the following general questions will help frame the process of determining compliance with the existing requirements of the CJIS Security Policy.

- Will access to Criminal Justice Information (CJI) within a cloud environment fall within the category of remote access? (AC-17 Remote Access and MA-4 Nonlocal Maintenance)
- Will multifactor authentication (MFA) be required for access to CJI within a cloud environment? (IA-2 Identification and Authentication (Organizational Users))
- Does/do any cloud service provider's datacenter(s) used in the transmission or storage of CJI meet all the requirements of a physically secure location? (Physical and Environmental Protection (PE))
- Are the encryption requirements being met? (SC-13 Cryptographic Protection and SC-28 Protection of Information at Rest)
 - Who will be providing the encryption as required in the CJIS Security Policy (client or cloud service provider)? *Note: individuals with access to the keys can decrypt the stored files and therefore have access to unencrypted CJI.*
 - Is the data encrypted while at rest and in transit?
- What are the cloud service provider's incident response procedures? (Incident Response (IR))
 - Will the cloud subscriber be notified of any incident?
 - If CJI is compromised, what are the notification and response procedures?
- Is the cloud service provider a private contractor/vendor?
 - If so, they are subject to the same screening and agreement requirements as any other private contractors hired to handle CJI? (CA-3 Information Exchange and SA-9 External System Services; PS-3 Personnel Screening)
- Will the cloud service provider allow the CSA and FBI to conduct compliance and security audits? *Note: Cloud facilities such as datacenters in which CJI will be stored or processed should be audited as would any other datacenter housing and processing CJI.*
- How will event and content logging be handled? (Auditing and Accountability (AU))

- Will the cloud service provider handle the events and content logging required by the CJIS Security Policy and provide that upon request?
- What are the cloud service provider's responsibilities with regard to media protection and destruction? (Media Protection (MP))

Ultimately, the goal is to remain committed to using technology in its information sharing processes, but not at the sacrifice of the security of the information with which it has been entrusted. As stated in the CJIS Security Policy, device and architecture independence permits the use of cloud computing, but the security requirements do not change.

Cloud Utilization Scenarios

1. Encrypted CJI in a Cloud Environment–Key Management Control, Security Awareness Training, and Personnel Controls

Prior to permitting CJI to be stored or traverse through a cloud environment, the agency should ensure proper encryption key management control procedures are implemented to determine who has access and control over the encryption keys. Proper key management control is vital to CJI security as those individuals (agency or cloud employees) with access to the keys can decrypt the stored files, and therefore, have unescorted access to unencrypted CJI. This means all those individuals must be subjected to security awareness training (CJIS Security Policy Awareness and Training (AT)) and must meet personnel security (CJIS Security Policy Section Personnel Security (PS)) requirements as individuals with unescorted access to unencrypted CJI.

Note: As a best security practice, the CJIS ISO Program does not recommend allowing the cloud service provider access to the encryption keys used to protect CJI. However, it may not always be reasonable to expect the agency, criminal justice or noncriminal justice, to accomplish this task.

Note: When selecting a cloud service provider, the CJIS ISO Program reminds agencies the CJIS Security Policy sets the minimum requirements for the protection of CJI. Additional security assurances from other authorizations such as FedRAMP, StateRAMP, SOC Type 2, etc., may be leveraged, however, they do not guarantee compliance with the CJIS Security Policy.

a. Scenario 1–Agency Stores CJI in a Cloud:

A CJA stores encrypted CJI (Backup files and drives) in a cloud service provider's environment. To access CJI, the agency will extract the CJI from the cloud to its local machine, and then decrypt the CJI. The CJI is processed, re-encrypted, and then re-uploaded to the cloud environment for storage. In this scenario, the agency always encrypts the CJI prior to placing it in the cloud and only authorized users of the agency have access to the encryption keys. Since the agency maintains the encryption keys, the cloud service provider employees would not need to undergo fingerprint-based background checks, nor have security awareness training. These requirements

are negated, because only authorized personnel with access to the keys have the ability to view this CJI in an unencrypted form.

b. Scenario 2–Agency Accesses CJI While in a Cloud:

A CJA stores CJI (files and drives) in a cloud service provider’s environment, but as part of daily operations authorized users will remotely access the encrypted CJI in the cloud. The user will decrypt the CJI while it is in the cloud’s virtual environment, process the data, and then re-encrypt the data prior to ending the remote session. The agency maintains the keys and the cloud service provider does not have access to the encryption keys. However, since the CJI is decrypted within the cloud’s virtual environment, any administrative personnel employed by the cloud provider having the ability to access the virtual environment must be identified and subjected to security awareness training and personnel security controls as described in the CJIS Security Policy.

c. Scenario 3–CJI Impact from a Cloud Datacenter Critical Systems Crash–Core Dump* Recovery:

A CJA utilizes a cloud service provider (IaaS or PaaS) to store CJI and remotely accesses the environment to process CJI. During normal operation, the cloud provider experiences systems outages within the datacenter in which CJI is processed and stored. The cloud provider’s administrators need to repair the systems and restore service using data from a core dump to return to normal operations. The cloud service provider as part of the Service Level Agreement (SLA) with the CJA has been authorized to maintain the encryption keys in order respond to such an event. The cloud administrators with such access have underwent fingerprint-based background checks and security awareness training. This allows the cloud administrators to decrypt CJI so that it is written to the core dump files for restoration following the system outage. CJI, however, is encrypted at all times except when part of the core dump files. As part of the SLA, the cloud service provider has agreed to treat the core dump files as CJI to ensure all protection are in place in compliance with the CJIS Security Policy.

Note: Writing encrypted data to a core dump corrupts the data and makes it unusable because the key no longer decrypts the data. This is problematic when attempting to recover encrypted data written to a core dump. The CJA could have ensured the cloud provider exclude encrypted data (CJI) from the core dump, but chose against it.

* Core Dump - A file of a computer’s documented memory of when a program or computer crashed. The file consists of the recorded status of the working memory at an explicit time, usually close to when the system crashed or when the program ended atypically as it presents the risk that the system failure would ensure the loss of the encrypted data.

The Cloud Model Explained:

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

The cloud model as defined by NIST consists of five essential characteristics, offers the option of three service models, and may be deployed via any of four deployment models as shown in Figure 1 below:

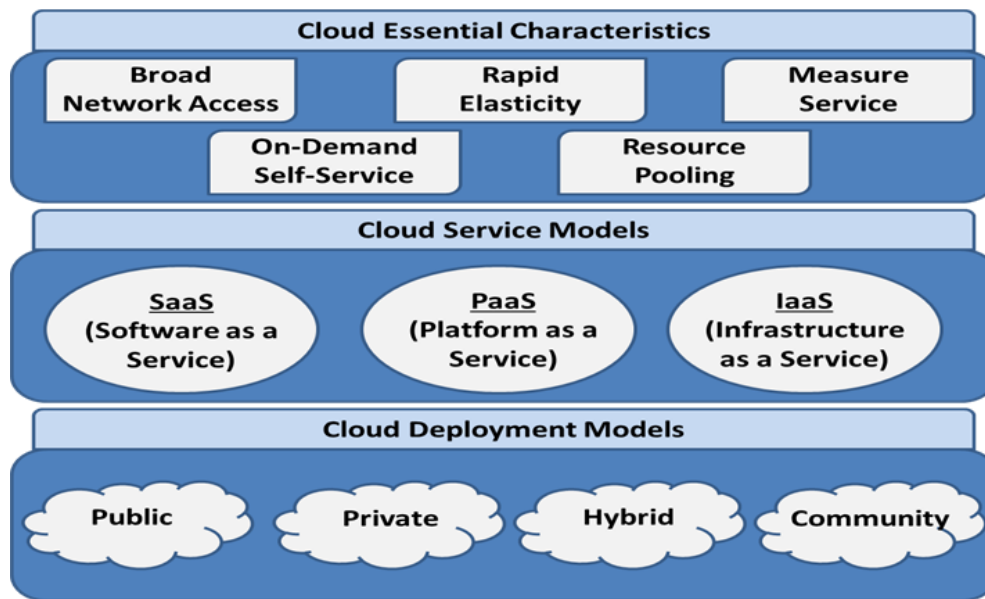


Figure 1 - Visual Depiction of the NIST Cloud Computing Definition

Essential Characteristics:

On-demand self-service

A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

Broad network access

Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

Resource pooling

The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence

in which the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter). Examples of resources include storage, processing, memory, and network bandwidth.

Rapid elasticity

Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.

Measured service

Cloud systems automatically control and optimize resource use by leveraging a metering capability* at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

** Typically this is done on a pay-per-use or charge-per-use basis.*

Deployment Models:

Private cloud

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

Community cloud

The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

Public cloud

The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.

Hybrid cloud

The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Service Models:

Cloud providers offer different levels of service, i.e.; Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). The way CJI is placed and accessed in the cloud determines if the Personnel Security (PS) requirements in apply. Access to encryption keys and management of the resources vary depending on what type of service is used. The SaaS offering is the most likely service wherein the cloud service provider may have access to unencrypted CJI due to software updates, patches, and management. However, through management and control of encryption keys, all service offerings may be implemented in an agency-controlled manner where the cloud service provider has no ability to access unencrypted CJI.

For cloud computing services that involve the storage, processing, or transmission of CJI, the Personnel Security (PS) terms and requirements apply to all personnel when their unescorted logical or physical access to any information system results in the ability, right, or privilege to view, modify, or make use of unencrypted CJI. It is critical for the agency to understand the level of service and access required for each cloud implementation.

Software as a Service (SaaS)

This model provides the consumer the capability to use the provider's applications running on a cloud infrastructure*.

** A cloud infrastructure is the collection of hardware and software that enables the five essential characteristics of cloud computing. The cloud infrastructure can be viewed as containing both a physical layer and an abstraction layer. The physical layer consists of the hardware resources that are necessary to support the cloud services being provided, and typically includes server, storage and network components. The abstraction layer consists of the software deployed across the physical layer, which manifests the essential cloud characteristics. Conceptually the abstraction layer sits above the physical layer.*

The SaaS service model is often referred to as “Software deployed as a hosted service and accessed over the Internet.”

The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface.

When using the SaaS service model it should be understood that the consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

Platform as a Service (PaaS)

This model provides the consumer the capability to deploy consumer-created or acquired applications* created using programming languages, libraries, services, and tools supported by the provider onto the cloud infrastructure.

** This capability does not necessarily preclude the use of compatible programming languages, libraries, services, and tools from other sources.*

When using the PaaS service model the consumer may have control over the deployed applications and possibly configuration settings for the application-hosting environment, but does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage.

Infrastructure as a Service (IaaS)

This model provides the consumer the capability to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, including operating systems and applications.

When using the IaaS service model the consumer may have control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls), but does not manage or control the underlying cloud infrastructure.

Key Security and Privacy Issues:

Although the emergence of cloud computing is a recent development, insights into critical aspects of security can be gleaned from reported experiences of early adopters and also from researchers analyzing and experimenting with available cloud provider platforms and associated technologies. The sections below highlight privacy and security-related issues that are believed to have long-term significance for public cloud computing and, in many cases, for other cloud computing service models.

Because cloud computing has grown out of an amalgamation of technologies, including service oriented architecture, virtualization, Web 2.0, and utility computing, many of the privacy and security issues involved can be viewed as known problems cast in a new setting. The importance of their combined effect in this setting, however, should not be discounted. Public cloud computing does represent a thought-provoking paradigm shift from conventional norms to an open organizational infrastructure—at the extreme, *displacing applications from one organization's infrastructure to the infrastructure of another organization, where the applications of potential adversaries may also operate.*

Governance

Governance implies control and oversight by the organization over policies, procedures, and standards for application development and information technology service acquisition, as well as the design, implementation, testing, use, and monitoring of deployed or engaged services. With the wide availability of cloud computing services, lack of organizational controls over employees engaging such services arbitrarily can be a source of problems. While cloud computing simplifies platform acquisition, it doesn't alleviate the need for governance; instead, it has the opposite effect, amplifying that need.

Dealing with cloud services requires attention to the roles and responsibilities involved between the organization and cloud provider, particularly with respect to managing risks and ensuring organizational requirements are met. Ensuring systems are secure and risk is managed is challenging in any environment and even more daunting with cloud computing. Audit mechanisms and tools should be in place to determine how data is stored, protected, and used, to validate services, and to verify policy enforcement. A risk management program should also be in place that is flexible enough to deal with the continuously evolving and shifting risk landscape.

Compliance

Compliance refers to an organization's responsibility to operate in agreement with established laws, regulations, standards, and specifications. Various types of security and privacy laws and regulations exist within different countries at the national, state, and local levels, making compliance a potentially complicated issue for cloud computing.

Law and Regulations

Cloud providers are becoming more sensitive to legal and regulatory concerns, and may be willing to commit to store and process data in specific jurisdictions and apply required safeguards for security and privacy. However, the degree to which they will accept liability in their service agreements, for exposure of content under their control, remains to be seen. Even so, organizations are ultimately accountable for the security and privacy of data held by a cloud provider on their behalf.

Data Location

One of the most common compliance issues facing an organization is data location. A characteristic of many cloud computing services is that data is stored redundantly in multiple physical locations and detailed information about the location of an organization's data is unavailable or not disclosed to the service consumer. This situation makes it difficult to ascertain whether sufficient safeguards are in place and whether legal and regulatory compliance requirements are being met. External audits and security certifications can alleviate this issue to some extent, but they are not a panacea.

When information crosses borders, the governing legal, privacy, and regulatory regimes can be ambiguous and raise a variety of concerns. Consequently, constraints on the trans-border flow of sensitive data, as well as the requirements on the protection afforded the data, have become the subject of national and regional privacy and security laws and regulations.

Electronic Discovery

The capabilities and processes of a cloud provider, such as the form in which data is maintained and the electronic discovery-related tools available, affect the ability of the organization to meet its obligations in a cost effective, timely, and compliant manner. A cloud provider's archival capabilities may not preserve the original metadata as expected, causing spoliation (i.e., the intentional, reckless, or negligent destruction, loss, material alteration, or obstruction of evidence that is relevant to litigation), which could negatively impact litigation.

Trust

Under the cloud computing paradigm, an organization relinquishes direct control over many aspects of security and privacy, and in doing so, confers a high level of trust onto the cloud provider. At the same time, federal agencies have a responsibility to protect information and information systems commensurate with the risk and magnitude of the harm resulting from unauthorized access, use, disclosure, disruption, modification, or destruction, regardless of whether the information is collected or maintained by or on behalf of the agency; or whether the information systems are used or operated by an agency or by a contractor of an agency or other organization on behalf of an agency

Insider Access

Data processed or stored outside the physical confines of an organization, its firewall, and other security controls bring with it an inherent level of risk. The insider security threat is a well-known issue for most organizations. Incidents may involve various types of fraud, sabotage of information resources, and theft of sensitive information.

Data Ownership

The organization's ownership rights over the data must be firmly established in the service contract to enable a basis for trust and privacy of data. The continuing controversy over privacy and data ownership rights for social networking users illustrates the impact that ambiguous terms can have on the parties involved.

Ideally, the contract should state clearly that the organization retains exclusive ownership over all its data; that the cloud provider acquires no rights or licenses through the agreement, including intellectual property rights or licenses, to use the organization's data for its own purposes; and that the cloud provider does not acquire and may not claim any interest in the data due to security. For these provisions to work as intended, the terms of data ownership must not be subject to unilateral amendment by the cloud provider.

Visibility

Continuous monitoring of information security requires maintaining ongoing awareness of security controls, vulnerabilities, and threats to support risk management decisions. Transition to public cloud services entails a transfer of responsibility to the cloud provider for securing portions of the system on which the organization's data and applications operate.

Ancillary Data

While the focus of attention in cloud computing is mainly on protecting application data, cloud providers also hold significant details about the accounts of cloud consumers that could be compromised and used in subsequent attacks.

Risk Management

Assessing and managing risk in systems that use cloud services can be a challenge. With cloud-based services, some subsystems or subsystem components fall outside of the direct control of a client organization. Many organizations are more comfortable with risk when they have greater control over the processes and equipment involved. Establishing a level of trust about a cloud service is dependent on the degree of control an organization is able to exert on the provider to provision the security controls necessary to protect the organization's data and applications, and also the evidence provided about the effectiveness of those controls. Ultimately, if the level of trust in the service falls below expectations and the organization is unable to employ compensating controls, it must either reject the service or accept a greater degree of risk.

Architecture

The architecture of the software and hardware used to deliver cloud services can vary significantly among public cloud providers for any specific service model. It is important to understand the technologies the cloud provider uses to provision services and the implications the technical controls involved have on security and privacy of the system throughout its lifecycle. With such information, the underlying system architecture of a cloud can be decomposed and mapped to a framework of security and privacy controls that can be used to assess and manage risk.

Identity and Access Management

Data sensitivity and privacy of information have become increasingly an area of concern for organizations. The identity proofing and authentication aspects of identity management entail the use, maintenance, and protection of PII collected from users. Preventing unauthorized access to information resources in the cloud is also a major consideration. One recurring issue is that the organizational identification and authentication framework may not naturally extend into a public cloud and extending or changing the existing framework to support cloud services may prove difficult.

Software Isolation

High degrees of multi-tenancy over large numbers of platforms are needed for cloud computing to achieve the envisioned flexibility of on-demand provisioning of reliable services and the cost benefits and efficiencies due to economies of scale. Regardless of the service model and multi-tenant software architecture used, the computations of different consumers must be able to be carried out in isolation from one another, mainly through the use of logical separation mechanisms.

Data Protection

Data stored in a public cloud typically resides in a shared environment collocated with data from other customers. Organizations placing sensitive and regulated data into a public cloud, therefore, must account for the means by which access to the data is controlled and the data is kept secure. Similar concerns exist for data migrated within or between clouds.

Value Concentration

Having data collocated with that of an organization with a high threat profile could also lead to a denial of service, as an unintended casualty from an attack targeted against that organization. Similarly, side effects from a physical attack against a high profile organization's cloud-based resources are also a possibility. For example, over the years, facilities of the Internal Revenue Service have attracted their share of attention from would-be attackers.

Data Isolation

Database environments used in cloud computing can vary significantly. Accordingly, various types of multi-tenant arrangements exist for databases. Each arrangement pools resources differently, offering different degrees of isolation and resource efficiency. Regardless of implementation decision, data must be secured while at rest, in transit, and in use, and access to the data must be controlled.

Data Sanitization

The data sanitization practices that a cloud provider implements have obvious implications for security. Sanitization involves the expunging of data from storage media by overwriting, degaussing, or other means, or the destruction of the media itself, to prevent unauthorized disclosure of information. Data sanitization also applies to backup copies made for recovery and restoration of service and residual data remaining upon termination of service.

In a public cloud computing environment, data from one consumer is physically collocated (e.g., in an IaaS data store) or commingled (e.g., in a SaaS database) with the data of other consumers, which can complicate matters. Service agreements should stipulate sufficient measures that are taken to ensure data sanitization is performed appropriately throughout the system lifecycle.

Encryption

Client end-to-end encryption (e.g., encryption/decryption occurs on the law enforcement controlled client prior to data entering the cloud and decryption occurs only on the client device after encrypted data is removed from the cloud service) with cryptographic keys managed solely by law enforcement would prevent exposure of sensitive data.

- May cause significant cloud service functionality limitations on available service types made available for sensitive data. This may also increase expenses to cover key items, such as key management and client software. Additionally, a number of specific SLA or contract clauses may be necessary for the implementation of client end-to end encryption.

Use of cloud services without end-to-end encryption implemented by the client is another option that would require cloud service provider participation in the encryption of data.

- This would require at least some cloud provider personnel to undergo personnel background screening and training.
- Specialized Service Level Agreements (SLA) and/or contractual clauses would be necessary to identify those personnel that may have access to unencrypted, sensitive data.
- Conducting the analysis and gaining approval of particular cloud service implementations not utilizing end-to-end encryption for sensitive law enforcement data may be costly and time consuming due to the high degree of technical complexity.

Availability

In simple terms, availability is the extent to which an organization's full set of computational resources is accessible and usable. Denial of service attacks, equipment outages, and natural disasters are all threats to availability. The concern is that most downtime is unplanned and can impact the mission of the organization. Some examples of unplanned service interruptions that cause concerns are:

- Temporary Outages
- Prolonged and Permanent Outages
- Denial of Service

Incident Response

The complexity of a cloud service can obscure recognition and analysis of incidents. Revising an organization's incident response plan to address differences between the organizational computing environment and a cloud computing environment is an important, but easy-to-overlook prerequisite to transitioning applications and data.

Data Availability

The availability of relevant data from event monitoring is essential for timely detection of security incidents. Cloud consumers are often confronted with extremely limited capabilities for detection of incidents in public cloud environments. The situation varies among cloud service models and cloud providers. For example, PaaS providers typically do not make event logs available to consumers, who are then left mainly with event data from self-deployed applications (e.g., via application logging). Similarly, SaaS consumers are completely dependent upon the cloud provider to provide event data such as activity logging, while IaaS consumers control more of the information stack and have access to associated event sources.

Incident Analysis and Resolution

An analysis to confirm the occurrence of an incident or determine the method of exploit needs to be performed quickly and with sufficient detail of documentation and care to ensure that traceability and integrity is maintained for subsequent use, if needed (e.g., a forensic copy of incident data for legal proceedings). Issues faced by cloud consumers when performing incident analysis include lack of detailed information about the architecture of the cloud relevant to an incident, lack of information about relevant event and data sources held by the cloud provider, ill-defined or vague incident handling responsibilities stipulated for the cloud provider, and limited capabilities for gathering and preserving pertinent data sources as evidence. Understanding and negotiating the provisions and procedures for incident response should be done before entering into a service contract, rather than as an afterthought.

General Recommendations:

A number of significant security and privacy issues were covered in the previous subsections. Table 1 summarizes those issues and related recommendations for organizations to follow when planning, reviewing, negotiating, or initiating a public cloud service outsourcing arrangement.

Table 1: Security and Privacy Issue Areas and Recommendations

Areas	Recommendations
Governance	• Extend organizational practices pertaining to the policies, procedures, and standards used for application development and service provisioning in the cloud, as well as the design, implementation, testing, use, and monitoring of deployed or engaged services. • Put in place audit mechanisms and tools to ensure organizational practices are followed throughout the system lifecycle.
Compliance	• Understand the various types of laws and regulations that impose security and privacy obligations on the organization and potentially impact cloud computing initiatives, particularly those involving data location, privacy and security controls, records management, and electronic discovery requirements. • Review and assess the cloud provider's offerings with respect to the organizational requirements to be met and ensure that the contract terms adequately meet the requirements. • Ensure that the cloud provider's electronic discovery capabilities and processes do not compromise the privacy or security of data and applications.
Trust	• Ensure that service arrangements have sufficient means to allow visibility into the security and privacy controls and processes employed by the cloud provider, and their performance over time. • Establish clear, exclusive ownership rights over data. • Institute a risk management program that is flexible enough to adapt to the constantly evolving and shifting risk landscape for the lifecycle of the system. • Continuously monitor the security state of the information system to support on-going risk management decisions.
Architecture	• Understand the underlying technologies that the cloud provider uses to provision services, including the implications that the technical controls involved have on the security and privacy of the system, over the full system lifecycle and across all system components.

Areas	Recommendations
Identity and Access Management	• Ensure that adequate safeguards are in place to secure authentication, authorization, and other identity and access management functions, and are suitable for the organization.
Software Isolation	• Understand virtualization and other logical isolation techniques that the cloud provider employs in its multi-tenant software architecture, and assess the risks involved for the organization.
Data Protection	• Evaluate the suitability of the cloud provider's data management solutions for the organizational data concerned and the ability to control access to data, to secure data while at rest, in transit, and in use, and to sanitize data. • Take into consideration the risk of collating organizational data with that of other organizations whose threat profiles are high or whose data collectively represent significant concentrated value. • Fully understand and weigh the risks involved in cryptographic key management with the facilities available in the cloud environment and the processes established by the cloud provider.
Availability	• Understand the contract provisions and procedures for availability, data backup and recovery, and disaster recovery, and ensure that they meet the organization's continuity and contingency planning requirements. • Ensure that during an intermediate or prolonged disruption or a serious disaster, critical operations can be immediately resumed, and that all operations can be eventually reinstituted in a timely and organized manner.
Incident Response	• Understand the contract provisions and procedures for incident response and ensure that they meet the requirements of the organization. • Ensure that the cloud provider has a transparent response process in place and sufficient mechanisms to share information during and after an incident. • Ensure that the organization can respond to incidents in a coordinated fashion with the cloud provider in accordance with their respective roles and responsibilities for the computing environment.

G.4 Mobile Appendix

Mobile Appendix

Introduction

Mobile devices present a unique security challenge with regard to the correct application of CJIS Security Policy requirements. This appendix is intended to provide best practices based on industry standards and on methods to achieve policy compliance in mobile device employment scenarios. The technical methods used to achieve compliance with CJIS Security Policy will typically be different within the mobile environment than those used in fixed locations. Many of the security features and capabilities inherited by endpoint devices from the fixed environment are either not present or present in a different form in the mobile environment. Additionally, the basic technologies used in some types of mobile devices may adequately fulfill some of the CJIS Security Policy requirements which would require additional software or added features in a traditional fixed computing environment. Due to the complexity and rapid evolvement of the mobile environment, this Appendix will remain as device and vendor agnostic as practical, however certain key requirements for specific mobile operating systems will be identified for the major mobile operating systems (e.g., Apple iOS, Android) as the underlying technologies are fundamentally different and offer different levels of built-in compliance to CJIS Security Policy.

Sections within this appendix will provide recommendations regarding priorities and level of effort versus value of applying certain security controls in the mobile environment. These recommendations do not supersede or modify the requirements listed in the CJIS Security Policy, and are intended to describe the effect of inherent security functions and inherent device limitations in many mobile platforms that impact the application of policy elements in the mobile environment.

Mobile Device Risk Scenarios

There are multiple risk scenarios that may apply to mobile devices depending on the category of device (e.g., Laptop, Tablet, and ‘Pocket sized’ devices such as smartphones) and the methods of device connectivity (e.g., cellular service, Wi-Fi + Cellular, Wi-Fi only). Device category and method of connection define the technology types within the device which inherently affects the total level of compliance with CJIS Security Policy that can be obtained by the mobile device.

It is advisable for acquiring agencies to review the mobile device guidance in this Appendix prior to completing selection and acquisition of particular devices. Both the device category and connectivity methods installed and configured on the device will impact the overall risk scenario associated with the device and may significantly affect the effective cost to bring use of the device in compliance with the CJIS Security Policy. For instance, inclusion of cellular radios with the ability to remotely control a device significantly changes the risk scenario by allowing remote tracking, file deletion, and device management which could provide a higher level of CJIS Security Policy compliance than a Wi-Fi only device that does not guarantee the ability to remotely manage the device. However, inclusion of cellular technology may significantly increase the initial device costs and incur ongoing subscription costs. Appropriate choices based on the intended use of the

device along with the types and methods of Criminal Justice Information (CJI) data to be accessed could greatly reduce agency cost and enhance security.

Device Categories

This appendix defines risk levels for three categories of devices. Prior to reading individual sections of this Appendix, the agency should identify which device categories will apply to their employment scenario. If multiple categories of devices are employed, individual technical configurations and local policy will likely need to be defined for each category of device based on the risk inherent in the technical characteristics associated with each device category.

Laptop devices

The laptop device category includes mobile devices in a larger format that are transported either in a vehicle mount or a carrying case and include a monitor with attached keyboard. This includes all traditional laptop computers that utilize a ‘traditional’, full-featured operating system (e.g., Windows or a Linux variant). Also included in this category are ‘tablet’ type full-featured computers running a traditional full-featured operating system but without an attached keyboard. The main defining factor is the use of a full-featured operating system and a form factor too large to be carried in a pocket. In general, devices of this type connect via Wi-Fi only, but may include an internal cellular access card in some cases.

The risks associated with this device type are similar to a standard desktop computer at the technical level, but are increased due to the potential to connect directly to the internet without the benefit of organizational network security layers (e.g., network firewall, IDS/IPS, network monitoring devices). There is also an increased risk of intentional device theft from vehicles or unsecure locations as these devices are too large to be carried on the authorized user’s body. There may be increased risk from the limited technical ability to wipe or track a lost/stolen device depending on the particular technical means used for remote device connectivity (e.g., cellular or Wi-Fi).

In general, the technical configurations for compliance with most of the CJIS Security Policy that is accomplished via the operating system (e.g., auditing, access control, etc) will remain consistent with normal fixed location computing systems for laptop devices, but some functions may operate in an unexpected manner due to lack of constant connectivity. Thorough testing of applied security policy elements within the expected mobile environments will help ensure the applied policy configurations remain effective and appropriate when applied to mobile laptop devices.

NOTE: Some newer devices running multi-function operating systems (e.g., Windows 8 or similar multi-mode operating systems) may exhibit technical features associated with both laptop and tablet device categories based on their current operating mode which may be reconfigured by the user on demand. If this is the case, it will be necessary to assess and configure multiple operating modes to be compliant with CJIS Security Policy on the device, or restrict the operating mode to one category of operation.

Tablet devices

The tablet device category includes larger format devices transported via vehicle mount or portfolio sized carry case that typically consist of a touch screen without attached keyboard. These devices utilize a limited-feature operating system (e.g., Apple iOS, Google Android, Windows mobile) that is inherently more resistant than a traditional operating system to certain types of network based technical attacks due to the limited-feature sets. Additionally, limited functionality

operating systems are designed specifically for the mobile environment where battery life and power efficiency are primary design drivers. This inherently limits the types of services that can function effectively on the devices (e.g., traditional real-time anti-virus software) as the base operating system may not be designed to allow installed applications enhanced execution priority in the background and or the ability to examine the contents or communications associated within another application. However, this same design methodology significantly limits the vectors available for malware transmission and the device or application data actually accessible to malware if a device becomes infected.

Tablet devices will have different risks associated depending on the installed and configured methods for network access (e.g., ‘always on cellular’ vs. Wi-Fi only). Physical risks associated with this category are similar to the laptop category for enhanced likelihood of intentional theft or device hijacking while unattended, while the technical risks are similar to the pocket device category.

Pocket devices/Handheld devices

The pocket/handheld device category is technically similar or identical to the tablet category and is primarily differentiated by device form factor. Pocket/handheld devices are characterized as having a limited functionality operating system and a small form factor intended for carry in a pocket or ‘holster’ attached to the body. The bulk of this category will be cellular ‘smartphones’ with integrated cellular data connectivity, however devices intended to be worn or carried on the body (e.g., portable fingerprint devices) may also be included in this category if they operate using a limited functionality operating system. Custom or specialty devices may meet the form factor distinction for this category, but operate using a full-feature operating system. In rare cases of this nature the employing agency should apply security guidance and principles in this appendix for both the laptop and pocket device categories.

Risks associated with this category are a reduced threat of theft to a stored devices (e.g., device left unattended in a vehicle) since these devices are typically carried continuously by the authorized user, but include a greater risk of temporary or permanent loss of control due to the device being misplaced by the authorized user.

Due to the installation of a limited functionality operating system, the technical threat to these devices via a network based attack is significantly lower than the laptop category, however, the threat of unauthorized access at the device level may be higher if the device is lost due to technical limits on multi-factor authentication to the operating system itself and practical limits to device passwords due to screen/software keyboard limitations.

NOTE: Data accessible on pocket or tablet devices simply through the entry of a single device PIN or password should not be considered secure due to the likelihood of enhanced password guessing based on fingerprints/smudges on the device touch screen. Any data stored on devices of these types should be protected within a separate secure container using Advanced Authentication.

Device Connectivity

There are three main categories of device connectivity that are associated with varying risk levels and threats to the devices. The Three categories are: Cellular Network Only (always on), Wi-Fi Only (includes ‘on demand’ cellular), and Cellular (always on) + Wi-Fi network. The risks associated with connectivity categories are general risks and may apply differently to any particular device at different points in its usage or lifecycle. Particular device configurations either

through the operating system or a third-party mobile device management (MDM) system may be able to significantly control and define which particular connectivity risks may be associated with a particular device.

Cellular Network Only (always on)

Cellular network connectivity is characterized by ‘always on’ network connection through the device internal radio to a cellular network provider. There is a reasonable assurance that devices with ‘always on’ cellular can be tracked, managed, or wiped remotely if lost or stolen. This will significantly reduce risks associated with loss of the device and attempted illicit access to the device. One important consideration for this risk category is characterization of the device as ‘always on’ or ‘on demand’. In effect the difference is typically a configuration setting, which in some cases may be changeable by the user. In particular most cellular smart phones contain ‘airplane’ mode settings that disable all internal radios allowing a user authenticated to the device operating system via password or personal identification number (PIN) to disable the cellular system. Access to this functionality may be disabled through the use of some MDM systems which would necessitate a complete power down of the device while carried on aircraft. Additionally, someone illicitly obtaining a device with properly configured password requirements and screen lock timeouts would be unlikely to guess the device password before the device was reported stolen in order for them to disable the cellular connection and prevent tracking or a remote wipe of the device.

Cellular networks do not allow for the same level of exposure of individual devices to random access from the internet. This significantly reduces the potential network based attack vectors that might reach a cellular connected device. The risk scenario in most cases from a network based attack would be similar to a device protected behind rudimentary network defenses (e.g., standard firewall but NOT advanced intrusion detection/prevention) Cellular device communications cannot typically be accessed by other ‘eavesdropping’ devices physically close to them without significant specialized equipment and can be considered well protected against network attacks below the nation/state level of technical capability by the hosting technical infrastructure and technology inherent in the device. However, network based attacks that utilize connections initiated by the user device may still succeed over the cellular infrastructure. For this reason, the technical protections inherent in the cellular infrastructure provide limited protection against user/device initiated actions (e.g., web surfing on a cellular connected web browser). Therefore, the protections provided by always on cellular connections are primarily in the ability to remotely access the mobile device for tracking or data deletion in case of device loss or compromise, which combined with a limited functionality device operating system, the protections are generally equivalent to a ‘personal firewall’ if properly configured and supported by a well-designed organizational infrastructure. However, that equivalency does not apply to full-featured operating systems connected through cellular infrastructure.

NOTE: It should be noted that a technically capable, intentional, thief knowingly obtaining an ‘always on’ cellular device for the purpose of data theft can physically disable the radio by utilizing a Faraday cage or similar external electromagnetic shield device while attempting to guess the device password. While technically possible these methods require specialized equipment and high technical expertise and would be very unlikely to be employed except for specifically targeted attacks. When always on cellular connectivity is combined with a robust incident reporting process and user training for rapid response to device loss or theft, the associated risks can be minimized.

Wi-Fi only (includes 'on-demand' cellular)

Wi-Fi only devices do not include cellular radios or include cellular radio that must be manually activated or 'connected' to the cellular network. They connect to the network or internet through Wi-Fi 'hotspots' or external access points or manually to cellular networks. Some MDM or device configurations may be able to limit the types and specific Wi-Fi access points the device can connect to, which may change the risk scenario of the device to a similar risk scenario as the Cellular Network Only scenario. However, if mobile devices are permitted (through technical and or policy decisions) to connect to any Wi-Fi access point designated by the device user, the overall device risk scenario is high and the device may be accessible to a large number of potential network based attack vectors. Unrestricted Wi-Fi access is not recommended on any agency owned device, but must be assumed to exist on any personally owned device authorized to access CJI. Significant compensating controls may be needed to ensure devices accessing CJI over 'public' Wi-Fi access points are not susceptible to communications network eavesdropping, credential hijacking or any of the various potential man-in-the-middle attacks possible through access point spoofing. The communications security risks can be significantly mitigated by mandatory device configurations (e.g., MDM based policy) that only allow devices to connect to cryptographically verified agency controlled Wi-Fi access points.

Wi-Fi only or devices with 'on-demand' cellular access (e.g., user or event driven cellular access initiated from the device and not from a centralized management location) are significantly more at risk from data loss subsequent to device loss or theft as there is no guarantee the tracking or remote wipe can be initiated once the device is out of agency control. This can be mitigated by utilizing tracking/anti-theft products that require a periodic network connection to authorize access and perform automated device locking ('bricking') or remote wipe if network connections are not made within a specified period. Software of this nature is generally available for full-featured laptops but may not be available for limited-feature mobile operating systems.

Cellular (always on) + Wi-Fi Network

This is a hybrid scenario that has become typical with most 'smartphones'. These devices contain both the always on cellular connection, but may also be configured to access local Wi-Fi networks for enhanced bandwidth. In considering devices with these technical characteristics, the theft/loss risks are similar to the cellular only scenario (due to tracking and remote access through the cellular connection), while the data and network based risks must be considered to be similar to the Wi-Fi scenario unless the capability of the device to connect to Wi-Fi networks is limited by technology or policy to agency owned Wi-Fi Access Points configured in accordance with the CJIS Security Policy. Careful consideration must be made to the particular configurations, management systems, and human oriented operational policies based on the particular technical capabilities and configurations of these types of devices.

Incident Handling (CJIS Security Policy Incident Response (IR))

Additional or enhanced incident reporting and handling procedures will need to be developed to cover mobile device operating scenarios. Various exploits and methods to compromise mobile devices require either specialized equipment or lengthy operations to implement. Rapid response to mobile device related incidents can significantly mitigate the risks associated with illicit data access either on the device itself or within online data resources associated with the device through an application or specialized interface. However, parallel or special incident handling procedures with associated equipment or systems may need to be put in place to properly respond to incidents

involving mobile devices. This section lists three areas where enhanced incident handling and response processes may need to be implemented to ensure mobile device compliance to the incident handling policy in the Incident Response (IR) control family.

If personally owned devices are utilized within the environment in a Bring Your Own device (BYOD) scenario, specialized and costly incident handling procedures and processes may need to be developed to support compliance for those devices. The costs associated with enhanced incident handling procedures may need to be incorporated in the cost and risk based analysis to allow personally owned devices in the BYOD scenario, as the technical methods and risk to achieve compliance under BYOD scenarios may exceed any cost savings potentially achieved through BYOD.

Loss of device Control

Mobile device users should be trained and provided with explicit user actions in case positive control of a mobile device is lost for any period of time. Loss of positive control means the device is in the physical control of non-CJIS authorized individual or the device is left unattended in an unsecure location (e.g., counter of the coffee shop). Even if the device is recovered quickly there is significant risk that either the device settings could be tampered with or data on the device could be illicitly accessed. The level of detail and particular scenarios identified in the agency incident response plan should be consistent with the presence of persistent CJI on the device or the technical means used to access CJI from the device (e.g., ask the question: “Is it reasonable to assume CJI could be accessed”) as well as the degree of device configuration control exercised by the user from the device main login. At a minimum, special incident handling procedures should be developed for the following scenarios:

- Device known to be locked, control loss of minimal duration
- Device lock state unknown at time of control loss, duration of loss minimal
- Device lock state unknown at time of control loss, duration of loss extended
- Device known to be unlocked at time of control loss, duration of loss more than momentary.

NOTE: Organizations should define appropriate time value criteria based on the operational environment for the above scenarios. For instance, a ‘momentary’ loss of control might be considered a matter of seconds in a situation where no one could reasonably have accessed the device, while ‘minimal’ durations might include a few minutes of time and ‘extended’ periods would be any time longer than a few minutes.

Other scenarios should be addressed as appropriate to the intended device employment, with explicit user and organizational actions identified based on the device technologies and any organizational management capabilities.

Total Loss of device

Incident response scenarios for the total loss of the device should be developed based on the methods/storage of CJI on the device, the lock state of the device at time of loss (known locked, known unlocked, or unknown), and the technical methods available for remote tracking or wiping of the device. It is critical to implement incident handling procedures quickly in this case. Remote wipe functions can be implemented for always on cellular devices with a high potential for success that may include positive confirmation from the device that the wipe was completed. However, for

Wi-Fi only and on demand cellular devices, incident handling procedures that lock the device out of accessing CJI may be necessary, while there would be no guarantee that any CJI stored on the device could not eventually be accessed. For this reason, CJI should not generally be stored directly on Wi-Fi only or on-demand cellular devices unless an extremely robust anti-tamper system is in place on the device itself.

Potential device Compromise (software/application)

Incident response scenarios for potential device compromise through intentional or unintentional user action should be developed to ensure compliance with policy. This includes rooting, jailbreaking or malicious application installation on the device during a loss of device control scenario or inappropriate user action in the installation of applications to the device (compromise can occur from either intentional threat agent actions or accidental user actions). Triggers for this incident handling process may be driven from either user notification or electronic detection of device tampering from an audit or MDM compliance check.

Audit and Accountability (CJIS Security Policy Audit and Accountability (AU))

The ability to implement some Audit and Accountability functions specified in the CJIS Security Policy on mobile devices with limited function operating systems (e.g., Android, Apple iOS) is not natively included within the operating system. Either additional device management systems, enterprise mobility management (EMM) or MDM, or auditing from systems accessed by the mobile device will be necessary to ensure appropriate levels of auditing exist.

Auditable Events (AU-2 Event Logging)

Some of the specific audit requirements in the CJIS Security Policy may not be technically relevant to the mobile operating system due to its internal functioning. To achieve compliance with the CJIS Security Policy it will be necessary in most cases to utilize some form of MDM or EMM system. Additional auditable events that compensate for the technical limitations of limited function mobile operating systems may be available through the use of MDM systems (e.g., association of event with global positioning system (GPS) location of the device). Specific auditable events of interest in the mobile environment will depend on the intended device usage, compartmentalization of data on the device, and options available with the specific technologies employed. For instance, item 2 in AU-2(c)(2) indicates an auditable event includes attempts to modify elements of user account modification. Due to the limited internal functions of mobile operating systems, this event type is not relevant to the operating system itself as they are generally provisioned with only a single non-modifiable user account on the device. To achieve compliance in a scenario where CJI is stored or accessed from a secure application on the device, auditing of access to the secure application either through application design, or third party MDM capability may provide an acceptable compensating control. For compliance with the policy each auditable event and event content must be compared to the particular technologies and applications employed to determine if adequate compensating controls are being met for audit items that either do not apply to mobile technologies or cannot be implemented within the technology itself.

Alternative and compensating controls that provide detailed audit of access to CJI either on the mobile device itself or through a controlled application to a central server may provide equivalent auditing capability to the events specified in the policy. However, multiple auditing systems may be required to replicate the auditing provided at the operating system level by a full function operating system. Therefore, the overall auditing design should take into account retrieval and

consolidation of events or audit data from multiple auditing systems as appropriate to comply with policy.

Audit Event Collection

Mobile devices without an ‘always-on’ cellular connection may pose technical challenges to ensure any audit records collected and stored on the mobile device itself can be retrieved for review and analysis per the CJIS Security Policy. Alternatively systems which explicitly require a network connection to a central server to access data or decrypt on-device storage may provide acceptable audit event collection and reporting since there is a guarantee that network connections must be in place for CJI to be accessed. Careful consideration should be made regarding the accessibility of audit records when developing the mobile audit scheme.

Access Control (CJIS Policy Access Control (AC))

Access control associated to limited functionality mobile operating systems will typically operate in a different manner than full function operating systems. For instance there is normally not a provision for multiple user accounts on many mobile operating systems which may mean the policy requirements for access control (e.g., regarding account management) would not be apply to the mobile operating system, but should rather be applied to a particular application, either stand-alone to the device or as part of a client server architecture. Application of access control policy identified in the CJIS Security Policy will often need to be applied to elements of the total system beyond the device operating system.

For example, CJI stored or accessed from a secure mobile application that requires connectivity to a CJIS authorized server architecture could potentially accomplish most or all of the access control policy elements based on user authorization via the secured application and be largely independent of the mobile operating system. Alternatively, if storing CJI in ‘general’ purpose data storage containers on a mobile device it may not be possible to achieve compliance with the CJIS Security Policy. Careful consideration and deliberate design of mobile applications or data storage will be required to achieve compliance on mobile devices.

Due to the inherent nature of limited function mobile operating systems, very tight access controls to specific data is actually implemented within the operating system. This effectively prevents applications from accessing or manipulating data associated with other applications to a very high degree of confidence as long as the device is not rooted or jailbroken. However, the device user is automatically granted access to all device data through the associated application unless the application itself has a secondary authentication and access control methodology. Additionally, since basic device functions (e.g., phone) are typically protected using the same password or PIN as the device level encryption, use of a weak PIN to allow easy access to basic device functions largely negates the value of the integrated device encryption.

If personally owned devices are utilized within the environment (BYOD scenario), specialized and costly access control methods may be required to reach compliance with CJIS Security Policy. The costs associated with enhanced access control procedures and technologies should be incorporated in the cost and risk based analysis to determine whether or not to allow personally BYOD, as the technical methods and compensating controls required for CJIS Security Policy compliance are likely to exceed any potential cost savings for implementing BYOD.

Device Control levels and access.

Limited function mobile operating systems are typically very constrained on the levels of access provided to the user. However, intentional user actions (e.g., installing an application and accepting inappropriate security access levels for that application) may bypass some of the built in security protections inherent in the limited functionality devices. Compliance with CJIS Security Policy may be difficult without the addition of strict device control policy. In a mixed environment (e.g., agency owned devices and BYOD), access control policy with BYOD systems may be impractical or impossible to fully implement.

Embedded passwords/login tied to device PIN.

Limited function mobile operating systems typically allow the association of multiple passwords and access credentials with particular applications. The system access provided by these embedded credentials will often be tied to the device password or PIN. An example would be access to device integrated email and calendar applications. Alternatively a 'corporate' email application may independently encrypt the data associated with the application and required a separate login from the device itself. Access to CJI utilizing only the device level password or PIN and device embedded credentials is not compliant with CJIS Security Policy unless protected with Advanced Authentication, which is not currently possible on most devices. Therefore, use of integrated device functions (e.g., built in email or chat) to store or transmit CJI would also not be compliant.

Access requirement specification

In general, due to weaknesses associated with password guessing based on analysis of fingerprints or swipes on the device touch screen, short (4-8 digit) device PIN numbers provide limited security to a determined password guessing attack. Conversely, utilization of a robust password at the device level may be inconsistent with quick access to basic device functions (e.g., phone). When developing specific CJIS compliant access control and authentication schemas a layered approach with the device PIN protecting only the basic device functions (e.g., phone, camera, non-secure applications) and a more robust password or multifactor authentication used to protect applications or data storage may achieve policy compliance where the device password/PIN would not. In a layered security deployment, careful attention must be placed on the capability to share data (e.g., cut and paste or screenshot functions) between secure applications with CJI or CJI access and basic device functions with limited security controls.

Special Login attempt limit

Depending on the access and authentication scheme applied to the mobile device, it may be appropriate to fully comply with the CJIS login attempt limits within a secure application or container and not solely at the device level. However, the device itself should have login attempt limits consistent with the risk associated to the data or configurations accessible on the device itself. Since mobile devices are inherently portable, and can easily be removed from a location. Brute force attempts to gain access to the system, especially when protected only by a short PIN, are likely to be successful given sufficient time. Special consideration should be made based on device connectivity methods (cellular, Wi-Fi, etc) on the appropriate number of unsuccessful login attempts that will be allowed and the resultant actions taken by the device. Most devices either natively allow for the device to wipe itself after a failed number of attempts, or allow the application of EMM/MDM applications to perform wiping actions after a predetermined number of failed login attempts.

Login failure actions

Mobile devices with or without MDM software can typically be configured to perform actions based on serial unsuccessful login attempts. Appropriate actions to configure may be dependent on the data resident on the device and the connectivity method employed by the device. Most devices can be configured to delete all data on the device and/or issue an alert to the network if a number of incorrect passwords are entered. This is a very advantageous feature, however specific configuration of the number of attempts and resultant action must be considered against the state of the device after an unsuccessful attempt action is triggered. A full device wipe will typically leave the device in a fully or partially non-functional state which could introduce risk if part of the intended use is time critical phone calls. Where possible, full device wipe associated with unsuccessful attempts at the device level password should be configured but the number of invalid attempts may exceed the CJIS Security Policy at the device level if all CJI on the device is protected by an additional layer of encryption protected by a subsequent secure application authentication method that is technically prevented (via complexity rules or entry rules) from being the same as the device level authentication and the secure application is configured in accordance with the policy and also contains a secure data wipe capability after a specified number of incorrect authentication attempts.

System use Notification (CJIS Policy AC-8 System Use Notification)

Agency policy should include specific mandatory language consistent with the CJIS Security Policy to identify the device restrictions and consent. However, due to screen size limits, some mobile devices may not be technically capable of displaying the full text used with traditional operating systems. To achieve compliance agencies should contact their legal department for appropriate wording of a short version of the system use notification that can be set to display within the constraints of the device lock screen. This may be accomplished through embedding the text into an image displayed on the lock screen or some other external device labeling method if the device does not permit sufficient text to be displayed.

In a BYOD environment or mixed (agency owned and BYOD), it may be necessary to develop or deploy custom applications that can achieve compliance with the system use notification upon access and prior to any CJI access being allowed.

Session Lock (CJIS Policy AC-11 Device Lock)

Due to the portable nature of mobile devices the session lock limit in the general CJIS Security Policy may be excessive in the mobile environment for certain device functions and insufficient for other functions based on intended device usage. Agencies should examine the minimum lock time practical for all mobile devices based on their employment scenario and ease for which a user can manually lock the device. The actual session lock times should be adjusted as appropriate to the device type, device operational location, and the data accessible on the device when unlocked. Pocket size devices are at greatest risk if screen lock times are insufficient, however, for devices used in emergency response or communication, extended lock times at the basic device level may be considered if CJI is subsequently protected by an application or web interface utilizing more stringent secure locking functions. A well designed solution may include multiple session lock settings at the device and individual application levels to ensure the CJIS Security Policy requirements are met for CJI access, while other device functions are accessible under different session lock configurations.

Device Wi-Fi Policy

Specific Wi-Fi configuration policy should be developed based on the intended use environment and data access requirements for the device. The policy should explicitly cover configuration of device connections. Technical methods specific to the mobile technologies may need to be implemented to ensure all mobile devices are compliant with CJIS Security Policy. Current CJIS Security Policy provides detailed configuration requirements for Wi-Fi connections, however it was originally intended for defining requirements for fixed infrastructure Wi-Fi (802.11) supporting wireless within a facility. The security requirements identified for fixed infrastructure installations are applicable to mobile usage, however there are several mobile specific scenarios where the requirements may not be clear. The following sections identify areas not specifically covered in the existing policy that will require special handling to ensure wireless connections are compliant.

Hotspot capability

Many mobile devices now include the capability to activate an internal Wi-Fi hotspot that allows other devices to connect through the hosting device to the internet over the devices cellular radio. While this is a potentially valuable capability when multiple law enforcement devices may need localized internet or network access, mobile hotspots should be configured as consistent with the CJIS Security Policy on wireless access points. Connections must only be accepted from known and approved devices in order to protect the integrity of the hosting device as well as the communications security of other connected devices. Since most mobile hotspots are not technically capable of providing the device authentication required for infrastructure wireless, use of mobile hotspot capability should assume the overall portable Wi-Fi network itself is not secure and CJI should not be transmitted or exposed on the network without appropriate encryption.

Connection to public hotspots

There are significant risks to connecting to public wireless access points. Rogue access points masquerading as legitimate public access points may allow for man-in-the-middle, eavesdropping, and session hijacking attacks. While not specifically prohibited in the current CJIS Security Policy, it is recommended that connection to public internet access points be technically restricted by device configuration or MDM systems if possible. CJI access mechanisms from mobile devices should include robust authentication methods specifically designed to prevent interception or hijacking of CJI or user information through the use of a rogue access point masquerading as a legitimate public wireless access point. Transmission encryption alone may not provide sufficient protections when device connections originate at public hotspots. Since the public hotspot controls access to all network services at the connection point (e.g., Domain Name System) attacks against the transmission path are possible that would not normally be feasible in a fixed environment where communications exist between two secured network enclaves.

Cellular Service abroad

If mobile devices are used outside of the United States, especially if connected to foreign cellular networks, specific handling procedures may need to be developed for the use of the device while abroad and the assessment or configuration check of the device state once the devices are returned to the United States. Certain device internal functions on cellular devices may be modified or compromised by the cellular carrier as the devices are intended to have certain parameters configured by the cellular service provider which is considered a 'trusted' entity by the device.

Cellular carriers within the United States are constrained by United States laws regarding acceptable modifications to devices. Similar legal constraints cannot be assumed to exist in some areas of the world where laws and regulations for data and personal privacy may allow cellular carriers significantly more leeway in changes made to devices on their networks.

Security plans involving cellular connected devices that will be connected to foreign cellular networks should include technical and policy controls to ensure device use while abroad, data resident on the device while abroad, and the software integrity of the device once returned to the United States are all appropriate to the specific device and threat levels associated with the expected foreign travel. This should explicitly include considerations for devices in which an internal subscriber identity module (SIM) card is inserted into the device to obtain Global System for Mobile (GSM) cellular connections abroad to ensure any residual data on the SIM card is properly purged. Additionally, incident handling procedures may need to specify more stringent responses to even momentary loss of device control, and it may not be possible to assume tracking, anti-theft, and remote data wipe functions that work in the United States would be functional in all potentially visited geographic and political regions.

Bluetooth

Mobile devices utilizing Bluetooth should be evaluated for their ability to comply with the CJIS Security Policy Bluetooth requirements prior to acquisition. This includes the data device itself and any authorized Bluetooth accessories which will be associated to the device. While the technical security in current versions of Bluetooth is significantly stronger than legacy versions, mis-configuration of devices can still pose a significant threat in the mobile environment. If not specifically utilized for a required purpose, it would likely be most cost effective to disable or restrict the device Bluetooth radio utilizing device configurations or an MDM product. Additionally, the using agency may need to develop technically extensive training or user awareness programs to ensure use of Bluetooth capability does not render the device out of compliance if device users have the ability to make Bluetooth associations to the device. Specific instructions or guidance for specific devices could be developed to ensure all implementations are compliant.

Voice/Voice over IP (VoIP)

Cellular voice transmissions are distinctly different at the technical level than Voice over IP (VoIP) transmissions using voice/video applications (e.g., FaceTime, Skype). The use of VoIP is not specifically granted the exemption identified in CJIS Security Policy Section 5.20.1.2.2. Agencies wishing to use capability of this type should ensure the specific technical implementation complies with the Policy on authentication and data encryption.

Chat/Text

Device integrated chat/texting applications and many common third party chat applications authenticate and are identified using embedded passwords or the device identifier only. These functions should not be considered secure or appropriate for transmission of CJI data. Texting functions that utilize a cellular service providers Short Message Service (SMS) or Multimedia Messaging Services (MMS) functions do not constitute a secure transmission medium. Third party applications utilizing appropriate encryption and authentication methods independent of the device password/PIN may provide a compliant solution where the device integrated utilities are will not provide a compliant solution.

Administrative Access

Local administrative access to the mobile device, regardless of device category should be restricted by some mechanism. For traditional operating systems, configuration of a separate administrative account other than that used for normal logins to the device is an acceptable method to ensure appropriate access permissions to the mobile user for which they are authorized. However for limited functionality mobile operating systems (e.g., Android, Apple iOS) internal permissions and accounts assume a single authorized device user with explicitly defined permissions. Those permissions may be modified through policy applied to the device, but are typically global to the device itself. As a result, to ensure appropriate separation of access permissions, it may be required to ensure specific applications or software on the device are configured with individual authentication methods to separate application data from ‘general user’ access. Without additional authentication at the application level, access to specific application data would be available to any user with the ability to unlock the device. This may be appropriate in some scenarios with a high degree of assurance that the device can only be accessed by a single user, but sufficiently stringent device passwords and short screen lock times may prove problematic for practical use of some device functions. An alternate method to ensure strict separation of ‘routine’ device functions which may be accessed by multiple individuals (e.g., phone function if loaned to someone for a critical call) is to ensure any method used to access or store CJI has a separate and more stringent authentication method configured with rules that make it impossible to use the same authentication credential (e.g., PIN/Password) on both the device authentication and the application or function with access to CJI.

Rooting/Jailbreaking

‘Rooting’ (Android OS) or ‘Jailbreaking’ (Apple iOS) refer to intentional modifications to the mobile device operating system in order to grant the device user or an installed application elevated control that would not normally exist on the device. The security model internal to the various mobile device architectures vary significantly, however the common effect of rooting or jailbreaking the devices is to bypass many or all of the built in security features. The security feature bypass may be universal to all device features and installed applications once completed. Intentionally rooting or jailbreaking mobile devices should be avoided in any scenario as it potentially defeats all built-in data access and segregation controls on the device. Additionally the rooting or jailbreaking process itself has a heightened risk of introducing malicious code as part of the process, and also substantially increases the risk for malware to infect the device through user action. Extreme caution should be used if software is being installed that requires the devices to be rooted or jailbroken for the software or application to function. This is inclusive of purported security software that requires a rooted or jailbroken device to function. For example, on both the Android and Apple iOS platforms, the built-in security features for data access and memory segmentation prevent the effective operation of ‘traditional’ anti-virus and intrusion detection/prevention software. Software or applications purporting to perform these functions but requiring rooting or jailbreaking of the device and may actually accomplish the anti-virus or IDS/IPS function but are also likely to significantly increase the overall risk associated to the device by effectively disabling most or all of the integrated security features. A careful risk-based assessment should be conducted by a trained security professional prior to allowing the operation of any rooted or jailbroken mobile devices regardless of intended use. Significant compensating controls would be required to return a rooted or jailbroken device to minimal compliance with most of the CJIS Security Policy and would likely not be a cost effective approach.

NOTE: There is a distinction between rooting a ‘stock’ Android installation vice the installation of a separately supported secure operating system. There are secure versions of Android available or that can be developed based on the open source Android source code and compiled for installation on a particular hardware device. Installation of a secure, supported mobile operating system that replaces the device original operating system may significantly enhance the security of the device and should not be confused with ‘rooting’ and Android installation. Due to the close integration of operating system security with hardware elements, and the proprietary nature of Apple source code, there are not currently separate ‘secure’ versions of the Apple iOS and it is unlikely they will be developed.

Identity and Authentication

Due to the technical methods used for identity and authentication on many limited functionality mobile operating systems, achieving compliance to CJIS Security Policy may require layering of identification and authentication mechanisms. With the complexity and large number of potential identity and authentication solutions in the mobile environment emphasis must be placed on designing secure identity management and authentication architecture prior to the selection of individual devices or applications. Failure to consider a robust identity and authentication scheme as part of system design or acquisition will significantly increase the risk of subsequent noncompliance with CJIS Security Policy and potential added costs for a remedial solution. Many identity and authentication schemes used by existing commercial applications may make claims that appear to be consistent with CJIS Security Policy Advanced Authentication requirements, however, extreme care must be taken to ensure the actual technical implementation is compliant with policy.

Utilizing Unique device Identification

Some commercial applications and features integrated with some mobile operating systems permit the mobile device to be uniquely identified in a cryptographically robust manner. Any authentication schema that considers the possession of the mobile device as a factor in uniquely identifying and authenticating a CJIS authorized user must also include factors beyond than mere possession of the device. Larger form factor devices that cannot be carried on the person of the authorized user should not rely on possession of the device as an identifying factor, but may still include identifying capability within the device to provide assurance that the device itself is an authorized device. This should still be coupled with multi-factor advanced authentication to the device itself or the application hosting CJI. Coupling unique device authentication with robust advanced authentication of the user provides a high degree of confidence that both the specific device and the operator of the device are correctly identified. Utilizing device unique identification in order to authorize initial connections from the remote device back to the CJI hosting system or enclave provides additional protection to the CJI hosting system to reduce the attack surface of the hosting system and should be considered a good practice, but not in itself an authentication mechanism for the device user.

Certificate Use

One method for uniquely identifying mobile devices is to place part of a public key pair on the device in the form of a public key certificate. While there is value to ensuring the device itself can authenticate to a system supplying CJI, and may provide a critical layer of identification or authentication in a larger scheme, a certificate alone placed on the device should not be considered valid proof that the device is being operated by an authorized CJIS user, only that the device itself is authorized to host CJIS users. Additional user identification and authentication should be used to supplement any device certificate installed. Using a PIN or password separate from the device login to ‘unlock’ the certificate from cryptographic storage within a secure application will provide an additional layer of security and may increase the confidence level the device is being used by the intended user. However, use of public/private key pairs or pre-shared encryption keys can be utilized as part of an architecture to protect against certain session hijacking or man-in-the-middle attacks a mobile device may be susceptible to if connected to public internet connections.

Certificate Protections

Any certificates or cryptographic keys stored on any mobile device should include protections against the certificate or key being extracted from the device. Additionally certificates or other keys stored on mobile devices that grant the device special access or unique identification should be configured for remote wipe on demand or self-deletion based on a number of unsuccessful login or access attempts. Alternatively, methods may be used to revoke or invalidate the unique certificate or keys associated with a device.

Minimum Password/Pin (Reference CJIS Security Policy IA-5(1)(a) Authenticator Management | Authenticator Types | Memorized Secret Authenticators and Verifiers)

The minimum password protections identified in the CJIS Security Policy may not be appropriate for the device PIN/password due to immediate access requirement for some device functions (e.g., phone function) secured by the device PIN/password and the difficulty to enter a complex password under emergency conditions on a small screen. In cases where the risk of a complex password on the device itself is deemed significant, a layered authentication approach may be necessary where CJI or access to CJI is protected via one or more additional layers of access control beyond the device PIN/password. In cases where the CJI or access to the CJI is cryptographically segregated from applications accessible using the device level PIN/Password (e.g., secure application or secure browser vice the built-in browser) the authentication mechanism for the secure application or browser may satisfy the CJIS Security Policy requirements if fully compliant as a stand-alone application.

Configuration Management

Due to the potential for inconsistent network access or monitoring capability on mobile devices, methods used to monitor and manage the configuration of traditional full-featured operating systems may not function properly on limited function mobile operating systems. Configuration Management systems in the mobile environment may be designed in order to duplicate some of the functions typically performed by traditional anti-malware systems that will not function properly on some mobile operating systems.

Mobile Device Management (MDM)/Enterprise Mobility Management (EMM)

MDM and EMM systems and applications coupled with device specific technical policy can provide a robust method for device configuration management if properly implemented. MDM

capabilities include the application of mandatory policy settings on the device, detection of unauthorized configurations or software/applications, detection of rooting/jailbreaking of the device, and many other security policy related functions. In many cases, the most cost effective way to achieve CJIS Security Policy compliance on mobile devices is the selection of MDM or EMM applications and infrastructure appropriate to the mobile operating systems and intended access to CJI on the mobile devices. MDM/EMM functions may be applicable to most of the CJIS Security Policy requirements and allow for significant compensating controls in areas where traditional methods of CJIS Security Policy compliance are not technically feasible. Section 5.20.2 of the CJIS Security Policy specifies the minimum functions required for MDM. However, careful selection of the MDM product will potentially provide a cost effective method for additional areas of compliance in the access, auditing, incident response, authentication, media protection and system integrity sections of the CJIS Security Policy.

Device Backups/Images

Device images and backups provide protection against data loss, but also provide a method to quickly recover a device after damage or potential compromise. Due to an inherently limited ability to access the internal file structure of mobile devices, it can be difficult to easily identify a device compromise or illicit modification of the device. Some device imaging and assessment software may provide a secondary forensic capability, especially if there is intent for the device to be used outside the United States.

Bring Your Own device (BYOD) employment

BYOD environments pose significant challenges to the management of secure device configurations. In many cases it may be impossible to apply effective security that is acceptable to the device owner or it may require extremely costly compensating controls to allow access to CJI on personally owned devices. While allowed by the CJIS Security Policy, agencies are advised to conduct a detailed cost analysis of the ancillary costs of compliance with CJIS Security Policy on personally owned devices when they are approved for use. In some cases, a BYOD user may agree to abide by the same device configurations and limitations as imposed on an agency owned device, but signed user agreements should still be in place to ensure the agency has a legal right to recover or clear the device of all data prior to device disposal or employee termination. In other cases, robust secure applications may provide acceptable levels of compliance in a BYOD environment for limited CJI access but application design and architecture should assume the device itself is un-trusted. If MDM/EMM software capable of detecting rooting or jailbreaking of the device is not installed, any CJIS or data access occurring from the device is at a substantially higher risk of compromise.

Configurations and tests

Common configurations specific to all employed mobile devices should be developed to ensure compliance. Configuration tests should be developed and executed on all versions of mobile devices under all possible connectivity scenarios to ensure CJIS Security Policy compliance under all expected operating conditions. Since mobile devices can expect to operate in different physical and network environments, testing and validating correct security functions is more critical than on fixed computing platforms. Additionally, security functions that function properly on one version of a mobile operating system on a particular device may not function in the same manner even on the same version on a different device or a different version on the same device.

Media Protection

Some mobile device hardware platforms include the ability to add removable storage in the form of memory cards. This function is primarily related to Android and Windows mobile platforms and is intentionally limited on Apple devices, but may be possible through certain application functions. While the Android platform performs robust cryptographic separation of data stores between applications within the 'internal' storage of the device, the Android OS does not provide secure separation of data stores on 'external' storage. Some Android hardware devices include additional storage hardwired inside the device that is classified by the operating system as external storage and the normal separation between applications accessing that storage is not applied. Each potential device considered for acquisition must be assessed regarding specific 'external' media protection requirements which may actually include built-in media or storage.

Protection of device connected media

As a result of the limited protection and encryption capabilities applied to device removable media and SIM cards for cellular provisioning that include onboard data storage, all externally removable media or memory should be handled consistently with the CJIS Security Policy on media protection.

Encryption for device media

While most mobile operating systems have the capability to encrypt internal storage, it may require specific device settings to be enabled. All mobile device storage should meet the encryption requirements identified for media in the CJIS Security Policy. Specific settings may need to be applied to ensure proper encryption is actually employed. Additionally, the device built-in encryption capability is typically tied to the device PIN or password. Depending on the device PIN or password requirements the integrated encryption may be easily bypassed by password guessing and appropriate consideration should be made to ensure additional encryption protected by advanced authentication methods be applied to all CJI.

Physical Protection

Due to small form factors and the fact that mobile devices are often stored in lower security areas and vehicles, physical protection of the devices must be considered in both policy and training. Physical protections will often be the responsibility of the assigned device user and physical protections typically inherited by individual information systems from a secure facility will not be available to mobile devices which will require compensating controls to achieve compliance.

Device Tracking/Recovery

MDM software as well as some integrated mobile operating system functions may allow tracking of stolen or lost devices via 'always-on' cellular data connections and the devices built-in GPS. Device tracking with Wi-Fi only or 'on-demand' cellular access may not be reliable. Enabling device tracking capabilities, while not a replacement for secure storage, could be a compensating control used to substantially reduce overall device risk in some scenarios. Device tracking is not currently required in the CJIS Security Policy but should be applied to agency owned devices where possible as a risk mitigation factor. Enabling of device tracking on personally owned devices in a BYOD environment may raise employee privacy concerns and should be considered only for critical systems with the full knowledge of the employee and concurrence of the legal department.

This is an enhanced risk that must be accepted for BYOD employments and should be considered when allowing BYOD employment. Device tracking is available for both limited function mobile operating systems as well as traditional operating systems installed on laptop devices.

Access to device tracking software or applications within the organization should be controlled with limits and formal processes required to initiate a tracking action. It is advisable to include appropriate clauses in user agreements under what conditions and controls the organization applies to device tracking.

Devices utilizing unique device identification/certificates

Devices utilizing unique device identification or have installed certificates may require additional physical protection and/or additional incident handling steps in case of device loss in order to ensure the device unique identifier or certificate is immediately revoked or disabled. Additional physical protection rules or policy would be appropriate for any device which contains access mechanisms tied to the device.

System and Information Integrity (SI)

Managing system integrity on limited function mobile operating systems may require methods and technologies significantly different from traditional full-feature operating systems. In many cases the requirements of System and Information Integrity of the CJIS Security Policy cannot be met with a mobile device without the installation of a third party MDM or EMM application and supporting server infrastructure.

Patching/Updates

MDM software may provide compliance to the SI-2 Flaw Remediation patch management requirements for particular platforms and software versions. However, devices without ‘always-on’ cellular connections may not be reachable for extended periods of time by the MDM or EMM solution either to report status or initiate patching. Supplementary or manual device accountability methods may need to be implemented to account for devices without persistent connections to ensure their patch and update state is current. Alternatively, some patches or system updates may not be practical over cellular connections and will require connection of devices to a Wi-Fi network. Compliance with CJIS Security Policy requirements through purely technical means may not be practical and considerations should be made for aggressive management of devices through training and mandatory periodic connection of devices to organizationally managed Wi-Fi networks.

TECHNOLOGY NOTE: Apple and Android based devices have different potential issues regarding device operating system updates. Apple maintains support for updating the operating system on Apple hardware for several device generations (typically 3-5 years) and provides a robust mechanism for system updates. However, updates to Android based systems are driven by the individual device manufacturer which may or may not support regular updates to current Android operating system versions. Additionally, different Android device vendors may offer updates/upgrades to the Android operating system on different schedules, which can complicate environments utilizing Android devices from multiple manufacturers.

Malicious code protection/Restriction of installed applications and application permissions

MDM or EMM software will typically allow restrictions on installed applications. One of the few effective attack vectors to compromise mobile operating systems is to manipulate the device user

to install a malicious application. Even though the application may be restricted from accessing other application data, it may have some access to common data stores on the device and access to device functions (e.g., GPS, microphone, and camera) that are undesirable. Unrestricted installation of applications by the device user could pose a significant risk to the device.

Malicious code protection using traditional virus scanning software is technically infeasible on most limited function mobile operating systems that are not rooted or jailbroken. The integrated data and program separations prevent any third party installed program from accessing or ‘scanning’ within another application data container. Even if feasible, power and storage limitations would be prohibitive in the effect on device battery life and storage capacity on most mobile devices. However, the cryptographic separation between applications and effective application virtualization technologies built into common mobile operating systems partially compensate for the lack of traditional virus scanning technologies. Appropriately configured MDM software is capable of checking the installed applications on the device and reporting the software inventory to a central management console in a matter analogous to traditional virus scan detection of unauthorized software. This behavior is analogous to the software inventory performed by anti-virus products and can provide a high degree of confidence that only known software or applications are installed on the device. While it is theoretically possible to bypass the application sandboxing and data segregation protections to compromise a mobile device through the web browser, the attack methods required are significantly more advanced than those required for a traditional full-featured operating system. Malicious code protections on the device web browser can be enforced through the use of a properly protected web proxy which the device is configured to use as a mandatory device policy. The most common method of malicious code installation is enticing the user to manually install the malicious app which can be mitigated on organizational devices using an MDM or other application installation restrictions which prevent the user from installing unauthorized or unknown applications. Mitigation of this issue within BYOD environments may not be possible and will present a significantly enhanced risk to the device.

TECHNOLOGY NOTE: In the particular area of application installation there is a significant difference between the behavior of Apple iOS and Android platforms. Apple cryptographically restricts the way applications will execute on the device and assigns mandatory application permissions when the application code is signed prior to release on the Apple App Store for distribution. Apps on the Apple platform must conform to Apple’s policy on app behavior and cannot exceed their design permissions on access to common device functions once the app has been signed and distributed. However, the Apple method does not typically advertise the precise internal permissions granted to the app to the user prior to installation. At runtime, the app is required to request user permission to access certain device functions, and the user may agree or not agree, which may introduce risk if they are unaware of what they are agreeing to allow. Unsigned or un-trusted apps are cryptographically prevented from executing on non-jailbroken iOS devices. Apple provides a mechanism for organizations to distribute custom apps within an organization with equivalent protections but all receiving devices must have a special certificate installed that will only allow official App Store and the organization custom apps to execute.

Conversely, the Android platform, while also requiring app code signing, allows for self-signed code which can be distributed by means other than an official app store and execute on any Android device. Application permissions are presented to the user once at app installation but ramifications of agreement to certain app permissions may not be obvious to a non-technical user. Permissions

in the Android model require user acceptance of all app requested permissions or the app is denied installation, which can result in unwise user acceptance of excessive permissions in order to gain functionality provided by the app.

On either platform user installation of applications can significantly change the security state of the device. Applications may be able to transmit and receive data or share device common data with other devices over the network or local Wi-Fi or Bluetooth connection. On either platform it is highly desirable to limit allowable applications to a pre-approved pool of apps via MDM or organizational App store structures and device policy. However, the risks associated with uncontrolled app installation is several orders of magnitude greater on Android based devices.

WARNING: Rooted or jailbroken devices are modified in such a manner that the built in protections against malicious code are effectively disabled. A rooted or jailbroken device would require significant and costly compensating controls to achieve compliance.

Firewall/IDS capability

Traditional device or “personal” firewalls as identified in CJIS Security Policy System and Communications Protection (SC) and System and Information Integrity (SI) may not be practical on limited function mobile device operating systems but significant compensating controls are available. By default, mobile device operating systems have a limited number of system services installed and carefully controlled network access. To a certain extent the mobile operating system performs similar effective functions as a personal firewall would perform on a general purpose operating system. Potential compensating controls for personal firewall requirements specified are listed below:

1. Manage Program Access to the Internet: On agency controlled devices with an MDM, limiting the apps installed on the device will effectively perform the same function. Since no software or apps can be installed without MDM approval a robust approval process can effectively ensure internet access is only granted to approved apps. Built-in apps and functions can also be limited on network access by the MDM.
2. Block unsolicited requests to connect to the user device: Default configurations for mobile operating system platforms typically block incoming requests. It is possible to install an app that may ‘listen’ on the network and accept connections, but the same compensating control identified in item 1 will mitigate the likelihood of that occurring.
3. Filter incoming traffic by IP address or protocol: Protocol filtering effectively occurs due to the limited function of the operating system as long as no installed application opens network access ports. The mitigations in 1 effectively compensate for this control as well.
4. Filter incoming traffic by destination ports: Same as 3.
5. Maintain an IP traffic log: This may not be technically feasible on most mobile operating system platforms as maintaining this log would require access to lower level operating system functions that are not accessible unless the device is rooted or jailbroken. However, individual Apps that communicate over the network or accept connections from the network may permit logs of IP traffic associated to that application to be stored.

Spam Protection

Spam guards installed on corporate or organizational email systems may effectively accomplish the spam protection requirements for the CJIS Security Policy on mobile devices if properly configured to block spam before delivery to the device. If no upstream spam guard is installed on the mail server the mobile devices accesses, the device may not have adequate spam protection. Additionally access to internet based email (web mail) would need to be restricted to web mail with appropriate spam and/or antivirus protections to ensure compliance.

Periodic system integrity checks

One method to compensate for the technical infeasibility of traditional anti-virus and malicious code protection is to install an MDM that performs periodic system integrity checks that validate device configuration and status against an approved baseline. Deviations may provide indicators of potential device compromise or mis-configuration.

G.5 Administrator Accounts for Least Privilege and Separation of Duties

Administrator Accounts for Least Privilege and Separation of Duties

PURPOSE:

This appendix is provided to describe industry best security practices for assigning separate administrator accounts to support the concept of Least Privilege.

ATTRIBUTION:

- SANS, “The Critical Security Controls for Effective Cyber Defense”, version 5.0
- NIST SP 800-53, “Security and Privacy Controls for Federal Information Systems and Organizations”, Revision 4 dated April 2013
- NIST SP 800-12, “An Introduction to Computer Security: The NIST Handbook” dated October 1995
- CNSSI-4009, “National Information Assurance (IA) Glossary”, dated April 2010

DEFINITIONS:

Least Privilege – The principle that security architecture be designed to grant individual users and processes only the minimum accesses to system resources and authorizations required to perform their official duties or function.

Separation of Duties – The security principle requiring the division of roles and responsibilities so that a single individual cannot subvert a critical process or function.

SUMMARY:

The implementation of least privilege is accomplished by assigning user or process access to system resources based on operational or business needs. Thus, access is granted to only those resources required to perform assigned duties. For individuals who have multiple roles within the organization requiring varying levels privileges, this assignment of access can be challenging. Often times the agency will assign a single userid to the individual and elevate the privileges for that account based on the different roles. While it may seem logical to allow the user access to all required resources using a single account, security vulnerabilities can be introduced into the system.

Associated with least privilege is separation of duties. This concept aids in maintaining the integrity of the system by preventing the abuse of elevated privileges for making unauthorized changes to the system. This objective essentially requires different individuals to perform separate functions with relation to (primarily) administrative duties. For instance, those with the ability to create and assign user access to system should not be able to access the audit logs that contain the evidence of the account actions.

USER ACCESS AND ACCOUNT MANAGEMENT:

Several factors influence the manner in which an agency implements and manages user access. Many times, the size of the agency and the technical expertise of the IT staff employed by the agency become primary drivers. Larger agencies with a broad base of technically savvy personnel normally have the ability to dedicate resources specifically to the administration and management of user access. This could translate to the use of multiple accounts for a single user performing duties requiring varying levels of access.

Smaller agencies with few or no technically experienced personnel will often assign single user accounts with the highest level of access required by users. Other smaller agencies may go as far as assigning every user an account with elevated privileges so there are no delays or problems requiring intervention by already overburdened system administrators. It is not uncommon for a smaller agency to outsource system administration duties.

Regardless of the size or resources of an organization, each agency should base the process for assigning access to system resources based on their operational requirements and a thorough risk assessment. To mitigate risk for accessing system resources, industry best security practices prescribe those individuals performing duties requiring elevated privileges be assigned a separate userid to be used in the performance of those duties. This account would be separate from a standard user account.

Why are some agencies unwilling to implement controls for least privilege? One common reason/perception is administrative overhead. There is a time factor for a system administrator to create user accounts and configure those accounts correctly based on the user's role. In larger agencies with many employees, this could add up to a significant impact on the system administrator(s) especially if there is a high level of turnover. Resources in some agencies may allow for a single system administrator dedicated strictly for account management. On the other end of the spectrum, in agencies with fewer employees, the impact may be more burdensome. While there are fewer user accounts to manage, a full-time system administrator for account management may not be feasible. Those duties then become shared between a few people or added to the duties of a lone person.

Another reason may be the burden on system administrators to remember multiple userids and passwords. This could result in the user using the same password for each account or the user writing down the credentials for ease of remembrance. Additionally, an administrator could get the credentials mixed up between accounts causing an account lockout. This could then require system administrator intervention to reset or unlock the account.

Some agencies may feel that creating additional accounts reduces system resources. Depending on the size of the agency, this could be a concern. In most cases, the number of individuals that would require a secondary account would be minimal. The impact could be limited to a slight increase in disk space usage on the systems accessed by the system administrators with the separate accounts and perhaps the server housing the account information.

THREATS:

A primary goal of attackers is to gain administrative or root privileges on a network or system. Therefore, protection of credentials with that level of access is a key to preventing unauthorized access. Attackers may use many methods in attempts to gain unauthorized, privileged access to computer or network systems. There are two common techniques that take advantage of improperly managed administrative privileges.

Phishing Attacks

In this first method, consider a small organization with limited system administrative resources. Each user is assigned an account with elevated privileges that allows them to perform a myriad of duties including gaining access to critical system security resources. Because this is the only account the user has, normal non-administrative duties are also performed with administrative rights. While checking their email, the user is fooled into reading a message and opening a malicious attachment. Because the user's account has elevated privileges, malware is now installed on the system with elevated privileges. The malware could now allow the attacker to take over the system and install other malicious software such as key loggers, sniffers, or remote control applications. Other key system resources such as firewalls, routers, switches, or intrusion detection systems are now also compromised.

Password Brute Force Guessing / Cracking

The second method may not be as easy as the first and involves the guessing or cracking of passwords on the part of the attacker. Based on human nature, we tend to develop passwords that are easy to remember and most likely contain some kind of information that is pertinent to us. Some passwords could be easily guessed with a minimal amount of social engineering or fact finding. Consider again an agency that assigns users a single account to perform all duties

including those requiring elevated privileges. A user has created a password that, while meeting the requirements of the CJIS Security Policy, is comprised of easily guessed information about the user. An attacker has previously determined the userid and is now able to begin guessing the password. Upon success, the attacker will have unauthorized access to critical system resources.

MITIGATION:

The first step to implementing least privilege is to create separate user accounts for those individuals that require elevated privileges for their duties. These duties could include system or security administration, reviewing audit logs, backup administration, or configuring network devices (e.g., firewalls, routers). The passwords associated with these accounts should have a higher level of complexity than an account without elevated privileges. By disassociating the access levels required for system administration functions from an individual's "everyday use account", should a password be compromised, access would be limited to that of a user with non-elevated privileges.

Second is to implement procedures to ensure accounts with elevated privileges are used only for those duties requiring the higher level of access. This would mean disabling or blocking access to email, web browsers, and other external facing connections. While technical processes are the preferred method of preventing the misuse of accounts with elevated privileges, written policies can be used in situations where technology does not support that type of account management.

Several governance organizations recognize the importance of the security value of Least Privilege. The Payment Card Industry (PCI) includes requirements in their Data Security Standards (DSS). The National Institute of Standards and Technology (NIST) addresses the concept of Least Privilege in its Special Publication (SP) 800-53 rev. 4. While not considered a governance organization, the System Administration, Networking, and Security (SANS) Institute publishes a list of the top 20 security controls which includes "Controlled Use of Administrator Privileges" at number 12. Although the actual security controls or required implementation may slightly differ, the concept is consistent across the groups. The actual controls from NIST and SANS are included here in this appendix.

NIST CONSIDERATIONS FOR LEAST PRIVILEGE:

NIST Special Publication 800-53 rev. 4 includes controls required for all systems under the Federal Information Security Management Act. The publication specifies the guidance for Least Privilege in the control catalog under the Access Control (AC) family and specifically as AC-6. While the NIST requirements are not enforceable under the CJIS Security Policy, they were the genesis of the Policy and do provide a sound security baseline that can be leveraged by the criminal and noncriminal justice community. AC-6 is a key control having several enhancements which, when

implemented, bolster the overall security of the information system by reducing the risk of compromise through the misuse or misconfiguration of access to system resources.

AC-6 Least Privilege

Control: The organization employs the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.

Supplemental Guidance: Organizations employ least privilege for specific duties and information systems. The principle of least privilege is also applied to information system processes, ensuring that the processes operate at privilege levels no higher than necessary to accomplish required organizational missions/business functions. Organizations consider the creation of additional processes, roles, and information system accounts as necessary, to achieve least privilege. Organizations also apply least privilege to the development, implementation, and operation of organizational information systems. Related controls: AC-2, AC-3, AC-5, CM-6, CM-7, PL-2.

Control Enhancements:

(1) *LEAST PRIVILEGE | AUTHORIZE ACCESS TO SECURITY FUNCTIONS*

The organization explicitly authorizes access to [Assignment: organization-defined security functions (deployed in hardware, software, and firmware) and security-relevant information].

Supplemental Guidance: Security functions include, for example, establishing system accounts, configuring access authorizations (i.e., permissions, privileges), setting events to be audited, and setting intrusion detection parameters. Security-relevant information includes, for example, filtering rules for routers/firewalls, cryptographic key management information, configuration parameters for security services, and access control lists. Explicitly authorized personnel include, for example, security administrators, system and network administrators, system security officers, system maintenance personnel, system programmers, and other privileged users. Related controls: AC-17, AC-18, AC-19.

Control Enhancements:

(2) *LEAST PRIVILEGE | NON-PRIVILEGED ACCESS FOR NONSECURITY FUNCTIONS*

The organization requires that users of information system accounts, or roles, with access to [Assignment: organization-defined security functions or security-relevant information], use non-privileged accounts or roles, when accessing nonsecurity functions.

Supplemental Guidance: This control enhancement limits exposure when operating from within privileged accounts or roles. The inclusion of roles addresses situations where organizations implement access control policies such as role-based access control and where a change of role provides the same degree of assurance in the change of access authorizations for both the user and all processes acting on behalf of the user as would be provided by a change between a privileged and non-privileged account. Related control: PL-4.

(3) *LEAST PRIVILEGE | NETWORK ACCESS TO PRIVILEGED COMMANDS*

The organization authorizes network access to [Assignment: organization-defined privileged commands] only for [Assignment: organization-defined compelling operational needs] and documents the rationale for such access in the security plan for the information system.

Supplemental Guidance: Network access is any access across a network connection in lieu of local access (i.e., user being physically present at the device). Related control: AC-17.

(4) *LEAST PRIVILEGE | SEPARATE PROCESSING DOMAINS*

The information system provides separate processing domains to enable finer-grained allocation of user privileges.

Supplemental Guidance: Providing separate processing domains for finer-grained allocation of user privileges includes, for example: (i) using virtualization techniques to allow additional privileges within a virtual machine while restricting privileges to other virtual machines or to the underlying actual machine; (ii) employing hardware and/or software domain separation mechanisms; and (iii) implementing separate physical domains. Related controls: AC-4, SC-3, SC-30, SC-32.

(5) *LEAST PRIVILEGE | PRIVILEGED ACCOUNTS*

The organization restricts privileged accounts on the information system to [Assignment: organization-defined personnel or roles].

Supplemental Guidance: Privileged accounts, including super user accounts, are typically described as system administrator for various types of commercial off-the-shelf operating systems. Restricting privileged accounts to specific personnel or roles prevents day-to-day users from having access to privileged information/functions. Organizations may differentiate in the

application of this control enhancement between allowed privileges for local accounts and for domain accounts provided organizations retain the ability to control information system configurations for key security parameters and as otherwise necessary to sufficiently mitigate risk. Related control: CM-6.

(6) *LEAST PRIVILEGE | PRIVILEGED ACCESS BY NON-ORGANIZATIONAL USERS*

The organization prohibits privileged access to the information system by non-organizational users.

Supplemental Guidance:

Related control: IA-8.

(7) *LEAST PRIVILEGE | REVIEW OF USER PRIVILEGES*

The organization:

(a) Reviews [Assignment: organization-defined frequency] the privileges assigned to [Assignment: organization-defined roles or classes of users] to validate the need for such privileges; and

(b) Reassigns or removes privileges, if necessary, to correctly reflect organizational mission/business needs.

Supplemental Guidance: The need for certain assigned user privileges may change over time reflecting changes in organizational missions/business function, environments of operation, technologies, or threat. Periodic review of assigned user privileges is necessary to determine if the rationale for assigning such privileges remains valid. If the need cannot be revalidated, organizations take appropriate corrective actions. Related control: CA-7.

(8) *LEAST PRIVILEGE | PRIVILEGE LEVELS FOR CODE EXECUTION*

The information system prevents [Assignment: organization-defined software] from executing at higher privilege levels than users executing the software.

Supplemental Guidance: In certain situations, software applications/programs need to execute with elevated privileges to perform required functions. However, if the privileges required for execution are at a higher level than the privileges assigned to organizational users invoking such applications/programs, those users are indirectly provided with greater privileges than assigned by organizations.

(9) LEAST PRIVILEGE | AUDITING USE OF PRIVILEGED FUNCTIONS

The information system audits the execution of privileged functions.

Supplemental Guidance: Misuse of privileged functions, either intentionally or unintentionally by authorized users, or by unauthorized external entities that have compromised information system accounts, is a serious and ongoing concern and can have significant adverse impacts on organizations. Auditing the use of privileged functions is one way to detect such misuse, and in doing so, help mitigate the risk from insider threats and the advanced persistent threat (APT). Related control: AU-2.

(10) LEAST PRIVILEGE | PROHIBIT NON-PRIVILEGED USERS FROM EXECUTING PRIVILEGED FUNCTIONS

The information system prevents non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.

Supplemental Guidance: Privileged functions include, for example, establishing information system accounts, performing system integrity checks, or administering cryptographic key management activities. Non-privileged users are individuals that do not possess appropriate authorizations. Circumventing intrusion detection and prevention mechanisms or malicious code protection mechanisms are examples of privileged functions that require protection from non-privileged users.

References: None.

Priority and Baseline Allocation:

P1	LOW Not Selected	MOD AC-6 (1) (2) (5) (9) (10)	HIGH AC-6 (1) (2) (3) (5) (9) (10)
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SYSTEM ADMINISTRATION, NETWORKING, AND SECURITY (SANS) CONSIDERATION FOR LEAST PRIVILEGE:

There are many negative factors that affect our cyber lives: massive data loss, intellectual property theft, credit card breaches, and identity theft just to name a few. Cyber defense is rapidly evolving

to address the plethora of challenges we face. Defenders have access to a wide array of resources to combat those wishing to do harm. Ranging from the collection of vast amounts of intelligence data to security standards to training and certifications, security practitioners are well armed.

But can information overload actually worsen the problem? Organizations must decide, hopefully based on risk analysis, how to wade through all available resources and select those best suited to their own operating environment. The threats continue to evolve, the attackers become smarter, and user access more mobile. The cloud beckons and can provide reduced cost and infrastructure at a price of less control and accountability for vital information.

The SANS Institute publishes the “20 Critical Security Controls for Effective Cyber Defense”. This list of controls is the combined result of work by an international community to create, adopt, and support the controls. The components of the community provide insight, tools, information, and solutions into threats and adversaries. This list includes the control titled “Controlled Use of Administrative Privileges”. SANS describes this control as: *The process and tools used to track/control/prevent/correct the use, assignment, and configuration of administrative privileges on computers, networks, and applications.*

Critical Security Control (CSC) 12: Controlled Use of Administrative Privileges

The processes and tools used to track/control/prevent/correct the use, assignment, and configuration of administrative privileges on computers, networks, and applications.

ID #	Description	Category
CSC 12---1	Minimize administrative privileges and only use administrative accounts when they are required. Implement focused auditing on the use of administrative privileged functions and monitor for anomalous behavior.	<i>Quick win (One of the “First Five”)</i>
CSC 12---2	Use automated tools to inventory all administrative accounts and validate that each person with administrative privileges on desktops, laptops, and servers is authorized by a senior executive	<i>Quick win</i>
CSC 12---3	Configure all administrative passwords to be complex and contain letters, numbers, and special characters intermixed, and with no dictionary words present in the password. Pass phrases containing multiple dictionary words, along with special characters, are acceptable if they are of a reasonable length.	<i>Quick win</i>

CSC 12---4	Before deploying any new devices in a networked environment, change all default passwords for applications, operating systems, routers, firewalls, wireless access points, and other systems to have values consistent with administration---level accounts.	<i>Quick win</i>
CSC 12---5	Ensure that all service accounts have long and difficult--- to--- guess passwords that are changed on a periodic basis, as is done for traditional user and administrative passwords.	<i>Quick win</i>
CSC 12---6	Passwords should be hashed or encrypted in storage. Passwords that are hashed should be salted and follow guidance provided in NIST SP 800---132 or similar guidance. Files containing these encrypted or hashed passwords required for systems to authenticate users should be readable only with super---user privileges.	<i>Quick win</i>
CSC 12---7	Utilize access control lists to ensure that administrative accounts are used only for system administration activities, and not for reading e---mail, composing documents, or surfing the Internet. Web browsers and e---mail clients especially must be configured to never run as administrator.	<i>Quick win</i>
CSC 12---8	Through policy and user awareness, require that administrators establish unique, different passwords for their administrative and non---administrative accounts. Each person requiring administrative access should be given his/her own separate account. Users should only use the Windows “administrator” or UNIX “root” accounts in emergency situations. Domain administration accounts should be used when required for system administration instead of local administrative accounts.	<i>Quick win</i>
CSC 12---9	Configure operating systems so that passwords cannot be re--- used within a timeframe of six months.	<i>Quick win</i>
CSC 12---10	Configure systems to issue a log entry and alert when an account is added to or removed from a domain administrators’ group, or when a new local administrator account is added on a system.	<i>Visibility/ Attribution</i>
CSC 12---11	Configure systems to issue a log entry and alert when unsuccessful login to an administrative account is attempted.	<i>Visibility/ Attribution</i>
CSC 12---12	Use multifactor authentication for all administrative access, including domain administrative access. Multi--- factor authentication can include a variety of techniques, to include the use of smart cards with certificates, One Time Password (OTP) tokens, and biometrics.	<i>Configuration/ Hygiene</i>

CSC 12---13 (NEW)	When using certificates to enable multi---factor certificate---based authentication, ensure that the private keys are protected using strong passwords or are stored in trusted, secure hardware tokens.	<i>Configuration/ Hygiene</i>
CSC 12---14	Block access to a machine (either remotely or locally) for administrator---level accounts. Instead, administrators should be required to access a system using a fully logged and non---administrative account. Then, once logged on to the machine without administrative privileges, the administrator should transition to administrative privileges using tools such as Sudo on Linux/UNIX, RunAs on Windows, and other similar facilities for other types of systems. Users would use their own administrative accounts and enter a password each time that is different than their user account.	<i>Configuration/ Hygiene</i>

Quick win: Implementation provides significant risk reduction without major financial, procedural, architectural, or technical changes to an environment, or that provide substantial and immediate risk reduction against very common attacks that most security-aware organizations prioritize these key controls.

Visibility / attribution: Measures to improve the process, architecture, and technical capabilities of organizations to monitor their networks and computer systems to detect attack attempts, locate points of entry, identify already-compromised machines, interrupt infiltrated attackers' activities, and gain information about the sources of an attack.

Configuration / hygiene: reduce the number and magnitude of security vulnerabilities and improve the operations of networked computer systems, with a focus on protecting against poor security practices by system administrators and end-users that could give an attacker an advantage.

SEPARATION OF DUTIES:

Separation of duties is another security control related to least privilege. Many of the same challenges faced by least privilege apply to this concept as well. Agency size and resources play a major in the implementation of separation of duties. As the name implies, some key functions should be separated between different individuals. The goal of this concept is to provide protection against a single individual's ability to circumvent system security controls to gain unauthorized access or perform unauthorized actions without colluding with other individuals.

Simply put separation of duties entails distributing certain critical mission oriented functions or system administrative support functions amongst different individuals or roles. It also includes

delineating information system support duties such as auditing, configuration control, or network security between different individuals.

As with least privilege, an agency's ability to implement separation of duties is typically based on financial and personnel resources. While a very large agency may have ready availability to those resources to ensure critical functions are spread across multiple individuals, a small agency probably does not have that luxury.

THREATS:

What effect can an individual with carte blanc access to all critical functions of a system have? Consider a single individual with the ability to install nefarious applications on a system (e.g., a keylogger). If this same individual also has the ability to edit any audit logs that would have recorded the actions of installing the software, those entries could be deleted and any evidence of the installation eliminated.

Perhaps a disgruntled system administrator wants to open a port on a firewall to allow a remote backdoor connection into the information system in order to siphon off criminal justice information. Because the perpetrator has access to the firewall and all logs, the port can be opened and the logs tampered with to eliminate records of the action.

As mentioned previously, the two concepts of least privilege and separation of duties are related. Additional threats are presented when a system administrator using a single account with unlimited elevated privileges across the information system uses that account to check email. In a successful phishing attack that compromises this account, the attacker now has unrestricted unauthorized access to all system resources and the ability to hide their tracks.

MITIGATION:

The primary method to avoid these situations is to configure system privileges and duties such that a single person is unable to effect questionable change to the system and then are able to erase any evidence of the change.

Technical configurations are most secure and sound enforceable policies compliment the technical solutions. When an information system does not support separating duties, strong policies help mitigate risk.

NIST CONSIDERATIONS FOR SEPARATION OF DUTIES:

NIST Special Publication 800-53 specifies the guidance for separation of duties in the control catalog under the Access Control (AC) family and specifically as AC-5. While the NIST requirements are not enforceable under the CJIS Security Policy, they were the genesis of the Policy and do provide a sound security baseline that can be leveraged by the criminal and noncriminal justice community. AC-5 is a relatively small control with no enhancements, but it is significant in protecting the integrity of an information system.

AC-5 Separation of Duties

Control: The organization:

- a. Separates [*Assignment: organization-defined duties of individuals*];
- b. Documents separation of duties of individuals; and
- c. Defines information system access authorizations to support separation of duties.

Supplemental Guidance: Separation of duties addresses the potential for abuse of authorized privileges and helps to reduce the risk of malevolent activity without collusion. Separation of duties includes, for example: (i) dividing mission functions and information system support functions among different individuals and/or roles; (ii) conducting information system support functions with different individuals (e.g., system management, programming, configuration management, quality assurance and testing, and network security); and (iii) ensuring security personnel administering access control functions do not also administer audit functions. Related controls: AC-3, AC-6, PE-3, PE-4, PS-2.

Control Enhancements: None.

References: None.

Priority and Baseline Allocation:

P1	LOW Not Selected	MOD AC-5	HIGH AC-5
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G.6 Encryption

Encryption

Purpose:

This paper was created to provide assistance and guidance on encryption types, methods, and to provide general best practices in the implementation of encryption.

Attribution:

- FIPS 140 – 2, Security Requirements for Cryptographic Modules (May 2001)
- FIPS 197, Advanced Encryption Standard (Nov 2001)
- NIST SP 800-111, Guide to Storage Encryption Technologies for End User Devices
- CNSSP-15, National Information Assurance Policy on the Use of Public Standards for Secure Sharing of Information among Security Systems
- CJIS Security Policy

Definitions and Terms:

Encryption – A form of cryptology that applies a cryptographic operation to provide confidentiality of (sensitive) information.

Decryption – The inverse cryptographic operation used to convert encrypted information back into a plaintext (readable) format.

Asymmetric Encryption – A type of encryption that uses key pairs for encryption. One key is used to encrypt a message and another key to decrypt the message. Asymmetric encryption is also commonly known as public key encryption.

Symmetric Encryption – A type of encryption where the same key is used to encrypt and decrypt a message. Symmetric encryption is also known as secret key encryption.

Hybrid encryption – A type of encryption where both asymmetric encryption and symmetric encryption keys are used creating what is referred to as cipher suites. In a hybrid solution the asymmetric encryption keys are used for client/server certificate exchange to provide session integrity while the symmetric encryption keys are used for bulk data encryption to provide data confidentiality.

Authorized User/Personnel - An individual, or group of individuals, who have been appropriately vetted through a national fingerprint-based record check and have been granted access to CJI.

Summary:

CJIS Security Policy encryption requirements are intended to provide protection of the sensitive data that is criminal justice information (CJI). The primary goal of encrypting CJI is to prevent unauthorized access to this sensitive data. Encryption is a great tool that can be applied to accomplish this protection and ensure compliance with the vast majority of the CJI requirements. CJIS Security Policy controls SC-13 Cryptographic Protection and SC-28 Protection of Information at Rest details when encryption is required and provides information on the exceptions to the encryption requirement.

Achieving CJIS Security Policy Compliance:

To determine when encryption is required one must first read and understand CJIS Security Policy Physical and Environmental Protection (PE). The reason for this is simple: encryption is not required while within a physically secure location. Conversely, whenever CJI is transmitted or stored (at rest) outside the boundaries of a physically secure location encryption may be required. The exact standards to which the data would be required to meet are detailed along with any exceptions in CJIS Security Policy controls SC-13 Cryptographic Protection and SC-28 Protection of Information at Rest.

Additionally, both security awareness training and personnel security requirements can be affected by whether or not CJI is encrypted. Requirements surrounding these Policy areas is determined by answering the following question: Who has unescorted access to unencrypted CJI?

Unless personnel are escorted, security awareness training is required as correlated with the access level needed by personnel as identified in CJIS Security Policy Awareness and Training (AT). Similarly, fingerprint-based background checks as detailed in CJIS Security Personnel Security (PS) may be required on individuals to permit unescorted access to CJI.

The intent of all these requirements is to limit access to CJI to only authorized personnel. CJIS Security Policy Appendix A: Terms and Definitions defines authorized user/personnel as an individual, or group of individuals, who have been appropriately vetted through a national fingerprint-based record check and have been granted access to CJI.

What is Encryption?

Encryption is the process of encoding messages or information in such a manner that only people with the knowledge or means to decrypt the message can do so. But how does this work?

In an encryption process, legible data, referred to as plaintext, is encrypted by applying a cipher (otherwise known as an encryption algorithm or crypto key) to the data. The data then becomes encrypted and is now referred to as ciphertext. The ciphertext is essentially unreadable until decrypted. The decryption process requires the process of applying the same algorithm (crypto key) to encrypt the data in an inverse manner to convert the data back into plaintext.

Encryption is important because it allows you to securely protect data that you don't want anyone else to have access to. Encryption has been used throughout history to send “secrets” securely by some form of obfuscation to a recipient. Businesses and enterprises use encryption to protect corporate secrets and sensitive employee data, such as payroll information and personally identifiable information (PII). Governments secure classified information with encryption. Additionally, individuals may use encryption to protect personal information, such as credit card data, banking information, and passwords to guard against things like identity theft.

It should be known that encryption may not always prevent the interception of data. If the stolen data is encrypted, though, it would be extremely difficult for any of the data to be decrypted without having the decryption key. While it may be possible to decrypt the message without possessing the key, it does require large computational resources, great skill, and lots of time to accomplish such a task. Exercising encryption along with key management policies is one of the best security practices that can be put into place with regard to sensitive data security and protection.

Types of Encryption:

Symmetric Encryption

Symmetric encryption is also commonly known as secret key encryption. Symmetric encryption is a form of cryptography utilizing a singular encryption key to guise an electronic message. Its data conversion uses a mathematical algorithm along with a secret key, which results in the inability to make sense out of a message. Symmetric encryption is a two-way algorithm because the mathematical algorithm is reversed when decrypting the message along with using the same secret key.

Symmetric encryption is most often used for data protection whether at rest or in transit, especially in bulk, due to the ease and speed with which the encryption can be implemented. The most common examples of symmetric algorithms are: AES and Triple-DES (3DES or TDEA).

How it works:

To encrypt and send a message to Jane, John does the following:

1. Generates a new symmetric key
2. Encrypts the message using this new symmetric key
3. Sends the message to Jane
4. Sends the encrypted symmetric key to Jane - out of band

To decrypt this ciphertext, Jane does the following:

1. Receives the encrypted message
2. Receives the symmetric key
3. Uses the symmetric key to decrypt the message

Asymmetric Encryption

Asymmetric encryption is also commonly known as public-key encryption. Asymmetric cryptography is cryptography in which a pair of keys, a public key and a private key, are used to encrypt and decrypt a message so that it arrives securely. Initially, a network user receives a public and private key pair from a certificate authority. Any other user who wants to send an encrypted message can get the intended recipient's public key from a public directory. They use this key to encrypt the message, and they send it to the recipient. When the recipient gets the message, they decrypt it with their private key, which no one else should have access to.

Creating Key Pairs:

Asymmetric encryption requires the use of algorithms of great computational complexity to create the key pairs. This is accomplished by using a large, random number that an algorithm is applied to which generates a pair of keys for use as asymmetric key algorithms (as shown in Figure 1 below).

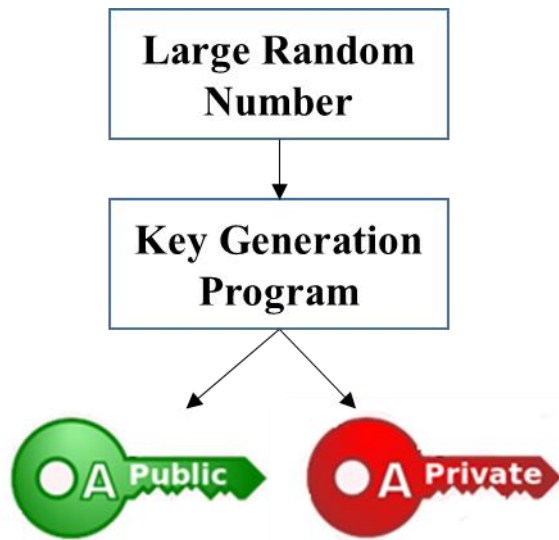


Figure 1 – Asymmetric key pair generation

Asymmetric encryption is most often used to encrypt a single message before transmission. The most common examples of asymmetric algorithms are: RSA and DSA.

How it works:

To encrypt and send a message to Jane, John does the following:

1. Obtains Jane's public key
2. Encrypts the message using Jane's public key
3. Sends the message to Jane

To decrypt this ciphertext, Jane does the following:

1. Receives the encrypted message
2. Uses her private key to decrypt the message

Advantages of Using Symmetric Encryption for Data Protection

Asymmetric encryption requires the use of algorithms with great computational complexity to create the key pairs, and therefore is not practical for large amounts of data. It is typically used for only for short messages. Also, asymmetric encryption must use a comparatively stronger key than symmetric key encryption to achieve the same level of protection as one key (public) will be published in the public directory for all to see.

Symmetric encryption is based on large, but simple algorithms which require less computation. Therefore, is much faster to create and use keys. This allows the same key to be used to encrypt and decrypt the message. So, data can be encrypted in real time. The (shared) key is sent to the recipient out of band so that it can be used to decrypt the data.

For the reasons stated above, symmetric key encryption is the preferred choice by both industry and government alike to encrypt large amounts of data (bulk encryption) simply due to the ease and real time encryption capabilities as detailed above. Additionally, a new key can be generated for every session, message transaction, etc., as desired. This means a sender won't have to use one key (public) to encrypt a message and have the recipient use another key (private) to decrypt the message.

Hybrid Encryption

Hybrid encryption solution exist where both asymmetric encryption and symmetric encryption keys are used to create what is referred to as cipher suites. In a hybrid solution the asymmetric encryption keys are used for client/server certificate exchange to provide session integrity while the symmetric encryption keys are used for bulk data encryption to provide data confidentiality.

Hybrid solutions are most often used by Internet browsers to protect data in transit. The most common examples of hybrid encryption are: TLS/SSL, PGP, IPSEC, and S/MIME.

How it works:

To encrypt a message to Jane in a hybrid cryptosystem, John does the following:

1. Obtains Jane's public key
2. Generates a new symmetric key
3. Encrypts the message using this new symmetric key
4. Encrypts the symmetric key using Jane's public key
5. Sends the message to Jane

To decrypt this hybrid cipher text, Jane does the following:

1. Receives the encrypted message
2. Receives the encrypted symmetric key
3. Uses her private key to decrypt the symmetric key
4. Uses the symmetric key to decrypt the message

Explaining Cipher Suites:

A cipher suite is a set of cryptographic algorithms used for the following:

- Protect information required to create shared keys (key exchange)
- Encrypt messages exchanged between clients and servers (bulk encryption)
- Generate message hashes and signatures to ensure the integrity of a message (message authentication)

Examples of Transport Layer Security (TLS) 1.2 Cipher Suites:

- TLS_RSA_WITH_AES_128_CBC_SHA256
- TLS_RSA_WITH_AES_256_CBC_SHA256

A cipher suite specifies one algorithm for each of the above tasks. For example, the TLS_RSA_WITH_AES_128_SHA256 cipher suite is used for TLS. The suite uses the RSA asymmetric algorithm for key exchange, AES with a 128-bit key for bulk data encryption, and SHA256 for message authentication.

Symmetric and Asymmetric Key Strength Comparison:

<u>Symmetric</u>		<u>Asymmetric</u>		
<u>Bits of security</u>	<u>Symmetric key algorithms</u>	<u>Finite-Field Cryptography (FFC)</u> <u>(e.g., DSA, D-H)</u> <u>Bits of security</u>	<u>Integer-Factorization Cryptography (IFC)</u> <u>(e.g., RSA)</u> <u>Bits of security</u>	<u>Elliptic-Curve Cryptography (ECC)</u> <u>(e.g., ECDSA)</u> <u>Bits of security</u>
<u>80</u>	<u>2TDEA18</u>	<u>Public key = 1024</u> <u>Private key = 160</u>	<u>Key size = 1024</u>	<u>Key size = 160-223</u>
<u>112</u>	<u>3TDEA</u>	<u>Public key = 2048</u> <u>Private key = 224</u>	<u>Key size = 2048</u>	<u>Key size = 224-255</u>
<u>128</u>	<u>AES-128</u>	<u>Public Key = 3072</u> <u>Private key = 256</u>	<u>Key size = 3072</u>	<u>Key size = 256-383</u>
<u>192</u>	<u>AES-192</u>	<u>Public key = 7680</u> <u>Private key = 384</u>	<u>Key size = 7680</u>	<u>Key size = 384-511</u>
<u>256</u>	<u>AES-256</u>	<u>Public key = 15360</u> <u>Private key = 512</u>	<u>Key size = 15360</u>	<u>Key size = 512+</u>

Figure 2 - Symmetric and asymmetric key strength comparison

As you can see in the chart provided above, the equivalent key strengths between symmetric and asymmetric key strengths do not necessarily correlate. There is a reason for this. As stated previously, asymmetric algorithms must use a comparatively stronger key than symmetric key encryption to achieve the same strength. The simplest explanation for this is because one of the keys is published to the public directory and can constantly be attacked by anyone with access to the directory. Therefore, the public key must be made of such strength that it can resist getting compromised while made public.

Federal Information Processing Standard (FIPS) 140-2 Explained

Origin of FIPS 140-2

On July 17, 1995, the National Institute of Standards and Technology (NIST) established the Cryptographic Module Validation Program (CMVP) to validate cryptographic modules to Federal Information Processing Standards (FIPS) Security Requirements for Cryptographic Modules, and other FIPS cryptography based standards. The CMVP is a joint effort between NIST and the Communications Security Establishment Canada (CSEC). FIPS 140-2, Security Requirements for Cryptographic Modules, was released on May 25, 2001 to supersede the original FIPS 140-1. Modules validated as conforming to FIPS 140-1 and FIPS 140-2 are accepted by the Federal Agencies of both countries for the protection of sensitive information.

What is FIPS 140-2?

Federal Information Processing Standard (FIPS) is a standard developed and recommended (often mandated) for use in federal-government-operated IT systems by the following two government bodies:

- The National Institute of Standards and Technology (NIST) in the United States
- The Communications Security Establishment (CSE) in Canada

FIPS 140-2 specifies the security requirements a cryptographic module must meet when utilized within a security system protecting sensitive information within information systems (computer and telecommunication systems). FIPS 140-2 specifies which encryption algorithms can be used and how encryption keys are to be generated and managed.

How does a product get certified?

Vendors of cryptographic modules can have their products tested by independent, accredited Cryptographic and Security Testing (CST) laboratories. The CST laboratories use the Derived Test

Requirements (DTR), Implementation Guidance (IG) and applicable CMVP programmatic guidance to test cryptographic modules against the applicable standards in a variety of implementations. The result of these tests are reported to NIST's Computer Security Division (CSD) and CSEC who jointly serve as the Validation Authorities for the program. These results are then reviewed and certificates would be issued if the results are determined to be acceptable.

What is the difference between being FIPS 140-2 compliant and being FIPS 140-2 certified?

It is common theme to discover a product is “FIPS compliant.” What does this mean, though? The difference between compliance and certification is not subtle. Certification requires a vast testing, verification, and validation process be performed by a CST laboratory as described in the previous section. Compliance is merely a claim stating the implementation of an encryption solution is done in accordance with the security policy related to the FIPS certification. Any claim of compliance would need to be validated and the corresponding certificate number would have to be known.

NIST has addressed related claims as shown below in their Frequently Asked Questions for the Cryptographic Module Validation Program:

A vendor makes the following claims of conformance to FIPS 140-2. Are they acceptable?

- The module has been designed for compliance to FIPS 140-2. <NO>
- Module has been pre-validated and is on the CMVP pre-validation list. <NO>
- The module will be submitted for testing. <NO>
- The module has been independently reviewed and tested to comply with FIPS 140-2. <NO>
- The module meets all the requirements of FIPS 140-2. <NO>
- The module implements FIPS Approved algorithms; including having algorithm certificates. <NO>
- The module follows the guidelines detailed in FIPS 140-2. <NO>
- The module has been validated and has received Certificate #XXXX. <YES>

A cryptographic module does not meet the requirements or conform to the FIPS 140-2 standard unless a reference can be made to the validation certificate number. The module used must also be the same version/part number as annotated on the validation certificate. Any other claims are not relevant.

To read more FAQs from NIST on FIPS certification, use the following NIST website link:
<http://csrc.nist.gov/groups/STM/cmvp/documents/CMVPFAQ.pdf><http://csrc.nist.gov/cryptval/140-2.htm>

Where can I learn more about FIPS 140-2?

For more information about the FIPS 140-2 standard, go to the following NIST website:
<http://csrc.nist.gov/cryptval/140-2.htm>

General Recommendations:

Encryption key management control is of paramount importance! Agencies should develop policies and procedures define and monitor the administrative tasks involved with protection, storage, organization, access controls and the lifecycle management of encryption keys. After all, encryption keys should not be accessible by just anyone. An encryption key management control process should ensure only authorized users have access to encryption keys. Key management is a best security practice and helps to ensure the confidentiality and integrity of CJI data and enforces key access control.

The CJIS Security Policy is a “living” document under constant review and receiving regular updates through the Advisory Policy Board (APB) process. Agencies need to always keep up to date on the latest requirements. These requirements can be found in CJIS Security Policy controls SC-13 Cryptographic Protection and SC-28 Protection of Information at Rest. Please contact the FBI CJIS ISO Program anytime to address any questions or concerns about CJIS Security Policy requirements, the current APB status of CJIS Security Policy requirements, or if seeking general information or guidance.

G.7 Incident Response

Incident Response

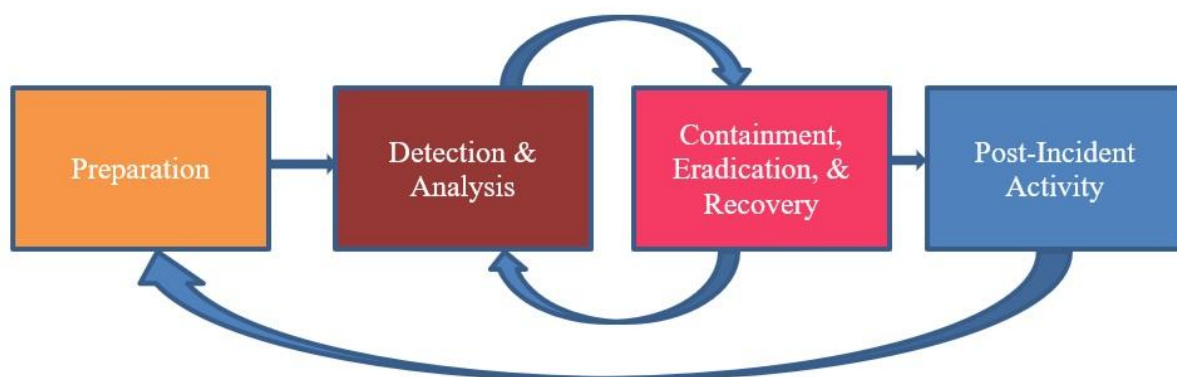
Introduction

Information technology (IT) security incident response is an important and critical component of information technology programs. Performing incident response effectively can be a complex undertaking – for that reason, establishing a successful incident response capability requires planning and resources. Everyone in an organization must be aware of IT security risks, threats, and actions to take in situations where an actual IT security incident has occurred. Even the best-secured and controlled environments can experience these security risks, threats, events, and incidents. This document provides guidelines for appropriate response to IT security incidents, and are independent of specific hardware platforms, operating systems, protocols, or applications.

The following example incidents are used to highlight appropriate actions during each phase:

- Malicious code execution
- Ransomware execution
- Denial of service attack
- Social Engineering
- Phishing

NIST Special Publication 800-61 rev. 2 outlines the “Incident Response Life Cycle” as a collection of phases – distinct sets of activities that will assist in the handling of a computer security incident, from start to finish. The following diagram explains the process flow of the incident response life cycle:



Preparation

The initial phase of the incident response life cycle, “Preparation”, involves establishing and training an incident response team, and acquiring the necessary tools and resources. A computer security incident may not have happened at this phase, but it is important to utilize all available knowledge and security measures to obtain the best posture for responding to potential future incidents. One of the most important preparation steps involves the collection, storage, and accessibility of event data and telemetry from hardware and software resources such as firewall logs, application logs, operating system logs, and other valuable sources of situational data, as well as the output of products that perform analysis on such data. Preventive measures to mitigate or eliminate future incidents are deployed during this phase, using industry best practices, data obtained from research and intelligence sources, and lessons learned from past incidents.

It is also imperative to prepare a list of contact information or notification methodologies to employ when an incident occurs, as well as notification and communication strategies within the team, with stakeholders, and with upper management and potentially other criminal justice and non-criminal justice agencies. This will help ensure that when incidents arise, the proper personnel and organizations are notified and kept informed of the circumstances regarding the incident.

Using the example incident categories outlined earlier, some overview into appropriate actions and activities for the Preparation phase can be given:

Malicious code execution

Preparation for incidents involving malicious code execution should initially involve user awareness of sources of malicious code. There are many potential sources of malicious code, such as web pages, emails, and removable media. The utilization and deployment of effective antivirus software, integrity-monitoring software, and intrusion detection and prevention software are effective measures to take to prepare for incidents involving malicious code execution.

Ransomware execution

Preparation phase activities for incidents involving ransomware execution are much the same as activities for malicious code execution, as ransomware is a specialized form of malware that encrypts potentially important or critical files, with the intention of coercing a victim to pay for a decryption key. Implementing a robust offline backup solution for these types of files is an important preparative action to take regarding the execution of ransomware. This will ensure that when ransomware attacks do happen, the mission impact is as minimal as possible and very little or no data is lost.

Denial of service attack

Denial of service attacks are given attention in the preparation phase. Defensive responses to denial of service attacks typically involve the use of a combination of attack detection

and traffic classification and response tools, aiming to block traffic identified as abusive denial of service activity. Deploying solutions such as IDS/IPS devices and software, network hardware with rate-limiting capabilities (routers, switches, and firewalls), and upstream filtering devices at the system perimeter can mitigate for denial of service attacks.

Social Engineering

Preparation for social engineering attacks starts with user awareness training. Understanding and identifying attempts to obtain information in an unauthorized manner is crucial to thwarting these types of scenarios. Social engineering is the art of manipulating people to obtain information they may not be authorized to handle. Training and routinely testing users on potential social engineering scenarios and tactics, and providing training regarding appropriate responses to requests involving personal or otherwise sensitive information (for example, passwords or criminal justice information), is an effective way to ensure social engineering attacks never traverse past the preparation phase of the incident response life cycle.

Phishing

Like social engineering, preparation for phishing attacks is imperative. Phishing is a social engineering technique attackers employ to deceive users, in a fraudulent attempt to obtain sensitive information, or to gain unauthorized access to systems. Phishing is extremely widespread, and attackers disguising fraudulent scenarios in electronic communication such as email and instant messages are the most common. User awareness of these types of tactics is paramount to prepare for phishing attacks and schemes.

Detection and Analysis

The detection and analysis phase begins when a security incident has occurred. To understand when this phase begins, there must be a capability for an intelligent determination of circumstances constituting a security incident. Specialized knowledge and highly trained personnel are necessary for this step to be effective. Many organizations employ teams of personnel who are specifically trained to handle the intricacies of the incident response life cycle. The determination of a security incident can arise from one or several circumstances simultaneously – for example:

- Trained personnel manually reviewing collected event data for evidence of compromise
- Software applications analyzing events, trends, and patterns of behavior
- The observation of suspicious or anomalous activity on a computer system

The goals of this phase are:

- To detect whether a security incident occurred
- To determine the vector (i.e., method) of attack
- To determine the impact of the incident to the mission, systems, and personnel involved in the incident

- To obtain or create intelligence products regarding attack vectors and methodologies, especially when dealing with malicious code

Prioritization of incidents is also an important decision point in the incident response life cycle, as the circumstances regarding an incident can bring the situation to a critical level. There are three major impacts to consider when addressing priority of incidents:

- **Functional Impact:** the impact to business functionality
- **Information Impact:** the impact to confidentiality, integrity, and/or availability of criminal justice information
- **Recoverability:** the amount of time and resources that must be spent on recovering from an incident

Documentation regarding an incident should be thorough and applicable to the incident. This can be crucial in incidents that may lead to legal prosecution, as well as being invaluable to efficiently document, track, handle, manage, and resolve one or more incidents at the same time.

Using the example incident categories outlined earlier, some overview into appropriate actions and activities for the Detection and Analysis phase are given:

Malicious code execution

Detection of malicious code execution is often a primary job of host-based antivirus software. Having a capable and up-to-date antivirus solution installed on a system can detect known malicious code, as well as detect potentially malicious behaviors. The delivery of malicious code to a system can be detected by network traffic analysis and protection tools and hardware. Additionally, some malicious code may produce network traffic that is indicative of successful execution, exploitation, and/or compromise of a system. Solutions such as intrusion detection/prevention systems, Security Information and Event Management (SIEM) tools, and file integrity monitoring software can provide the necessary level of fidelity to make a determination of malicious code execution.

Knowing if or when a system is infected is not always immediately evident. Security controls may have been bypassed or even disabled by the malicious code. However, systems infected by malicious code or software (i.e., malware) can exhibit several indicators. These indicators include, but are not limited to:

- Unexpected pop-up windows
- Slow start up and/or slow performance
- Suspicious hard drive activity including an unexpected lack of storage space
- Missing files
- Crashes and/or error messages
- Unexplained network activity
- Hijacked email

Analysis of malicious code can be performed in several ways. Static analysis of malicious code can be performed to determine the capabilities of the malicious code and generate actionable intelligence. Dynamic analysis of malicious code can be used to observe how the malicious code interacts with the system and what actions it performs and can often more rapidly determine the capabilities of malicious code. Both static and dynamic analysis can be performed manually, as well as in an automated fashion. Trained specialized personnel are crucial to the analysis of malicious code.

Ransomware execution

The detection of ransomware is identical to the detection of malicious code. Ransomware is specialized malicious code that encrypts potentially valuable files, generally with the intent to coerce a victim to pay a ransom for the possibility of the decryption of those files. Host-based antivirus solutions can also detect these threats, and network traffic analysis and protection tools and hardware can be used to prevent the successful execution of ransomware. SIEM tools and file integrity monitoring software can also detect the execution of ransomware.

Analysis of ransomware is identical to the analysis of malicious code, and the same intelligence can be determined in the same fashion as with the analysis of malicious code. The most obvious sign that ransomware has affected a system is the existence of encrypted files, the disappearance of certain types of files, and/or the presence of “ransom notes” on the system, which contain instructions for payment to obtain a decryption key, which may or may not be legitimate.

Denial of service attack

Denial of service (DoS) attacks are often detected at the perimeter of an organization but can also be detected within the organization as well. Often, from a user’s perspective, the signs of a DoS attack appear to be network performance or administrative maintenance related issues such as slow or broken network connections or down websites. Additionally, an administrator may notice ping time outs, event logs overflowing or alerts from network monitoring systems as issues that may identify a DoS attack. Intrusion detection and prevention software and platforms can detect denial of service attacks, as well as some network monitoring hardware and appliances, such as web application filters, routers, firewalls, and switches. Devices targeted by denial of service attacks can also detect the attacks in some instances, if they have the capabilities to determine explicit attack activity versus normal network traffic.

Analysis of denial of service attacks include the determination of the source traffic, the protocols used to generate the traffic, the service(s) targeted by the attack, and the potential impacts of the attack. Network monitoring devices can often provide these types of data, with the exception of potential impacts of denial of service attacks on systems.

Social Engineering

Detection of social engineering attacks is primarily based on the situational awareness of the individual targeted by social engineering. Given that social engineering is a broad topic that can involve the manipulation and exploitation of people in control of an information system, user awareness of social engineering attempts is crucial. If the target has security awareness training in detecting attempts to gain information or access in an unapproved manner, social engineering is easier to detect.

Analysis of social engineering attacks will generally rely on the recollection abilities of or documentation taken by the targets of the attack. Social engineering may not occur on an information system and may be completely carried out in-person. If the target can recollect or produce documentation regarding the social engineering attempt, the motivation and desired access can potentially be determined. For successful social engineering attempts, recollection and documentation of the attempt is crucial to determining the level of unauthorized access that was obtained.

Phishing

Detection of phishing attacks generally will first occur at an organization's email point of presence. Some organizations still run their own email servers, and many have migrated to cloud solutions. Having an on-premise email server or server farm or cluster will require additional functionality to detect phishing attempts. For example, the header content of the email will need to be read, as well as the content inside the body of the email, to check for potentially malicious content and potentially falsified data that may indicate a phishing email. Many cloud email providers have built this capability into their email solutions, but it is still possible for users to receive phishing emails, as attacker tactics and capabilities evolve daily. The most effective detection of phishing comes from heightened situational awareness of potential attacks. Validating the source of the email can uncover potential phishing attempts.

Analysis of phishing attacks involves examination of email headers, as well as contents of the body of the email. The body of the email may contain malicious content, attachments, or links to suspicious or malicious content. Manual or automated analysis activities can be performed on the email content. Analysis of these elements should be performed by trained specialized personnel to generate intelligence and aid with the determination of indicators of compromise.

Containment, Eradication, and Recovery

Containment activities for computer security incidents involve decision-making and the application of strategies to help control attacks and damage, cease attack activities, or reduce the impact or damage caused by the incident. Often, this requires intelligence gathered by the detection

and analysis phases of the incident – for example, identification of affected hosts, identification of attacking hosts or attackers, identification of malware and its capabilities, and identification and monitoring of attacker communication channels can be invaluable to the implementation of containment activities. In most cases, it is important to introduce containment solutions all at once, as attackers may escalate their attack activity if deployment of the strategy is delayed.

Eradication efforts for a computer security incident involve removal of latent threats from systems (such as malware on the system and user accounts that may have been created), identifying and mitigating potential vulnerabilities or misconfigurations that may have been exploited, and identification of other hosts that may have been affected within the organization.

Recovery efforts for incidents involve restoration of affected systems to normal operation. This may include actions like restoring systems from backups, rebuilding systems from an agency-approved baseline, replacing compromised files with clean versions, installing patches, changing passwords, and increasing network perimeter and host-based security.

Compromised hosts are often attacked during these phases, as attackers try to regain their foothold on compromised systems or systems on the same network or others in the logical vicinity.

Malicious code execution

Containment activities for malicious code execution involve the logical or physical isolation of the host from the attacker's control and from any mission services or systems that would be impacted by the compromised host. This may include putting the host in a restricted VLAN, using firewalls to block traffic, disconnecting it from the network completely, shutting it down, or disabling functionality. Exercise caution as malicious code may have capabilities to take further actions on a host in case communications with a command and control server are severed. It is important to understand the capabilities of the malicious code before taking containment actions.

Eradication activities include the removal of malicious code from the system. This may be as simple as removing files, configuration rules, accounts, and other persistent items that the malicious code utilizes to function and maintain a presence on the system. This phase also involves the discovery and removal of indicators of compromise on other systems, if applicable. It is imperative to remediate vulnerabilities that may have been exploited during eradication as well.

Recovery from malicious code execution generally is similar across many environments. Rebuilding the system from a clean baseline or restoring files from backup are typical activities that help restore the functionality of the system to continue the mission. Changing system passwords, installing patches, implementing tighter network access control, and

ensuring appropriate levels of logging fidelity of the information system are integral parts of the recovery process.

Ransomware execution

Containment for ransomware execution should be as swift and immediate as possible, as ransomware can execute and spread to accessible media at a rapid pace. Considering files are being encrypted or have already been encrypted, immediate action should be taken to logically or physically isolate the system by disconnecting network connectivity. It is up to the system owner whether to take the risk in powering off the system, as valuable forensic artifacts may be destroyed in the process, but it will halt the execution of the ransomware and protect potentially valuable files. Please note that containment of active ransomware execution is one of the only circumstances where measures such as immediate shutdown are recommended.

Eradication of ransomware does not need to occur in most circumstances, as the entire goal of ransomware is to encrypt files and leave “recovery” instructions to extort victims. The vast majority of ransomware will delete itself once encryption of files is complete, but it is possible that some ransomware is persistent and can remain on the system. If this is the case, analysis should be performed on the ransomware to determine its capabilities, and eradication activities will proceed in an identical fashion to malicious code execution eradication activities.

Recovery from ransomware execution involves restoring encrypted files from backup and may involve the rebuilding of an entire system depending on the extent of the encryption from the ransomware. If a robust offline backup solution for hosts is not present or not utilized on a regular basis, the loss of potentially valuable data may be incredibly costly in several areas to repair, to include man-hours, revenue, and business products, data, and intelligence.

Denial of service attack

Containment of denial of service attacks involve the modification of access control where the attack is occurring. For example, if a web or application server is experiencing a denial of service attack, the system itself, as well as network monitoring devices, should be examined to determine the source of the attack traffic. Once the source of the traffic is identified, modifications to access controls or rate-limiting features such as firewall access controls lists (ACLs) and web application filters can be employed to block the traffic. Care must be taken to determine if the observed traffic is actually intentional malicious denial of service traffic, versus heavy legitimate network traffic. Implementing access control mechanisms or rate-limiting features may negatively affect the mission of the system. It is also important to note that manual containment in this fashion may not be entirely effective, as attackers can circumvent the ACL by changing the attacking IP address, protocol, or other attribute of the connection.

Eradication is not necessarily applicable in denial of service scenarios, unless a vulnerability or misconfiguration is being exploited to cause the denial of service condition. If this is the case, take steps to remediate the vulnerability or misconfiguration.

Recovery actions depend on the available resources of the information system. For example, on-premise load balancers can be used to distribute the traffic, whether legitimate or malicious, to other less-burdened systems. Many cloud providers and content delivery networks also have denial of service mitigation capabilities. It may also be prudent to increase the resources (memory, processing capacity) of internet-facing systems so that they can handle larger amounts of traffic simultaneously.

Social Engineering

Containment regarding social engineering attacks is dependent upon the information or access that was provided to the attacker. For example, if an attacker gained access to an account on a system following a social engineering attempt, the account should be administratively disabled and all sources of event data regarding that account should be immediately collected. If sensitive data was divulged to the attacker, the impact of the exposure of that data should be examined and mitigating activity should be initiated to determine or reduce the damage of the spread of the information.

Eradication regarding social engineering attacks also depends on the information or access provided to the attacker. Removing or limiting the provided access is a pertinent eradication action. If the information provided is a credential to a system, disable and remove the credential from the system. Eradication may also involve the physical detainment or removal of personnel from a site.

Recovery actions for social engineering attacks are dependent on the information or access provided to the attacker. Additionally, security awareness training is an appropriate recovery action to ensure staff understands the threats of social engineering.

Phishing

Containment of phishing activity is tied very closely to the identification and analysis of the phishing activity. Understanding the tactics of the phishing attacker is paramount to deploying containment activities. Activities include, but are not limited to, administratively blocking sender email addresses and IPs, blocking potential malicious content in email via a web proxy, communicating with potential recipients, and implementation of email content or hyperlink blacklisting if possible. Phishing attacks can also include attempts to have users execute malicious code on systems, where containment activities regarding malicious code will be applicable.

Eradication of phishing attacks include the administrative removal of the emails from email systems, as well as eradication actions for malicious code if applicable.

Recovery from phishing attacks can include:

- Implementation and enforcement of the Domain Keys Identified Mail (DKIM) email authentication method, which can mitigate the possibility that attackers can send spoofed email
- Implementation and enforcement of Sender Policy Framework (SPF) to control and stop sender forgeries
- Implementation and enforcement of Domain-based Message Authentication, Reporting, and Conformance (DMARC), which enables message senders to indicate that their messages are protected with SPF and/or DKIM

Additionally, if malicious code is present in the phishing attack, recovery actions regarding malicious code may be applicable.

Post-Incident Activity

Post-incident activities occur after the detection, analysis, containment, eradication, and recovery from a computer security incident. Arguably one of the most important phases of incident response, post-incident activities involve the reflection, compilation, and analysis of the activities that occurred leading to the security incident, and the actions taken by those involved in the security incident, including the incident response team. Some of the important items to consider:

- Exactly what happened, and at what times?
- How well did staff and management perform in dealing with the incident?
- What information was needed sooner?
- Were any steps or actions taken that might have inhibited the recovery?
- What would the staff and management do differently the next time a similar incident occurs?
- How could information sharing with other organizations have been improved?
- What corrective actions can prevent similar actions in the future?
- What precursors or indicators should be watched for in the future to detect similar incidents?
- What additional tools or resources are needed to detect, analyze, and mitigate future incidents?

Smaller incidents, and those that are similar to others that have been well documented, do not necessarily need much focus in this phase of incident response. Larger and less-understood security incidents should be the focus of a comprehensive post-mortem evaluation that outlines

many of the items listed above and should include personnel that can have a direct impact on or are directly affected or responsible for the involved systems.

Post-incident activities such as these also help to serve as training opportunities for all parties involved in the incident, from victims, to system administration personnel, to incident responders.

Malicious code execution

Post-incident activities for malicious code execution generally will follow similar patterns. A timeline of activity should have been prepared using digital forensic data collected during the detection and analysis phases of the incident. This timeline should include all affected systems and times of all activities and actions taken during the incident. Steps that victims and system administrators may have taken during the course of the incident, as well as in close proximity to the time range of the incident, are valuable items to document and discuss. Any deviation from organizational policy should be noted and taken as training items or assigned consequences in accordance with organizational policies. It may also be pertinent to ensure that appropriate information and intelligence sharing was performed during and after the incident occurred. Corrective actions that may have prevented the execution of malicious code, such as antivirus solutions, restrictions on where executables can run, tightened permissions, and script blockers for browsers, should be considered as a mitigation for the risks posed by malicious code threats. Web proxy blocks from information discovered during analysis can be utilized to ensure that malicious hosts are not contacted.

Ransomware execution

Post-incident activities for ransomware execution include all the activities involved with malicious code execution, with the addition of ensuring the functionality of a robust offline backup solution. An offline backup solution ensures that backup data is kept inaccessible to ransomware threats and is available if ransomware is successfully executed. A functional and frequent (such as daily incremental and weekly full) backup process helps ensure that business continuity is maintained in the event of issues and incidents.

Denial of service attack

Denial of service post-incident activities should include a timeline of traffic activities, as well as organizational responses to the attack traffic as well as the timeline of any business impacts and the damage associated with the impacts. Any attack precursors should be investigated and noted, and intelligence implemented to notify personnel and potentially take action as soon as attack traffic is observed. Impacts on affected systems should be noted, and a consensus should be reached on whether the systems should be upgraded or supplemented with load-balancing capabilities.

Social Engineering

Post-incident activities for social engineering incidents should include a timeline that includes all applicable activities, such as points of contact, narratives from the parties involved, CCTV footage (if applicable), system and network log files, and physical access control logging data. If unauthorized access was obtained, the impact of the access should be assessed and mitigating factors should be identified for inclusion to reduce the risk of future incidents (such as multifactor authentication, physical locks, greater CCTV coverage, improved physical access control, etc.). Security awareness training should be imperative if policy was breached, and information or access was given to unauthorized parties.

Phishing

Phishing post-incident activities should also include a timeline of actions taken since the phishing email was received, to include descriptions of the type of phishing campaign observed (malicious code, financial exploitation, credential harvesting, etc.), malicious attachments contained (if any), malicious or suspicious links in the body of emails, as well as narratives from recipients of the email and any potential victims, either self-reported or discovered through email, network, or host-based monitoring. If malicious code was included in the campaign, typical post-incident activities involving malicious code should be considered as well. Training opportunities can often arise from phishing attacks, whether successful or not, that can be valuable in giving employees better situational awareness regarding phishing.

The CJIS Security Policy requires each agency with access to CJI to establish operational incident handling procedures (i.e., a local policy). Gleaning from the requirements in the Incident Response (IR) control family, the local policy may include the following elements:

- Overall incident handling procedures. This section describes and identifies the processes used locally how the agency successfully prepares for, manages, and recovers from an incident. It includes sections on:
 - Preparation
 - Detection and Analysis
 - Containment
 - Recovery
 - User response activities
- How the agency performs incident reporting. This section describes the process of notifying internal and external partners when an incident has occurred and how the incident is documented. It includes sections on:
 - Internal and external points of contact
 - Required tracking and reporting documents
 - Escalation procedures
- Incident management procedures. This section describes the agency's approach to a consistent and repeatable approach to managing incidents. It includes sections on:
 - Roles and responsibilities

- Incident-related information collection
- Updating policies with lessons learned
- Collection of evidence
- Incident response training
- Document and artifact retention

G.8 Secure Coding

Secure Coding

This appendix documents a source of information on best practices and standards for secure coding. With the increased use of software products and the rapid pace of modern software development, it is essential to discover and resolve insecure software risks. The mitigations and guidelines to reduce these common risks can be found in secure coding best practices.

Understanding how software applications work can be a daunting thing; however, it could be key to know if data security is in jeopardy. Awareness of secure coding practices allows an agency to review potential vendors and their products prior to purchase and implementation. It also empowers the agency with the knowledge of the questions to ask a vendor of how the software was developed and whether the vendor uses secure coding practices or standards.

Additionally, the information in this appendix can provide a path forward for agencies with the internal capability to produce “in-house” software applications. By implementing security during the code writing process, security is “baked in” and there is more trust the software will aid in protecting the information it processes.

Open Web Application Security Project (OWASP) Foundation

The OWASP Foundation is a not-for-profit charitable organization focused on improving the security of software. OWASP operates as a community of like-minded professionals to provide unbiased and practical information about application security (AppSec) through software tools and documentation. These materials are available under a free and open software license, which can be located at the link below.

https://www.owasp.org/index.php/Main_Page

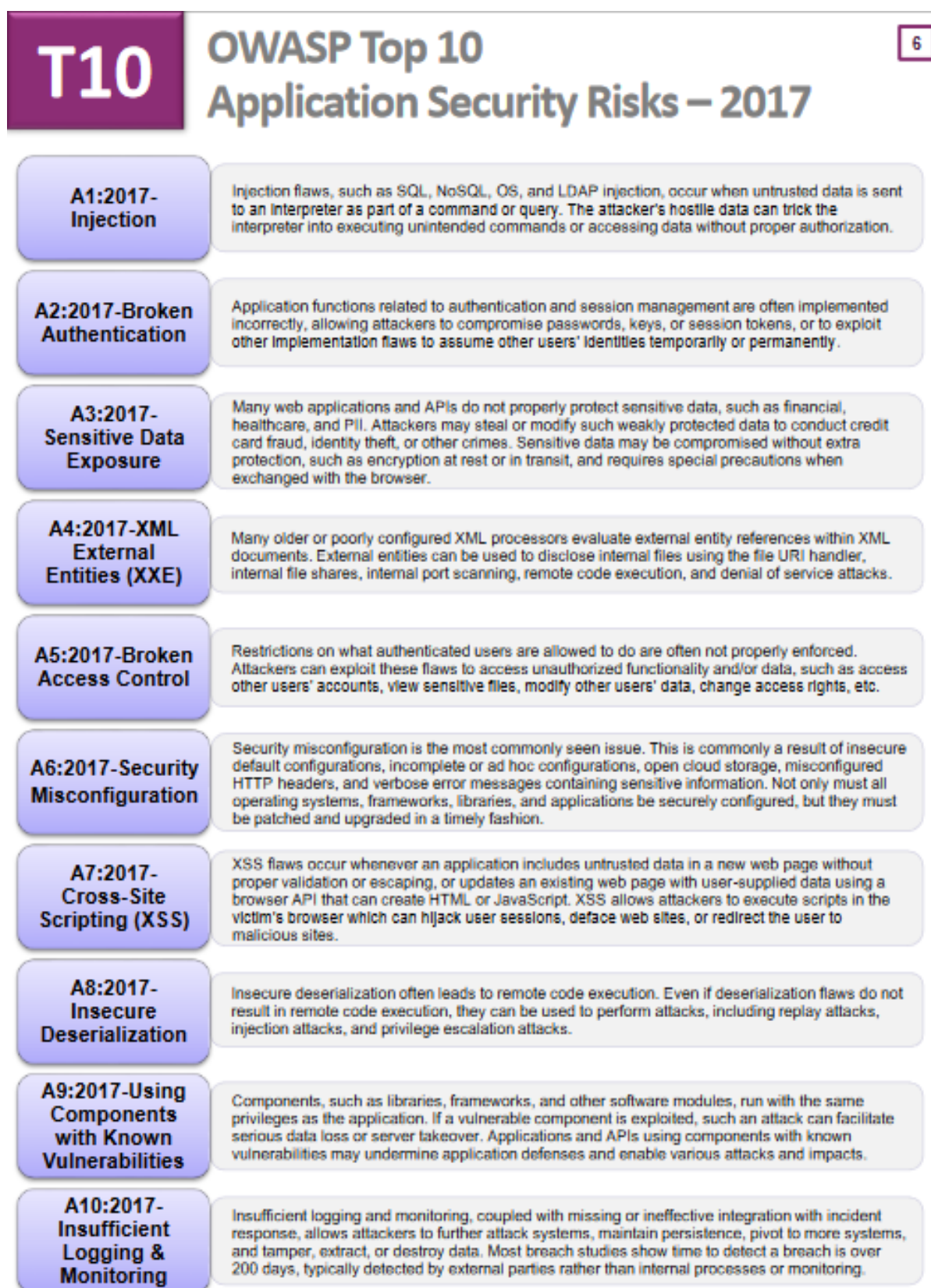
Software is becoming increasingly complex and connected, and the difficulty of achieving application security increases exponentially. The rapid pace of modern software development processes makes the most common risks essential to discover and resolve quickly and accurately.

The OWASP Foundation publishes the Top 10 Application Security Risks, which focus on the most serious web application security risks. The OWASP Top 10 is based primarily on 40 plus data submissions from firms that specialize in application security and an industry survey that was completed by over 500 individuals. This data spans vulnerabilities gathered from hundreds of organizations and over 100,000 real world applications and application program interfaces (API). The Top 10 items are selected and prioritized according to this data, in combination with consensus estimates of exploitability, detectability, and impact.

A primary aim of the OWASP Top 10 is to educate developers, designers, architects, managers, and organizations about the consequences of the most common and most important web application security weaknesses. The Top 10 provides basic techniques to protect against these high risks problem areas, and provides guidance on a path forward.

The OWASP Top 10 focuses on identifying the most serious web application security risks for a broad array of organizations. For each of these risks, generic information about likelihood and technical impact using the following simple ratings scheme, which is based on the OWASP Risk Rating Methodology.

Figure G.8-A



Each organization is unique, and so are the threat actors for that organization, their goals, and the impact of any breach. It is critical to understand the risk to your organization based on applicable threat agents and business impacts.

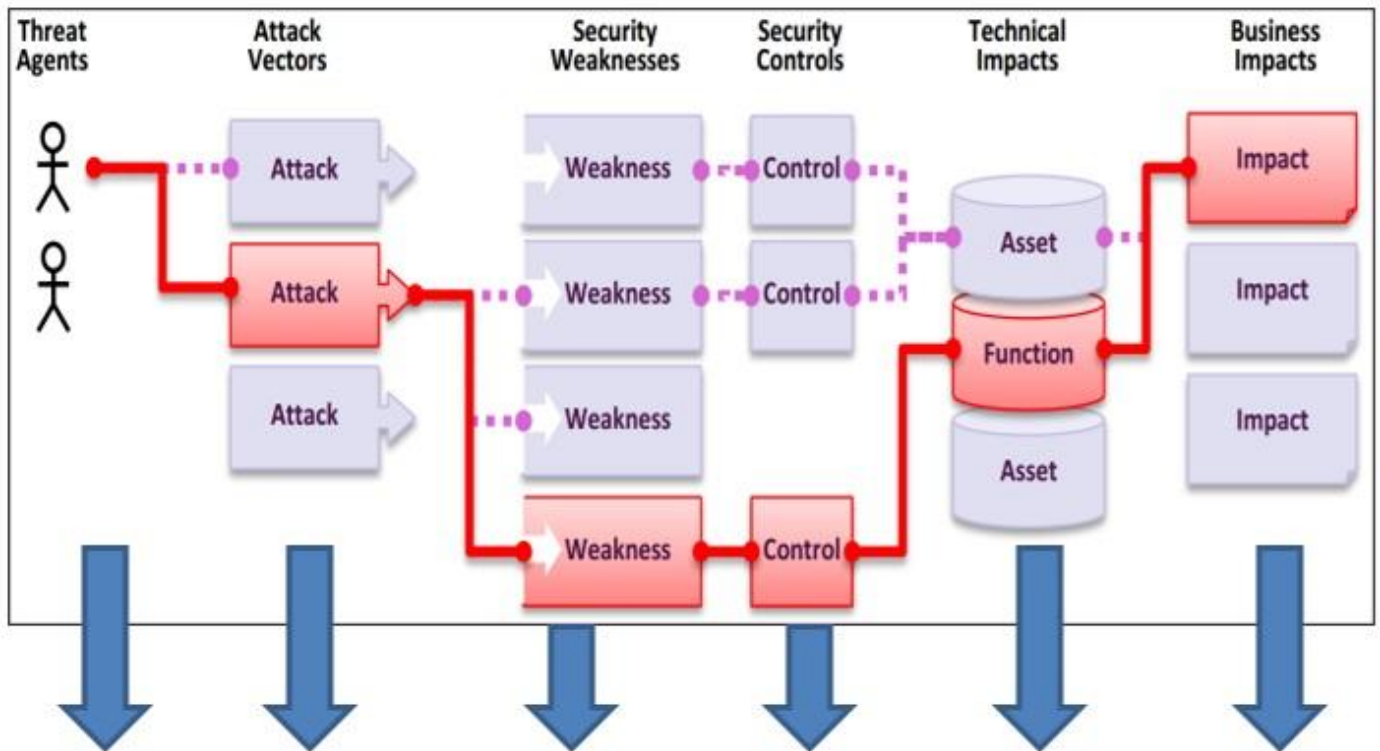
Application Security Risks

The figures immediately below illustrate the path of a sample threat beginning with the threat agent and ending with the target or affected business resource. Various paths are available but the agent would normally select the path of least resistance which would be the most vulnerable and with the fewest number of effective security controls.

The sample risk matrix can be used to assign in the various aspects of potential vulnerability. Each column corresponds to a phase in the attack process. In the matrix, a lower value represents less risk and is more desirable.

Concerning secure coding practices, when security is built-in during code development, vulnerabilities can be identified and controls included reducing the overall risk to information processed by the code.

Figure G.8-B Sample Threat Path



Threat Agents	Exploitability	Weakness Prevalence	Weakness Detectability	Technical Impacts	Business Impacts
App Specific	EASY: 3	WIDESPREAD: 3	EASY: 3	SEVERE: 3	App / Business Specific
	AVERAGE: 2	COMMON: 2	AVERAGE: 2	MODERATE: 2	
	DIFFICULT: 1	UNCOMMON: 1	DIFFICULT: 1	MINOR: 1	

Figure G.8-C General Risk Matrix

To understand these risks for a particular application or organization, you must consider your own specific threat agents and business impacts. Even severe software weaknesses may not present a serious risk if there are no threat agents in a position to perform the necessary attack or the business impact is negligible for the assets involved. The following table presents a summary of the 2017 Top 10 Application Security Risks, and the risk factors that have been assigned to each risk.

Figure G.8-D Top 10 Risk Factor Summary

RISK							Score
	Threat Agents	Exploitability	Prevalence	Detectability	Technical	Business	
A1:2017-Injection	App Specific	EASY: 3	COMMON: 2	EASY: 3	SEVERE: 3	App Specific	8.0
A2:2017-Authentication	App Specific	EASY: 3	COMMON: 2	AVERAGE: 2	SEVERE: 3	App Specific	7.0
A3:2017-Sens. Data Exposure	App Specific	AVERAGE: 2	WIDESPREAD: 3	AVERAGE: 2	SEVERE: 3	App Specific	7.0
A4:2017-XML External Entities (XXE)	App Specific	AVERAGE: 2	COMMON: 2	EASY: 3	SEVERE: 3	App Specific	7.0
A5:2017-Broken Access Control	App Specific	AVERAGE: 2	COMMON: 2	AVERAGE: 2	SEVERE: 3	App Specific	6.0
A6:2017-Security Misconfiguration	App Specific	EASY: 3	WIDESPREAD: 3	EASY: 3	MODERATE: 2	App Specific	6.0
A7:2017-Cross-Site Scripting (XSS)	App Specific	EASY: 3	WIDESPREAD: 3	EASY: 3	MODERATE: 2	App Specific	6.0
A8:2017-Insecure Deserialization	App Specific	DIFFICULT: 1	COMMON: 2	AVERAGE: 2	SEVERE: 3	App Specific	5.0
A9:2017-Vulnerable Components	App Specific	AVERAGE: 2	WIDESPREAD: 3	AVERAGE: 2	MODERATE: 2	App Specific	4.7
A10:2017-Insufficient Logging&Monitoring	App Specific	AVERAGE: 2	WIDESPREAD: 3	DIFFICULT: 1	MODERATE: 2	App Specific	4.0

Whether you are new to web application security or already very familiar with these risks, the task of producing a secure web application or fixing an existing one can be difficult. If you have to manage a large application portfolio, this task can be daunting.

To help organizations, developers, testers and managers reduce their application security risks in a cost-effective manner; OWASP has produced numerous free and open resources that you can use to address application security in your organization. The following are some of the many resources OWASP has produced to help organizations produce secure web applications and APIs.

Get Started:

- Document all applications and associated data assets.
- Larger organizations should consider implementing a Configuration Management Database (CMDB).
- Establish an application security program to conduct analysis to define key improvement areas and an execution plan.

Risk Based Portfolio Approach:

- Identify the protection needs of your application portfolio from a business perspective.
- Establish a common risk-rating model with a consistent set of likelihood and impact factors reflective of your organization's tolerance for risk.
- Measure and prioritize all applications and APIs and add results to CMDB.

Enable with a Strong Foundation:

- Establish a set of policies and standards that provide an application security baseline for all development teams to adhere too.
- Define a common set of reusable security controls that complement these policies and standards and provide design and development guidance on their use.

Integrate Security into Existing Processes:

- Define and integrate secure implementation and verification activities into existing development and operational processes.
 - Activities include threat modeling, secure design and design review, secure coding and code review, penetration testing, and remediation.

Application Security Requirements - to produce a secure web application, you must define what secure means for that application.

- [Application Security Verification Standard \(ASVS\):](https://www.owasp.org/index.php/ASVS)
<https://www.owasp.org/index.php/ASVS>
- [OWASP Secure Software Contract Annex:](https://www.owasp.org/index.php/OWASP_Secure_Software_Contract_Annex)
https://www.owasp.org/index.php/OWASP_Secure_Software_Contract_Annex

Application Security Architecture - retrofitting security into your applications and APIs, it is far more cost effective to design the security in from the start.

- OWASP Prevention Cheat Sheets:

https://www.owasp.org/index.php/OWASP_Cheat_Sheet_Series

Standard Security Controls - building strong and usable security controls is difficult. Using a set of standard security controls radically simplifies the development of secure applications and APIs.

- OWASP Proactive Controls:
https://www.owasp.org/index.php/OWASP_Proactive_Controls

Secure Development Lifecycle - to improve the process your organization follows when building applications and APIs, organizations formulate and implement a strategy for software security that is tailored to the specific risks facing their organization.

- OWASP Software Assurance Maturity Model (SAMM):
https://www.owasp.org/index.php/OWASP_SAMM_Project
- OWASP Application Security Guide for CISOs:
https://www.owasp.org/index.php/Application_Security_Guide_For_CISOs

Application Security Education – hands-on learning about vulnerabilities to help educate developers on web application security.

- OWASP Education Project:
https://www.owasp.org/index.php/Category:OWASP_Education_Project
- OWASP WebGoat:
<https://www.owasp.org/index.php/WebGoat>
- OWASP Broken Web Application Project:
https://www.owasp.org/index.php/OWASP_Broken_Web_Applications_Project

Understand the Threat Model – be sure to understand the priorities when it comes to threat model.

- OWASP Testing Guide:
https://www.owasp.org/index.php/OWASP_Testing_Project
- [Application Security Verification Standard \(ASVS\):](https://www.owasp.org/index.php/ASVS)
<https://www.owasp.org/index.php/ASVS>

Testing Strategies – choose the simplest, fastest, most accurate technique to verify each requirement.

- OWASP Security Knowledge Framework:
https://www.owasp.org/index.php/OWASP_Security_Knowledge_Framework

- [Application Security Verification Standard \(ASVS\):
https://www.owasp.org/index.php/ASVS](https://www.owasp.org/index.php/ASVS)

G.9 NIST Authentication Primer

NIST Authentication Primer

Attribution:

The following information was extracted from NIST Special Publications 800-63-4 Digital Identity Guidelines and 800-63B Digital Identity Guidelines: Authentication and Authenticator Management. It was created by the NIST National Cybersecurity Center of Excellence (NCCoE).

Definitions:

Authentication is verifying the identity of a user, process, or device, often as a prerequisite to allowing access to resources in an information system.

Multi-Factor Authentication (MFA) is a security control that requires a user to present a combination of two or more different authenticators to verify their identity.

Summary:

This authentication primer serves as a quick reference guide for information sourced from the source NIST documents. It is presented in a summarized form specific to the criminal and noncriminal justice communities.

Important Note

This **Authentication Primer** is being provided as a quick-reference guide for those attending the **2022 CJIS ISO Symposium**. It presents content in a summarized form, however for additional detail the **CJIS Security Policy** takes precedence when making decisions about procuring products or implementation these requirements. The update to the CJIS Security Policy relating to the content in this primer will be available later this year. Much of these requirements have been derived from NIST Special Publication 800-63B and 800-63 which can be downloaded from <https://pages.nist.gov/800-63-3/>.

Background

Authentication is the process of “verifying the digital identity of a user engaged in an online transaction” and involves a combination of one or more of the following factors: something you have (e.g., a hardware token), something you know (e.g., a password), or something you are (e.g., biometrics).

Multi-Factor Authentication (MFA) uses a memorized secret (e.g., a password) or biometric + possession- based verification factor (e.g., a hardware token).

NIST Special Publication 800-63 defines three **Authenticator Assurance Levels**:

- AAL1 – single-factor authentication, any permitted authentication process, may be user ID/PW only
- AAL2 – multi-factor authentication required using permitted authentication processes
- AAL3 – MFA + phishing resistant + replay resistant + secure cryptographic authentication protocol required

The risk profile for **CJIS** is **AAL2**. AAL2 allows for the following multi-factor authenticators to be used:

- Multi-Factor OTP Device
- Multi-Factor Cryptographic Software
- Multi-Factor Cryptographic Device

If implementing AAL2 using two single-factor authenticators, one authenticator must be a memorized secret and other must be a possession-based (i.e., “something you have”) authenticator from the following list:

- Look-Up Secret
- Out-of-Band Device
- Single-Factor OTP Device
- Single-Factor Cryptographic Software
- Single-Factor Cryptographic Device

The below tables provide information about each of the 9 authenticator types as defined by NIST in 800-63. Using this primer and the additional details in the CJIS security policy, readers will have enough information to evaluate how vendor technology meets



authentication requirements set by their own agencies and the CJIS Security Policy. As a starting point these questions could include:

- What are the authentication factors?
- What is the secret(s) used in the authentication system?
- How is the secret protected?
- How is the secret verified?



	1. Memorized Secret Authenticator - commonly referred to as a password or, if numeric, a PIN.
What are the factors?	The memorized secret is a single factor authenticator (something you know).
What is the secret?	A secret string of characters including letters, numbers, and/or symbols at least 8 characters in length (if chosen by the user) or at least 6 characters in length (if chosen by the Verifier).
How is the secret protected?	The secret is protected by the user by memorizing it, not writing it down, not disclosing it to others, etc. The secret is protected by the Verifier by not storing the secret itself but rather by salting and hashing the secret to derive a hash of the password. This hash is stored when the secret is reset (registered) by or issued to the user.
How is the secret verified?	When the user visits the Verifier to be authenticated, the Verifier challenges the user to enter the memorized secret. The Verifier uses the same algorithm that was used during registration or issuance to generate a hash value. If this generated hash value matches the stored hash value, the secret is successfully verified, and the user is authenticated. An important property of hash functions that makes this possible is their deterministic property (the same secret always results in the same unique hash value and given a hash value it is computationally infeasible to derive the secret value that resulted in the hash value).



	2. Look-Up Secret Authenticator - a physical or electronic record that stores a set (or list) of secrets shared between the user and the Verifier.
What are the factors?	The look-up secret is a single factor authenticator (something you have).
What is the secret?	An example of a look-up secret is a list, with each item on the list identified by a sequence number followed by a secret string of characters that might include letters, numbers, and / or symbols. A common application of a look-up secret authenticator is its use as a “backup” authenticator if the primary authenticator is lost or malfunctions.
How is the secret protected?	The user protects the look-up secret by safeguarding the physical or electronic record in a physical safe or a secure electronic device. The distribution of the look-up secret is protected by in-person delivery to the user, distribution over a secure online channel or by postal mail to the user’s address of record. The Verifier protects each look-up secret on a list by not storing the secret itself but storing a salted hash value like the method used in the description of the memorized secret.
How is the secret verified?	Each look-up secret on a list can only be used once. Verifiers use approved encryption and an authenticated protected channel when requesting look-up secrets from users. When the user attempts to authenticate, the Verifier prompts for the next secret on the list. The secret is verified by comparing its hash value the stored hash value like the method used in the description of the memorized secret.



	3. Out-of-Band Device Authenticator - a physical device that communicates securely with the Verifier over a communications channel that is different and distinct from the primary channel over which the user is attempting to authenticate to the Verifier. As an example, the user may be attempting to authenticate to the Verifier from a laptop and the authenticator may be an app on a smartphone. The session between the user and the Verifier on the laptop is the primary channel and the session between the user and the Verifier on the smartphone app is the secondary channel.
What are the factors?	The out-of-band device is a single factor authenticator (something you have).
What is the secret?	Depending on the out-of-band device, the secret could be different. It could be a QR Code, push notification, etc. In this example the user transfers a secret (typically a 6-digit code) received via the secondary channel to the primary channel. The secret's purpose is to securely bind the authentication operation on the primary and secondary channel. When the response is via the primary communication channel, the secret also establishes the user's control of the out-of-band device.
How is the secret protected?	The secret is protected because the authenticator app uniquely authenticates itself and establishes a protected channel to the Verifier using approved cryptography. When the secret is sent by the Verifier to the out-of-band device, the app does not display the authentication secret while it is locked by the owner (i.e., requires an entry of a PIN, passcode, or biometric to view). However, the app does notify the user of the receipt of an authentication secret on a locked device to alert the user that an authentication attempt has been initiated on the user's behalf.
How is the secret verified?	The secret is verified by the Verifier when the user transfers the secret to the Verifier via the primary channel.



	4. Single-Factor OTP Device – hardware and software-based devices that generate OTPs (One Time Passcodes). This category includes hardware devices and software-based OTP generators installed on devices such as smartphones. These devices have an embedded secret that is used as the seed for generation of OTPs and does not require activation through a second factor. The OTP is displayed on the device and manually input for transmission to the Verifier, thereby proving possession and control of the device. An OTP device may, for example, display 6 characters at a time.
What are the factors?	A single-factor OTP device is something you have.
What is the secret?	The secret is the OTP displayed on the device and manually input for transmission to the Verifier. For example, an OTP device may display 6 characters at a time. The secret is computed based on a nonce that may be time-based or from a counter on the authenticator and Verifier. The secret is cryptographically and independently generated by the authenticator and Verifier and compared by the Verifier.
How is the secret protected?	Single-factor OTP authenticators contain two persistent values. The first is a symmetric key that persists for the device's lifetime. The second is a nonce that is either changed each time the authenticator is used or is based on a real-time clock. The Verifier protects the secret OTP using approved encryption and an authenticated protected channel when collecting the OTP from the user, to provide resistance to eavesdropping and MitM attacks. Time-based OTPs have a defined lifetime that is determined by the expected clock drift — in either direction — of the authenticator over its lifetime, plus allowance for network delay and user entry of the OTP. To provide replay resistance, Verifiers accept a given time-based OTP only once during the validity period. The symmetric key used by the authenticator is also present in the Verifier and is strongly protected against compromise.
How is the secret verified?	Single-factor OTP Verifiers effectively duplicate the process of generating the OTP used by the authenticator. The Verifier compares the value of the generated OTP with the value transmitted by the user to verify the secret.



	5. Multi-Factor OTP Device – hardware and software-based devices that generate OTPs (One Time Passcodes) for use in authentication after activation through an additional authentication factor. This includes hardware devices and software-based OTP generators installed on devices such as smartphones.
What are the factors?	A multi-factor OTP device is something you have, activated by either something you know or something you are.
What is the secret?	The secret is the OTP displayed on the device when the user activates the second factor of authentication that may be achieved through an integral entry pad, an integral biometric (e.g., fingerprint) reader, or a direct interface (e.g., USB port). The user then manually submits the secret for transmission to the Verifier. For example, an OTP device may display 6 characters at a time. The secret is computed based on a nonce that may be time-based or from a counter on the authenticator and Verifier. The secrets are cryptographically and independently generated by the authenticator and Verifier and compared by the Verifier.
How is the secret protected?	Multi -factor OTP authenticators contain two persistent values. The first is a symmetric key that persists for the device’s lifetime. The second is a nonce that is either changed each time the authenticator is used or is based on a real-time clock. Multi-factor OTP authenticators protect the secret by requiring the user to either submit a memorized secret or use a biometric to obtain the secret from the authenticator. The Verifier protects the secret OTP using approved encryption and an authenticated protected channel when collecting the OTP from the user, to provide resistance to eavesdropping and MitM attacks. Time-based OTPs have a defined lifetime that is determined by the expected clock drift — in either direction — of the authenticator over its lifetime, plus allowance for network delay and user entry of the OTP. To provide replay resistance Verifiers accept a given time-based OTP only once during the validity period. The symmetric key used by authenticator and Verifier to generate the secret OTP is strongly protected against compromise.
How is the secret verified?	Single-factor OTP Verifiers effectively duplicate the process of generating the OTP used by the authenticator. The Verifier compares the value of the generated OTP with the value transmitted by the user to verify the secret.



	6. Single-Factor Cryptographic Software – is a cryptographic key stored on disk or other “soft” media where authentication is accomplished by proving possession and control of the key. The authenticator output is highly dependent on the specific cryptographic protocol, but it is generally some type of signed message.
What are the factors?	Single-factor Cryptographic software is something you have.
What is the secret?	Single-factor cryptographic software authenticators encapsulate one or more secret keys unique to the authenticator.
How is the secret protected?	In the case of a single factor crypto software native app on a smartphone, the secret is stored in some form of secure storage, such as keychain storage or a trusted execution environment on the smartphone. The secret is protected by limiting key access to only the native single factor crypto software app on the smartphone requiring access. Single-factor cryptographic software authenticators should discourage and shall not facilitate the cloning of the secret key onto multiple devices.
How is the secret verified?	Single-factor cryptographic software Verifiers generate a challenge nonce, send it to the corresponding authenticator, and use the authenticator output to verify possession of the device. The authenticator output is highly dependent on the specific cryptographic device and protocol, but it is generally some type of signed message. The Verifier has either symmetric or asymmetric cryptographic keys corresponding to each authenticator. While both types of keys are protected against modification, symmetric keys are additionally be protected against unauthorized disclosure.



	7. Single-Factor Cryptographic Device – is a hardware device that performs cryptographic operations using protected cryptographic key(s) and provides the authenticator output via direct connection to the user endpoint.
What are the factors?	Single-factor Cryptographic device is something you have.
What is the secret?	Single-factor cryptographic device authenticators encapsulate one or more secret keys (symmetric or asymmetric) unique to the authenticator.
How is the secret protected?	Single-factor cryptographic device authenticators encapsulate one or more secret keys unique to the device that cannot be removed from the device. The authenticator operates by signing a challenge nonce presented through a direct computer interface (e.g., a USB port). Alternatively, the authenticator could be a suitably secure processor integrated with the user endpoint itself (e.g., a hardware Trusted Platform module). Although cryptographic devices contain software, they differ from cryptographic software authenticators in that all embedded software is under control of the issuer and that the entire authenticator is subject to all applicable FIPS 140 Level 1 requirements.
How is the secret verified?	Single-factor cryptographic software Verifiers generate a challenge nonce, send it to the corresponding authenticator. The Verifier uses the authenticator response to verify possession of the device using the applicable cryptographic key depending on the selected protocol. The authenticator response is highly dependent on the specific cryptographic device and protocol, but it is generally some type of signed message.



	8. Multi-Factor Cryptographic Software Authenticator – is a cryptographic key stored on disk or other "soft" media that requires activation through a second authentication factor.
What are the factors?	The multi-factor cryptographic software authenticator is something you have, and it is activated by either something you know or something you are.
What is the secret?	Multi-factor cryptographic software authenticators encapsulate one or more secret cryptographic keys unique to the authenticator.
How is the secret protected?	In the case of a multi factor cryptographic software native app on a smartphone, the secret is stored in some form of secure storage, such as keychain storage or a trusted execution environment on the smartphone. The secret is protected by limiting key access to only the single factor cryptographic software native app on the smartphone requiring access and is accessible only when the user has submitted an additional factor (either a memorized secret or a biometric). Multi-factor cryptographic software authenticators should discourage and shall not facilitate the cloning of the secret key onto multiple devices. Each authentication operation using the authenticator requires distinct and separate submission of both factors.
How is the secret verified?	Multi-factor cryptographic software Verifiers generate a challenge nonce and send it to the corresponding authenticator. The Verifier confirms possession of the device using the applicable cryptographic key to validate the authenticator response depending on the selected protocol. Verification of the response from the multi-factor cryptographic software authenticator also proves use of the activation factor. The authenticator response is highly dependent on the specific cryptographic device and protocol, but it is generally some type of signed message.



An icon showing a blue smartphone with a fingerprint sensor and a blue security card with a chip, both representing multi-factor authentication factors.	9. Multi-Factor Cryptographic Device – is a hardware device that performs cryptographic operations using one or more protected cryptographic keys and requires activation through a second authentication factor.
What are the factors?	The multi-factor cryptographic device is something you have, and it is activated by either something you know or something you are.
What is the secret?	Multi-factor cryptographic device authenticators encapsulate one or more secret cryptographic keys unique to the authenticator.
How is the secret protected?	Multi-factor cryptographic device authenticators use tamper-resistant hardware to encapsulate one or more secret keys unique to the authenticator and accessible only through the input of an additional factor, either a memorized secret or a biometric. The authenticator operates by using the cryptographic key that was unlocked by the additional factor to sign a challenge nonce presented through a direct computer interface (e.g., a USB port). Alternatively, the authenticator could be a suitably secure processor integrated with the user endpoint itself (e.g., a hardware TPM). Although cryptographic devices contain software, they differ from cryptographic software authenticators in that all embedded software is under control of the issuer and that the entire authenticator is subject to all applicable FIPS 140 Level 1 requirements.
How is the secret verified?	Multi-factor cryptographic device Verifiers generate a challenge nonce and send it to the corresponding authenticator. The Verifier confirms possession of the device using the applicable cryptographic key to validate the authenticator response depending on the selected protocol. Verification of the response from the multi-factor cryptographic device authenticator also proves use of the activation factor. The authenticator response is highly dependent on the specific cryptographic device and protocol, but it is generally some type of signed message.

APPENDIX H: SECURITY ADDENDUM

The following pages contain:

The legal authority, purpose, and genesis of the Criminal Justice Information Services Security Addendum (H-2-H-4);

An example of a contract addendum (H-5);

The Security Addendum itself (H-6-H-7);

The Security Addendum Certification page (H-8).

**FEDERAL BUREAU OF INVESTIGATION
CRIMINAL JUSTICE INFORMATION SERVICES
SECURITY ADDENDUM**

**Legal Authority for and Purpose and Genesis of the
Security Addendum**

Traditionally, law enforcement and other criminal justice agencies have been responsible for the confidentiality of their information. Accordingly, until mid-1999, the Code of Federal Regulations Title 28, Part 20, subpart C, and the National Crime Information Center (NCIC) policy paper approved December 6, 1982, required that the management and exchange of criminal justice information be performed by a criminal justice agency or, in certain circumstances, by a noncriminal justice agency under the management control of a criminal justice agency.

In light of the increasing desire of governmental agencies to contract with private entities to perform administration of criminal justice functions, the FBI sought and obtained approval from the United States Department of Justice (DOJ) to permit such privatization of traditional law enforcement functions under certain controlled circumstances. In the Federal Register of May 10, 1999, the FBI published a Notice of Proposed Rulemaking, announcing as follows:

1. Access to CHRI [Criminal History Record Information] and Related Information, Subject to Appropriate Controls, by a Private Contractor Pursuant to a Specific Agreement with an Authorized Governmental Agency To Perform an Administration of Criminal Justice Function (Privatization). Section 534 of title 28 of the United States Code authorizes the Attorney General to exchange identification, criminal identification, crime, and other records for the official use of authorized officials of the federal government, the states, cities, and penal and other institutions. This statute also provides, however, that such exchanges are subject to cancellation if dissemination is made outside the receiving departments or related agencies. Agencies authorized access to CHRI traditionally have been hesitant to disclose that information, even in furtherance of authorized criminal justice functions, to anyone other than actual agency employees lest such disclosure be viewed as unauthorized. In recent years, however, governmental agencies seeking greater efficiency and economy have become increasingly interested in obtaining support services for the administration of criminal justice from the private sector. With the concurrence of the FBI's Criminal Justice Information Services (CJIS) Advisory Policy Board, the DOJ has concluded that disclosures to private persons and entities providing support services for criminal justice agencies may, when subject to appropriate controls, properly be viewed as permissible disclosures for purposes of compliance with 28 U.S.C. 534.

We are therefore proposing to revise 28 CFR 20.33(a)(7) to provide express authority for such arrangements. The proposed authority is similar to the authority that already exists in 28 CFR 20.21(b)(3) for state and local CHRI systems. Provision of CHRI under this authority would only be permitted pursuant to a specific agreement with an authorized governmental agency for the purpose of providing services for the administration of criminal justice. The agreement would be required to incorporate a security addendum approved

by the Director of the FBI (acting for the Attorney General). The security addendum would specifically authorize access to CHRI, limit the use of the information to the specific purposes for which it is being provided, ensure the security and confidentiality of the information consistent with applicable laws and regulations, provide for sanctions, and contain such other provisions as the Director of the FBI (acting for the Attorney General) may require. The security addendum, buttressed by ongoing audit programs of both the FBI and the sponsoring governmental agency, will provide an appropriate balance between the benefits of privatization, protection of individual privacy interests, and preservation of the security of the FBI's CHRI systems.

The FBI will develop a security addendum to be made available to interested governmental agencies. We anticipate that the security addendum will include physical and personnel security constraints historically required by NCIC security practices and other programmatic requirements, together with personal integrity and electronic security provisions comparable to those in NCIC User Agreements between the FBI and criminal justice agencies, and in existing Management Control Agreements between criminal justice agencies and noncriminal justice governmental entities. The security addendum will make clear that access to CHRI will be limited to those officers and employees of the private contractor or its subcontractor who require the information to properly perform services for the sponsoring governmental agency, and that the service provider may not access, modify, use, or disseminate such information for inconsistent or unauthorized purposes.

Consistent with such intent, Title 28 of the Code of Federal Regulations (C.F.R.) was amended to read:

§ 20.33 Dissemination of criminal history record information.

- a) Criminal history record information contained in the Interstate Identification Index (III) System and the Fingerprint Identification Records System (FIRS) may be made available:
 - 1) To criminal justice agencies for criminal justice purposes, which purposes include the screening of employees or applicants for employment hired by criminal justice agencies.
 - 2) To noncriminal justice governmental agencies performing criminal justice dispatching functions or data processing/information services for criminal justice agencies; and
 - 3) To private contractors pursuant to a specific agreement with an agency identified in paragraphs (a)(1) or (a)(6) of this section and for the purpose of providing services for the administration of criminal justice pursuant to that agreement. The agreement must incorporate a security addendum approved by the Attorney General of the United States, which shall specifically authorize access to criminal history record information, limit the use of the information to the purposes for which it is provided, ensure the security and confidentiality of the information consistent with these regulations, provide for sanctions, and contain such other provisions as the Attorney General may require. The power

and authority of the Attorney General hereunder shall be exercised by the FBI Director (or the Director's designee).

This Security Addendum, appended to and incorporated by reference in a government-private sector contract entered into for such purpose, is intended to insure that the benefits of privatization are not attained with any accompanying degradation in the security of the national system of criminal records accessed by the contracting private party. This Security Addendum addresses both concerns for personal integrity and electronic security which have been addressed in previously executed user agreements and management control agreements.

A government agency may privatize functions traditionally performed by criminal justice agencies (or noncriminal justice agencies acting under a management control agreement), subject to the terms of this Security Addendum. If privatized, access by a private contractor's personnel to NCIC data and other CJIS information is restricted to only that necessary to perform the privatized tasks consistent with the government agency's function and the focus of the contract. If privatized the contractor may not access, modify, use or disseminate such data in any manner not expressly authorized by the government agency in consultation with the FBI.

EXAMPLE OF A CONTRACT ADDENDUM

AMENDMENT NO. ____ TO THE CONTRACT BETWEEN
[PARTY NO. 1] AND [PARTY NO. 2], ENTERED INTO [DATE]

[Name of Law Enforcement Agency] and [Party No. 2], upon notification and pursuant to Paragraph/Section No. ____ [the amendment clause of the original contract] of that certain contract entered into by these parties on [date][and entitled “____”], hereby amend and revise the contract to include the following:

1. Access to and use of criminal history record information and other sensitive information maintained in [state and] FBI-managed criminal justice information systems by [private party] are subject to the following restrictions:

- a.
- b.
- c.

and

- d. The Security Addendum appended hereto, which is incorporated by reference and made a part thereof as if fully appearing herein.

This amendment is effective the ____ day of _____, 20__.

On behalf of [Party No. 1]:

_____ Printed Name	_____ Title
_____ Signature	_____ Date

On behalf of [Party No. 2]:

_____ Printed Name	_____ Title
_____ Signature	_____ Date

FEDERAL BUREAU OF INVESTIGATION
CRIMINAL JUSTICE INFORMATION SERVICES
SECURITY ADDENDUM

The goal of this document is to augment the CJIS Security Policy to ensure adequate security is provided for criminal justice systems while (1) under the control or management of a private entity or (2) connectivity to FBI CJIS Systems has been provided to a private entity (contractor). Adequate security is defined in Office of Management and Budget Circular A-130 as “security commensurate with the risk and magnitude of harm resulting from the loss, misuse, or unauthorized access to or modification of information.”

The intent of this Security Addendum is to require that the Contractor maintain a security program consistent with federal and state laws, regulations, and standards (including the CJIS Security Policy in effect when the contract is executed), as well as with policies and standards established by the Criminal Justice Information Services (CJIS) Advisory Policy Board (APB).

This Security Addendum identifies the duties and responsibilities with respect to the installation and maintenance of adequate internal controls within the contractual relationship so that the security and integrity of the FBI’s information resources are not compromised. The security program shall include consideration of personnel security, site security, system security, and data security, and technical security.

The provisions of this Security Addendum apply to all personnel, systems, networks and support facilities supporting and/or acting on behalf of the government agency.

1.00 Definitions

1.01 Contracting Government Agency (CGA) – the government agency, whether a Criminal Justice Agency or a Noncriminal Justice Agency, which enters into an agreement with a private contractor subject to this Security Addendum.

1.02 Contractor – a private business, organization or individual which has entered into an agreement for the administration of criminal justice with a Criminal Justice Agency or a Noncriminal Justice Agency.

2.00 Responsibilities of the Contracting Government Agency.

2.01 The CGA will ensure that each Contractor employee receives a copy of the Security Addendum and the CJIS Security Policy and executes an acknowledgment of such receipt and the contents of the Security Addendum. The signed acknowledgments shall remain in the possession of the CGA and available for audit purposes. The acknowledgement may be signed by hand or via digital signature (see glossary for definition of digital signature).

3.00 Responsibilities of the Contractor.

3.01 The Contractor will maintain a security program consistent with federal and state laws, regulations, and standards (including the CJIS Security Policy in effect when the contract is executed and all subsequent versions), as well as with policies and standards established by the Criminal Justice Information Services (CJIS) Advisory Policy Board (APB).

4.00 Security Violations.

4.01 The CGA must report security violations to the CJIS Systems Officer (CSO) and the Director, FBI, along with indications of actions taken by the CGA and Contractor.

4.02 Security violations can justify termination of the appended agreement.

4.03 Upon notification, the FBI reserves the right to:

- a. Investigate or decline to investigate any report of unauthorized use;
- b. Suspend or terminate access and services, including telecommunications links. The FBI will provide the CSO with timely written notice of the suspension. Access and services will be reinstated only after satisfactory assurances have been provided to the FBI by the CGA and Contractor. Upon termination, the Contractor's records containing CHRI must be deleted or returned to the CGA.

5.00 Audit

5.01 The FBI is authorized to perform a final audit of the Contractor's systems after termination of the Security Addendum.

6.00 Scope and Authority

6.01 This Security Addendum does not confer, grant, or authorize any rights, privileges, or obligations on any persons other than the Contractor, CGA, CJA (where applicable), CSA, and FBI.

6.02 The following documents are incorporated by reference and made part of this agreement: (1) the Security Addendum; (2) the NCIC 2000 Operating Manual; (3) the CJIS Security Policy; and (4) Title 28, Code of Federal Regulations, Part 20. The parties are also subject to applicable federal and state laws and regulations.

6.03 The terms set forth in this document do not constitute the sole understanding by and between the parties hereto; rather they augment the provisions of the CJIS Security Policy to provide a minimum basis for the security of the system and contained information and it is understood that there may be terms and conditions of the appended Agreement which impose more stringent requirements upon the Contractor.

6.04 This Security Addendum may only be modified by the FBI, and may not be modified by the parties to the appended Agreement without the consent of the FBI.

6.05 All notices and correspondence shall be forwarded by First Class mail to:

Information Security Officer

Criminal Justice Information Services Division, FBI

1000 Custer Hollow Road

Clarksburg, West Virginia 26306

**FEDERAL BUREAU OF INVESTIGATION
CRIMINAL JUSTICE INFORMATION SERVICES
SECURITY ADDENDUM**

CERTIFICATION

I hereby certify that I am familiar with the contents of (1) the Security Addendum, including its legal authority and purpose; (2) the NCIC Operating Manual; (3) the CJIS Security Policy; and (4) Title 28, Code of Federal Regulations, Part 20, and agree to be bound by their provisions.

I recognize that criminal history record information and related data, by its very nature, is sensitive and has potential for great harm if misused. I acknowledge that access to criminal history record information and related data is therefore limited to the purpose(s) for which a government agency has entered into the contract incorporating this Security Addendum. I understand that misuse of the system by, among other things: accessing it without authorization; accessing it by exceeding authorization; accessing it for an improper purpose; using, disseminating or re-disseminating information received as a result of this contract for a purpose other than that envisioned by the contract, may subject me to administrative and criminal penalties. I understand that accessing the system for an appropriate purpose and then using, disseminating or re-disseminating the information received for another purpose other than execution of the contract also constitutes misuse. I further understand that the occurrence of misuse does not depend upon whether or not I receive additional compensation for such authorized activity. Such exposure for misuse includes, but is not limited to, suspension or loss of employment and prosecution for state and federal crimes.

Printed Name/Signature of Contractor Employee

Date

Printed Name/Signature of Contractor Representative

Date

Organization and Title of Contractor Representative

APPENDIX I: REFERENCES

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